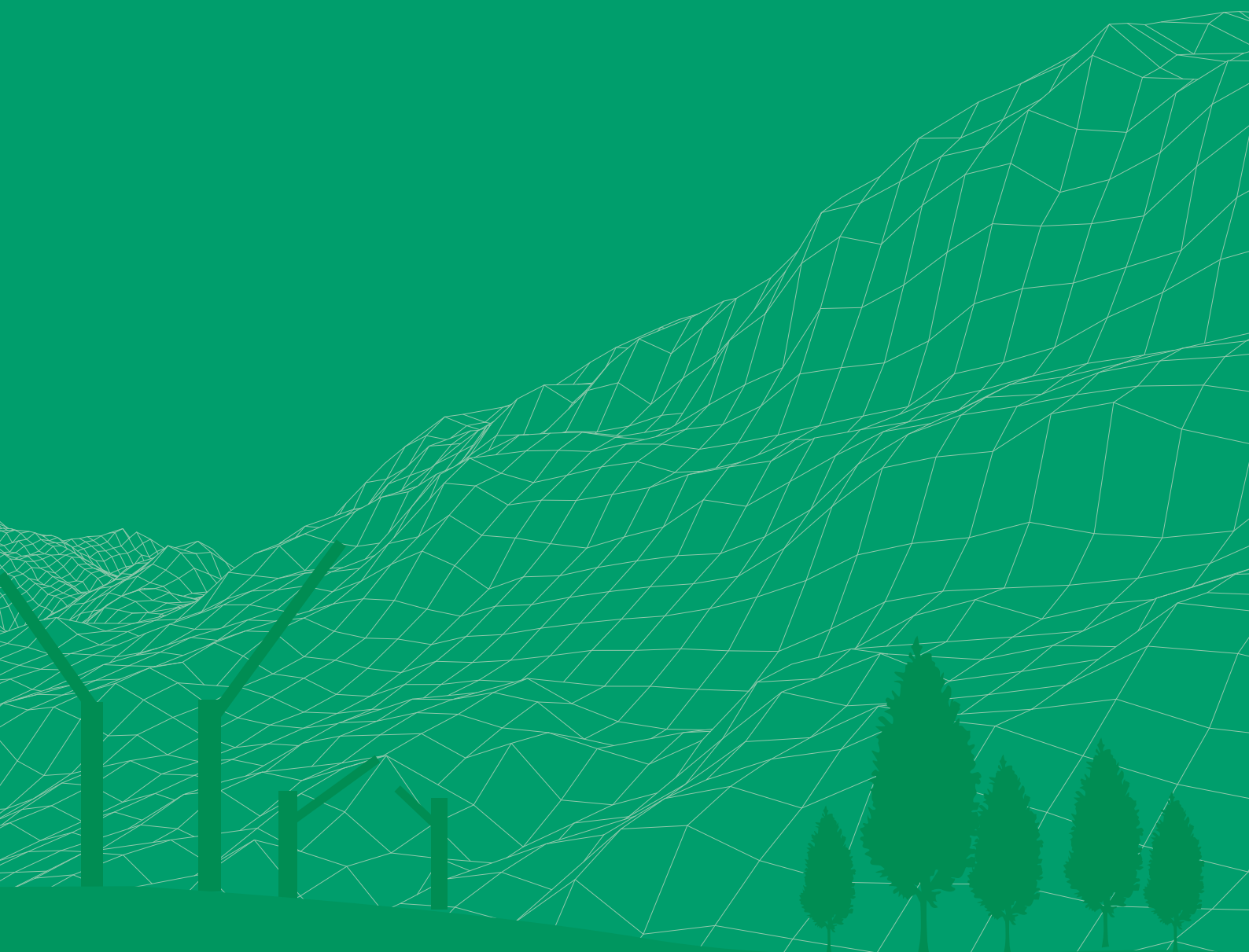


Julien Rosset

Inference and model selection for fuzzy regression methods



DOI : <https://doi.org/10.51363/unifr.sth.2026.069>

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DEPARTMENT OF INFORMATICS
UNIVERSITY OF FRIBOURG (SWITZERLAND)

INFERENCE AND MODEL SELECTION FOR FUZZY REGRESSION METHODS THESIS

PRESENTED TO THE FACULTY OF SCIENCE AND MEDICINE OF THE UNIVERSITY OF FRIBOURG (SWITZERLAND) IN CONSIDERATION FOR THE AWARD OF THE ACADEMIC GRADE OF DOCTOR OF PHILOSOPHY IN COMPUTER SCIENCE

BY

Julien Rosset

FROM

PRÉZ-VERS-NORÉAZ (FR)

THESIS No: 8026

UNIPRINT

2026

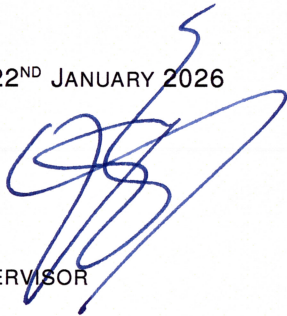
ACCEPTED BY THE FACULTY OF SCIENCE AND MEDICINE OF THE UNIVERSITY OF FRIBOURG
(SWITZERLAND) UPON THE RECOMMENDATION OF

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Statistics is the art and science of learning from data.

A. Agresti and C. Franklin, 2013



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Acknowledgements

First and foremost, I would like to express my deepest gratitude to my supervisor, Laurent Donzé, for his invaluable guidance, encouragement, and unwavering support throughout the course of this thesis. His expertise, insightful feedback, and high standards were crucial in shaping both the content and the quality of this work. It is also through his guidance that I have begun to spot the little devils hiding in the details, yet I must admit that I am still learning to perfect this craft.

I extend my gratitude to Professors Christian Mazza and João Miguel da Costa Sousa for their insightful exchanges and constructive feedback, which greatly helped me refine this thesis. I am also thankful to Professor Philippe Cudré-Mauroux, who presided over the defense with both rigor and warmth, fostering an engaging and thoughtful discussion.

I am also sincerely thankful to the members of the research group and colleagues at the University of Fribourg, whose constructive discussions and camaraderie created a stimulating environment that greatly enriched my research experience.

Special thanks go to Rédina Berkachy, Alizée Lombard, Laurent Bugnon and Emmanuel di Bernardo for their collaboration, suggestions, and the fruitful exchanges that helped me refine important aspects of this work.

I am grateful to the administrative and technical staff of the Department of Informatics also known as Diuf, whose assistance and kindness helped make the daily academic life smooth and pleasant.

On a personal note, I would like to thank my friends and family for their unwavering support, patience, and encouragement, especially during the most demanding moments of this journey. Their belief in me gave me the strength to persevere.

Finally, to everyone who contributed to this work in ways large or small, thank you.

Abstract

Classical statistical modeling relies on probabilistic frameworks to represent uncertainty, assuming data are precise and complete. However, many real-world situations involve imprecision, vagueness, or incomplete information, types of uncertainty that traditional models struggle to accommodate. Fuzzy set theory, introduced by Zadeh in 1965, provides a powerful alternative for representing such uncertainty, offering tools to model and reason with imprecise data.

This thesis investigates the use of fuzzy logic in statistical modeling, with an emphasis on fuzzy linear regression, statistical inference, and model selection. It begins by formalising key concepts in fuzzy logic, including fuzzy sets and fuzzy numbers, and by equipping them with arithmetic and logical operations via Zadeh's extension principle. These foundational tools enable rigorous treatment of fuzzy-valued data and support the development of more advanced methodologies.

Building on classical approaches to fuzzy regression, the work introduces new models that accommodate fuzziness in both explanatory and response variables. A central contribution is the use of fuzzy orthogonality to demonstrate that regression parameters can be estimated independently, i.e., one parameter at a time, without loss of generality.

To enable inference in fuzzy settings, the thesis defines fuzzy random variables and constructs empirical fuzzy distributions. From these, it develops fuzzy analogues of confidence intervals, hypothesis tests, and p -values, providing interpretable measures of statistical significance under epistemic uncertainty.

The problem of comparing fuzzy quantities, crucial in both inference and decision-making, is addressed through the introduction of a new ranking method, the **Polygon ranking method (PRM)**. This approach captures directional asymmetries and generalizes well beyond common fuzzy number shapes, offering a geometrically intuitive and consistent comparison framework.

Finally, model selection is reimagined under fuzziness by extending classical criteria such as the **Akaike information criterion (AIC)** and its empirical likelihood variant. These fuzzy versions rely on resampling and ranking strategies informed by the **PRM**, allowing meaningful model comparison when crisp criteria are unreliable or inapplicable.

Overall, this thesis presents a unified and practical approach to fuzzy statistical modeling, inference, and model comparison. Through theoretical innovation and algorithmic development, it provides a robust framework for analysing uncertain data in a wide range of applications.

Résumé

La modélisation statistique classique s'appuie sur des cadres probabilistes pour représenter l'incertitude, en supposant que les données sont précises et complètes. Or, de nombreuses situations réelles se caractérisent par de l'imprécision, des informations incomplètes ou des formes d'incertitude que les modèles traditionnels peinent à intégrer. Introduite par Zadeh en 1965, la théorie des ensembles flous constitue une alternative particulièrement adaptée à ce contexte, en proposant des outils permettant de représenter et de traiter rigoureusement des données imprécises.

Cette thèse explore l'apport de la logique floue à la modélisation statistique, en mettant l'accent sur la régression linéaire floue, l'inférence statistique et la sélection de modèles. Elle s'ouvre par une formalisation des notions fondamentales de la logique floue, notamment les ensembles flous et les nombres flous, ainsi que par la définition des opérations arithmétiques et logiques associées à ces objets via le principe d'extension de Zadeh. Ces bases théoriques permettent un traitement cohérent des données à valeurs floues et constituent le socle du développement de méthodologies plus avancées.

Dans le prolongement des approches classiques de la régression floue, le travail propose de nouveaux modèles capables de prendre en compte la présence d'incertitude floue tant dans les variables explicatives que dans la variable de réponse. Une contribution majeure repose sur l'introduction du concept d'orthogonalité dans le cadre des nombres flous, permettant de montrer que les paramètres de régression peuvent être estimés de manière indépendante, c'est-à-dire un paramètre à la fois, sans perte de généralité.

Afin de rendre possible l'inférence statistique dans un cadre flou, la thèse définit la notion de variable aléatoire floue et construit des distributions empiriques floues. À partir de ces objets, elle développe des analogues flous des intervalles de confiance, des tests d'hypothèses et des valeurs de p , fournissant ainsi des mesures interprétables de la significativité statistique en présence d'incertitude épistémique.

La comparaison de quantités floues, enjeu central tant pour l'inférence que pour l'aide à la prise de décision, est abordée par l'introduction d'une nouvelle méthode de classement, la **Polygon ranking method (PRM)**. Cette approche prend en compte les asymétries directionnelles et se généralise à une large variété de formes de nombres flous, offrant un cadre de comparaison cohérent, stable et intuitif d'un point de vue géométrique.

Enfin, la sélection de modèles est repensée dans un contexte flou par l'extension de critères classiques tels que l' **AIC** et sa variante fondée sur la vraisemblance empirique. Ces versions floues reposent sur des stratégies de rééchantillonnage et de classement s'appuyant sur la **PRM**, permettant une comparaison pertinente des modèles lorsque les critères fondés sur des valeurs précises deviennent peu fiables ou inapplicables.

Dans l'ensemble, ce travail fournit un cadre à la fois théorique et pratique pour la modélisation statistique floue, l'inférence et la comparaison de modèles dans des contextes d'incertitude. Par des contributions théoriques originales et le développement d'outils algorithmiques, elle fournit un cadre robuste pour l'analyse de données incertaines dans un large éventail d'applications.

Introduction

In classical statistical modeling, uncertainty is typically represented using probabilistic tools, most notably random variables, probability distributions, and statistical inference. While this probabilistic framework is well suited to many practical applications, it can be proved inadequate or overly restrictive in contexts where uncertainty arises not only from randomness but also from vagueness, imprecision, or incomplete information. Traditional statistical methods, which typically rely on precise and complete numerical inputs, may therefore struggle to capture the kind of uncertainty that characterizes human reasoning or arises from imperfect observations. In such cases, the theory of fuzzy sets and fuzzy numbers, first introduced by Zadeh (1965), provides a powerful and flexible alternative for representing and handling imprecise data.

Such forms of imprecision are pervasive in many real-world applications. In the social sciences and psychology, survey responses frequently involve qualitative terms such as “often” or “moderately,” which cannot be precisely quantified using classical numerical scales; Neff (2013) illustrate how fuzzy approaches can be used to model these subjective assessments more realistically. Similar issues arise in engineering and chemical systems, where quantities such as concentrations, temperatures, pH levels, or reaction rates are affected by measurement uncertainty, environmental variability, or limited sensor resolution. In these settings, fuzzy modelling has been successfully applied to process control, quality assessment, and system description, as shown by Fileti et al. (2007) and C.-H. Chen, Jeng, and Lin (2022), including applications to pollutant concentration dynamics in fluids proposed by Alcino et al. (2022). In environmental sciences, fuzzy logic has also been employed for air quality assessment and pollution modelling, where pollutant levels are spatially and temporally variable and rarely known with precision, as displayed in the work of Atacak, Arıcı, and Guner (2017). Across these domains, fuzzy logic offers a structured way to incorporate imprecision directly into statistical and modelling frameworks. From a practical standpoint, it is worth noting that fuzzy data are most often constructed rather than directly observed; accordingly, and to the best of our knowledge, publicly available repositories of real-world fuzzy datasets remain limited.

Motivated by these considerations, this thesis explores the application of fuzzy logic within statistical modelling, with a particular focus on fuzzy linear regression, statistical inference, and model selection. It begins by defining essential concepts in fuzzy logic, such as fuzzy sets and fuzzy numbers, and introduces the associated arithmetic and logical operations using

Zadeh's extension principle. These foundational principles provide a structured approach to handling fuzzy-valued data and lay the groundwork for the development of more sophisticated modelling techniques, including linear fuzzy regression, fuzzy statistical inference, and model selection criteria suitable for such models.

Building on these foundations, the thesis pursues three main objectives: first, to formalise essential concepts in fuzzy logic and their associated operations; second, to estimate fuzzy linear regression models that integrate fuzzy information in the inputs, the outputs, or both; and finally, to develop fuzzy statistical inference and model selection criteria adapted to fuzzy-valued data.

We begin in Chapter 1 with a review of fundamental concepts from fuzzy logic, including fuzzy sets, fuzzy numbers, and fuzzy arithmetic, all defined through the extension principle. These foundations are critical for understanding the structure and behaviour of fuzzy-valued data. Special attention is paid to distinguishing fuzzy vectors from vectors of fuzzy numbers, a nuance with important implications for modelling.

Chapter 2 is devoted to fuzzy linear regression. We begin by revisiting two classical approaches: the fuzzy least squares method introduced by Diamond (1988), which minimizes the squared distance between fuzzy numbers using a dedicated metric, and the possibilistic regression framework of Tanaka, Hayashi, and Watada (1989), which minimizes the spread of fuzzy outputs under constraint-based formulations. Building on these foundations, we propose an extended fuzzy least squares model that accommodates both fuzzy independent and dependent variables (Rosset and Donzé (2023)).

In a complementary direction in Chapter 3, we investigate the notion of fuzzy orthogonality. The motivation for introducing orthogonality is to demonstrate that, under an orthogonal design matrix of fuzzy covariates, regression coefficients can be estimated independently, one at a time, by regressing the dependent variable on a single covariate, yielding the same results as estimating all regression coefficients simultaneously by regressing the dependent variable on all covariates. This insight leads to the development of two distinct orthogonal fuzzy regression frameworks: one based on Itoh (2017)'s definition of orthogonal fuzzy numbers, and another, more general approach derived directly from the extension principle (Rosset and Donzé (2023, 2024c)). These methods offer new perspectives on parameter estimation and provide a tool for handling orthogonal fuzzy data in regression analysis.

In Chapter 4, we turn to the challenge of statistical inference in fuzzy environments. We introduce fuzzy random variables and empirical fuzzy distributions, from which we derive fuzzy confidence intervals and fuzzy hypothesis tests. We further extend the classical p -value concept to the fuzzy setting, enabling a more nuanced interpretation of statistical significance.

Ranking fuzzy numbers, a crucial task for both decision-making and model evaluation, is addressed in Chapter 5. After surveying the limitations of existing methods, many of which fail to account for asymmetries or are restricted to specific fuzzy number types. We then introduce a ranking method to address these issues. This new bilateral ranking procedure compares directional non-overlapping areas between fuzzy numbers and generalises well beyond triangular shapes, offering an intuitive and geometrically meaningful ranking mechanism.

Finally, Chapter 6 is devoted to model selection under fuzziness. We propose fuzzy extensions of the **AIC** and its empirical likelihood-based variant, integrating uncertainty directly into model comparison. These fuzzy criteria are derived using resampling techniques based on Zadeh's extension principle, and evaluated through the previously introduced ranking method. This framework provides a basis for fuzzy model selection.

Together, these chapters build unified and practical tools for modelling, inferring, comparing, and selecting statistical models in the presence of fuzziness. Through theoretical developments and algorithmic innovations, this thesis lays the groundwork for deeper analyses in fields where imprecision is not an anomaly but a central feature of the data.

Chapter 1

Notation and theoretical reminders

This chapter is devoted to the fundamental notions of fuzzy logic. We start with the definition of fuzzy sets and fuzzy numbers which are a particular case of the former. Important properties of fuzzy sets are reminded and with the help of the so called t-norm and t-conorm, an extension of the usual logical operations performed on such sets is described. Given that fuzzy numbers are a generalisation of real numbers, they also have their own arithmetic. Through the use of Zadeh's extension principle, such arithmetic is defined. The notion of fuzzy numbers leads to the definition of fuzzy vectors which should not be confused with a vector of fuzzy numbers.

The book of Kruse and Meyer (1987) provides an excellent overview of all the fundamental notions concerning fuzzy logic. Thorough explanations regarding t-norms and conorms can be found in Klement, Mesiar, and Pap (2000). Gil, López-Díaz, and Dan A. Ralescu (2006), Couso, Dubois, and Sánchez (2014) and Berkachy (2021) clarify the two different perceptions of fuzzy randomness, the so called epistemic and ontic views.

1.1 Fuzzy sets and fuzzy numbers

We begin with the notion of fuzzy sets which is the fundamental concept upon which fuzzy logic is based. It is important to view a fuzzy set $\tilde{x}$ as an object giving a membership degree to each value x contained in a set I , thus providing additional information on the elements constituting it.

Definition 1.1.1 (Fuzzy set).

Let $I \neq \emptyset$ be a set with elements denoted by x . A fuzzy set $\tilde{x}$ is a set of ordered pairs

$$\tilde{x} = \{(x, \mu_{\tilde{x}}(x)) | x \in I\}, \quad (1.1)$$

where $\mu_{\tilde{x}}$ is a real valued function such that

$$\begin{aligned}\mu_{\tilde{x}} : \mathbb{R} &\rightarrow [0, 1] \\ x &\mapsto \mu_{\tilde{x}}(x).\end{aligned}$$

The function $\mu_{\tilde{x}}$ is commonly called membership function because it represents the grade of membership of the value x . Sometimes, this function is also called generalized characteristic function.

When dealing with fuzzy sets, it is very common to find some of their elements associated with a grade of membership of 1. Formally, we call such fuzzy sets normal.

Definition 1.1.2 (Normal fuzzy set).

A fuzzy set $\tilde{x}$ is called normal if there exists an element $x \in \tilde{x}$ such that

$$\mu_{\tilde{x}}(x) = 1. \quad (1.2)$$

Moreover, the notion of convexity is also a very desirable property that constrains the possible shapes of a fuzzy set.

Definition 1.1.3 (Convex fuzzy set).

A fuzzy set $\tilde{x}$ is said to be convex if and only if, for $x_1, x_2 \in \tilde{x}$, $t \in [0, 1]$,

$$\mu_{\tilde{x}}(tx_1 + (1 - t)x_2) \geq \min(\mu_{\tilde{x}}(x_1), \mu_{\tilde{x}}(x_2)). \quad (1.3)$$

Definition 1.1.4 (Strongly convex fuzzy set).

A fuzzy set $\tilde{x}$ is said to be strongly convex if and only if, for $x_1, x_2 \in \tilde{x}$, $t \in [0, 1]$,

$$\mu_{\tilde{x}}(tx_1 + (1 - t)x_2) > \min(\mu_{\tilde{x}}(x_1), \mu_{\tilde{x}}(x_2)). \quad (1.4)$$

Example 1.1.1. A normal and convex fuzzy set

Let us consider a fuzzy set $\tilde{x}$ with membership function $\mu_{\tilde{x}} : \mathbb{R} \rightarrow [0, 1]$ defined by

$$\mu_{\tilde{x}}(z) = \begin{cases} \frac{1}{16}z^2 & \text{for } z \in [0, 4) \\ 1 & \text{for } z \in [4, 5] \\ 1 - (z - 5) & \text{for } z \in (5, 6] \\ 0 & \text{elsewhere.} \end{cases} \quad (1.5)$$

Since $\mu_{\tilde{x}}(z) = 1$ for $z \in [4, 5]$, $\tilde{x}$ is normal. By construction, $\mu_{\tilde{x}}(z)$ is also convex. Indeed, since $\frac{1}{16}z^2$ is a monotonic increasing function for $z \in [0, 4)$ and $1 - (z - 5)$ is monotonic decreasing on $(5, 6]$, we have that $\mu_{\tilde{x}}(tz_1 + (1 - t)z_2) \in [\min\{\mu_{\tilde{x}}(z_1), \mu_{\tilde{x}}(z_2)\}, 1]$, thus $\mu_{\tilde{x}}(tz_1 + (1 - t)z_2) \geq \min\{\mu_{\tilde{x}}(z_1), \mu_{\tilde{x}}(z_2)\}$ for $t \in [0, 1]$ which shows the convexity of $\tilde{x}$.

Definition 1.1.5 (Compact fuzzy set).

A fuzzy set $\tilde{x}$ is said to be *compact* if its membership function $\mu_{\tilde{x}}$ is upper semi-continuous and its support

$$\text{Supp}(\tilde{x}) = \overline{\{x \in \mathbb{R} \mid \mu_{\tilde{x}}(x) > 0\}},$$

is a compact subset of $\mathbb{R}$, where the overline denotes the closure in $\mathbb{R}$. Remember that the closure of a set is the set together with all its limit points.

Let us observe that, in practical applications, fuzzy sets are in general compact, normal and convex. Indeed, supports of membership functions are usually compact subsets $\mathbb{R}$. Additionally, the grade 1 of membership corresponds to the most precise information available, the assumption of normality is met. Lastly, when the grade of membership becomes lower, the information is getting less precise, thus more values are eligible to have such grade of membership, which implies the convexity. Let us accurately describe them below.

Definition 1.1.6 (Class of non-empty, normal and convex fuzzy sets).

The class $\mathcal{F}(\mathbb{R})$ represents non-empty, normal and convex fuzzy sets with an upper semi-continuous membership function. If all elements of $\mathcal{F}(\mathbb{R})$ are compact (see Definition 1.1.5), let us write it as $\mathcal{F}_c(\mathbb{R})$.

By using the class $\mathcal{F}_c(\mathbb{R})$, we can easily define fuzzy numbers.

Definition 1.1.7 (Fuzzy number).

A fuzzy set $\tilde{x} \in \mathcal{F}_c(\mathbb{R})$ is a fuzzy number. That is, a normal and convex fuzzy set with compact α -cuts, $\forall \alpha \in [0, 1]$, is a fuzzy number if its membership function $\mu_{\tilde{x}}$ is upper semi-continuous. Sometimes, fuzzy sets with a non compact support are still called fuzzy numbers (e.g. gaussian fuzzy numbers).

Note that, throughout this thesis, we will use fuzzy numbers unless specified otherwise.

Definition 1.1.8 (Support and core of a fuzzy number).

The support $\text{Supp}(\tilde{x})$ and the core $\text{Core}(\tilde{x})$ of a fuzzy number $\tilde{x}$ are respectively defined by

$$\text{Supp}(\tilde{x}) = \overline{\{x \in \mathbb{R} \mid \mu_{\tilde{x}} > 0\}}, \quad (1.6)$$

$$\text{Core}(\tilde{x}) = \{x \in \mathbb{R} \mid \mu_{\tilde{x}} = 1\}, \quad (1.7)$$

where the overline denotes the closure of a set.

Example 1.1.2.

The support of the fuzzy number from example 1.1.1 is $\text{Supp}(\tilde{x}) = [0, 6]$ and its core is given by $\text{Core}(\tilde{x}) = [4, 5]$.

Fuzzy numbers can also be called "negative" or "positive" when all values of their support are negative or positive.

Definition 1.1.9 (Negative and positive fuzzy numbers).

A fuzzy number $\tilde{x}$ is said to be negative when all values of its support are negative,

$$\max\{z \in \text{Supp}(\tilde{x})\} < 0. \quad (1.8)$$

In the same way, a fuzzy number $\tilde{x}$ is said to be positive when all values of its support are positive,

$$\min\{z \in \text{Supp}(\tilde{x})\} > 0. \quad (1.9)$$

Definition 1.1.10 (α -cut).

The so called α -cut of a fuzzy number $\tilde{x} \in \mathcal{F}(\mathbb{R})$ noted by $\tilde{x}^\alpha$ is the set of elements with membership value greater than a threshold $\alpha \in (0, 1]$:

$$\tilde{x}^\alpha = \{x \in \mathbb{R} \mid \mu_{\tilde{x}}(x) \geq \alpha\}. \quad (1.10)$$

The left and right α -cuts of $\tilde{x}$ are respectively denoted by $\tilde{x}^{L,\alpha}$, $\tilde{x}^{R,\alpha}$ and defined as

$$x^{L,\alpha} = \inf\{x \in \mathbb{R} \mid \mu_{\tilde{x}}(x) \geq \alpha\} \quad (1.11)$$

$$x^{R,\alpha} = \sup\{x \in \mathbb{R} \mid \mu_{\tilde{x}}(x) \geq \alpha\}. \quad (1.12)$$

In order to have closed α -cuts, for $\alpha = 0$, we sometimes define the 0-cut $\tilde{x}^0$ by the support of $\tilde{x}$.

In most practical applications, three types of fuzzy numbers are commonly used to model vague data: triangular fuzzy numbers, trapezoidal fuzzy numbers and gaussian fuzzy numbers.

Definition 1.1.11 (Triangular fuzzy number).

A **Triangular fuzzy number (TFN)** $\tilde{x}$ is defined by a triplet (x^L, x, x^R) , $x^L < x < x^R \in \mathbb{R}$ and is called "triangular" because it possesses a triangular membership function defined by

$$\mu_{\tilde{x}}(z) = \begin{cases} \frac{z-x^L}{x-x^L} & \text{for } z \in [x^L, x], \\ \frac{x^R-z}{x^R-x} & \text{for } z \in (x, x^R], \\ 0 & \text{elsewhere.} \end{cases} \quad (1.13)$$

For $\alpha \in [0, 1]$, its α -cuts are given by

$$x^{L,\alpha} = x^L + \alpha(x - x^L), \quad (1.14)$$

$$x^{R,\alpha} = x^R - \alpha(x^R - x). \quad (1.15)$$

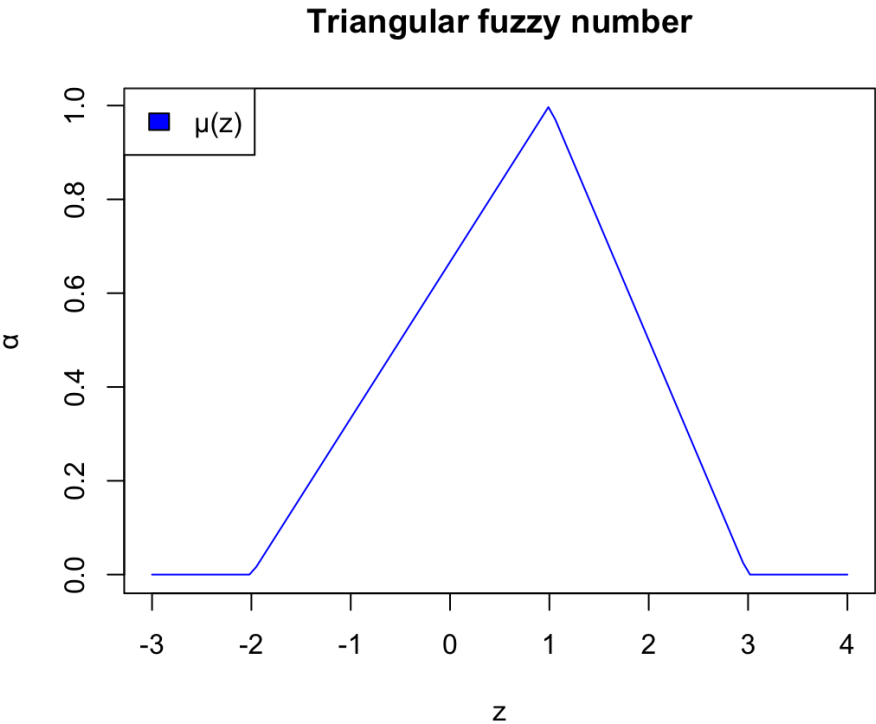


Figure 1.1: Example of a triangular fuzzy number. The figure depicts the triangular fuzzy number $\tilde{x} = (-2, 1, 3)$.

Definition 1.1.12 (Trapezoidal fuzzy number).

A **Trapezoidal fuzzy number (TRFN)** $\tilde{x}$ is defined by a quadruplet (x^L, x^l, x^r, x^R) , $x^L < x^l \leq x^r < x^R \in \mathbb{R}$ and, as its name suggests, its membership function

$$\mu_{\tilde{x}}(z) = \begin{cases} \frac{z-x^L}{x^l-x^L} & \text{for } z \in [x^L, x^l) \\ 1 & \text{for } z \in [x^l, x^r] \\ \frac{x^R-z}{x^R-x^r} & \text{for } z \in (x^r, x^R] \\ 0 & \text{elsewhere,} \end{cases} \quad (1.16)$$

has a trapezoidal shape. In the special case of $x^l = x^r$, one retrieves the definition of a **TFN**. Lastly, for $\alpha \in [0, 1]$, its α -cuts are given by

$$x^{L,\alpha} = x^L + \alpha(x^l - x^L), \quad (1.17)$$

$$x^{R,\alpha} = x^R - \alpha(x^R - x^r). \quad (1.18)$$

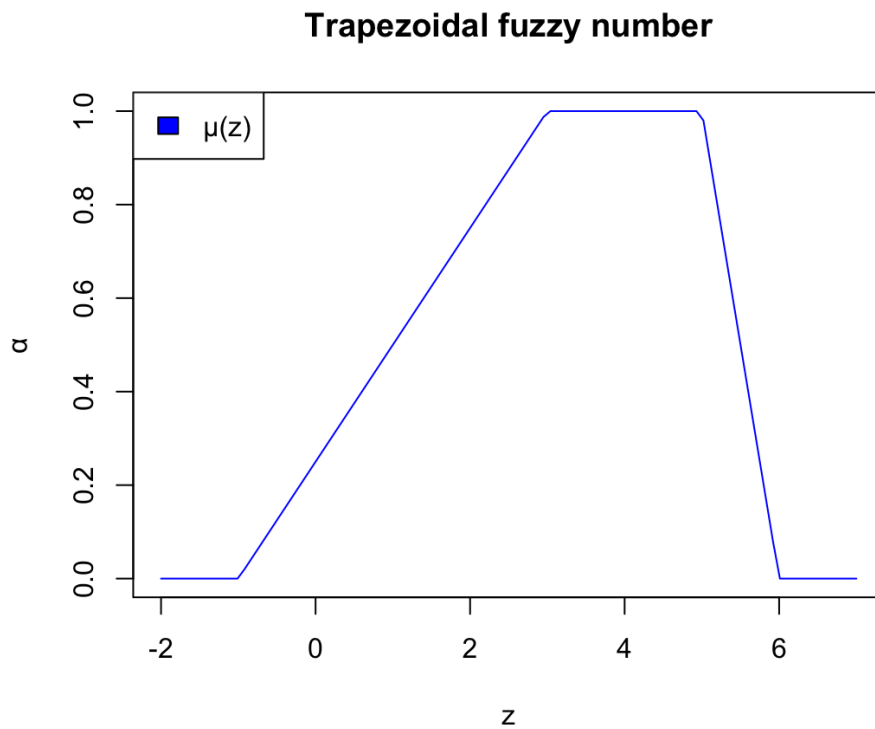


Figure 1.2: Example of a trapezoidal fuzzy number. The figure depicts the trapezoidal fuzzy number $\tilde{x} = (-1, 3, 5, 6)$.

Definition 1.1.13 (LR fuzzy numbers).

Let L and R be two shape functions, $L : (0, \infty) \rightarrow [0, 1]$ $R : (0, \infty) \rightarrow [0, 1]$ satisfying

1. L and R are monotonic non-increasing functions;
2. L and R have finite support;
3. L and R are both left continuous functions.

Then, an LR fuzzy number $\tilde{x} = (x^L, x^l, x^r, x^R)$ with $x^L < x^l < x^r < x^R$ is completely determined by its membership function $\mu_{\tilde{x}}$ given by

$$\mu_{\tilde{x}}(z) = \begin{cases} L(\frac{x^l - z}{x^l - x^L}) & \text{for } z < x^l, \\ 1 & \text{for } x^l \leq z \leq x^r, \\ R(\frac{z - x^r}{x^R - x^r}) & \text{for } z > x^r. \end{cases} \quad (1.19)$$

With $u = \frac{x^l - z}{x^l - x^L}$, when $z < x^l$, $u = 1$ when $x^l \leq z \leq x^r$ and $u = \frac{z - x^r}{x^R - x^r}$ when $z > x^r$, if $L(u) = R(u) = \max(0, 1 - u)$, $\forall u \in [0, \infty)$, one retrieves **TRFNs**. Moreover, the left and right endpoints of an LR fuzzy number $\tilde{x}$, at a given α -cut, $\alpha \in [0, 1]$, are given by

$$x^{L,\alpha} = x - L^{-1}(\alpha), \quad x^{R,\alpha} = x + R^{-1}(\alpha).$$

Definition 1.1.14. Symmetric fuzzy number

A symmetric fuzzy number $\tilde{a} = (a, \gamma)_S$ is defined by its centre a , spread $\gamma \geq 0$ and shape function S (where $S = L = R$). Its membership function is defined as

$$\mu_{\tilde{a}}(z) = S(\frac{z - a}{\gamma}).$$

Note that, for all α -cut $\in [0, 1]$, the left and right spreads are equal, i.e. $a - a^{L,\alpha} = a^{R,\alpha} - a = \gamma^\alpha$. Additionally, $\gamma^a \geq 0$, meaning that $\gamma^{\alpha=1}$ may be greater than zero. Lastly, observe that $\gamma^{a=0} = \gamma$.

Definition 1.1.15 (Gaussian fuzzy set).

A gaussian fuzzy set $\tilde{x}$ is defined by the couple (μ, σ) , $\mu, \sigma \in \mathbb{R}$, $\sigma > 0$. Its membership function defined through the well known gaussian function,

$$\mu_{\tilde{x}}(z) = \exp \frac{-(z - \mu)^2}{2\sigma^2}. \quad (1.20)$$

Its α -cuts can be retrieved using the logarithm,

$$x^{L,\alpha} = \mu - \sqrt{-2\sigma \ln \alpha}, \quad (1.21)$$

$$x^{R,\alpha} = \mu + \sqrt{-2\sigma \ln \alpha}. \quad (1.22)$$

Since for $\alpha \in [0, 1]$, we have $\ln \alpha \in (-\infty, 0]$, the quantity $-2\sigma \ln \alpha$ is strictly positive and unbounded. Consequently, the support of a gaussian fuzzy set is $\text{Supp}(\tilde{x}) = (-\infty, +\infty)$. For this reason, it is technically incorrect to refer to a gaussian fuzzy set as a gaussian fuzzy number, although they are commonly treated as fuzzy numbers in practice.

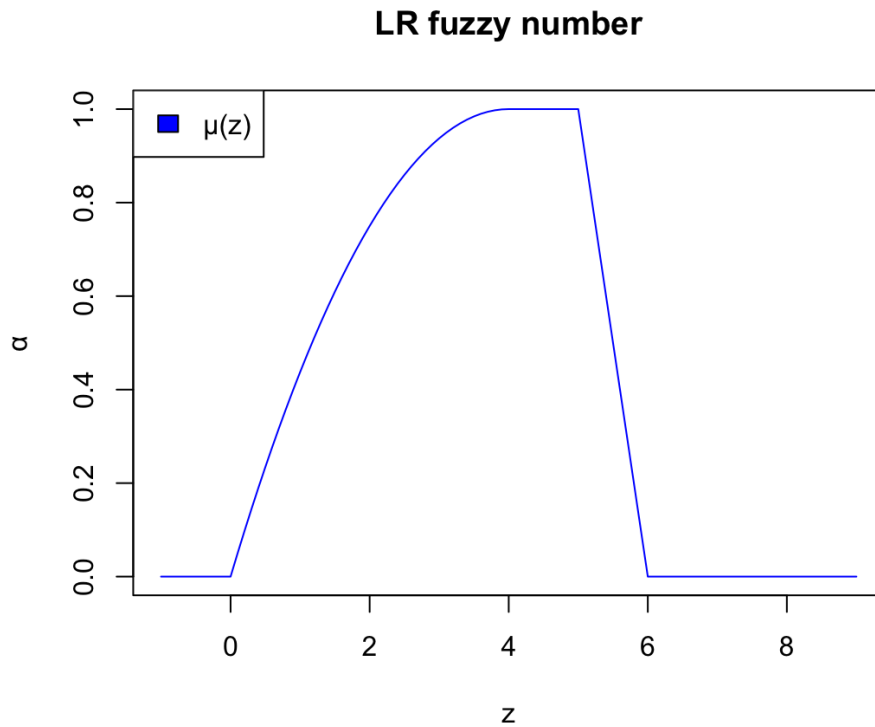


Figure 1.3: Example of an LR fuzzy number. The figure depicts the LR fuzzy number $\tilde{x} = (x^L = 0, x^l = 4, x^r = 5, x^R = 6)$ with the functions $L(z) = \max(0, 1 - ((4 - z)/4)^2)$ for $z \leq 4$ and $R(z) = \max(0, 1 - (z - 5))$ for $z \geq 5$.

Definition 1.1.16. Two sided gaussian fuzzy set

The left and right sides of a two sided gaussian fuzzy set are respectively defined by a gaussian function of parameters (μ_1, σ_1) , (μ_2, σ_2) . Hence, a two sided gaussian fuzzy is denoted by $\tilde{x} = (\mu_1, \sigma_1, \mu_2, \sigma_2)$ and its membership function is defined $\forall z \in \mathbb{R}$ as

$$\mu_{\tilde{x}}(z) = \begin{cases} \exp\left(\frac{-(z-\mu_1)^2}{2\sigma_1}\right) & \text{for } z < \mu_1, \\ 1 & \text{for } \mu_1 \leq z \leq \mu_2, \\ \exp\left(\frac{-(z-\mu_2)^2}{2\sigma_2}\right) & \text{for } z > \mu_2. \end{cases} \quad (1.23)$$

Its left and right α -cuts are expressed by

$$x^{L,\alpha} = \mu_1 - \sqrt{-2\sigma_1 \ln \alpha}, \quad (1.24)$$

$$x^{R,\alpha} = \mu_2 + \sqrt{-2\sigma_2 \ln \alpha}. \quad (1.25)$$

To be complete, let us define the notion of a *generalized fuzzy number* as introduced by Dubois and Prade (1978). Definition 1.1.17 will be especially useful when considering triangular fuzzy numbers with an arbitrary positive height $h > 0$ in Chapter 5.

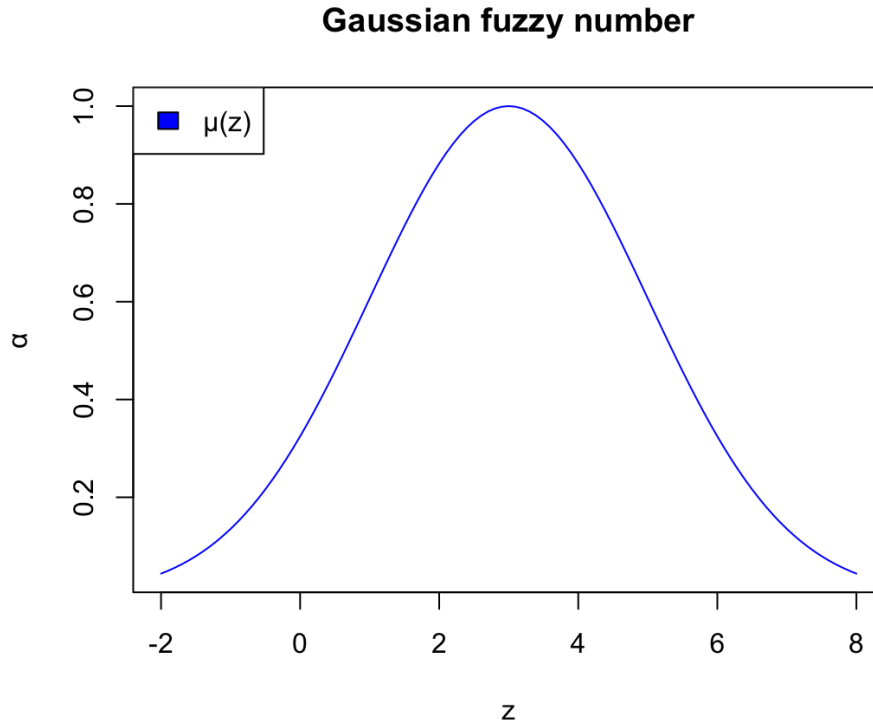


Figure 1.4: Example of a gaussian fuzzy set. The figure depicts the gaussian fuzzy number $\tilde{x} = (3, 2)$.

Definition 1.1.17 (Generalized fuzzy number).

A **generalized fuzzy number** $\tilde{x}$ is defined via its membership function $\mu_{\tilde{x}}(x)$ as follows:

$$\mu_{\tilde{x}}(x) = \begin{cases} L_{\tilde{x}}(x), & a \leq x \leq b, \\ h, & b \leq x \leq c, \\ R_{\tilde{x}}(x), & c \leq x \leq d, \\ 0, & \text{otherwise,} \end{cases}$$

where $h > 0$ denotes the height of the fuzzy number $\tilde{x}$, and

$$L_{\tilde{x}} : [a, b] \rightarrow [0, h], \quad R_{\tilde{x}} : [c, d] \rightarrow [0, h],$$

are strictly monotonic and continuous functions.

If $h = 1$, then $\tilde{x}$ is said to be a normal fuzzy number. If $L_{\tilde{x}}$ and $R_{\tilde{x}}$ are linear, one can then denote trapezoidal fuzzy numbers by

$$\tilde{x} = (a, b, c, d; h), \quad \text{or simply } (a, b, c, d) \text{ when } h = 1.$$

When $b = c$, the trapezoidal fuzzy number simplifies to a **triangular** fuzzy number, written as

$$\tilde{x} = (a, b, d; h), \quad \text{or } (a, b, d) \text{ if } h = 1.$$

Since $L_{\tilde{x}}$ and $R_{\tilde{x}}$ are continuous and strictly monotonic, their inverse functions

$$L_{\tilde{x}}^{-1} : [0, h] \rightarrow [a, b], \quad R_{\tilde{x}}^{-1} : [0, h] \rightarrow [c, d],$$

also exist and are themselves continuous and strictly monotonic. These inverse functions are thus integrable over the interval $[0, h]$, that is:

$$\int_0^h L_{\tilde{x}}^{-1}(r) dr < \infty, \quad \int_0^h R_{\tilde{x}}^{-1}(r) dr < \infty.$$

Now that we introduced the most common fuzzy numbers used in practical and theoretical applications, it is important to observe that fuzzy numbers are generalised real numbers in the sense of Remark 1.1.1.

Remark 1.1.1.

Fuzzy numbers can be regarded as a generalisation of real numbers. Indeed, when the membership function $\mu_{\tilde{x}}$ of a fuzzy number $\tilde{x}$ is just the characteristic function

$$\mu_{\tilde{x}}(z) = I_x(z) := \begin{cases} 1 & \text{if } z = x \\ 0 & \text{otherwise} \end{cases}$$

$\tilde{x}$ reduces to the real number $x \in \mathbb{R}$.

1.2 Logical operations on fuzzy sets

As fuzzy sets are a generalisation of crisp sets, logical operations performed on them need to be generalised as well. Through the use of a t-norm $\top : [0, 1] \times [0, 1] \rightarrow [0, 1]$ and a t-conorm $\perp : [0, 1] \times [0, 1] \rightarrow [0, 1]$, their union and intersection can be easily defined.

Definition 1.2.1 (t-norm and t-conorm).

A function $\top : [0, 1] \times [0, 1] \rightarrow [0, 1]$ is a t-norm when the four following conditions are met $\forall x, y, z \in [0, 1]$:

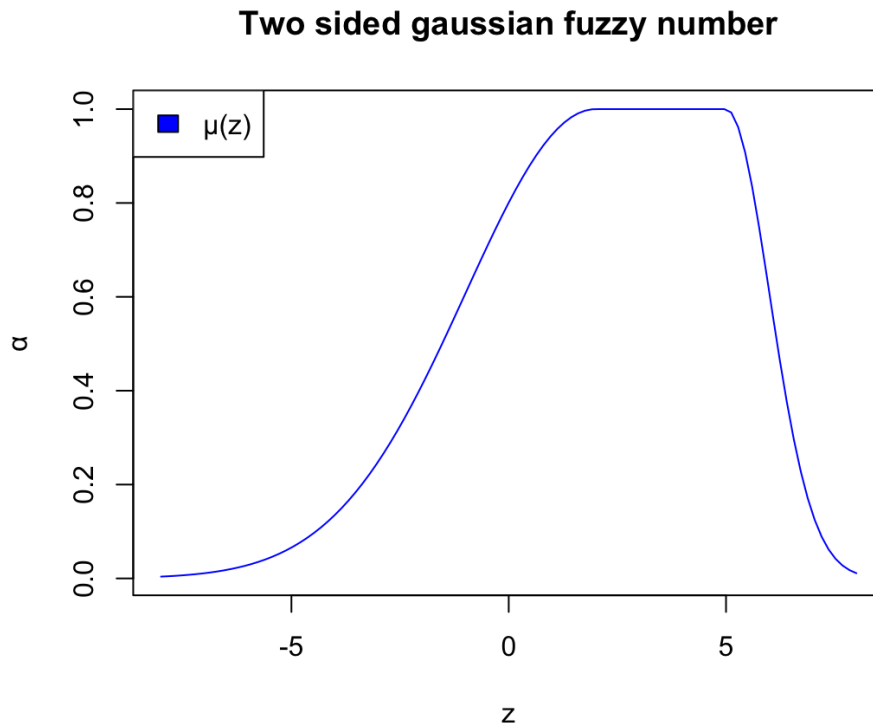


Figure 1.5: Example of a two sided gaussian fuzzy set. The figure depicts the two sided gaussian fuzzy number $\tilde{x} = (\mu_1 = 2, \sigma_1 = 3, \mu_2 = 5, \sigma_2 = 1)$.

1. $\top(x, y) = \top(y, x)$ (commutativity);
2. $\top(\top(x, y), z) = \top(x, \top(y, z))$ (associativity);
3. $\top(x, 1) = x$ (neutral element 1);
4. $x < y$ implies $\top(x, z) < \top(y, z)$ (monotonicity).

A t-conorm is a function $\perp : [0, 1] \times [0, 1] \rightarrow [0, 1]$ satisfying the following four properties $\forall x, y, z \in [0, 1]$:

1. $\perp(x, y) = \perp(y, x)$ (commutativity);
2. $\perp(\perp(x, y), z) = \perp(x, \perp(y, z))$ (associativity);
3. $\perp(x, 0) = x$ (neutral element 0);
4. $x < y$ implies $\perp(x, z) < \perp(y, z)$ (monotonicity).

Moreover, we can also define a t-conorm $\perp : [0, 1] \times [0, 1] \rightarrow [0, 1]$ with a t-norm $\top : [0, 1] \times [0, 1] \rightarrow [0, 1]$ by

$$\perp(x, y) := 1 - \top(1 - x, 1 - y). \quad (1.26)$$

Further useful mathematical properties of t-norms and t-conorms can be found in Kruse and Meyer (1987)[p.26-27], Klement, Mesiar, and Pap (2000), Boixader and Recasens (2022).

Name	t-norm	Name	t-conorm
Minimum	$\top(x, y) = \min(x, y)$	Maximum	$\perp(x, y) = \max(x, y)$
Product	$\top(x, y) = x \cdot y$	Probabilistic sum	$\perp(x, y) = x + y - x \cdot y$
Łukasiewicz	$\top(x, y) = \max(0, x + y - 1)$	Bounded sum	$\perp(x, y) = \min(1, x + y)$
Hamacher product	$\top(x, y) = \begin{cases} 0 & \text{if } x = y = 0 \\ \frac{xy}{x+y-xy} & \text{otherwise} \end{cases}$	Einstein sum	$\perp(x, y) = \frac{x+y}{1+xy}$

Table 1.1: Common t-norms and t-conorms from Boixader and Recasens (2022), Klement, Mesiar, and Pap (2000).

In practical applications, the so called *minimum* t-norm and *maximum* t-conorm are commonly used to define the intersection and union of fuzzy sets as presented in Definitions 1.2.2 and 1.2.3.

Definition 1.2.2 (Intersection of two fuzzy sets).

Given a t-norm $\top : [0, 1] \times [0, 1] \rightarrow [0, 1]$, the intersection of two fuzzy sets $\tilde{X}$ and $\tilde{Y}$ is defined, for $x \in \mathbb{R}$ and $y \in \mathbb{R}$ as

$$\mu_{\tilde{X} \cap \tilde{Y}} = \top(\mu_{\tilde{X}}(x), \mu_{\tilde{Y}}(y)). \quad (1.27)$$

With the use of the minimum t-norm, the intersection becomes

$$\mu_{\tilde{X} \cap \tilde{Y}} = \min(\mu_{\tilde{X}}(x), \mu_{\tilde{Y}}(y)). \quad (1.28)$$

Definition 1.2.3 (Union of two fuzzy sets).

Given a t-conorm $\perp : [0, 1] \times [0, 1] \rightarrow [0, 1]$, the union of two fuzzy sets $\tilde{X}$ and $\tilde{Y}$ is defined, for $x \in \mathbb{R}$ and $y \in \mathbb{R}$ as

$$\mu_{\tilde{X} \cup \tilde{Y}} = \perp(\mu_{\tilde{X}}(x), \mu_{\tilde{Y}}(y)). \quad (1.29)$$

By using the maximum t-conorm, the union writes as

$$\mu_{\tilde{X} \cup \tilde{Y}} = \max(\mu_{\tilde{X}}(x), \mu_{\tilde{Y}}(y)). \quad (1.30)$$

Definition 1.2.4 (Complement of a fuzzy set).

The complement of a fuzzy set $\tilde{X}$ is denoted by $\tilde{X}^c$ and is defined through the use of the membership function by

$$\mu_{\tilde{X}^c} = 1 - \mu_{\tilde{X}}. \quad (1.31)$$

1.3 Fuzzy vectors

Similarly to real numbers, real-valued vectors also can be extended to so called fuzzy vectors. One need to be careful not to confuse fuzzy vectors with vectors of fuzzy numbers. A k -dimensional fuzzy vector, noted $\tilde{\mathbf{x}}^*$ associates a membership degree to a k -dimensional real vector via a single membership function $\mu_{\tilde{\mathbf{x}}^*} : \mathbb{R}^k \rightarrow [0, 1]$ while a k -dimensional vector of fuzzy numbers, noted by $\tilde{\mathbf{x}}$, is defined by assigning to each of its k element a membership function $\mu_{\tilde{x}_i} : \mathbb{R} \rightarrow [0, 1]$, $i = 1, \dots, k$, which specifies a membership degree to the corresponding one-dimensional real-valued component.

Definition 1.3.1. Vector of fuzzy numbers

A k -dimensional vector of fuzzy numbers is a k -components column vector of fuzzy numbers $\tilde{y}_i$, each having their own associated membership function $\mu_{\tilde{y}_i} : \mathbb{R} \rightarrow [0, 1]$, $i = 1, \dots, k$. Vector of fuzzy numbers are denoted by lowercase bold letters with no index, e.g. $\tilde{\mathbf{y}} = (\tilde{y}_1, \dots, \tilde{y}_k)'$.

Definition 1.3.2. k -dimensional fuzzy vector

A k -dimensional fuzzy vector $\tilde{\mathbf{x}}^*$ with membership function $\mu_{\tilde{\mathbf{x}}^*}$ is such that:

1. $\mu_{\tilde{\mathbf{x}}^*} : \mathbb{R}^k \rightarrow [0, 1]$.
2. The support of $\mu_{\tilde{\mathbf{x}}^*}$ is a bounded set.
3. $\forall \alpha \in (0, 1]$ the so-called α -cut $C_\alpha(\tilde{\mathbf{x}}^*) := \{\mathbf{x} \in \mathbb{R}^k \mid \mu_{\tilde{\mathbf{x}}^*}(\mathbf{x}) \geq \alpha\}$ is non-empty, bounded, and a finite union of simply connected and closed bounded sets.

The set of all k -dimensional fuzzy vectors is denoted by $\mathcal{F}(\mathbb{R}^k)$. A k -dimensional fuzzy vector is called a k -dimensional fuzzy interval if all α -cuts are connected compact sets.

1.4 Zadeh's extension principle

Zadeh's extension principle (Zadeh (1965)) is a fundamental result of fuzzy set theory that gives a way to extend algebraic operations on fuzzy quantities. In particular, it allows to define the summation, difference and product of fuzzy sets, fuzzy orthogonal matrices, fuzzy model selection criteria etc.

Theorem 1.4.1 (Zadeh's extension principle).

Let I_1 and I_2 two sets (generally subspaces of $\mathbb{R}$) and $f : I_1 \rightarrow I_2$ a function. Let $\tilde{x} \in \mathcal{F}(I_1)$ be a fuzzy set and define $\tilde{y} = f(\tilde{x})$. Zadeh's extension principle states that the membership function of $\tilde{y}$ is

$$\mu_{\tilde{y}}(z) = \begin{cases} \sup(\mu_{\tilde{x}}(x) \mid x \in I_1, f(x) = z) & \text{if } \exists x : f(x) = z \\ 0 & \text{if } \nexists x : f(x) = z \end{cases} \quad \forall z \in I_2, \quad (1.32)$$

or equivalently,

$$\mu_{\tilde{y}}(z) = \begin{cases} \sup(\mu_{\tilde{x}}(x) \mid x \in I_1, f(x) = z) & \text{if } f^{-1}(z) \neq \emptyset \\ 0 & \text{if } f^{-1}(z) = \emptyset \end{cases} \quad \forall z \in I_2. \quad (1.33)$$

To extend the summation, difference and product operations to fuzzy numbers, we need to consider Zadeh's extension principle for functions of two variables.

Corollary 1.4.1.

Let $f : I_1 \times I_2 \rightarrow I_3$ be a function of two variables and let $\tilde{x}_1 \in \mathcal{F}(I_1)$, $\tilde{x}_2 \in \mathcal{F}(I_2)$ be fuzzy sets of I_1 and I_2 respectively. The membership function of $f(\tilde{x}_1, \tilde{x}_2) = \tilde{y}$ is defined by

$$\mu_{\tilde{y}}(z) = \begin{cases} \sup\{\min(\mu_{\tilde{x}_1}(x_1), \mu_{\tilde{x}_2}(x_2))\} & \text{if } \exists (x_1, x_2) \in I_1 \times I_2 : f(x_1, x_2) = z \\ 0 & \text{if } \nexists (x_1, x_2) \in I_1 \times I_2 : f(x_1, x_2) = z \end{cases} \quad \forall z \in I_3. \quad (1.34)$$

1.5 Fuzzy arithmetic

Now that we enunciated Zadeh's extension principle, we are able to extend the classical arithmetic operators $+$, $-$, $\times$, $\div$ to fuzzy numbers. This fuzzy extension of the classical arithmetic is the so called fuzzy arithmetic. We will note by $\oplus$, $\ominus$, $\otimes$, $\oslash$ respectively the fuzzy summation, subtraction, multiplication and division operators. Since arithmetic operators can be seen as functions $\mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$, their extension will be defined by using Corollary 1.4.1.

$\oplus$ Sum of fuzzy numbers

Definition 1.5.1 (sum of two fuzzy numbers).

Let $\tilde{x}_1$ and $\tilde{x}_2$ two fuzzy numbers belonging to $\mathcal{F}(\mathbb{R})$ with membership functions $\mu_{\tilde{x}_1}$, $\mu_{\tilde{x}_2}$. Using Zadeh's extension principle, the membership function of the sum of $\tilde{x}_1$ and $\tilde{x}_2$, noted by $\tilde{x}_1 \oplus \tilde{x}_2$ is

$$\mu_{\tilde{x}_1 \oplus \tilde{x}_2}(z) = \begin{cases} \sup\{\min(\mu_{\tilde{x}_1}(x_1), \mu_{\tilde{x}_2}(x_2))\} & \text{if } \exists x_1, x_2 \in \mathbb{R} : x_1 + x_2 = z \\ 0 & \text{if } \nexists x_1, x_2 \in \mathbb{R} : x_1 + x_2 = z \end{cases} \quad \forall z \in \mathbb{R} \quad (1.35)$$

which can be rewritten as

$$\mu_{\tilde{x}_1 \oplus \tilde{x}_2}(z) = \sup\{\min(\mu_{\tilde{x}_1}(x_1), \mu_{\tilde{x}_2}(x_2)) : x_1, x_2 \in \mathbb{R} \text{ s.t. } x_1 + x_2 = z\}, \quad \forall z \in \mathbb{R}. \quad (1.36)$$

The sum of two (bounded) fuzzy numbers $\tilde{x}_1$ and $\tilde{x}_2$, i.e. $\tilde{y} = \tilde{x}_1 \oplus \tilde{x}_2$ can also be defined with the α -cuts as

$$\begin{aligned} [\tilde{y}^{L,\alpha}, \tilde{y}^{R,\alpha}] &= [(\tilde{x}_1 \oplus \tilde{x}_2)^{L,\alpha}, (\tilde{x}_1 \oplus \tilde{x}_2)^{R,\alpha}] \\ &= [\tilde{x}_1^{L,\alpha} + \tilde{x}_2^{L,\alpha}, \tilde{x}_1^{R,\alpha} + \tilde{x}_2^{R,\alpha}]. \end{aligned} \quad (1.37)$$

The latter, (1.37), is computationally-wise more efficient than (1.36) to compute the sum of two fuzzy numbers. The sum of two TFNs or two TRFNs can be computed even faster since we only need to know their three, respectively four components as depicted in Examples 1.5.1 and 1.5.2. Additionally, Figure 1.6 displays geometrically the result of the sum between $\tilde{x}_1 = (-2, 1, 3)$ and $\tilde{x}_2 = (4, 6, 7)$.

Example 1.5.1 (Sum of two triangular fuzzy numbers).

The sum of two triangular fuzzy numbers $\tilde{x}_1 = (x_1^L, x_1, x_1^R)$ and $\tilde{x}_2 = (x_2^L, x_2, x_2^R)$ is

$$\tilde{x}_1 \oplus \tilde{x}_2 = (x_1^L + x_2^L, x_1 + x_2, x_1^R + x_2^R). \quad (1.38)$$

With $\tilde{x}_1 = (-2, 1, 4)$ and $\tilde{x}_2 = (-3, -2, -1)$, we find

$$\tilde{x}_1 \oplus \tilde{x}_2 = (-5, -1, 3).$$

Example 1.5.2 (Sum of two trapezoidal fuzzy numbers).

The sum of two trapezoidal fuzzy numbers $\tilde{x}_1 = (x_1^L, x_1^l, x_1^r, x_1^R)$ and $\tilde{x}_2 = (x_2^L, x_2^l, x_2^r, x_2^R)$ is

$$\tilde{x}_1 \oplus \tilde{x}_2 = (x_1^L + x_2^L, x_1^l + x_2^l, x_1^r + x_2^r, x_1^R + x_2^R). \quad (1.39)$$

Numerically, if we pick $\tilde{x}_1 = (-3, -1, 2, 5)$ and $\tilde{x}_2 = (2, 4, 7, 8)$, we find

$$\tilde{x}_1 \oplus \tilde{x}_2 = (-1, 3, 9, 13).$$

⊗ Product of fuzzy numbers

Definition 1.5.2. Product of two fuzzy numbers

Let $\tilde{x}_1$ and $\tilde{x}_2$ be two fuzzy numbers belonging to $\mathcal{F}(\mathbb{R})$ with membership functions $\mu_{\tilde{x}_1}$, $\mu_{\tilde{x}_2}$. Using Zadeh's extension principle, the membership function of the fuzzy product between $\tilde{x}_1$, $\tilde{x}_2$, noted by $\tilde{x}_1 \otimes \tilde{x}_2$, is

$$\mu_{\tilde{x}_1 \otimes \tilde{x}_2}(z) = \begin{cases} \sup\{\min(\mu_{\tilde{x}_1}(x_1), \mu_{\tilde{x}_2}(x_2))\} & \text{if } \exists x_1, x_2 \in \mathbb{R} : x_1 \cdot x_2 = z \\ 0 & \text{if } \nexists x_1, x_2 \in \mathbb{R} : x_1 \cdot x_2 = z \end{cases} \quad \forall z \in \mathbb{R}. \quad (1.40)$$

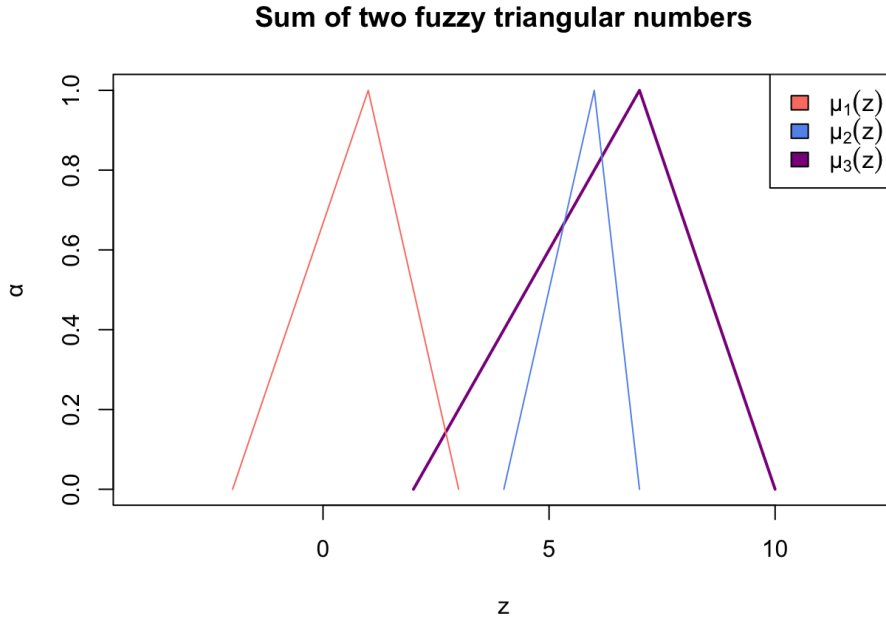


Figure 1.6: Sum of two triangular fuzzy numbers $\tilde{x}_1 = (-2, 1, 3)$, $\tilde{x}_2 = (4, 6, 7)$ given by $\tilde{x}_3 = \tilde{x}_1 \oplus \tilde{x}_2 = (2, 7, 10)$. Additionally, μ_1 , μ_2 and μ_3 are the respective membership functions of $\tilde{x}_1$, $\tilde{x}_2$ and $\tilde{x}_3$.

The product of two fuzzy numbers can also be expressed α -cut-wise by

$$\begin{aligned}
 [(\tilde{x}_1 \otimes \tilde{x}_2)^{L,\alpha}, (\tilde{x}_1 \otimes \tilde{x}_2)^{R,\alpha}] &= \left[\min_{(z_1, z_2) \in [x_1^{L,\alpha}, x_1^{R,\alpha}] \times [x_2^{L,\alpha}, x_2^{R,\alpha}]} z_1 \cdot z_2, \right. \\
 &\quad \left. \max_{(z_1, z_2) \in [x_1^{L,\alpha}, x_1^{R,\alpha}] \times [x_2^{L,\alpha}, x_2^{R,\alpha}]} z_1 \cdot z_2 \right] \\
 &= \left[\min\{x_1^{L,\alpha} \cdot x_2^{L,\alpha}, x_1^{L,\alpha} \cdot x_2^{R,\alpha}, x_1^{R,\alpha} \cdot x_2^{L,\alpha}, x_1^{R,\alpha} \cdot x_2^{R,\alpha}\} \right. \\
 &\quad \left. \max\{x_1^{L,\alpha} \cdot x_2^{L,\alpha}, x_1^{L,\alpha} \cdot x_2^{R,\alpha}, x_1^{R,\alpha} \cdot x_2^{L,\alpha}, x_1^{R,\alpha} \cdot x_2^{R,\alpha}\} \right].
 \end{aligned} \tag{1.41}$$

Moreover, the multiplication of two positive fuzzy numbers $\tilde{x}$ and $\tilde{y}$ can be reduced to

$$[\tilde{x} \otimes \tilde{y}]^\alpha = [x^{L,\alpha} \cdot y^{L,\alpha}, x^{R,\alpha} \cdot y^{R,\alpha}] \tag{1.42}$$

for all $\alpha \in [0, 1]$.

By applying Definition 1.5.2, Examples 1.5.3 and 1.5.4 show how to compute the product of two **TFNs** and two **TRFNs**. Figure 1.7 depicts in purple the membership function of the product of two **TFNs**, (2, 5, 10) and (3, 7, 12).

Example 1.5.3 (multiplication of two triangular fuzzy numbers).

The product of two triangular fuzzy numbers $\tilde{x}_1 = (2, 3, 4)$ and $\tilde{x}_2 = (1, 5, 9)$ is given at $\alpha = 0$ by

$$\begin{aligned} [(\tilde{x}_1 \otimes \tilde{x}_2)^{L,0}, (\tilde{x}_1 \otimes \tilde{x}_2)^{R,0}] &= [\min\{2 \cdot 1, 2 \cdot 9, 4 \cdot 1, 4 \cdot 9\} \\ &\quad \max\{2 \cdot 1, 2 \cdot 9, 4 \cdot 1, 4 \cdot 9\}] \\ &= [2, 36]. \end{aligned} \quad (1.43)$$

Thus, at $\alpha = 0$, $(\tilde{x}_1 \otimes \tilde{x}_2)^{\alpha=0} = (2, 36)$. Let us calculate it for another α – cut to see the difference. For $\alpha = 0.5$, we compute $\tilde{x}_1^{\alpha=0.5} = (2.5, 3.5)$, $\tilde{x}_2^{\alpha=0.5} = (3, 7)$ which yields

$$\begin{aligned} [(\tilde{x}_1 \otimes \tilde{x}_2)^{L,0.5}, (\tilde{x}_1 \otimes \tilde{x}_2)^{R,0.5}] &= [\min\{2.5 \cdot 3, 2.5 \cdot 7, 3.5 \cdot 3, 3.5 \cdot 7\} \\ &\quad \max\{2.5 \cdot 3, 2.5 \cdot 7, 3.5 \cdot 3, 3.5 \cdot 7\}] \\ &= [7.5, 24.5]. \end{aligned} \quad (1.44)$$

Therefore, we find at $\alpha = 0.5$, $(\tilde{x}_1 \otimes \tilde{x}_2)^{\alpha=0.5} = (7.5, 24.5)$. For the $\alpha = 1$ –cut, the product is simply calculated as $(\tilde{x}_1 \otimes \tilde{x}_2)^{\alpha=1} = x_1 \cdot x_2 = 3 \cdot 5 = 15$.

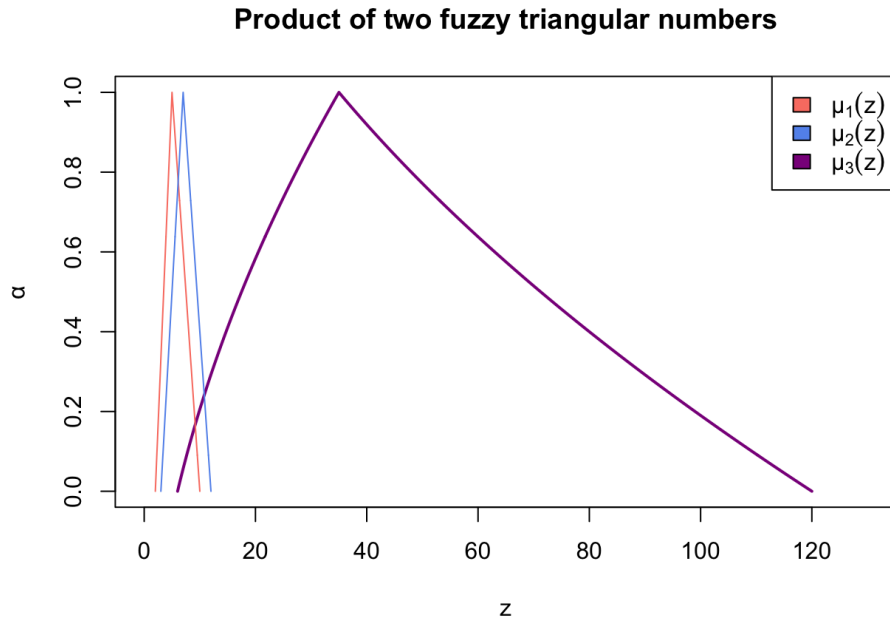


Figure 1.7: Product of the two fuzzy numbers, $\tilde{x}_1 = (2, 5, 10)$ and $\tilde{x}_2 = (3, 7, 12)$. Their multiplication is depicted by the membership function μ_3 associated to the product $\tilde{x}_1 \otimes \tilde{x}_2$.

Example 1.5.4 (multiplication of two TRFNs).

The product of two TRFNs $\tilde{x}_1 = (2, 3, 4, 5)$ and $\tilde{x}_2 = (1, 5, 7, 9)$ is defined for $\alpha = 0$ by

$$\begin{aligned} [(\tilde{x}_1 \otimes \tilde{x}_2)^{L,0}, (\tilde{x}_1 \otimes \tilde{x}_2)^{R,0}] &= [\min\{2 \cdot 1, 2 \cdot 9, 5 \cdot 1, 5 \cdot 9\} \\ &\quad \max\{2 \cdot 1, 2 \cdot 9, 5 \cdot 1, 5 \cdot 9\}] \\ &= [2, 45]. \end{aligned} \quad (1.45)$$

and for $\alpha = 1$ by

$$\begin{aligned} [(\tilde{x}_1 \otimes \tilde{x}_2)^{L,1}, (\tilde{x}_1 \otimes \tilde{x}_2)^{R,1}] &= [\min\{3 \cdot 5, 3 \cdot 7, 4 \cdot 5, 4 \cdot 7\} \\ &\quad \max\{3 \cdot 5, 3 \cdot 7, 4 \cdot 5, 4 \cdot 7\}] \\ &= [15, 28]. \end{aligned} \quad (1.46)$$

For $\alpha = 0.5$, we compute $\tilde{x}_1^{\alpha=0.5} = (2.5, 3, 4, 4.5)$, $\tilde{x}_2^{\alpha=0.5} = (3, 5, 7, 8)$ which yields

$$\begin{aligned} [(\tilde{x}_1 \otimes \tilde{x}_2)^{L,0.5}, (\tilde{x}_1 \otimes \tilde{x}_2)^{R,0.5}] &= [\min\{2.5 \cdot 3, 2.5 \cdot 8, 4.5 \cdot 3, 4.5 \cdot 8\} \\ &\quad \max\{2.5 \cdot 3, 2.5 \cdot 8, 4.5 \cdot 3, 4.5 \cdot 8\}] \\ &= [7.5, 36]. \end{aligned} \quad (1.47)$$

Note that in general, as it can be seen on Figure 1.8, the multiplication of two TFNs does not yield back a TFN. The same is true for TRFNs and gaussian fuzzy numbers. Additionally, on Figure 1.8, we can observe that, on the upper left and lower right, when two positive or negative triangular fuzzy numbers are multiplied, the left side of the resulting fuzzy number is concave and its right side is convex (situation 1). However, notice, on the upper right, that the situation is reversed for the product between a negative and positive triangular fuzzy number (situation 2). Moreover, with the help of the lower left graph, we see that the left side of μ_3 is first convex and then concave. This effect is in fact a combination of the two above observations. The triangular fuzzy number $\tilde{x}_1 = (-2, 2, 3)$ will have a negative left α -cut $x^{L,\alpha}$ for $\alpha \in [0, \frac{1}{2})$ and a positive left α -cut $x^{L,\alpha}$ for $\alpha \in [\frac{1}{2}, 1]$. Thus, being in the second situation when $\alpha \in [0, \frac{1}{2}]$ and then in the first situation when $\alpha \in [\frac{1}{2}, 1]$.

⊖ Substraction of fuzzy numbers

Definition 1.5.3. Substraction of two fuzzy numbers

Let $\tilde{x}_1$ and $\tilde{x}_2$ be two fuzzy numbers belonging to $\mathcal{F}(\mathbb{R})$ with membership functions $\mu_{\tilde{x}_1}$, $\mu_{\tilde{x}_2}$. Using Zadeh's extension principle, the membership function of the substraction of $\tilde{x}_1$ by $\tilde{x}_2$, noted $\tilde{x}_1 \ominus \tilde{x}_2$ is

$$\mu_{\tilde{x}_1 \ominus \tilde{x}_2}(z) = \begin{cases} \sup \{ \min(\mu_{\tilde{x}_1}(x_1), \mu_{\tilde{x}_2}(x_2)) \} & \text{if } \exists x_1, x_2 \in \mathbb{R} : x_1 - x_2 = z \\ 0 & \text{if } \nexists x_1, x_2 \in \mathbb{R} : x_1 - x_2 = z \end{cases} \quad \forall z \in \mathbb{R} \quad (1.48)$$

Additionally, the substraction of $\tilde{x}_1$ by $\tilde{x}_2$, $\tilde{x}_1 \ominus \tilde{x}_2$ can be defined using the fuzzy summation and multiplication operators in the following way :

$$\tilde{x}_1 \ominus \tilde{x}_2 := \tilde{x}_1 \oplus (-1) \otimes \tilde{x}_2. \quad (1.49)$$

Using the above definition, let us show with Example 1.5.5 how to subtract a triangular fuzzy number from another.

Example 1.5.5.

Let $\tilde{x} = (-2, -1, 0)$ and $\tilde{y} = (-1, 2, 4)$ be two **TFN**. Then $\tilde{x} \ominus \tilde{y}$ is calculated as

$$\begin{aligned}\tilde{x} \ominus \tilde{y} &= \tilde{x} \oplus (-1) \otimes \tilde{y} \\ &= (-2, -1, 0) \oplus (\min\{(-1) \cdot (-1), (-1) \cdot 4\}, -1 \cdot 2, \max\{(-1) \cdot (-1), (-1) \cdot 4\}) \\ &= (-2, -1, 0) \oplus (-4, -2, 1) \\ &= (-6, -3, 1).\end{aligned}\tag{1.50}$$

○ Quotient of fuzzy numbers

The quotient or division of two fuzzy numbers is defined similarly to the multiplication of two fuzzy numbers.

Definition 1.5.4. Quotient of two fuzzy numbers

Let $\tilde{x}_1$ and $\tilde{x}_2$ be two fuzzy numbers belonging to $F(\mathbb{R})$ with membership functions $\mu_{\tilde{x}_1}$, $\mu_{\tilde{x}_2}$. If the support of $\tilde{x}_2$ does not contain 0, $0 \notin \text{Supp}(\tilde{x}_2)$, using Zadeh's extension principle, the membership function of the quotient between $\tilde{x}_1$ and $\tilde{x}_2$, noted by $\tilde{x}_1 \oslash \tilde{x}_2$, is

$$\mu_{\tilde{x}_1 \oslash \tilde{x}_2}(z) = \begin{cases} \sup (\min (\mu_{\tilde{x}_1}(x_1), \mu_{\tilde{x}_2}(x_2))) & \text{if } \exists x_1, x_2 \in \mathbb{R} : x_1 \div x_2 = z \\ 0 & \text{if } \nexists x_1, x_2 \in \mathbb{R} : x_1 \div x_2 = z \end{cases} \quad \forall z \in \mathbb{R}.\tag{1.51}$$

The quotient of two fuzzy numbers $\tilde{x}_1$, $\tilde{x}_2$ can also be expressed α -cut-wise by (under the condition that $0 \notin \text{Supp}(\tilde{x}_2)$)

$$\begin{aligned}[(\tilde{x}_1 \oslash \tilde{x}_2)^{L,\alpha}, (\tilde{x}_1 \oslash \tilde{x}_2)^{R,\alpha}] &= \left[\min_{(z_1, z_2) \in [x_1^{L,\alpha}, x_1^{R,\alpha}] \times [x_2^{L,\alpha}, x_2^{R,\alpha}]} z_1 \div z_2, \right. \\ &\quad \left. \max_{(z_1, z_2) \in [x_1^{L,\alpha}, x_1^{R,\alpha}] \times [x_2^{L,\alpha}, x_2^{R,\alpha}]} z_1 \div z_2 \right] \\ &= \left[\min\{x_1^{L,\alpha} \div x_2^{L,\alpha}, x_1^{L,\alpha} \div x_2^{R,\alpha}, x_1^{R,\alpha} \div x_2^{L,\alpha}, x_1^{R,\alpha} \div x_2^{R,\alpha}\}, \right. \\ &\quad \left. \max\{x_1^{L,\alpha} \div x_2^{L,\alpha}, x_1^{L,\alpha} \div x_2^{R,\alpha}, x_1^{R,\alpha} \div x_2^{L,\alpha}, x_1^{R,\alpha} \div x_2^{R,\alpha}\} \right].\end{aligned}\tag{1.52}$$

To better understand the quotient between two fuzzy numbers, given by Definition 1.5.4, let us compute in Example 1.5.6 the quotient between two **TFNs**. Concerning a geometric view of the fuzzy quotient, Figure 1.9 displays the quotients between two positive **TFNs**, two negative **TFNs** and between a negative **TFN** and a positive **TFN**. On this figure, $\mu_3(z)$ (in purple) is the membership function of the quotient between two triangular fuzzy numbers with membership functions $\mu_1(z)$ (in orange) and $\mu_2(z)$ (in blue). Notice that, when the **TFNs** are both positive or negative, displayed by the upper graphs on 1.9, the left part of the membership function is concave while its right part is convex. In the case of a fuzzy quotient between a negative and positive **TFN**, the left part of the membership function is this time convex while the right part is concave.

Example 1.5.6 (Quotient of two triangular fuzzy numbers).

Similarly to the fuzzy multiplication of two fuzzy numbers, numerically, the quotient is calculated at a given α -cut. Let us compute the quotient of two triangular fuzzy numbers

$\tilde{x}_1 = (2, 3, 4)$ and $\tilde{x}_2 = (1, 5, 9)$ at $\alpha = 0$. Since $0 \notin \text{Supp}(\tilde{x}_2)$ we can calculate it. The left and right 0– cuts $(\tilde{x}_1 \oslash \tilde{x}_2)^{L,0}$, $(\tilde{x}_1 \oslash \tilde{x}_2)^{R,0}$ are

$$\begin{aligned} [(\tilde{x}_1 \oslash \tilde{x}_2)^{L,0}, (\tilde{x}_1 \oslash \tilde{x}_2)^{R,0}] &= [\min\{2 \div 1, 2 \div 9, 4 \div 1, 4 \div 9\} \\ &\quad \max\{2 \div 1, 2 \div 9, 4 \div 1, 4 \div 9\}] \\ &= [\frac{2}{9}, 4]. \end{aligned} \quad (1.53)$$

The centre is calculated as $x_1 \div x_2 = 3 \div 5 = \frac{3}{5}$. At $\alpha = 0$, $(\tilde{x}_1 \oslash \tilde{x}_2)^{\alpha=0} = (\frac{2}{9}, 4)$. Let us calculate it for $\alpha = 0.5$. With $\tilde{x}_1^{\alpha=0.5} = (2.5, 3.5)$, $\tilde{x}_2^{\alpha=0.5} = (3, 7)$, it yields

$$\begin{aligned} [(\tilde{x}_1 \oslash \tilde{x}_2)^{L,0.5}, (\tilde{x}_1 \oslash \tilde{x}_2)^{R,0.5}] &= [\min\{2.5 \div 3, 2.5 \div 7, 3.5 \div 3, 3.5 \div 7\} \\ &\quad \max\{2.5 \div 3, 2.5 \div 7, 3.5 \div 3, 3.5 \div 7\}] \\ &= [\frac{5}{14}, \frac{7}{6}]. \end{aligned} \quad (1.54)$$

Therefore, we find at $\alpha = 0.5$, $(\tilde{x}_1 \oslash \tilde{x}_2)^{\alpha=0.5} = (\frac{5}{14}, \frac{7}{6})$.

It is also important to underline that, in general, $1 \oslash \tilde{x}$ is **not** the inverse of $\tilde{x}$, $\tilde{x}^{-1}$. Indeed, one can observe that, given a TFN $\tilde{x} = (3, 4, 5)$, at $\alpha = 0$, $(1 \oslash (3, 4, 5))^0 = (\frac{1}{5}, \frac{1}{3})$. However, $(1 \oslash (3, 4, 5))^0 \otimes (3, 4, 5)^0 = (\frac{1}{5}, \frac{1}{3}) \otimes (3, 5) = (\frac{3}{5}, \frac{5}{3}) \neq (1, 1)$. Thus, the property $x_1 \cdot x_2 = x_1 \div (1 \div x_2)$ is in general not true for fuzzy numbers, that is $\tilde{x}_1 \otimes \tilde{x}_2 \neq \tilde{x}_1 \oslash (1 \oslash \tilde{x}_2)$. On Figure 1.10 observe that, while the results are very close to each other, $(-4, -3, -1) \oslash (\frac{1}{8}, \frac{1}{5}, \frac{1}{3}) \neq (-4, -3, -1) \otimes (3, 5, 8)$. As stated earlier, this comes from the fact that $1 \oslash (\frac{1}{8}, \frac{1}{5}, \frac{1}{3}) \neq (3, 5, 8)$, since the result of a division between two triangular fuzzy numbers is not a triangular fuzzy number. Thus, the fuzzy quotient $\oslash$ is in general not the inverse operator of the fuzzy product $\otimes$. Moreover, we can see that the curvature yielded by the fuzzy quotient is greater than the one given by the fuzzy product.

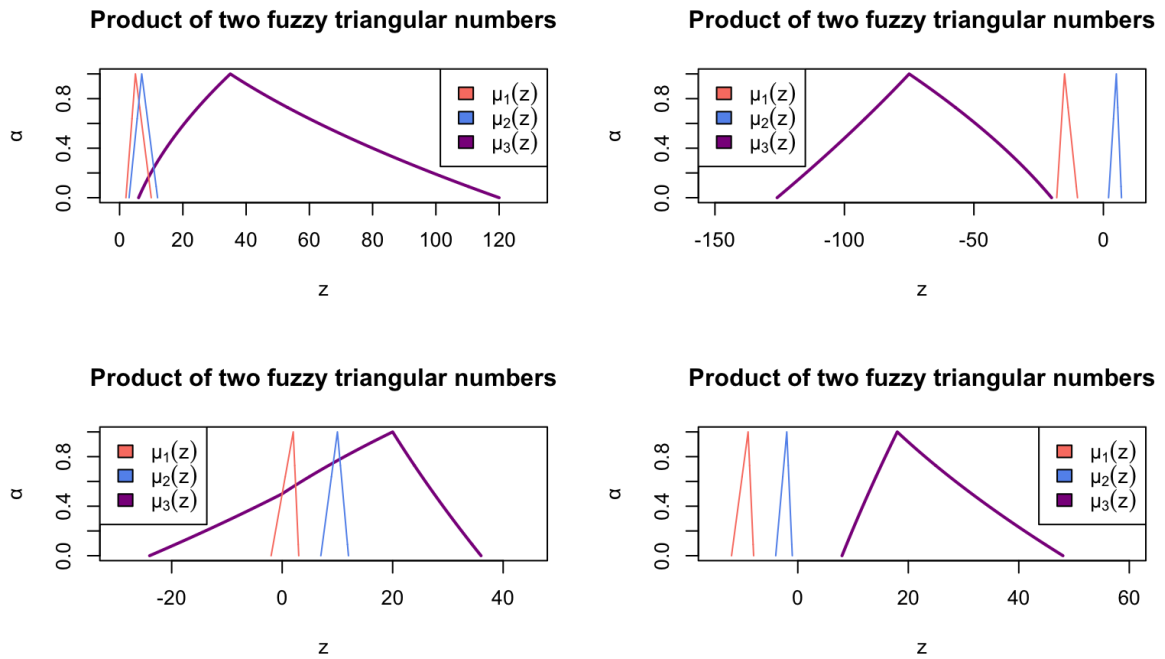


Figure 1.8: Four cases of multiplication between triangular fuzzy numbers $\tilde{x}_1$ and $\tilde{x}_2$ are displayed. μ_3 is the membership function of the product $\tilde{x}_1 \otimes \tilde{x}_2$ while μ_1, μ_2 are the membership functions of $\tilde{x}_1, \tilde{x}_2$. The upper left graph depicts the product between $\tilde{x}_1 = (2, 5, 10)$ and $\tilde{x}_2 = (3, 7, 12)$, the upper right graph shows the multiplication of $\tilde{x}_1 = (-18, -15, -10)$ with $\tilde{x}_2 = (2, 5, 7)$. The lower left and lower right graphs display the products between $\tilde{x}_1 = (-2, 2, 3)$ and $\tilde{x}_2 = (7, 10, 12)$ respectively $\tilde{x}_1 = (-12, -9, -8)$ and $\tilde{x}_2 = (-4, -2, -1)$.

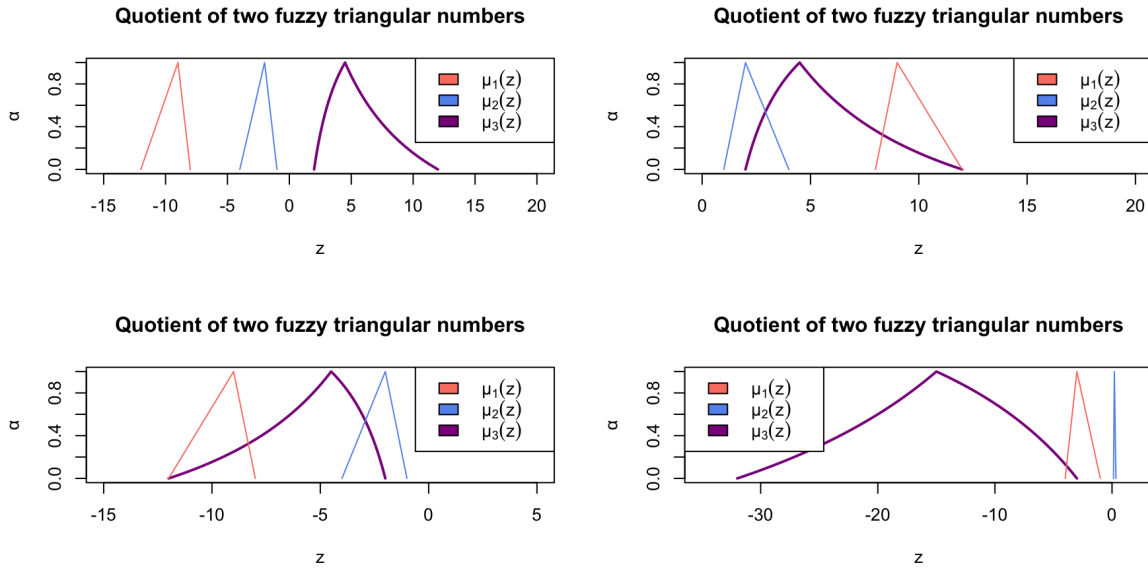


Figure 1.9: Four cases of division between triangular fuzzy numbers $\tilde{x}_1$ and $\tilde{x}_2$ are displayed. μ_3 is the membership function of the quotient $\tilde{x}_1 \oslash \tilde{x}_2$ while μ_1, μ_2 are the membership functions of $\tilde{x}_1, \tilde{x}_2$. The upper left graph depicts the quotient between $\tilde{x}_1 = (-12, -9, -8)$ and $\tilde{x}_2 = (-4, -2, -1)$, the upper right graph shows the division of $\tilde{x}_1 = (12, 9, 8)$ by $\tilde{x}_2 = (1, 2, 4)$. The lower left and lower right graphs display the quotients between $\tilde{x}_1 = (-12, -9, -8)$ and $\tilde{x}_2 = (-4, -2, -1)$ respectively $\tilde{x}_1 = (-4, -3, -1)$ and $\tilde{x}_2 = (\frac{1}{8}, \frac{1}{5}, \frac{1}{3})$.

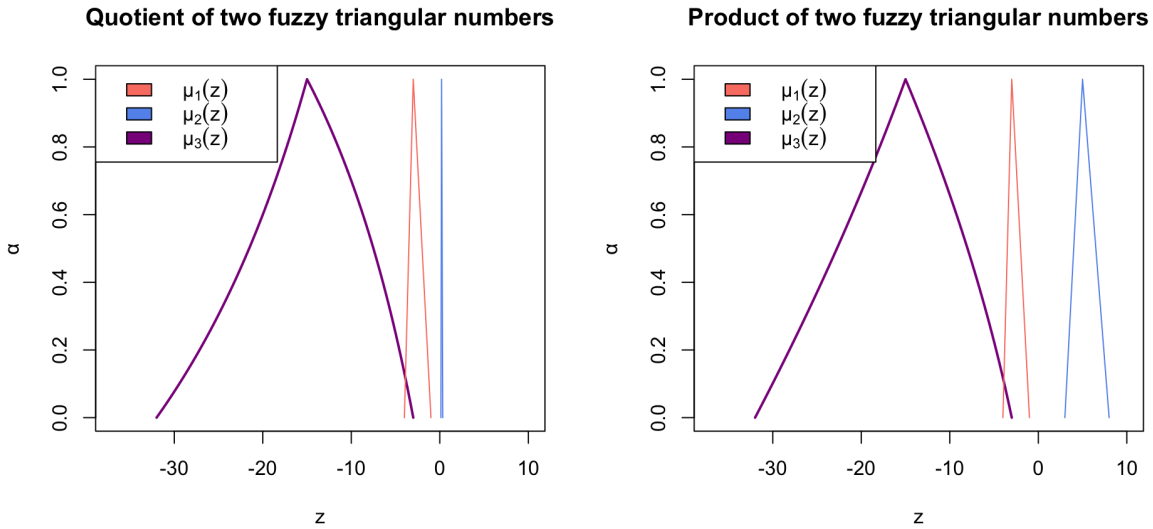


Figure 1.10: On the left : in purple, display of the division between the triangular fuzzy numbers $\tilde{x}_1 = (-4, -3, -1)$ and $\tilde{x}_2 = (\frac{1}{8}, \frac{1}{5}, \frac{1}{3})$. On the right : in purple, depiction of the multiplication between the triangular fuzzy numbers $\tilde{x}_1 = (-4, -3, -1)$ and $\tilde{x}_2 = (3, 5, 8)$.

Chapter 2

Fuzzy regression methods

This chapter is devoted to the study of fuzzy regression methods. Around 1990, two main approaches, the fuzzy least squares by Diamond (1988) and the possibilistic linear regression by Tanaka, Hayashi, and Watada (1989) emerged to perform fuzzy regression. The method developed by Diamond (1988) aims at minimising the squared distance between two fuzzy numbers, the fuzzy dependent variable and its fitted estimation, the fuzzy predicted values. To do so, Diamond first defined a proper metric and then seek parameters that minimise the sum of squared residuals. Diamond regression method is thus based on the use of a metric to handle the fuzziness occurring from the regression parameters and the observations. This metric allows to consider crisp residuals despite the use of fuzzy dependent variables and fitted values.

The possibilistic linear regression analysis introduced by Tanaka, Hayashi, and Watada (1989) aims at minimising the sum of spreads of estimated fuzzy dependent variables through three different problem formulations. A decade later, Tanaka and Lee (1997) will conciliate both the properties of the possibilistic linear regression and fuzzy least squares approaches. Their idea is to find a quadratic objective function that expresses both the spreads and sum of squared residuals and then, with the use of quadratic programming, to minimise it under a set of given constraints. However, in their approaches, Tanaka, Hayashi, and Watada (1989) only considered crisp covariates. To address this problem, D'Urso and Massari (2013) proposed a fuzzy least squares regression method that accounts for both fuzzy dependent and independent variables. However, due to the computational complexity and iterative nature of D'Urso and Massari (2013)'s approach, Rosset and Donzé (2023) extended the formulation of Tanaka and Lee (1997) and introduced a simplified fuzzy least squares regression technique, while maintaining the assumption that both the response variable and the predictors are fuzzy. Using their proposed method, Rosset and Donzé (2023) showed that, when the design matrix of fuzzy covariates is orthogonalised according to the notion of fuzzy orthogonality introduced by Itoh (2017), regression coefficients can be estimated independently by regressing the dependent variable on each covariate separately, yielding the same results as joint estimation using all covariates. Lastly, Rosset and Donzé (2024c) also proposed another orthogonal fuzzy least squares method to estimate the k -th fuzziest estimate by least squares. This approach seeks

to find fuzzy estimates through the use of Zadeh's extension principle. The idea is to generate many matrices having their components bounded by the left and right endpoints of the fuzzy matrix of observations and to orthogonalise them. Then, they estimate the regression problems associated to these orthogonalised matrices of observations and the fuzziest solution is chosen. Although this method is based on an alternative definition of fuzzy orthogonality derived from Zadeh's extension principle, it preserves the independent estimation of regression coefficients observed in Rosset and Donzé (2023)'s approach.

The approach proposed by Rosset and Donzé (2023) offers a significant gain in computational efficiency; however, this improvement comes at the cost of relying on an approximation of fuzzy multiplication and on classical orthogonality, rather than fuzzy orthogonality derived from Zadeh's extension principle. Moreover, the resulting estimates take the form of fuzzy numbers only under additional constraints. By contrast, the method introduced by Rosset and Donzé (2024c) strictly follows Zadeh's extension principle, thereby ensuring a coherent treatment of fuzzy arithmetic and fuzzy orthogonality. In addition, this approach always yields estimates in the form of fuzzy numbers, without requiring any supplementary constraints.

2.1 Fuzzy least squares by Diamond (1988)

The fuzzy least squares regression method proposed by Diamond (1988) relies on an L^2 -metric for measuring the distance between fuzzy numbers. This distance, defined for fuzzy numbers $\tilde{x}$ and $\tilde{y}$, with supports $\text{Supp}(\tilde{x}) = [x^L, x^R]$ and $\text{Supp}(\tilde{y}) = [y^L, y^R]$, is given by

$$d^2 : \mathcal{F}_c(\mathbb{R}) \times \mathcal{F}(\mathbb{R})_c \rightarrow \mathbb{R}^+ \quad (2.1)$$

$$d^2(\tilde{x}, \tilde{y}) = (x^R - y^R)^2 + (x^L - y^L)^2 + (x - y)^2, \quad (2.2)$$

where $\mathcal{F}_c(\mathbb{R})$ denotes the space of fuzzy numbers. One can find a detailed summary concerning the construction of d^2 in A.1. This distance is crucial as it enables the computation of the sum of squares that arises in fuzzy linear least squares regression.

Diamond (1988) considered a simple regression problem with y as the dependent variable and x as the single independent variable. We treat the case where the data is fuzzy. Let $\tilde{x}_i = (x_i^L, x_i, x_i^R)$, $\tilde{y}_i = (y_i^L, y_i, y_i^R)$ be data pairs of fuzzy triangular observations, $i = 1, \dots, n$. Diamond (1988) assumed, by means of translating the data if necessary, that all $\tilde{x}_i$ are positive TFNs. $\tilde{\mathbf{x}}$ will denote the column vector consisting of the observations $\tilde{x}_i$. Similarly, $\tilde{\mathbf{y}}$ denotes the vector consisting of the observations $\tilde{y}_i$. Additionally, $\bar{x} = \frac{1}{n} \sum_{i=1}^n (x_i)$.

The following two models,

$$\text{Model 1:} \quad \tilde{\mathbf{y}} = \beta_0 + \beta_1 \tilde{\mathbf{x}} \quad \beta_0, \beta_1 \in \mathbb{R};$$

$$\text{Model 2:} \quad \tilde{\mathbf{y}} = \tilde{\beta}_0 + \beta_1 \tilde{\mathbf{x}} \quad \beta_0 \in \mathcal{F}_c(\mathbb{R}), \beta_1 \in \mathbb{R};$$

will be considered along with their respective fuzzy least squares optimisation problems,

$$\text{Problem 1:} \quad \text{Minimise} \quad f_1(\beta_0, \beta_1) := \sum_i^n \left(d^2(\beta_0 + \beta_1 \otimes \tilde{x}_i, \tilde{y}_i) \right); \quad (2.3)$$

$$\text{Problem 2:} \quad \text{Minimise} \quad f_2(\tilde{\beta}_0, \beta_1) := \sum_i^n \left(d^2(\tilde{\beta}_0 + \beta_1 \otimes \tilde{x}_i, \tilde{y}_i) \right). \quad (2.4)$$

To solve the two minimisation problems defined by equations (2.3) and (2.4), it is essential to recognise that the distance $d^2(\beta_0 + \beta_1 \otimes \tilde{x}_i, \tilde{y}_i)$ depends on the sign of β_1 . Since the sign of β_1 is unknown a priori, the minimisation problem must be solved twice: first under the assumption that $\beta_1 \geq 0$, and then assuming $\beta_1 < 0$. This procedure results in two distinct solutions.

Problem 1 Let us begin by considering the first problem. Recall that for a positive real number $a \in \mathbb{R}$ with $a \geq 0$, the fuzzy product for a triangular fuzzy number $\tilde{x} = (x^L, x, x^R)$ is defined as

$$a \otimes \tilde{x} = a \otimes (x^L, x, x^R) = (ax^L, ax, ax^R).$$

On the other hand, when $a < 0$, the fuzzy product is given by

$$a \otimes \tilde{x} = a \otimes (x^L, x, x^R) = (ax^R, ax, ax^L),$$

where the positions of the left and right components of the triangular fuzzy number $\tilde{x}$ are permuted in the resulting fuzzy product.

Consider the case where $\beta_1 \geq 0$. Using equation (2.1), the squared distance d^2 is written as

$$d^2(\beta_0 + \beta_1 \otimes \tilde{x}_i, \tilde{y}_i) = \left(\beta_0 + \beta_1 x_i^L - y_i^L \right)^2 + \left(\beta_0 + \beta_1 x_i^R - y_i^R \right)^2 + (\beta_0 + \beta_1 x_i - y_i)^2. \quad (2.5)$$

Conversely, if $\beta_1 < 0$, the squared distance d^2 becomes

$$d^2(\beta_0 + \beta_1 \otimes \tilde{x}_i, \tilde{y}_i) = \left(\beta_0 + \beta_1 x_i^L - y_i^R \right)^2 + \left(\beta_0 + \beta_1 x_i^R - y_i^L \right)^2 + (\beta_0 + \beta_1 x_i - y_i)^2. \quad (2.6)$$

For the case $\beta_1 \geq 0$, (2.5) allows to find a potential solution to Problem 1 by solving the first order condition. Specifically, the partial derivatives $\partial f_1(\beta_0, \beta_1)/\partial \beta_0$, $\partial f_1(\beta_0, \beta_1)/\partial \beta_1$ are computed as follows.

First, for $\frac{\partial f_1(\beta_0, \beta_1)}{\partial \beta_0}$ we have

$$\begin{aligned}
 \frac{\partial f_1(\beta_0, \beta_1)}{\partial \beta_0} &= \frac{\partial}{\partial \beta_0} \sum_i^n \left(d^2(\beta_0 + \beta_1 \otimes \tilde{x}_i, \tilde{y}_i) \right) \\
 &= \sum_i^n \left(\frac{\partial}{\partial \beta_0} d^2(\beta_0 + \beta_1 \otimes \tilde{x}_i, \tilde{y}_i) \right) \\
 &= \sum_i^n \frac{\partial}{\partial \beta_0} \left((\beta_0 + \beta_1 x_i^L - y_i^L)^2 + (\beta_0 + \beta_1 x_i^R - y_i^R)^2 + (\beta_0 + \beta_1 x_i - y_i)^2 \right) \\
 &= \sum_i^n \left(2\beta_0 + 2\beta_1 x_i^L - 2y_i^L + 2\beta_0 + 2\beta_1 x_i^R - 2y_i^R + 2\beta_0 + 2\beta_1 x_i - 2y_i \right) \\
 &= \sum_i^n \left(6\beta_0 + 2\beta_1 (x_i^L + x_i^R + x_i) - 2y_i^L - 2y_i^R - 2y_i \right) \\
 &= 6n\beta_0 + 2\beta_1 \sum_i^n (x_i^L + x_i^R + x_i) - 2 \sum_i^n (y_i^L + y_i^R + y_i), \tag{2.7}
 \end{aligned}$$

Next, for $\frac{\partial f_1(\beta_0, \beta_1)}{\partial \beta_1}$, we have

$$\begin{aligned}
 \frac{\partial f_1(\beta_0, \beta_1)}{\partial \beta_1} &= \frac{\partial}{\partial \beta_1} \sum_i^n \left(d^2(\beta_0 + \beta_1 \otimes \tilde{x}_i, \tilde{y}_i) \right) \\
 &= \sum_i^n \frac{\partial}{\partial \beta_1} \left((\beta_0 + \beta_1 x_i^L - y_i^L)^2 + (\beta_0 + \beta_1 x_i^R - y_i^R)^2 + (\beta_0 + \beta_1 x_i - y_i)^2 \right) \\
 &= \sum_i^n \left(2(\beta_0 + \beta_1 x_i^L - 2y_i^L)x_i^L + 2(\beta_0 + \beta_1 x_i^R - 2y_i^R)x_i^R + 2(\beta_0 + \beta_1 x_i - 2y_i)x_i \right) \\
 &= \sum_i^n \left(2\beta_0 x_i^L + 2\beta_1 (x_i^L)^2 - 2x_i^L y_i^L + 2\beta_0 x_i^R + 2\beta_1 (x_i^R)^2 - 2x_i^R y_i^R + 2\beta_0 x_i + 2\beta_1 x_i^2 - 2x_i y_i \right) \\
 &= 2\beta_0 \sum_i^n (x_i^L + x_i^R + x_i) + 2\beta_1 \sum_i^n \left(x_i^2 + (x_i^L)^2 + (x_i^R)^2 \right) - 2 \sum_i^n \left(x_i^L y_i^L + x_i^R y_i^R + x_i y_i \right). \tag{2.8}
 \end{aligned}$$

Setting the above partial derivatives to zero yields the following system of equations:

$$\begin{aligned}
 3n\beta_0 + \beta_1 \sum_i^n (x_i^L + x_i^R + x_i) &= \sum_i^n (y_i^L + y_i^R + y_i) \tag{2.9} \\
 \beta_0 \sum_i^n (x_i^L + x_i^R + x_i) + \beta_1 \sum_i^n \left(x_i^2 + (x_i^L)^2 + (x_i^R)^2 \right) &= \sum_i^n \left(x_i^L y_i^L + x_i^R y_i^R + x_i y_i \right).
 \end{aligned}$$

A solution to (2.9), if it exists, will be denoted by $\beta_{0,+}, \beta_{1,+}$.

Similarly, for $\beta_1 < 0$, using equation (2.6), a potential solution $\beta_{0,-}, \beta_{1,-}$ to Problem 1 is found by solving the system of equations :

$$\begin{aligned} 3n\beta_0 + \beta_1 \sum_i^n (x_i^L + x_i^R + x_i) &= \sum_i^n (y_i^L + y_i^R + y_i) \quad (2.10) \\ \beta_0 \sum_i^n (x_i^L + x_i^R + x_i) + \beta_1 \sum_i^n (x_i^2 + (x_i^L)^2 + (x_i^R)^2) &= \sum_i^n (x_i^R y_i^L + x_i^L y_i^R + x_i y_i). \end{aligned}$$

Note that the difference between (2.9) and (2.10) lies in the last sum, where the terms $x_i^L y_i^L + x_i^R y_i^R$ in (2.9) are replaced by $x_i^R y_i^L + x_i^L y_i^R$ in (2.10).

Diamond (1988) [pp. 148-149] demonstrated that if both $\beta_{1,-}$ and $\beta_{1,+}$ are either positive or negative, then the solution to the minimisation problem (2.3) is unique. Specifically, if $\beta_{1,-} \geq 0$ and $\beta_{1,+} \geq 0$, the unique solution corresponds to $\beta_{1,+}$, while, if $\beta_{1,-} < 0$ and $\beta_{1,+} < 0$, it is given by $\beta_{1,-}$. However, in general, $\beta_{1,-}$ and $\beta_{1,+}$ will differ in sign, resulting in two distinct solutions. This highlights a limitation in Diamond (1988)'s approach.

Problem 2 When the intercept $\tilde{\beta}_0 = (\beta_0^L, \beta_0, \beta_0^R)$ is fuzzy, the minimisation problem is described by Problem 2. Similarly to Problem 1, we proceed by deriving the potential solutions for the cases $\beta_1 \geq 0$ or $\beta_1 < 0$. The first step is to calculate the distance $d^2(\tilde{\beta}_0 + \beta_1 \tilde{x}_i, \tilde{y}_i)$ appearing in Problem 2. Thus, when $\beta_1 \geq 0$, the distance becomes

$$d^2(\tilde{\beta}_0 + \beta_1 \tilde{x}_i, \tilde{y}_i) = (\beta_0^L + \beta_1 x_i^L - y_i^L)^2 + (\beta_0^R + \beta_1 x_i^R - y_i^R)^2 + (\beta_0 + \beta_1 x_i - y_i)^2, \quad (2.11)$$

and the function f_2 in (2.4) writes

$$f_2(\beta_0^L, \beta_0, \beta_0^R, \beta_1) = \sum_i^n \left((\beta_0^L + \beta_1 x_i^L - y_i^L)^2 + (\beta_0^R + \beta_1 x_i^R - y_i^R)^2 + (\beta_0 + \beta_1 x_i - y_i)^2 \right). \quad (2.12)$$

If $\beta_1 < 0$, the distance d^2 is

$$d^2(\tilde{\beta}_0 + \beta_1 \tilde{x}_i, \tilde{y}_i) = (\beta_0^L + \beta_1 x_i^R - y_i^L)^2 + (\beta_0^R + \beta_1 x_i^L - y_i^R)^2 + (\beta_0 + \beta_1 x_i - y_i)^2, \quad (2.13)$$

which gives

$$f_2(\beta_0^L, \beta_0, \beta_0^R, \beta_1) = \sum_i^n \left((\beta_0^L + \beta_1 x_i^R - y_i^L)^2 + (\beta_0^R + \beta_1 x_i^L - y_i^R)^2 + (\beta_0 + \beta_1 x_i - y_i)^2 \right). \quad (2.14)$$

For $\beta_1 \geq 0$, the first order conditions, $\frac{\partial f_2(\beta_0^L, \beta_0, \beta_0^R, \beta_1)}{\partial \beta_0^L} = 0$, $\frac{\partial f_2(\beta_0^L, \beta_0, \beta_0^R, \beta_1)}{\partial \beta_0^R} = 0$, $\frac{\partial f_2(\beta_0^L, \beta_0, \beta_0^R, \beta_1)}{\partial \beta_0} = 0$ and $\frac{\partial f_2(\beta_0^L, \beta_0, \beta_0^R, \beta_1)}{\partial \beta_1} = 0$, give the system of equations

$$\begin{aligned}
\beta_0 + \beta_1 \bar{x} &= \bar{y} \\
\beta_0^L + \beta_1 \bar{x}^L &= \bar{y}^L \\
\beta_0^R + \beta_1 \bar{x}^R &= \bar{y}^R
\end{aligned} \tag{2.15}$$

$$\begin{aligned}
\beta_0 (\bar{x} + \bar{x}^L + \bar{x}^R) + (\beta_0 - \beta_0^L) \bar{x}^L + (\beta_0^R - \beta_0) \bar{x}^R + \frac{\beta_1}{n} \sum_i^n ((x_i^L)^2 + (x_i)^2 + (x_i^R)^2) \\
= \frac{1}{n} \sum_i^n (x_i^L y_i^L + x_i^R y_i^R + x_i y_i).
\end{aligned}$$

On the other hand, for $\beta_1 < 0$, the system is

$$\begin{aligned}
\beta_0 + \beta_1 \bar{x} &= \bar{y} \\
\beta_0^L + \beta_1 \bar{x}^R &= \bar{y}^L \\
\beta_0^R + \beta_1 \bar{x}^L &= \bar{y}^R
\end{aligned} \tag{2.16}$$

$$\begin{aligned}
\beta_0 (\bar{x} + \bar{x}^L + \bar{x}^R) + (\beta_0 - \beta_0^L) \bar{x}^L + (\beta_0^R - \beta_0) \bar{x}^R + \frac{\beta_1}{n} \sum_i^n ((x_i^L)^2 + (x_i)^2 + (x_i^R)^2) \\
= \frac{1}{n} \sum_i^n (x_i^L y_i^L + x_i^R y_i^R + x_i y_i).
\end{aligned}$$

When $\beta_1 \geq 0$, we note by $\tilde{\beta}_{0,+} = (\beta_{0,+}^L, \beta_{0,+}, \beta_{0,+}^R)$ and $\beta_{1,+}$ the solutions to (2.15). Conversely, for $\beta_1 < 0$, we instead note by $\tilde{\beta}_{0,-} = (\beta_{0,-}^L, \beta_{0,-}, \beta_{0,-}^R)$ and $\beta_{1,-}$ the solutions to (2.16).

For Problem 2, Diamond (1988) [pp. 150-152] established that the minimisation problem (2.4) admits a unique solution if $\beta_{1,-}$ and $\beta_{1,+}$ are either both positive or both negative, and if the product

$$((x_i^R - x_i^L) - (\bar{x}^R - \bar{x}^L))((y_i^R - y_i^L) - (\bar{y}^R - \bar{y}^L)) \geq 0,$$

where $\bar{x}^R = 1/n \sum_i^n x_i^R$, $\bar{x}^L = 1/n \sum_i^n x_i^L$ and similarly for $\bar{y}^R$ and $\bar{y}^L$. Under these conditions, the solution is uniquely determined. Specifically, when $\beta_{1,-} \geq 0$ and $\beta_{1,+} \geq 0$, the minimisation leads to the solution $\beta_{1,+}$. Conversely, if $\beta_{1,-} < 0$ and $\beta_{1,+} < 0$, the solution corresponds to $\beta_{1,-}$. However, in more general cases where $\beta_{1,-}$ and $\beta_{1,+}$ have opposite signs, the problem results in two distinct solutions. This distinction reveals the same limitation as before in Problem 1 in Diamond (1988)'s method, where the existence of multiple solutions complicates the interpretation of the model.

Note that a decade later, Diamond and Körner (1997)[p.20] extended Diamond (1988)'s approach and provided a method to always select an optimal solution.

2.2 Possibilistic linear regression analysis

Let us now introduce the possibilistic linear regression analysis proposed by Tanaka, Hayashi, and Watada (1989). Instead of using a metric to deal with the fuzziness arising from the observations and regression parameters, Tanaka, Hayashi, and Watada (1989) minimise the spreads of the fuzzy dependent variables by considering three distinct problems, namely the minimum, maximum and conjunction problems. A key element in understanding their approach is the concept of inclusion for fuzzy numbers.

Definition 2.2.1. Inclusion of fuzzy numbers

We say that a fuzzy number $\tilde{b}_2$ is included in the fuzzy number $\tilde{b}_1$ at a given α -cut $0 \leq \alpha < 1$ if $\tilde{b}_2^\alpha \subseteq \tilde{b}_1^\alpha$, where $\tilde{b}_1^\alpha = [b_1^L, b_1^R]$ and $\tilde{b}_2^\alpha = [b_2^{L,\alpha}, b_2^{R,\alpha}]$. To simplify some notation, this inclusion will be denoted as $\tilde{b}_2 \subseteq^\alpha \tilde{b}_1$.

In their approach, Tanaka, Hayashi, and Watada (1989) assumed that all fuzzy numbers are **symmetrical**. Recall that, a **symmetric** fuzzy number $\tilde{a}_i = (a_i, \gamma_i)$ is defined at any given α -cut by its centre a_i and the relation $\gamma_i^\alpha = a_i - a_i^{L,\alpha} = a_i^{R,\alpha} - a_i \geq 0$ for all $\alpha \in [0, 1]$, ensuring that the left and right spreads $a_i - a_i^{L,\alpha}$, $a_i^{R,\alpha} - a_i$ are equal for all α forcing $\tilde{a}_i = (a_i, \gamma_i)$ to be symmetrical. Moreover, observe that the spread $\gamma_i^{\alpha=1}$ may be greater than 0. See definition 1.1.14 for further information about symmetric fuzzy numbers. Moreover, the most suitable representation, either with the spreads or left/right endpoints will be of use.

Three formulations of possibilistic linear regressions

With the inclusion of fuzzy number defined above in Definition 2.2.1, three formulations of possibilistic linear regressions can be proposed. Assuming the data is organised in a $n \times (m+1)$ matrix $(\tilde{\mathbf{y}}, \mathbf{X})$, where the first column $\tilde{\mathbf{y}}$ is the fuzzy dependent variable and $\mathbf{X}$ is an $n \times m$ matrix consisting of the crisp covariates $(\mathbf{x}^1, \dots, \mathbf{x}^m)$. The i -th component of the fuzzy dependent variable $\tilde{\mathbf{y}}$ is $\tilde{y}_i = (y_i, \eta_i)$. The i -th line of the matrix of observations $\mathbf{X}$ is written as $\mathbf{x}_i = (x_{i1}, \dots, x_{im})'$. Three different possibilistic linear systems can be enunciated:

$$\text{Minimum Problem} \quad \tilde{y}_i \subseteq^\alpha \hat{\tilde{y}}_i = (\hat{\beta}_1 \otimes x_{i1}) \oplus \dots \oplus (\hat{\beta}_m \otimes x_{im}), \quad (2.17)$$

$$\text{Maximum Problem} \quad \tilde{y}_i \supseteq^\alpha \check{\tilde{y}}_i = (\check{\beta}_1 \otimes x_{i1}) \oplus \dots \oplus (\check{\beta}_m \otimes x_{im}), \quad (2.18)$$

$$\text{Conjunction Problem} \quad \tilde{y}_i^\alpha \cap \check{\tilde{y}}_i^\alpha = (\check{\beta}_1 \otimes x_{i1}) \oplus \dots \oplus (\check{\beta}_m \otimes x_{im}) \neq \emptyset, \quad (2.19)$$

where $\hat{\beta}_j = (\hat{\beta}_j, \hat{\gamma}_j)$, $\check{\beta}_j = (\check{\beta}_j, \check{\gamma}_j)$, $\check{\beta}_j = (\check{\beta}_j, \check{\gamma}_j)$, $j = 1, \dots, m$, are symmetric fuzzy numbers and parameters of the above three possibilistic linear regression problems. Equations (2.17) and (2.18) should be verified for all α -cuts, $\alpha \in [0, 1]$. The first one, equation (2.17) means that the observed fuzzy dependent variable $\tilde{y}_i$ is contained in the fuzzy predicted value $\hat{\tilde{y}}_i$ while equation (2.18) means the converse, that is, the predicted value $\check{\tilde{y}}_i$ lies inside the observed value $\tilde{y}_i$. Lastly, equation (2.19) tells that, each α -cut of the estimated fuzzy number $\check{\tilde{y}}_i^\alpha$

must intersect the observed fuzzy α -cut $\tilde{y}_i^\alpha$. The first problem is called *Minimum Problem* since it aims at finding the smallest sum of spreads $\sum \gamma_j^\alpha$ such that the fuzzy observation $\tilde{y}_i$ is fully covered by the estimated fuzzy number $\hat{\tilde{y}}_i$ (true for all α -cuts), i.e. $\tilde{y}_i \subseteq^\alpha \hat{\tilde{y}}_i$ but can also be understood as finding the minimal interval described by $\hat{\tilde{y}}_i^\alpha$ fully containing the interval expressed by $\tilde{y}_i^\alpha$. By using the fact that the condition (2.17) is expressed in terms of α -cuts, one can rewrite it as the following linear programming :

$$\begin{aligned} \min_{\dot{\gamma}_j^\alpha} \sum_{i=1}^n \sum_{j=1}^m \dot{\gamma}_j^\alpha |x_{ij}| &= \min_{\dot{\gamma}_j^\alpha} J(\dot{\gamma}^\alpha, \mathbf{X}), \\ y_i^{R,\alpha} &\leq \sum_{j=1}^m (\dot{\beta}_j \otimes x_{ij})^R, \\ y_i^{L,\alpha} &\geq \sum_{j=1}^m (\dot{\beta}_j \otimes x_{ij})^L. \end{aligned}$$

Note that, $\gamma = (\gamma_1, \dots, \gamma_m)$ and $\mathbf{X}$ refer to the full data set of observations x_{ij} . Because of the symmetry of $\dot{\beta}_j$, the fuzzy product can be computed as

$$\dot{\beta}_j \otimes x_{ij} = (\dot{\beta}_j x_{ij} - \dot{\gamma}_j^\alpha |x_{ij}|, x_{ij}, \dot{\beta}_j x_{ij} + \dot{\gamma}_j^\alpha |x_{ij}|), \quad (2.20)$$

which yields the linear programming

$$\begin{aligned} \min_{\dot{\gamma}_j^\alpha} \sum_{i=1}^n \sum_{j=1}^m \dot{\gamma}_j^\alpha |x_{ij}| &= \min_{\dot{\gamma}_j^\alpha} J(\dot{\gamma}^\alpha, \mathbf{X}), \\ y_i^{R,\alpha} &\leq \sum_{j=1}^m \dot{\beta}_j x_{ij} + \dot{\gamma}_j^\alpha |x_{ij}|, \\ y_i^{L,\alpha} &\geq \sum_{j=1}^m \dot{\beta}_j x_{ij} - \dot{\gamma}_j^\alpha |x_{ij}|. \end{aligned} \quad (2.21)$$

Conversely, the second problem, is named *Maximum Problem* since it aims at finding the greatest sum of spreads $\sum \gamma_j^\alpha$ such that $\tilde{y}_i \subseteq^\alpha \tilde{\tilde{y}}_i$. Similarly to the *Minimum Problem*, with the use of (2.20), it can be translated to a linear programming:

$$\begin{aligned} \max_{\check{\gamma}_j^\alpha} \sum_{i=1}^n \sum_{j=1}^m \check{\gamma}_j^\alpha |x_{ij}| &= \max_{\check{\gamma}_j^\alpha} \check{J}(\check{\gamma}^\alpha, \mathbf{X}), \\ y_i^{R,\alpha} &\geq \sum_{j=1}^m \check{\beta}_j x_{ij} + \check{\gamma}_j^\alpha |x_{ij}|, \\ y_i^{L,\alpha} &\leq \sum_{j=1}^m \check{\beta}_j x_{ij} - \check{\gamma}_j^\alpha |x_{ij}|. \end{aligned} \quad (2.22)$$

The goal of the third problem, *Conjunction Problem*, is to find the smallest spread at a given α -cut such that the intersection $\tilde{y}_i^\alpha \cap \check{y}_i^\alpha$ holds for all $\alpha \in [0, 1]$. Writing this last problem as a linear programming gives

$$\begin{aligned} \min_{\check{\gamma}_j^\alpha} \sum_{i=1}^n \sum_{j=1}^m \check{\gamma}_j^\alpha |x_{ij}| &= \min_{\check{\gamma}^\alpha} \check{J}(\check{\gamma}^\alpha, \mathbf{X}) \\ y_i^{R,\alpha} &\geq \sum_{j=1}^m \check{\beta}_{ij} - \check{\gamma}_j^\alpha |x_{ij}| \\ y_i^{L,\alpha} &\leq \sum_{j=1}^m \check{\beta}_{ij} + \check{\gamma}_j^\alpha |x_{ij}|. \end{aligned} \quad (2.23)$$

The mathematical results concerning the existence and uniqueness of solutions $\dot{\beta}_j^* = (\dot{\beta}_j^*, \dot{\gamma}_j^*)$, $\ddot{\beta}_j^* = (\ddot{\beta}_j^*, \ddot{\gamma}_j^*)$, $\check{\beta}_j^* = (\check{\beta}_j^*, \check{\gamma}_j^*)$, for $j = 1, \dots, m$ arising from solving the three linear programming problems (2.21), (2.22) and (2.23), as well as their relationships, are presented in Tanaka, Hayashi, and Watada (1989).

2.3 Linear least squares and quadratic programming

It is of common practice to rewrite problems in a familiar way to use a well established method to solve them. Starting from the realisation that constrained weighted linear least squares problems and quadratic programming problems share a similar form, it would be interesting to know if one can reshape a linear least squares problem into a quadratic programming problem and to solve it as such. This question has already been investigated and Caron (2020) presents a detailed answer from the perspective of computer science. By taking inspiration from Caron (2020), let us analyse the connection between a least squares problem and quadratic programming.

2.3.1 Least squares problem

The so called constrained weighted linear least squares problem can be enunciated as follow:

$$\begin{aligned} &\text{minimise}_{\mathbf{x} \in \mathbb{R}^m} && \frac{1}{2} \|\mathbf{R}\mathbf{x} - \mathbf{y}\|_W = \frac{1}{2} (\mathbf{R}\mathbf{x} - \mathbf{y})' \mathbf{W} (\mathbf{R}\mathbf{x} - \mathbf{y}) \\ &\text{subject to} && \mathbf{G}\mathbf{x} \leq \mathbf{h} \\ &&& \mathbf{A}\mathbf{x} = \mathbf{b}, \end{aligned} \quad (2.24)$$

where $\mathbf{x} \in \mathbb{R}^m$ is the column vector of the regression coefficients, $\mathbf{R}$ is an $n \times m$ matrix of covariates, $\mathbf{y} \in \mathbb{R}^n$ is the vector of the dependent variable, $\mathbf{W}$ is a positive semi-definite

$n \times n$ matrix and $\mathbf{G}$, $\mathbf{A}$, $\mathbf{h}$, $\mathbf{b}$ denote the matrices, respectively the vectors defining the set of inequality and equality constraints. The aim of this problem is to minimise the objective function (or cost function) $f : \mathbb{R}^m \rightarrow \mathbb{R}$, $f = \frac{1}{2} \|\mathbf{R}\mathbf{x} - \mathbf{y}\|_{\mathbf{W}}$ subject to constraints. More precisely, one wants to find a vector $\mathbf{x} \in \mathbb{R}^m$ such that the distance between $\mathbf{R}\mathbf{x}$ and $\mathbf{y}$ defined by the weighted norm $\|\cdot\|_{\mathbf{W}}$ is as little as possible under the set of linear constraints expressed by $\mathbf{G}\mathbf{x} \leq \mathbf{h}$ and $\mathbf{A}\mathbf{x} = \mathbf{b}$.

2.3.2 Quadratic programming

A quadratic program is usually written in standard form as

$$\begin{aligned} & \underset{\mathbf{x} \in \mathbb{R}^m}{\text{minimise}} && \frac{1}{2} \mathbf{x}' \mathbf{P} \mathbf{x} + \mathbf{q}' \mathbf{x} \\ & \text{subject to} && \mathbf{G}\mathbf{x} \leq \mathbf{h} \\ & && \mathbf{A}\mathbf{x} = \mathbf{b}, \end{aligned} \tag{2.25}$$

where $\mathbf{P}$ is a symmetric positive semidefinite $m \times m$ matrix and $\mathbf{q} \in \mathbb{R}^m$ is a vector. This problem aims at finding a vector $\mathbf{x}$ that minimises the quadratic objective function $\mathbf{x} \mapsto \frac{1}{2} \mathbf{x}' \mathbf{P} \mathbf{x} + \mathbf{q}' \mathbf{x}$ under the same set of linear constraints as before in (2.24).

By comparing (2.24) and (2.25), notice again that their constraints are the same and that the difference lies in the objective function. We'd like to know if it's possible to write a least square problem as a quadratic programming problem.

Definition 2.3.1. Two objective functions f and g are said to be equivalent if they share the same optimum. Mathematically,

$$f \equiv g \Leftrightarrow \underset{\mathbf{x} \in \mathbb{R}^m}{\operatorname{argmin}} \{f(\mathbf{x}) : \mathbf{G}\mathbf{x} \leq \mathbf{h}, \mathbf{A}\mathbf{x} = \mathbf{b}\} = \underset{\mathbf{x} \in \mathbb{R}^m}{\operatorname{argmin}} \{g(\mathbf{x}) : \mathbf{G}\mathbf{x} \leq \mathbf{h}, \mathbf{A}\mathbf{x} = \mathbf{b}\}.$$

Equivalently, adding a constant or multiplying an objective function by a strictly positive scalar does not alter its set of optimal solutions. That is, two objective functions f and g are equivalent if there exist $k > 0$ and $c \in \mathbb{R}$ such that $g(\mathbf{x}) = k \cdot f(\mathbf{x}) + c$, since such transformations preserve the set of optimal solutions. This notion allows us to simplify objective functions or reformulate optimisation problems while preserving the set of optimal solutions.

2.3.3 Least squares to quadratic programming

Let us show that, it is indeed possible to write a constrained weighted linear least squares problem as a quadratic programming problem. The goal is to write the least squares objective function $f(\mathbf{x}) = \frac{1}{2} \|\mathbf{R}\mathbf{x} - \mathbf{y}\|_{\mathbf{W}}^2$ under the form of a quadratic programming objective function $g(\mathbf{x}) = \frac{1}{2} \mathbf{x}' \mathbf{P} \mathbf{x} + \mathbf{q}' \mathbf{x}$ such that $f \equiv g$. To do so, we first expand the weighted norm,

$$\frac{1}{2} \|\mathbf{R}\mathbf{x} - \mathbf{y}\|_{\mathbf{W}}^2 = \frac{1}{2} (\mathbf{R}\mathbf{x} - \mathbf{y})' \mathbf{W} (\mathbf{R}\mathbf{x} - \mathbf{y}) = \frac{1}{2} (\mathbf{x}' \mathbf{R}' \mathbf{W} \mathbf{R} \mathbf{x} - \mathbf{x}' \mathbf{R}' \mathbf{W} \mathbf{y} - \mathbf{y}' \mathbf{W} \mathbf{R} \mathbf{x} + \mathbf{y}' \mathbf{W} \mathbf{y})$$

Since the term $\mathbf{y}'\mathbf{W}\mathbf{y}$ does not depend on $\mathbf{x}$ or the covariate matrix $\mathbf{R}$, it is constant in the optimisation problem; therefore, removing it does not affect the location of the optimum (see Definition 2.3.1). Hence we can write

$$\frac{1}{2}\|\mathbf{R}\mathbf{x} - \mathbf{y}\|_{\mathbf{W}}^2 \equiv \frac{1}{2}((\mathbf{R}\mathbf{x} - \mathbf{y})'\mathbf{W}(\mathbf{R}\mathbf{x} - \mathbf{y}) = \mathbf{x}'\mathbf{R}'\mathbf{W}\mathbf{R}\mathbf{x} - \mathbf{x}'\mathbf{R}'\mathbf{W}\mathbf{y} - \mathbf{y}'\mathbf{W}\mathbf{R}\mathbf{x}).$$

Given that $\mathbf{W}$ is a symmetric positive semidefinite matrix, we can use the fact that $(\mathbf{y}'\mathbf{W}\mathbf{R}\mathbf{x})' = \mathbf{x}'\mathbf{R}'\mathbf{W}\mathbf{y}$, yielding

$$\frac{1}{2}\|\mathbf{R}\mathbf{x} - \mathbf{y}\|_{\mathbf{W}}^2 \equiv \frac{1}{2}(\mathbf{x}'\mathbf{R}'\mathbf{W}\mathbf{R}\mathbf{x} - 2\mathbf{x}'\mathbf{R}'\mathbf{W}\mathbf{y}).$$

To make it more obvious, with $2\mathbf{x}'\mathbf{R}'\mathbf{W}\mathbf{y} = 2(\mathbf{R}'\mathbf{W}\mathbf{y})'\mathbf{x}$, we can further write

$$\frac{1}{2}\|\mathbf{R}\mathbf{x} - \mathbf{y}\|_{\mathbf{W}}^2 \equiv \frac{1}{2}\mathbf{x}'(\mathbf{R}'\mathbf{W}\mathbf{R})\mathbf{x} - (\mathbf{R}'\mathbf{W}\mathbf{y})'\mathbf{x},$$

which finally gives

$$\frac{1}{2}\mathbf{x}'\mathbf{P}\mathbf{x} + \mathbf{q}'\mathbf{x},$$

where $\mathbf{P} = \mathbf{R}'\mathbf{W}\mathbf{R}$ and $\mathbf{q} = -\mathbf{R}'\mathbf{W}\mathbf{y}$.

2.4 Fuzzy linear regression combining two approaches

Recall that, in section 2.1, similar to the idea behind classical linear regression analysis, Diamond (1988) proposed a fuzzy regression method based on minimising the sum of squared residuals. Then, in section 2.2, Tanaka, Hayashi, and Watada (1989) introduced a fuzzy regression procedure based on minimising the sum of spreads of the fuzzy regression coefficients, which is different than minimising the sum of squared residuals as done by Diamond. Thus, a decade later, Tanaka and Lee (1997) developed another regression method aiming at minimising both the spreads of fuzzy regression coefficients and the sum of squared residuals simultaneously.

2.4.1 Spreads and residuals

Let us consider again the fuzzy regression model expressed by

$$\tilde{y}_i = \tilde{\beta}_1 x_{i1} \oplus \tilde{\beta}_2 x_{i2} \oplus \dots \oplus \tilde{\beta}_m x_{im}$$

where the dependent variable $\tilde{y}_i = (y_i, \eta_i)$, $i = 1, \dots, n$ and regression coefficients $\tilde{\beta}_j = (\beta_j, \gamma_j)$, $j = 1, \dots, m$, are fuzzy symmetric numbers while the observations x_{ij} are real numbers. Previously, in section 2.2, this problem has been solved by minimising the following objective function consisting of the sum of spreads,

$$J = \sum_{i=1}^n \sum_{j=1}^m \gamma_j |x_{ij}|. \quad (2.26)$$

However, by using a quadratic version of the above objective function,

$$J = \sum_{i=1}^n \left(|\mathbf{x}_i|' \boldsymbol{\gamma} \right)^2 = \sum_{i=1}^n \left(\boldsymbol{\gamma}' (|\mathbf{x}_i| |\mathbf{x}_i|') \boldsymbol{\gamma} \right) \quad (2.27)$$

where $\boldsymbol{\gamma}' = (\gamma_1, \dots, \gamma_m)$, $\mathbf{x}_i' = (x_{i1}, \dots, x_{im})$ and $\mathbf{x}_i \mathbf{x}_i'$ is a positive definite matrix. By minimising (2.27) by means of quadratic programming, Tanaka and Lee empirically realised that the found regression coefficients remained fuzzy more often than the ones found by minimising (2.26) using linear programming. From this empirical realisation, they got the idea to combine the central tendency expressed by the minimisation of the sum of squared residuals,

$$\sum_{i=1}^n (\mathbf{x}_i' \boldsymbol{\beta} - y_i)^2,$$

with the previous quadratic objective function (2.27). To achieve it, consider the following quadratic objective function, where $\boldsymbol{\beta}' = (\beta_1, \dots, \beta_m)$

$$J = k_1 \sum_{i=1}^n (\mathbf{x}_i' \boldsymbol{\beta} - y_i)^2 + k_2 \sum_{i=1}^n \left(\boldsymbol{\gamma}' (|\mathbf{x}_i| |\mathbf{x}_i|') \boldsymbol{\gamma} \right), \quad (2.28)$$

which intertwines both the central tendency of the least squares and the fuzzy aspect of the possibilistic linear regression 2.2. Parameters k_1 and k_2 control the amount of weight given to the central tendency, respectively to the minimisation of the spreads. Generally, $k_1 = k_2 = 1$. If $k_2 = 0$, (2.28) reduces to a least square problem and if $k_1 = 0$, (2.28) becomes again (2.27). Our goal is to determine the regression coefficients $\tilde{\beta}_j = (\beta_j, \gamma_j)$ that minimise the quadratic objective function (2.28). This optimisation is performed while maintaining both the regression coefficients and the dependent variables $\tilde{y}_i$ as fuzzy numbers. Specifically, the following constraints must hold to ensure that both $\tilde{y}$ and $\tilde{\beta}_j$ remain fuzzy numbers:

$$\hat{y}_i^L \leq y_i \leq \hat{y}_i^R, \quad \forall i = 1, \dots, n \quad \text{and} \quad \gamma_j \geq 0, \quad \forall j = 1, \dots, m.$$

By recognising that $\hat{y}_i^R = \boldsymbol{\beta}' \mathbf{x}_i + \boldsymbol{\gamma}' |\mathbf{x}_i|$ and $\hat{y}_i^L = \boldsymbol{\beta}' \mathbf{x}_i - \boldsymbol{\gamma}' |\mathbf{x}_i|$, (see Tanaka and Lee (1997)[p.64]), these constraints can be reformulated as:

$$\boldsymbol{\beta}' \mathbf{x}_i + \boldsymbol{\gamma}' |\mathbf{x}_i| \geq y_i, \quad (2.29)$$

$$\boldsymbol{\beta}' \mathbf{x}_i - \boldsymbol{\gamma}' |\mathbf{x}_i| \leq y_i \quad \forall i = 1, \dots, n, \quad (2.30)$$

$$\gamma_j \geq 0 \quad \forall j = 1, \dots, m. \quad (2.31)$$

Here, (2.29) and (2.30) ensure that $y_i^{L,\alpha} \leq y_i \leq y_i^{R,\alpha}$, while (2.31) guarantees that the spreads remain non-negative.

This problem can thus be written as

$$\text{Minimise} \quad k_1 \sum_{i=1}^n (\beta \mathbf{x}_i - y_i)^2 + k_2 \sum_{i=1}^n \gamma' (|\mathbf{x}_i| |\mathbf{x}_i'|) \gamma \quad (2.32)$$

$$\text{Subject to} \quad \beta' \mathbf{x}_i + \gamma' |\mathbf{x}_i| \geq y_i \quad (2.33)$$

$$\beta' \mathbf{x}_i - \gamma' |\mathbf{x}_i| \leq y_i \quad \forall i = 1, \dots, n \quad (2.34)$$

$$\gamma_j \geq 0 \quad \forall j = 1, \dots, m. \quad (2.35)$$

Due to the quadratic nature of the objective function, it can easily be solved using a quadratic solver. In relation to subsection 2.3.2, to obtain the vector of solutions $(\beta_1, \dots, \beta_m, \gamma_1, \dots, \gamma_m)'$, it is necessary to compute the matrices $\mathbf{P}$, $\mathbf{G}$ and the vectors $\mathbf{q}$, $\mathbf{h}$ as defined in (2.25), while ensuring the constraints (2.33), (2.34), and (2.35) are satisfied. The matrix $\mathbf{P}$ and the vector $\mathbf{q}$ are given by:

$$\mathbf{P} = \begin{bmatrix} \mathbf{X}'\mathbf{X} & \mathbf{0}_{m \times m} \\ \mathbf{0}_{m \times m} & \mathbf{X}'\mathbf{X} \end{bmatrix} \quad \text{and} \quad \mathbf{q} = \begin{bmatrix} -\mathbf{X}'\mathbf{y} \\ \mathbf{0}_m \end{bmatrix}, \quad (2.36)$$

where $\mathbf{X}$ is an $n \times m$ matrix containing the observations x_{ij} , for $i = 1, \dots, n$ and $j = 1, \dots, m$, $\mathbf{0}_{m \times m}$ is an $m \times m$ zero matrix, $\mathbf{y} = (y_1, \dots, y_n)'$, and $\mathbf{0}_m$ is an m -dimensional zero vector.

For the constraints, the matrix $\mathbf{G}$ and the vector $\mathbf{h}$ can be written as follows, where $\mathbf{I}_{m \times m}$ represents the identity matrix of size $m \times m$:

$$\mathbf{G} = \begin{bmatrix} \mathbf{X} & \mathbf{X} \\ \mathbf{X} & -\mathbf{X} \\ \mathbf{0}_{m \times m} & -\mathbf{I}_{m \times m} \end{bmatrix} \quad \text{and} \quad \mathbf{h} = \begin{bmatrix} \mathbf{y} \\ \mathbf{y} \\ \mathbf{0}_m \end{bmatrix}. \quad (2.37)$$

2.5 Fuzzy least squares

While the method presented by Tanaka and Lee (1997) is able to combine the central tendency and possibilistic aspects of the first two regression methods discussed in this chapter, one problem persists: The approach does not permit to study sets of fully fuzzy observations. Indeed, one can observe that, only crisp covariates are considered. Rosset and Donzé (2023) attempted to face this problem and to further extend the fuzzy least squares by proposing regression models involving the use of fuzzy covariates. One of the main issue appearing when attempting to generalise fuzzy least squares to fully fuzzy observations is the emergence of the fuzzy product operator in the regression model. A key innovation of Rosset and Donzé (2023) is to replace the minimisation of the spreads of the fuzzy fitted values with the minimisation of the left, central, and right components of the sum of squared fuzzy residuals. This strategy not only provides a more accurate treatment of fuzzy multiplication but also naturally allows for non-symmetrical fuzzy numbers, thereby overcoming a critical limitation of previous methods.

2.5.1 Two fuzzy regression models

Fuzzy vectors and matrices containing fuzzy elements will be widely used in this section. To avoid any confusion, let us first remind some important notation. First of all, in what follows, triangular fuzzy numbers of the form $\tilde{y} = (y^L, y, y^R)$ will be of concern. Lowercase bold letters with no index will designate n -components column vectors of fuzzy numbers, e.g. $\tilde{\mathbf{y}} = (\tilde{y}_1, \dots, \tilde{y}_n)'$. A $(n \times m)$ -matrix is noted in capital bold letters, e.g. $\tilde{\mathbf{X}}$, with the i -th line and j -th column given respectively by $\tilde{\mathbf{x}}'_i = (\tilde{x}_{i1}, \dots, \tilde{x}_{im})$, $i = 1, \dots, n$, and $\tilde{\mathbf{x}}^j = (\tilde{x}_{1j}, \dots, \tilde{x}_{nj})'$, $j = 1, \dots, m$. A specific element of $\tilde{\mathbf{X}}$ is denoted by $(x_{ij}^L, x_{ij}, x_{ij}^R)$. The j -th fuzzy regression parameter is expressed by $\tilde{\beta}_j = (\beta_j^L, \beta_j, \beta_j^R)$, $j = 1, \dots, m$. $\tilde{\beta}$ denotes the fuzzy vector of fuzzy regression coefficients. Assuming crisp observations are stored in an $(n \times m)$ -matrix $\mathbf{X}$ and the fuzzy ones in an $(n \times m)$ -matrix $\tilde{\mathbf{X}}$, by using the above notation, let us express the following two fuzzy regression models

$$\text{Model 1:} \quad \tilde{y}_i = \sum_{j=1}^m \tilde{\beta}_j x_{ij}, \quad i = 1, \dots, n, \quad (2.38)$$

$$\text{Model 2:} \quad \tilde{y}_i = \sum_{j=1}^m \tilde{\beta}_j \otimes \tilde{x}_{ij}, \quad i = 1, \dots, n, \quad (2.39)$$

where the indices i and j denote respectively the i -th observation and the j -th covariate, $i = 1, \dots, n$, $j = 1, \dots, m$. Additionally, assume that the first covariate is an n -vector of ones, $(1, 1, \dots, 1)$, representing the intercept.

Remark 2.5.1.

Observe that, in Model 1 the covariates are crisp while the dependent variable and regression coefficients are fuzzy. By assuming positive observations x_{ij} , the product of a fuzzy number with a positive real number is simple to compute, $\tilde{\beta}_j x_{ij} = (\beta_j^L x_{ij}, \beta_j x_{ij}, \beta_j^R x_{ij})$, hence why the fuzzy product operator is not appearing in this case. Model 2 goes further by assuming fuzzy independent variables. When both the independent variables and regression coefficients are fuzzy, the use of the fuzzy product operator $\otimes$ is mandatory.

Model 1 When the observations are crisp, we come back to the regression problem discussed in sections 2.1, 2.2 and 2.4. Therefore, the solution of this problem depends on the choice of the objective function. Tanaka and Lee (1997) observed that the solution involving a quadratic cost function was giving better results in the sense that the regression parameters were less often becoming crisp which allowed to retain more of the fuzzy information. With this in mind, to solve this problem, let us propose a slightly modified version of Tanaka and Lee's quadratic objective function 2.28 which is inspired from the cost function introduced by Donoso, Marín, and Villa (2006):

$$\begin{aligned} J &= \sum_{i=1}^n \left(k_1 (y_i - \sum_{j=1}^m \beta_j x_{ij})^2 + k_2 (y_i^L - \sum_{j=1}^m \beta_j^L x_{ij})^2 + k_3 (y_i^R - \sum_{j=1}^m \beta_j^R x_{ij})^2 \right) \\ &= k_1 \sum_{i=1}^n (y_i - \mathbf{x}'_i \boldsymbol{\beta})^2 + k_2 \sum_{i=1}^n (y_i^L - \mathbf{x}'_i \boldsymbol{\beta}^L)^2 + k_3 \sum_{i=1}^n (y_i^R - \mathbf{x}'_i \boldsymbol{\beta}^R)^2. \end{aligned} \quad (2.40)$$

Remark 2.5.2.

Notice that, unlike (2.28) which was minimising both the central residuals,

$$\sum_{i=1}^n (\beta \mathbf{x}_i - y_i)^2$$

and the spreads

$$\sum_{i=1}^n (|\mathbf{x}'_i| \gamma)^2 = \sum_{i=1}^n \gamma' (|\mathbf{x}_i| |\mathbf{x}'_i|) \gamma,$$

the cost function (2.40) focuses on the minimising the left, central and right residuals. The parameters k_1 , k_2 and k_3 , usually equal to 1, control the amount of weight we give to the left, central and right quadratic errors respectively.

Minimising (2.40) under the usual constraints of having non-negative spreads,

$$(\beta_j - \beta_j^L), (\beta_j^R - \beta_j) \geq 0 \text{ for } j = 1, \dots, m, \quad (2.41)$$

can be easily solved by quadratic programming. The details can be found in Appendix A.2.

Model 2 Let us now discuss a more general case, when all the observations are fuzzy, that is $\tilde{y}_i = (y_i^L, y_i, y_i^R)$, $\tilde{x}_{ij} = (x_{ij}^L, x_{ij}, x_{ij}^R)$, $i = 1, \dots, n$, $j = 1, \dots, m$, which yields Model 2:

$$\tilde{y}_i = \sum_{j=1}^m \tilde{\beta}_j \otimes \tilde{x}_{ij}, \quad i = 1, \dots, n.$$

Now that the covariates are fuzzy, the objective function (2.40) needs to be rewritten in consequence. Since both the regression parameters and observations are fuzzy, the cost function associated to Model 2 becomes

$$\begin{aligned} J = k_1 \sum_{i=1}^n \left(y_i - \sum_{j=1}^m (x_{ij} \beta_j) \right)^2 &+ k_2 \sum_{i=1}^n \left(y_i^L - \sum_{j=1}^m (\tilde{x}_{ij} \otimes \tilde{\beta}_j) \right)^2 \\ &+ k_3 \sum_{i=1}^n \left(y_i^R - \sum_{j=1}^m (\tilde{x}_{ij} \otimes \tilde{\beta}_j) \right)^2. \end{aligned} \quad (2.42)$$

Using the fuzzy multiplication operator's rule, (2.42) becomes

$$\begin{aligned} J = \sum_{i=1}^n \left(k_1 (y_i - \sum_{j=1}^m \beta_j x_{ij})^2 \right. \\ + k_2 (y_i^L - \sum_{j=1}^m \min(\beta_j^L x_{ij}^L, \beta_j^R x_{ij}^L, \beta_j^L x_{ij}^R, \beta_j^R x_{ij}^R))^2 \\ \left. + k_3 (y_i^R - \sum_{j=1}^m \max(\beta_j^L x_{ij}^L, \beta_j^R x_{ij}^L, \beta_j^L x_{ij}^R, \beta_j^R x_{ij}^R))^2 \right). \end{aligned} \quad (2.43)$$

The following problem arises. One has to find the minimum and maximum while the parameters $\hat{\beta}_j$ are unknown. A way to solve this problem is to consider the estimates of the regression coefficients given by Model 1, $\hat{\beta}_j$ and retrieving, for all $i = 1, \dots, n$, $j = 1, \dots, m$, the observations $x_{ij}^{\min}$, $x_{ij}^{\max}$ that minimise, respectively maximise

$$(\hat{\beta}_j^L x_{ij}^L, \hat{\beta}_j^R x_{ij}^L, \hat{\beta}_j^L x_{ij}^R, \hat{\beta}_j^R x_{ij}^R)$$

Then, build two matrices $X^{\min}$ and $X^{\max}$ whose elements are respectively $x_{ij}^{\min}$ and $x_{ij}^{\max}$. With these reorganised observations, the cost function finally writes as

$$J = \sum_{i=1}^n \left(k_1 (y_i - \sum_{j=1}^m \beta_j x_{ij})^2 + k_2 (y_i^L - \sum_{j=1}^m \beta_j^{\min} x_{ij}^{\min})^2 + k_3 (y_i^R - \sum_{j=1}^m \beta_j^{\max} x_{ij}^{\max})^2 \right). \quad (2.44)$$

The minimisation problem thus writes as

$$\min_{\beta^{\min}, \beta, \beta^{\max}} J = \sum_{i=1}^n \left(k_1 (y_i - \sum_{j=1}^m \beta_j x_{ij})^2 + k_2 (y_i^L - \sum_{j=1}^m \beta_j^{\min} x_{ij}^{\min})^2 + k_3 (y_i^R - \sum_{j=1}^m \beta_j^{\max} x_{ij}^{\max})^2 \right)$$

under the constraints

$$(\beta_j - \beta_j^{\min}), (\beta_j^{\max} - \beta_j) \geq 0 \text{ for } j = 1, \dots, m.$$

The methodology to solve it with quadratic programming is shown in Appendix A.2. Let us point out that, while there may be cases where

$$\begin{aligned} \min(\hat{\beta}_j^L x_{ij}^L, \hat{\beta}_j^R x_{ij}^L, \hat{\beta}_j^L x_{ij}^R, \hat{\beta}_j^R x_{ij}^R) &\neq \min(\beta_j^L x_{ij}^L, \beta_j^R x_{ij}^L, \beta_j^L x_{ij}^R, \beta_j^R x_{ij}^R) \\ \max(\hat{\beta}_j^L x_{ij}^L, \hat{\beta}_j^R x_{ij}^L, \hat{\beta}_j^L x_{ij}^R, \hat{\beta}_j^R x_{ij}^R) &\neq \max(\beta_j^L x_{ij}^L, \beta_j^R x_{ij}^L, \beta_j^L x_{ij}^R, \beta_j^R x_{ij}^R) \end{aligned} \quad (2.45)$$

this rearrangement of observations helps to better approximate the fuzzy multiplication operator.

2.5.2 Numerical examples

To better understand the two regression models we introduced, let us apply them in a small numerical application. Consider the left, middle and right components of the fuzzy dependent

variable $\tilde{y}$ given by the three vectors,

$$\begin{aligned} \mathbf{y}^L &= (15, 20, 15, 25, 25, 40, 55, 70, 80, 90, 115, 140, 155, 175, 200, 240, 270, 300, \\ &\quad 340, 380, 420, 465)', \\ \mathbf{y} &= (22.50, 28.75, 25.00, 42.50, 40.00, 52.50, 75.00, 85.00, 105.00, 120.00, 145.00, 167.50, \\ &\quad 187.50, 212.50, 240.00, 275.00, 305.00, 342.50, 380.00, 420.00, 460.00, 507.50)', \\ \mathbf{y}^R &= (30, 37.5, 35, 60, 55, 65, 95, 100, 130, 150, 175, 195, 220, 250, 280, 310, 340, \\ &\quad 385, 420, 460, 500, 550)'. \end{aligned}$$

Then, consider the covariate

$$\mathbf{x}^1 = (1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22)'.$$

To fuzzify $\mathbf{x}^1$, we need to generate its left and right components. To do so, denote by u_{1i} , u_{2i} , for $i = 1, \dots, 22$, the realisations of the vector of random uniform variables $\mathbf{u}_1 \sim U(0.3, 0.7)$ and $\mathbf{u}_2 \sim U(0.4, 0.9)$. The left and right components are then given by $\mathbf{x}^{1,L} = \mathbf{x}^1 - \mathbf{u}_1$, $\mathbf{x}^{1,R} = \mathbf{x}^1 + \mathbf{u}_2$. This way, we created a vector of **TFNs** $\tilde{\mathbf{x}}_1 = (\mathbf{x}^{1,L}, \mathbf{x}^1, \mathbf{x}^{1,R})$. Then, by making use of the fuzzy product, we can build a second covariate $\tilde{\mathbf{x}}^2 = (\mathbf{x}^{2,L}, \mathbf{x}^2, \mathbf{x}^{2,R})$, where

$$\mathbf{x}^{2,L} = ((\tilde{x}_{1,1} \otimes \tilde{x}_{1,1})^L, \dots, (\tilde{x}_{i,1} \otimes \tilde{x}_{i,1})^L, \dots, (\tilde{x}_{22,1} \otimes \tilde{x}_{22,1})^L)', \quad (2.46)$$

$$\mathbf{x}^2 = (x_{1,1}^2, \dots, x_{i,1}^2, \dots, x_{22,1}^2)', \quad (2.47)$$

$$\mathbf{x}^{2,R} = ((\tilde{x}_{1,1} \otimes \tilde{x}_{1,1})^R, \dots, (\tilde{x}_{i,1} \otimes \tilde{x}_{i,1})^R, \dots, (\tilde{x}_{22,1} \otimes \tilde{x}_{22,1})^R)'. \quad (2.48)$$

Consider first the fuzzy linear regression model based on the observations presented in Table 2.1, formulated as:

$$\tilde{y}_i = \tilde{\beta}_0 \oplus (\tilde{\beta}_1 \otimes x_{i1}) \oplus (\tilde{\beta}_2 \otimes x_{i2}).$$

where we used crisp positive covariates x_{i1} and x_{i2} . The fuzzy multiplication writes as:

$$\tilde{\beta}_1 \otimes x_{i1} = (\beta_1^L x_{i1}, \beta_1 x_{i1}, \beta_1^R x_{i1}),$$

$$\tilde{\beta}_2 \otimes x_{i2} = (\beta_2^L x_{i2}, \beta_2 x_{i2}, \beta_2^R x_{i2}).$$

Thus, the regression model can be rewritten as:

$$\tilde{y}_i = \tilde{\beta}_0 \oplus \tilde{\beta}_1 \otimes x_{i1} \oplus \tilde{\beta}_2 \otimes x_{i2}. \quad (2.49)$$

Using the estimation procedure corresponding to Model 1 (2.38) and incorporating the constraints (2.41), we obtain the constrained fuzzy parameter estimates:

$$\hat{\tilde{\beta}} = \begin{pmatrix} (15.673701, 22.605519, 29.537338) \\ (-2.160238, -0.376641, 1.406956) \\ (1.010993, 1.010993, 1.010993) \end{pmatrix}. \quad (2.50)$$

The fitted fuzzy responses $\hat{y}_i = (\hat{y}_i^L, \hat{y}_i, \hat{y}_i^R)$, computed as $\hat{y}_i = \hat{\beta}_0 \oplus \hat{\beta}_1 \otimes x_{i1} \oplus \hat{\beta}_2 \otimes x_{i2}$, are depicted in Figure 2.3. As expected, it holds that $\hat{y}_i^L \leq \hat{y}_i \leq \hat{y}_i^R$ for all i .

If the constraints (2.41) are relaxed, the resulting unconstrained solution is:

$$\hat{\beta}^* = \begin{pmatrix} (20.032468, 22.605519, 25.178571) \\ (-3.249929, -0.376641, 2.496647) \\ (1.058371, 1.010993, 0.963615) \end{pmatrix}. \quad (2.51)$$

Note that in $\hat{\beta}_2^* = (1.058371, 1.010993, 0.963615)$, the left component exceeds the right. This is acceptable if we interpret (2.51) as a fuzzy vector (see Definition 1.3.2), since the fuzzy multiplication preserves the order:

$$(1.058371, 1.010993, 0.963615) \otimes x_{i2} = (0.963615 \cdot x_{i2}, 1.010993 \cdot x_{i2}, 1.058371 \cdot x_{i2}).$$

Observe that, since $x_{i2} \geq 0$, after the fuzzy multiplication, the left component $0.963615 \cdot x_{i2}$ is smaller than the right component $1.058371 \cdot x_{i2}$. The resulting fitted values, computed as

$$\hat{y}_i = \hat{\beta}_0^* + \hat{\beta}_1^* \otimes x_{i1} + \hat{\beta}_2^* \otimes x_{i2},$$

are shown in Figure 2.2. Note that, although the left and right bounds of the fuzzy regression lines still enclose the observations adequately, the constrained solution provides a better fit than the unconstrained solution interpreted as a fuzzy vector. A tighter fit is attainable by explicitly using the initial left and right endpoints of $\hat{\beta}_2$, even though its left part is greater than its right part. That is, we define:

$$\begin{aligned} \hat{y}_i^L &= \hat{\beta}_0^{*,L} + \hat{\beta}_1^{*,L} x_{i1} + \hat{\beta}_2^{*,L} x_{i2}, \\ \hat{y}_i^R &= \hat{\beta}_0^{*,R} + \hat{\beta}_1^{*,R} x_{i1} + \hat{\beta}_2^{*,R} x_{i2}. \end{aligned}$$

As shown in Figure 2.1, these produce regression bounds that better capture the spread of fuzzy observations at the cost of not preserving the structure of the fuzzy number $\hat{\beta}_2$. Nevertheless, the constrained and above unconstrained solutions are minimal, suggesting that the constrained version guaranteeing that all regression coefficients are proper fuzzy numbers is generally preferable.

Next, consider the case where the covariates themselves are fuzzy. The regression model becomes:

$$\tilde{y}_i = \tilde{\beta}_0 + (\tilde{\beta}_1 \otimes \tilde{x}_{i1}) + (\tilde{\beta}_2 \otimes \tilde{x}_{i2}). \quad (2.52)$$

Using the procedure described for Model 2 (2.39), the constrained solution is:

$$\hat{\beta} = \begin{pmatrix} (18.002572, 22.796708, 27.097999) \\ (-1.386091, -0.424438, 0.257017) \\ (1.013071, 1.013071, 1.013071) \end{pmatrix}, \quad (2.53)$$

and the corresponding unconstrained solution is:

$$\hat{\beta}^* = \begin{pmatrix} (21.094245, 22.605519, 24.063150) \\ (-2.016111, -0.376641, 0.941183) \\ (1.041281, 1.010993, 0.984880) \end{pmatrix}. \quad (2.54)$$

Compared to model (2.49), the coefficients in model (2.52) exhibit reduced fuzziness. For analysis, we consider two forms of fitted values:

1. Using the endpoints of the crisp values:

$$\hat{y}_i^L = \hat{\beta}_0^L + \hat{\beta}_1^L x_{i1} + \hat{\beta}_2^L x_{i2}, \quad (2.55)$$

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_{i1} + \hat{\beta}_2 x_{i2}, \quad (2.56)$$

$$\hat{y}_i^R = \hat{\beta}_0^R + \hat{\beta}_1^R x_{i1} + \hat{\beta}_2^R x_{i2}, \quad (2.57)$$

2. Using fuzzy arithmetic directly:

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 \otimes \tilde{x}_{i1} + \hat{\beta}_2 \otimes \tilde{x}_{i2}. \quad (2.58)$$

In the constrained case, because $\tilde{x}_{i1}$ and $\tilde{x}_{i2}$ are positive fuzzy numbers, the results from (2.55) and (2.58) are equivalent. Figure 2.4 presents the fitted values obtained via the constrained solution. For comparison, Figure 2.5 shows the fitted values derived from the unconstrained solution (2.54), following the decomposition in (2.55). Once again, the difference between the constrained and unconstrained solutions is barely noticeable. In both cases, however, the fitted values fail to adequately capture the left and right bounds of the fuzzy observations, represented by the light blue rectangles with vertices at (x_{i1}^L, y_i^L) , (x_{i1}^L, y_i^R) , (x_{i1}^R, y_i^L) , and (x_{i1}^R, y_i^R) .

By contrast, applying the fuzzy product to compute the fitted values, as in (2.58), yields a significantly improved result, illustrated in Figure 2.6. This improvement aligns with the intuition that, in the context of fuzzy regression with both fuzzy independent and dependent variables, the fuzzy product should play a central role in estimating fitted values.

Previously, in the solution of Model 1, we observed that the fitted values $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 \otimes x_{i1} + \hat{\beta}_2 \otimes x_{i2}$ were less accurate than those obtained from the constrained solution that respects the conditions in (2.41). In contrast, for Model 2, the opposite holds true: the fuzzy product-based fitted values provided the best approximation.

In summary, when working with crisp covariates and a fuzzy dependent variable, the constrained solution is generally more appropriate. However, when both covariates and the dependent variable are fuzzy, the unconstrained approach using a fuzzy vector representation proves to be more effective.

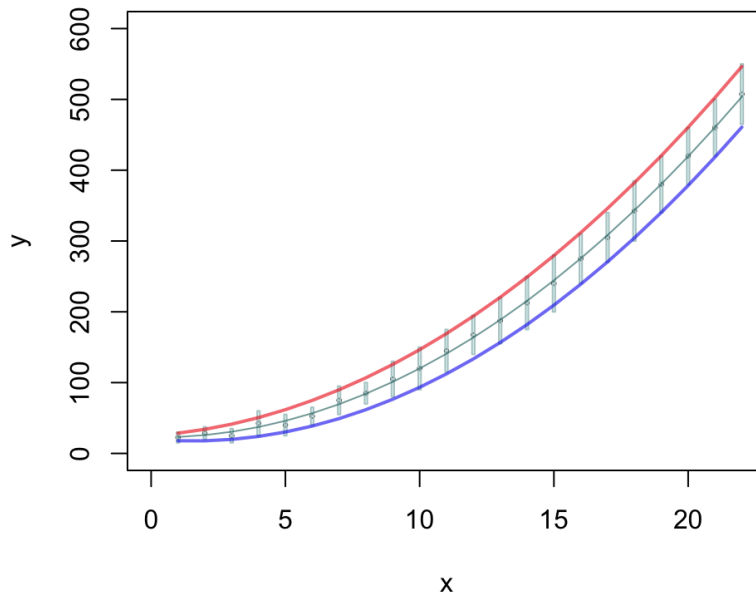


Figure 2.1: Plot of the fitted values $\hat{y}_i = (\hat{y}_i^L, \hat{y}_i, \hat{y}_i^R)$ given by the unconstrained solution (2.1) where $y_i^L = \hat{\beta}_0^{*,L} + \hat{\beta}_1^{*,L}x_{i1} + \hat{\beta}_2^{*,L}x_{i2}$, $y_i = \hat{\beta}_0^* + \hat{\beta}_1^*x_{i1} + \hat{\beta}_2^*x_{i2}$ and $y_i^R = \hat{\beta}_0^{*,R} + \hat{\beta}_1^{*,R}x_{i1} + \hat{\beta}_2^{*,R}x_{i2}$. The left, middle and right regression lines are respectively coloured in blue, turquoise and red. Fuzzy observations are represented by light blue segments between the points $(x_{i1}; y_i^L)$ and $(x_{i1}; y_i^R)$.

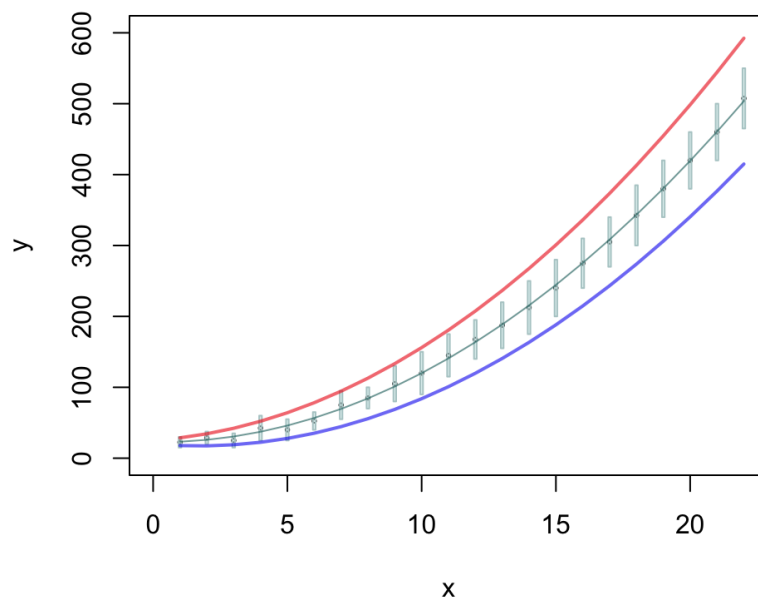


Figure 2.2: Plot of the fitted values $\hat{\mathbf{y}}_i = \hat{\beta}_0^* + (\hat{\beta}_1^* \otimes \mathbf{x}^1) + (\hat{\beta}_2^* \otimes \mathbf{x}^2)$ given by the unconstrained solution (2.51). The left, middle and right regression lines are respectively coloured in blue, turquoise and red. Fuzzy observations are represented by light blue segments between the points $(x_{i1}; y_i^L)$ and $(x_{i1}; y_i^R)$.

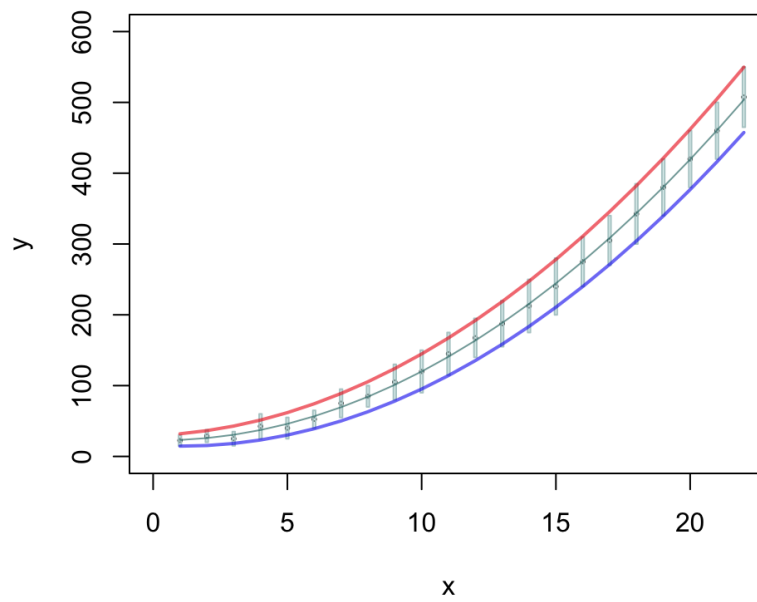


Figure 2.3: Plot of the fitted values $\hat{y}_i$ given by the constrained solution (2.50). The left, middle and right regression lines given by the left, middle and right parts of $\hat{y}_i = \hat{\beta}_0 \oplus \hat{\beta}_1 \otimes x_{i1} \oplus \hat{\beta}_2 \otimes x_{i2}$ are respectively coloured in blue, turquoise and red. Fuzzy observations are represented by light blue segments between the points $(x_{i1}; y_i^L)$ and $(x_{i1}; y_i^R)$.

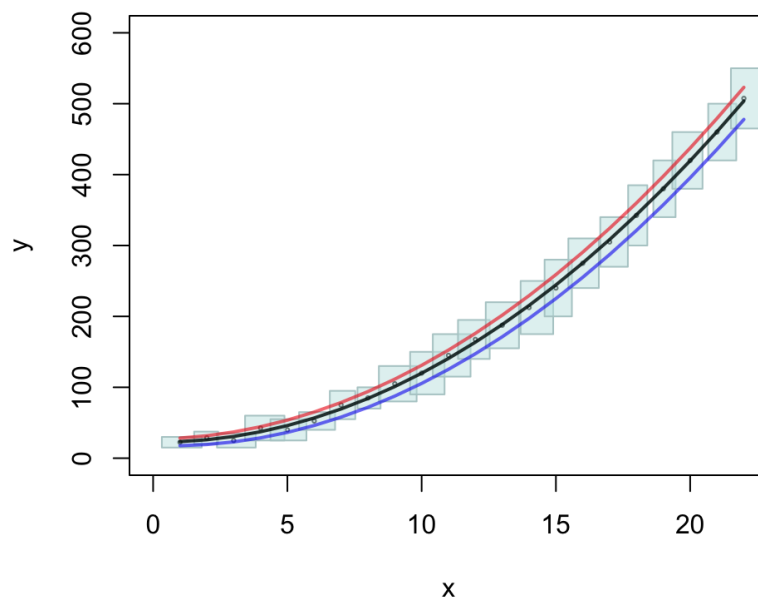


Figure 2.4: Plot of the fitted values $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 \otimes \tilde{x}_{i1} + \hat{\beta}_2 \otimes \tilde{x}_{i2}$. The left, middle and right regression lines are respectively coloured in blue, dark grey and red. Fuzzy observations are represented by light blue rectangles whose vertices are $(x_{i1}^L; y_i^L)$, $(x_{i1}^L; y_i^R)$, $(x_{i1}^R; y_i^L)$, $(x_{i1}^R; y_i^R)$.

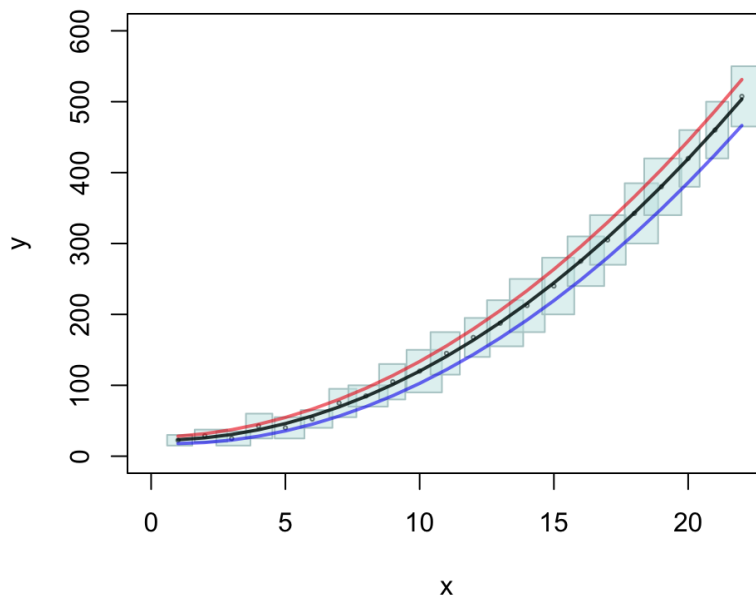


Figure 2.5: Plot of the fitted values $\hat{y}_i^L = \hat{\beta}_0^L + \hat{\beta}_1^L x_{i1}^L + \hat{\beta}_2^L x_{i2}^L$, $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_{i1} + \hat{\beta}_2 x_{i2}$, $\hat{y}_i^R = \hat{\beta}_0^R + \hat{\beta}_1^R x_{i1}^R + \hat{\beta}_2^R x_{i2}^R$. The left, middle and right regression lines are respectively coloured in blue, dark grey and red. Fuzzy observations are represented by light blue rectangles whose vertices are $(x_{i1}^L; y_i^L)$, $(x_{i1}^L; y_i^R)$, $(x_{i1}^R; y_i^L)$, $(x_{i1}^R; y_i^R)$.

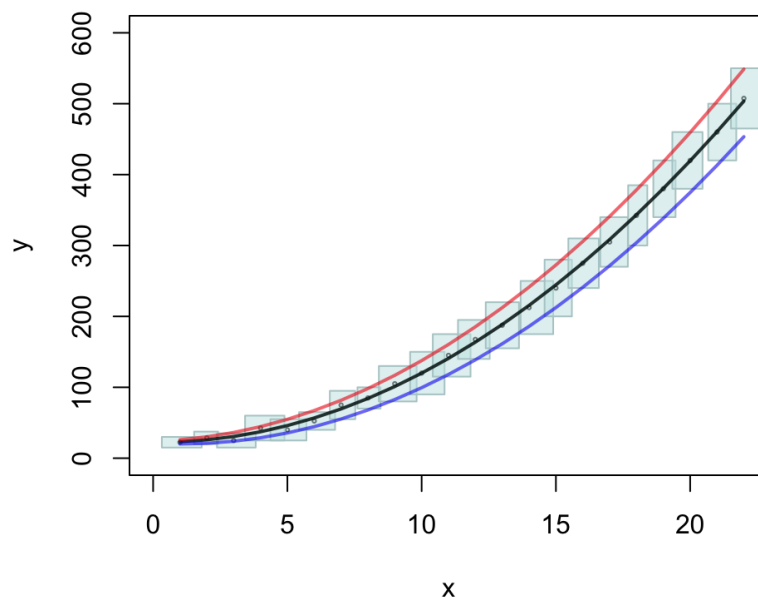


Figure 2.6: Plot of the fitted values $\hat{\mathbf{y}} = \hat{\beta}_0^* + (\hat{\beta}_1^* \otimes \mathbf{x}^1) + (\hat{\beta}_2^* \otimes \mathbf{x}^2)$. The left, middle and right regression lines are respectively coloured in blue, dark grey and red. Fuzzy observations are represented by light blue rectangles whose vertices are $(x_{i1}^L; y_i^L)$, $(x_{i1}^L; y_i^R)$, $(x_{i1}^R; y_i^L)$, $(x_{i1}^R; y_i^R)$.

i	y^L	y	y^R	$x^{1,L}$	x^1	$x^{1,R}$	$x^{2,L}$	x^2	$x^{2,R}$
1	15	22.50	30	1.09	1	1.72	1.188	1.000	2.958
2	20	28.75	37.50	2.35	2	2.50	5.528	4.000	6.250
3	15	25.00	35	3.12	3	3.61	9.718	9.000	13.072
4	25	42.50	60	4.18	4	4.48	16.982	16.000	20.070
5	25	40.00	55	5.13	5	5.78	26.309	25.000	30.112
6	40	52.50	65	6.09	6	6.41	37.128	36.000	41.056
7	55	75.00	95	7.39	7	7.63	54.530	49.000	58.303
8	70	85.00	100	8.28	8	8.63	68.430	64.000	74.477
9	80	105.00	130	9.18	9	9.73	83.130	81.000	94.352
10	90	120.00	150	10.20	10	10.55	104.095	100.000	111.699
11	115	145.00	175	11.25	11	11.67	126.651	121.000	136.421
12	140	167.50	195	12.12	12	12.73	144.996	144.000	162.883
13	155	187.50	220	13.20	13	13.65	175.460	169.000	186.976
14	175	212.50	250	14.32	14	14.89	197.052	196.000	221.286
15	200	240.00	280	15.18	15	15.68	224.476	225.000	247.902
16	240	275.00	310	16.02	16	16.79	267.649	256.000	281.614
17	270	305.00	340	17.11	17	17.68	296.296	289.000	314.791
18	300	342.50	385	18.21	18	18.75	335.152	324.000	353.625
19	340	380.00	420	19.23	19	19.51	368.905	361.000	379.986
20	380	420.00	460	20.33	20	20.70	398.303	400.000	421.000
21	420	460.00	500	21.15	21	21.58	441.347	441.000	451.276
22	465	507.50	550	22.23	22	22.77	484.464	484.000	497.373

Table 2.1: Components of the fuzzy dependent variable $\tilde{y} = (y^L, y, y^R)$ and of the fuzzy covariates $\tilde{x}^1 = (x^{1,L}, x^1, x^{1,R})$, $\tilde{x}^2 = (x^{2,L}, x^2, x^{2,R})$.

Chapter 3

Orthogonality and fuzzy regression methods

One may ask themselves : what does "fuzzy orthogonality" mean and how does it apply to fuzzy regression? It is indeed two good questions and, this chapter will hopefully shed light on this topic. Keeping the regression problem in mind, fuzzy orthogonality will be investigated through two different perspectives, one using Itoh (2017)'s definition of fuzzy orthogonality and another relying on the well known Zadeh's extension principle. To be more precise regarding the regression analysis, this chapter is devoted in solving the following two fuzzy regression models present in Rosset and Donzé (2023) and Rosset and Donzé (2024c):

$$\text{Model 3:} \quad \tilde{y}_i = \sum_{j=1}^m \tilde{\beta}_j x_{ij}^{\perp}, \quad i = 1, \dots, n, \quad (3.1)$$

$$\text{Model 4:} \quad \tilde{y}_i = \sum_{j=1}^m \tilde{\beta}_j \otimes \tilde{x}_{ij}^{\perp}, \quad i = 1, \dots, n, \quad (3.2)$$

where $x_{ij}^{\perp}$ and $\tilde{x}_{ij}^{\perp}$ are orthogonalised covariates. Notice that, in contrast with the previous regression models, these ones deal with orthogonalised observations. The main difference between the two models lies in the consideration of crisp (Model 3) or fuzzy (Model 4) covariates. Let us also again remark that, when the observations are fuzzy, one needs to deal with the fuzzy product operator $\otimes$, leading to an increased complexity as previously discussed with Model 2. While for Model 3, due to the use of crisp independent variables, a classical orthogonalisation procedure can be performed, Model 4 involves the orthogonalisation of fuzzy numbers which has yet to be defined. For that reason, Model 3 is simpler to solve and its solution will be presented first. Then, two different fuzzy regression procedures will be presented to overcome the difficulty of Model 4, namely the orthogonal fuzzy least squares and the fuzziest estimate by linear orthogonal least squares methods.

3.1 Orthogonal fuzzy least squares with crisp covariates and a fuzzy dependent variable

Due to the crisp nature of the covariates of Model 3 (3.1), we are allowed to use a standard orthonormalisation procedure to project the matrix of observation $\mathbf{X}$ onto an orthonormal basis. This is performed through the transformation

$$\mathbf{X}^\perp = (\mathbf{X} - \iota_n \bar{\mathbf{x}}') \mathbf{G}, \quad (3.3)$$

where n is the sample size, $\bar{\mathbf{x}} = (\bar{x}_1, \dots, \bar{x}_m)'$ is the vector of means associated to the m covariates $\mathbf{x}^1, \dots, \mathbf{x}^m$; $\mathbf{G}$ is the matrix of all the eigenvectors of $\mathbf{V}$; $\mathbf{V}$ is the estimated variance-covariance matrix; $\iota_n = (1, \dots, 1)'$ is a $(n \times 1)$ vector. Denote the elements of $\mathbf{X}^\perp$ by $x_{ij}^\perp$. Then, similarly to Model 1, the objective function to minimise under the constraint (2.41) is given by

$$\begin{aligned} J = & k_1 \sum_{i=1}^n (y_i - \sum_{j=1}^m \beta_j x_{ij}^\perp)^2 + k_2 \sum_{i=1}^n (y_i^L - \sum_{j=1}^m \beta_j^L x_{ij}^\perp)^2 \\ & + k_3 \sum_{i=1}^n (y_i^R - \sum_{j=1}^m \beta_j^R x_{ij}^\perp)^2, \end{aligned} \quad (3.4)$$

where k_1 , k_2 and k_3 are weights given respectively to the central, left and right tendencies. Usually, $k_1 = k_2 = k_3 = 1$. The solution of this minimisation problem is given by a quadratic solver. It can be solved with the same procedure used to minimise Model 1's objective function (2.40) under (2.41). Refer to subsection 2.4.1 for more information.

More importantly, the orthogonalisation procedure occurring in Model 3 enables the emergence of a nice property : regression coefficients can be estimated one at a time by regressing the dependent variable over a single covariate. That is, to estimate $\tilde{\beta}_j = (\beta_j^L, \beta_j, \beta_j^R)$, one only needs to minimise

$$\begin{aligned} J = & k_1 \sum_{i=1}^n (y_i - \beta_j x_{ij}^\perp)^2 + k_2 \sum_{i=1}^n (y_i^L - \beta_j^L x_{ij}^\perp)^2 \\ & + k_3 \sum_{i=1}^n (y_i^R - \beta_j^R x_{ij}^\perp)^2. \end{aligned} \quad (3.5)$$

In other words, estimating regression coefficients individually for all $j = 1, \dots, m$ using (3.5) or estimating them altogether with (3.4) yield the same solution.

3.2 Fuzzy orthogonality

To be able to estimate Model 4, one has to understand what fuzzy orthogonal covariates are. While it is not clear what "fuzzy orthogonality" truly means one could define it using Zadeh's

extension principle (1.32) to have a mathematically accurate description of a fuzzy orthogonal matrix in the sense of Zadeh's theory. While the approach we will ultimately adopt for estimating regression problems involving fuzzy orthogonal matrices will differ, it will remain conceptually aligned with the following interpretation of a fuzzy orthogonal matrix. To orthogonalise matrices, one can consider the following orthogonal transformation. If $f : \mathbb{R}^{n \times m} \rightarrow \mathbb{R}^{n \times m}$, with $n > m$, is an orthogonal transformation of the form (3.3) and $\tilde{\mathbf{X}} \in \mathcal{F}(\mathbb{R}^{n \times m})$ is the fuzzy version of a $n \times m$ matrix, that is a fuzzy subset of $\mathbb{R}^{n \times m}$. Then, with Zadeh's extension principle (1.32), one may define the membership function of $f(\tilde{\mathbf{X}}) = \tilde{\mathbf{X}}^\perp$ in accordance with the notion of a fuzzy vector as introduced in Definition 1.3.2. The orthogonalisation of a fuzzy matrix $\tilde{\mathbf{X}}$ gives an orthogonalised fuzzy matrix $\tilde{\mathbf{X}}^\perp$ with membership function given by

$$\mu_{\tilde{\mathbf{X}}^\perp}(\mathbf{Z}) = \begin{cases} \sup(\mu_{\tilde{\mathbf{X}}}(\mathbf{X}) \mid \mathbf{X} \in \mathbb{R}^{n \times m}, f(\mathbf{X}) = \mathbf{Z}) & \text{if } \exists \mathbf{X} : f(\mathbf{X}) = \mathbf{Z} \\ 0 & \text{if } \nexists \mathbf{X} : f(\mathbf{X}) = \mathbf{Z} \end{cases} \quad \forall \mathbf{Z} \in \mathbb{R}^{n \times m}. \quad (3.6)$$

To construct a fuzzy orthogonal matrix $\tilde{\mathbf{X}}^\perp$ following the procedure outlined above, one must consider the set of all crisp matrices $\mathbf{X}$ contained within the fuzzy matrix $\tilde{\mathbf{X}}$. Each such matrix $\mathbf{X}$ consists of elements x_{ij} bounded by the left and right endpoints of the corresponding fuzzy number $\tilde{x}_{ij}$, that is, $x_{ij}^L \leq x_{ij} \leq x_{ij}^R$, for $i = 1, \dots, n$ and $j = 1, \dots, m$.

The orthogonal projection is then applied to each matrix $\mathbf{X}$ to obtain its orthogonal counterpart $\mathbf{X}^\perp$. The fuzzy matrix $\tilde{\mathbf{X}}^\perp$ is subsequently defined component-wise by aggregating the projections: for each pair (i, j) , the corresponding fuzzy element is determined by the minimum and maximum values attained across all projected matrices, i.e., $\min x_{ij}^\perp$ and $\max x_{ij}^\perp$. In practice, a finite sample of such matrices $\mathbf{X}$ is used to approximate $\tilde{\mathbf{X}}^\perp$. However, a significant drawback of this approach lies in its high computational cost.

To be conscious of the problem, let us say we would like to orthogonalise a 2×2 fuzzy matrix $\tilde{\mathbf{X}}$ with entries described by TFNs,

$$\tilde{\mathbf{X}} = \begin{bmatrix} \tilde{2} & \tilde{3} \\ \tilde{4} & \tilde{5} \end{bmatrix} = \begin{bmatrix} (1, 2, 3) & (2, 3, 4) \\ (3, 4, 5) & (4, 5, 6) \end{bmatrix}.$$

Considering 1000 potential values for each x_{ij} , $x_{ij}^L \leq x_{ij} \leq x_{ij}^R$, $i, j = 1, 2$, the total amount is $1000 \times 4 = 4000$. If it were a 3×3 matrix, this number would raise to 9000. This is why, we will investigate two other methods that are more efficient to estimate equation (3.2). The first one, described in Section 3.3, involves Itoh (2017)'s definition of fuzzy orthogonality. The second one, explained in Section 3.4 will take inspiration from the above orthogonalisation procedure and make use of resampling to find so called "fuzziest estimates" of regression coefficients.

To define the concept of "fuzzy orthogonality" developed by Itoh (2017), we first need to introduce the concept of a fuzzy point and fuzzy inner product.

Definition 3.2.1 (Fuzzy point).

A fuzzy set $\tilde{x}$ as defined in Definition 1.1.1 is called a fuzzy point if and only if $\mu_{\tilde{x}}(y) = 0$ for all $y \in \tilde{x}$ except one, say, $x \in \tilde{x}$ where $\mu_{\tilde{x}}(x) = \lambda$, $0 < \lambda \leq 1$. We then denote this fuzzy point by x_λ and call x its support.

Definition 3.2.2 (Fuzzy inner product).

Let X be a nonzero vector space over the field $\mathbb{R}$ and $\tilde{X} = \{x_\lambda | x \in X, \lambda \in (0, 1]\}$ be the set of all fuzzy points in X . A function $\langle \cdot, \cdot \rangle : \tilde{X} \times \tilde{X} \rightarrow \mathbb{R}$ is said to be a **fuzzy inner product** on $\tilde{X}$ if

1. a) $\langle x_\lambda, x_\mu \rangle \geq 0$; b) $\langle x_\lambda, x_\mu \rangle = 0$ if and only if x_λ or $x_\mu = 0$;
2. $\langle kx_\lambda, y_\mu \rangle = k \langle x_\lambda, y_\mu \rangle$;
3. $\langle x_\lambda + y_\lambda, z_\mu \rangle = \langle x_\lambda, z_\mu \rangle + \langle y_\lambda, z_\mu \rangle$;
4. $\langle x_\lambda, y_\mu \rangle = \langle y_\mu, x_\lambda \rangle$;
5. $\langle x_\lambda, y_\mu \rangle \leq \langle x_\lambda, y_\nu \rangle$ if $0 < \nu \leq \mu \leq 1$;
6. for every $x_\lambda, y_\mu \in \tilde{X}$ and $\epsilon > 0$, there exists $0 < \delta < \mu$ such that $\langle x_\lambda, y_{\mu-\delta} \rangle < \langle x_\lambda, y_\mu \rangle + \epsilon$;
7. for every $x_\lambda, y_\mu \in \tilde{X}$ and $\epsilon > 0$, there exists $0 < \delta < 1 - \mu$ such that $\langle x_\lambda, y_{\mu+\delta} \rangle > \langle x_\lambda, y_\mu \rangle - \epsilon$.

The pair $(X, \langle \cdot, \cdot \rangle)$ is called a **strong fuzzy inner product space**.

Remark 3.2.1.

1. Let $(X, \langle \cdot, \cdot \rangle)$ be a strong fuzzy inner product space. If

$$\begin{aligned} \langle x_{\lambda_1}, y_{\mu_1} \rangle &= \langle x_{\lambda_2}, y_{\mu_2} \rangle & x_{\lambda_i}, y_{\mu_i} &\in \tilde{X}, \\ & & \lambda_i, \mu_i &\in (0, 1], \quad i = 1, 2, \end{aligned} \quad (3.7)$$

then $(X, \langle \cdot, \cdot \rangle)$ becomes a usual inner product space;

2. If we use a stronger condition,

$$\langle x_\lambda + y_\mu, z_\nu \rangle = \langle x_\lambda, y_\mu \rangle + \langle y_\mu, z_\nu \rangle, \quad (3.8)$$

then $(X, \langle \cdot, \cdot \rangle)$ is just a usual inner product space.

The concept of fuzzy orthogonality is then defined as:

Definition 3.2.3 (Fuzzy orthogonality).

Let $\tilde{x}$ and $\tilde{y}$ be vectors in a fuzzy inner product space. One says that $\tilde{x}$ is **fuzzy orthogonal** to $\tilde{y}$ if $\langle x_\lambda, y_\mu \rangle = 0$, for some $\lambda, \mu \in (0, 1]$.

The fuzzy orthogonality between two vectors of a fuzzy inner product space $\tilde{x}$ and $\tilde{y}$ is denoted by $\tilde{x} \perp_F \tilde{y}$. In case x_λ is orthogonal to y_μ for these specific values of $\lambda, \mu \in (0, 1]$, i.e. $\langle x_\lambda, y_\mu \rangle = 0$, we note it $x_\lambda \perp y_\mu$.

From these two distinct perspectives on fuzzy orthogonality emerge two corresponding regression methodologies. The *k-th fuzziest estimate by least squares* by Rosset and Donzé (2024c), introduced in Section 3.4, is derived from Zadeh's extension principle, whereas the *orthogonal fuzzy least squares* approach from Rosset and Donzé (2023), presented in Section 3.3, is based on Itoh's definition of fuzzy orthogonality.

3.3 Orthogonal fuzzy least squares with fuzzy covariates and a fuzzy dependent variable

We now present an original approach proposed by Rosset and Donzé (2023). As with Model 2, Rosset and Donzé (2023) start with the objective function (2.43) and, with the help of the regression parameter estimates from Model 1, we construct two matrices, $\mathbf{X}^{\min}$ and $\mathbf{X}^{\max}$ whose elements $x_{ij}^{\min}$ and $x_{ij}^{\max}$ are calculated as

$$x_{ij}^{\min} = \min(\hat{\beta}_j^L x_{ij}^L, \hat{\beta}_j^R x_{ij}^L, \hat{\beta}_j^L x_{ij}^R, \hat{\beta}_j^R x_{ij}^R) \quad x_{ij}^{\max} = \max(\hat{\beta}_j^L x_{ij}^L, \hat{\beta}_j^R x_{ij}^L, \hat{\beta}_j^L x_{ij}^R, \hat{\beta}_j^R x_{ij}^R).$$

Then, with the orthogonal transformation (3.3), project $\mathbf{X}^{\min}$ and $\mathbf{X}^{\max}$ onto an orthonormal basis to find $\mathbf{X}^{\perp \min}$ and $\mathbf{X}^{\perp \max}$, whose elements are respectively $x_{ij}^{\perp \min}$ and $x_{ij}^{\perp \max}$. This yield the following objective function :

$$\begin{aligned} J = & k_1 \sum_{i=1}^n (y_i - \sum_{j=1}^m \beta_j x_{ij}^{\perp})^2 + k_2 \sum_{i=1}^n (y_i^L - \sum_{j=1}^m \beta_j^L x_{ij}^{\perp \min})^2 \\ & + k_3 \sum_{i=1}^n (y_i^R - \sum_{j=1}^m \beta_j^R x_{ij}^{\perp \max})^2. \end{aligned} \quad (3.9)$$

Minimising (3.9) under the constraints (2.41) is done with a quadratic solver, following the same procedure given by (A.54).

By considering fuzzy orthogonal covariates $\tilde{\mathbf{x}}^{j, \perp} = (\mathbf{x}^{j, \perp, L}, \mathbf{x}^{j, \perp}, \mathbf{x}^{j, \perp, R})$ whose left, middle and right parts are $\mathbf{x}^{j, \perp, L} = (x_{1j}^{\perp \min}, \dots, x_{nj}^{\perp \min})'$, $\mathbf{x}^{j, \perp} = (x_{1j}^{\perp}, \dots, x_{nj}^{\perp})'$, $\mathbf{x}^{j, \perp, R} = (x_{1j}^{\perp \max}, \dots, x_{nj}^{\perp \max})'$, one can observe that, by the way they have been constructed, the following holds:

$$\begin{aligned} \langle \tilde{\mathbf{x}}^{j, \perp, L}, \tilde{\mathbf{x}}^{k, \perp, L} \rangle &= 0, \\ \langle \mathbf{x}^{j, \perp}, \mathbf{x}^{k, \perp} \rangle &= 0, \\ \langle \tilde{\mathbf{x}}^{j, \perp, R}, \tilde{\mathbf{x}}^{k, \perp, R} \rangle &= 0, \quad \forall j, k = 1, \dots, m, j \neq k. \end{aligned} \quad (3.10)$$

Thus, $\tilde{\mathbf{x}}^{j, \perp}$ is satisfying Itoh (2017)'s definition of fuzzy orthogonality.

Note that the orthogonalisation of the fuzzy observations has been handled with a classical orthonormalisation procedure (3.3). Hence, the regression parameters $\tilde{\beta}_j$ solutions of minimising

$$\begin{aligned} J = & k_1 \sum_{i=1}^n (y_i - \beta_j x_{ij}^{\perp})^2 + k_2 \sum_{i=1}^n (y_i^L - \beta_j^L x_{ij}^{\perp \min})^2 \\ & + k_3 \sum_{i=1}^n (y_i^R - \beta_j^R x_{ij}^{\perp \max})^2, \end{aligned} \quad (3.11)$$

individually for $j = 1, \dots, m$ are the same solutions as minimising (3.9).

Another interesting observation is to realise that there exists a linear transformation linking the vector of estimates $\tilde{\beta} = (\tilde{\beta}_1, \dots, \tilde{\beta}_m)'$ associated to the set of observations $\tilde{\mathbf{X}}$ and $\tilde{\mathbf{X}}^\perp$. With the matrix $\mathbf{G}$ from the transformation (3.3), one can retrieve the regression coefficients found with the non orthogonally projected observations $\tilde{\mathbf{X}}$:

$$\begin{aligned}\mathbf{G}\beta^{L,\perp} &= \beta^L, \\ \mathbf{G}\beta^\perp &= \beta, \\ \mathbf{G}\beta^{R,\perp} &= \beta^R.\end{aligned}$$

Where $\beta^L = (\beta_1^L, \dots, \beta_m^L)'$, $\beta = (\beta_1, \dots, \beta_m)'$ and $\beta^R = (\beta_1^R, \dots, \beta_m^R)'$. However, this transformation has a caveat since it can only be used for unconstrained solutions, that is minimising the objective functions of models 3 and 4, (3.4), (3.9) **under no constraint**. In the constrained case, in particular with the use of constraints (2.41), this is in general not true.

3.4 (k-th) fuzziest estimate by orthogonal least squares

Let us introduce another novel approach developed by Rosset and Donzé (2024c). As mentioned above with equation (3.11), the orthogonalisation of fuzzy numbers is performed through a classical procedure and thus the term "fuzzy orthogonalisation" cannot be used to describe it. Ideally, the aim is to extend orthogonalisation via Zadeh's extension principle to define a "fuzzy orthogonalisation", as described by (3.6). We propose the following algorithm. Among all possible matrices $\mathbf{X}_\chi$ we draw the ones having their components x_{ij_χ} bounded by the left and right components of a fuzzy matrix at a given α -cut $\tilde{\mathbf{X}}^\alpha$, $x_{ij}^{L,\alpha} \leq x_{ij_\chi} \leq x_{ij}^{R,\alpha}$, $\alpha \in [0, 1]$ and χ an index. Then, with an orthogonal transformation, for example described by (3.3), one computes all matrices $\mathbf{X}_\chi^\perp$ and, for each value of i, j , among all orthonormalised components $x_{ij_\chi}^\perp$, one chooses the smallest, respectively greatest components:

$$x_{ij}^{\perp, \min} = \inf_{\chi} \{x_{ij_\chi}^\perp\} \quad x_{ij}^{\perp, \max} = \sup_{\chi} \{x_{ij_\chi}^\perp\}. \quad (3.12)$$

With those, one can then construct a fuzzy orthogonalised matrix $\tilde{\mathbf{X}}^\perp$. Remark that, in practice, this would require to generate and orthogonally project an infinite amount of matrices. Therefore, this problem has to be tackled with optimisation strategies. To come back to our fuzzy regression problem, our aim is slightly different. Indeed, we seek to find, the estimates $\tilde{\beta}_j$ found by minimising:

$$J = \sum_{i=1}^n \left(\tilde{y}_i \ominus \sum_{j=1}^m \tilde{\beta}_j \otimes \tilde{x}_{ij}^\perp \right), \quad i = 1, \dots, n. \quad (3.13)$$

Seeing it with Zadeh's extension principle in mind, to estimate Model 4, the idea is to first generate a great number S of crisp matrices out of the fuzzy matrix of observations. Then,

orthogonalising these S matrices that will be used as S new matrices of crisp observations to estimate S regression problems of the form of Model 3. This then allows to find, the infimum and supremum among all S regression parameters estimated this way. Formally, to solve this problem, we will generate S crisp matrices $\mathbf{X}_s$, $s \in \{1, 2, \dots, S\}$ with components $x_{s,ij} \in [x_{ij}^L, x_{ij}^R]$, (x_{ij}^L, x_{ij}^R) being the left and right endpoints of the ij -th component of $\tilde{\mathbf{X}}$. Then, we orthogonalise $\mathbf{X}_s$ to find the matrix $\mathbf{X}_s^\perp$. Thus, for each $\mathbf{X}_s^\perp$, we are now interested in solving the following regression model

$$\tilde{y}_i = \sum_{j=1}^m \tilde{\beta}_{s,j} x_{s,ij}^\perp, \quad i = 1, \dots, n, \quad (3.14)$$

where $x_{s,ij}^\perp$ is the ij -th element of $\mathbf{X}_s^\perp$. Observe that (3.14) is actually Model 3. Hence to estimate (3.14), we minimise the objective function (3.15) for each s :

$$J = \sum_{i=1}^n \left(\tilde{y}_i - \sum_{j=1}^m \tilde{\beta}_{s,j} x_{s,ij}^\perp \right)^2, \quad i = 1, \dots, n. \quad (3.15)$$

This yields S fuzzy estimates on the form of fuzzy intervals as defined in 1.3 :

$$\hat{\tilde{\beta}}_s = \begin{pmatrix} \hat{\tilde{\beta}}_{s,1} \\ \vdots \\ \hat{\tilde{\beta}}_{s,k} \\ \vdots \\ \hat{\tilde{\beta}}_{s,m} \end{pmatrix} = \begin{pmatrix} (\hat{\beta}_{s,1}^L, \hat{\beta}_{s,1}, \beta_{s,1}^R) \\ \vdots \\ (\hat{\beta}_{s,k}^L, \hat{\beta}_{s,k}, \beta_{s,k}^R) \\ \vdots \\ (\hat{\beta}_{s,m}^L, \hat{\beta}_{s,m}, \beta_{s,m}^R) \end{pmatrix}. \quad (3.16)$$

Then, we would like to find the smallest, respectively greatest value among all S left and right components of the k -th estimated fuzzy parameters $\hat{\tilde{\beta}}_{s,k}$ represented by the sets $B_k^L = \{\hat{\beta}_{1,k}^L, \dots, \hat{\beta}_{s,k}^L, \dots, \hat{\beta}_{S,k}^L\}$, $B_k^R = \{\hat{\beta}_{1,k}^R, \dots, \hat{\beta}_{s,k}^R, \dots, \hat{\beta}_{S,k}^R\}$. To do so, let $B_k := B_k^L \cup B_k^R$. Then, find the infimum and supremum of B_k , $\hat{\beta}_{s_{\inf},k}^{\inf} = \inf(B_k)$ and $\hat{\beta}_{s_{\sup},k}^{\sup} = \sup(B_k)$, where $s_{\inf}$ and $s_{\sup}$ are indices associated to the position of the infimum and supremum in the set B_k . The solution is thus given by

$$(\hat{\beta}_{s_{\inf},k}^{\inf}, \hat{\beta}_k, \hat{\beta}_{s_{\sup},k}^{\sup}). \quad (3.17)$$

Note that the $\tilde{\beta}_j$ can be computed individually, by minimising

$$J = \sum_{i=1}^n \left(\tilde{y}_i \ominus (\tilde{\beta}_j \otimes \tilde{x}_{ij}^\perp) \right)^2, \quad j = 1, \dots, m, \quad (3.18)$$

and yielding the same solutions as when computed all together with the methodology described above. This procedure is summarised in algorithm 3.1.

Fuzzy regression problems described by Models 1-4 are addressed by minimizing an objective function subject to the constraints given in Equation (2.41). The resulting solutions satisfy,

Algorithm 3.1 k -th fuzziest estimate by orthogonal least squares

-
- 1: **for** $s = 1, \dots, S$ **do**:
 - 2: Generate $\mathbf{X}_s$ with components $x_{s,ij} \in [x_{ij}^L, x_{ij}^R]$, where (x_{ij}^L, x_{ij}^R) are the left and right endpoints of the components of the fuzzy matrix of observations.
 - 3: With respect to 3.3, orthonormalise $\mathbf{X}_s$ to find $\mathbf{X}_s^\perp$.
 - 4: Minimise the objective function

$$J = \sum_{i=1}^n \left(\tilde{y}_i - \sum_{j=1}^m \tilde{\beta}_{s,j} x_{s,ij}^\perp \right)^2,$$

to find the regression coefficients $\tilde{\beta}_{s,j}$, $j = 1, \dots, m$.

- 5: **end for**
 - 6: **for** $j = 1, \dots, m$ **do**:
 - 7: Find the infimum and supremum among all regression coefficients $\tilde{\beta}_{s,j}$.
 - 8: **end for**
 - 9: **Output:** $(\hat{\beta}_{s_{\inf},j}^{\inf}, \hat{\beta}_j, \hat{\beta}_{s_{\sup},j}^{\sup})$.
-

for all $j = 1, \dots, m$, the condition $\beta_j^L \leq \beta_j \leq \beta_j^R$, thereby ensuring that each estimated coefficient is expressed as a valid fuzzy number.

While obtaining fuzzy-valued solutions is desirable, constraint-based formulations described in sections 2.5 and 3.3 can introduce certain limitations. As illustrated in Section 3.4.1, solving a regression problem under these constraints may result in a crisp estimate for a given coefficient β_j . In contrast, removing the constraints may yield a fuzzy estimate for the same coefficient. However, unconstrained estimates can sometimes violate the basic property of a fuzzy number, namely, the ordering $\beta_j^L \leq \beta_j \leq \beta_j^R$. One may encounter situations where $\beta_j^L \geq \beta_j \geq \beta_j^R$, thereby violating the structure of a fuzzy number.

It is worth noting that, although the minimisation of the objective function (3.13) is performed in an unconstrained setting, the k -th fuzziest estimate by least squares method presented in this section inherently produces fuzzy solutions satisfying

$$\beta_j^L \leq \beta_j \leq \beta_j^R, \quad j = 1, \dots, m.$$

This desirable property arises from the incorporation of the fuzziness into the estimation process via Zadeh's extension principle, ensuring that the solutions are fuzzy numbers. To clarify this, consider the function $f : \mathbb{R}^{n \times m} \rightarrow \mathbb{R}$, which maps a matrix of observations $\mathbf{X}$ to the k -th regression coefficient $\tilde{\beta}_k$, i.e., $\mathbf{X} \xrightarrow{f} \tilde{\beta}_k$. According to Algorithm 3.1, the membership function of $\tilde{\beta}_k$ is given by

$$\mu_{\tilde{\beta}_k}(z) = \begin{cases} \sup\{\mu_{\mathbf{X}}(\mathbf{X}) \mid \mathbf{X} \in \mathbb{R}^{n \times m}, f(\mathbf{X}) = z\} & \text{if } \exists \mathbf{X} : f(\mathbf{X}) = z, \\ 0 & \text{otherwise} \end{cases} \quad \forall z \in \mathbb{R}. \quad (3.19)$$

This formulation clearly demonstrates that the estimated fuzzy coefficient $\tilde{\beta}_k$ is derived in accordance with Zadeh's extension principle. The resulting coefficient is inherently fuzzy. However, the computational cost of this method is significantly higher than that of the constraint-based approaches, which may limit its practical applicability in large-scale problems.

3.4.1 Numerical examples

By considering the fuzzy observations from Table 2.1, let us consider the regression problem given by

$$\tilde{y}_i = \tilde{\beta}_0 + (\tilde{\beta}_1 \otimes x_{i1}^\perp) + (\tilde{\beta}_2 \otimes x_{i2}^\perp).$$

By observing that this regression problem is of the form of equation (3.1), we can solve it with the procedure described in section 3.1. Under the constraints (2.41), the solution is written as

$$\hat{\beta} = \begin{pmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \\ \hat{\beta}_2 \end{pmatrix} = \begin{pmatrix} (165.2272727, 192.6704545, 220.1136364) \\ (-0.9947141, -0.9947141, -0.9947141) \\ (0.4177431, 0.4177431, 0.4177431) \end{pmatrix}. \quad (3.20)$$

On the other hand, the unconstrained solution is given by

$$\hat{\beta} = \begin{pmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \\ \hat{\beta}_2 \end{pmatrix} = \begin{pmatrix} (165.2272727, 192.6704545, 220.1136364) \\ (-0.9243395, -0.9947141, -1.065089) \\ (3.2905603, 0.4177431, -2.455074) \end{pmatrix}. \quad (3.21)$$

The plot of the fitted values $\hat{y}_i = (\hat{y}_i^L, \hat{y}_i, \hat{y}_i^R)$, where $\hat{y}_i = \hat{\beta}_0 \oplus (\hat{\beta}_1 \otimes x_{i1}^\perp) \oplus (\hat{\beta}_2 \otimes x_{i2}^\perp)$, respectively given by the constrained and unconstrained solutions are depicted in Figures 3.1 and 3.2. We can observe that the unconstrained solution yield fuzzy estimates for the regression coefficients $\hat{\beta}_1$ and $\hat{\beta}_2$ while the constrained solution give crisp estimates. It is also clear that, in both cases, $\hat{y}_i^L \leq \hat{y}_i \leq \hat{y}_i^R$.

Let us now consider fuzzy covariates with the following regression model corresponding to Model 4 described in equation (3.2) :

$$\tilde{y}_i = \tilde{\beta}_0 + (\tilde{\beta}_1 \otimes \tilde{x}_{i1}^\perp) + (\tilde{\beta}_2 \otimes \tilde{x}_{i2}^\perp). \quad (3.22)$$

There are two possible ways to estimate the regression coefficients as we have seen in sections 3.3 and 3.4. If we solve it with the method proposed in section 3.3, the constrained solution is given by

$$\hat{\beta} = \begin{pmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \\ \hat{\beta}_2 \end{pmatrix} = \begin{pmatrix} (165.2272727, 192.6704545, 220.1136364) \\ (-0.9916071, -0.9916071, -0.9916071) \\ (0.5586351, 0.5586351, 0.5586351) \end{pmatrix} \quad (3.23)$$

and the unconstrained solution writes as

$$\hat{\hat{\beta}} = \begin{pmatrix} \hat{\hat{\beta}}_0 \\ \hat{\hat{\beta}}_1 \\ \hat{\hat{\beta}}_2 \end{pmatrix} = \begin{pmatrix} (165.2272727, 192.6704545, 220.1136364) \\ (-0.9640743, -0.9947141, -1.011659) \\ (1.9816861, 0.4177431, -1.072702) \end{pmatrix}. \quad (3.24)$$

Surprisingly, the unconstrained solution (3.21) is fuzzier than the unconstrained solution (3.24). This effect may come from the added fuzziness in the covariates. Indeed, because this time the fitted values are $\hat{y}_i = \hat{\hat{\beta}}_0 + \hat{\hat{\beta}}_1 \otimes \tilde{x}_{i1}^\perp + \hat{\hat{\beta}}_2 \otimes \tilde{x}_{i2}^\perp$, the regression coefficients $\hat{\hat{\beta}}_1$ and $\hat{\hat{\beta}}_2$ do not need to be as fuzzy to estimate $\hat{y}_i$ since $\tilde{x}_1^\perp$ and $\tilde{x}_2^\perp$ are fuzzy. The constrained and unconstrained solutions are shown graphically in Figures 3.3 and 3.4.

Lastly, if we solve (3.22) with respect to the method of section 3.4, one finds the following solution :

$$\hat{\hat{\beta}} = \begin{pmatrix} \hat{\hat{\beta}}_0 \\ \hat{\hat{\beta}}_1 \\ \hat{\hat{\beta}}_2 \end{pmatrix} = \begin{pmatrix} (165.2272727, 192.6704545, 220.1136364) \\ (-1.109607, -0.9684732, -0.8762916) \\ (-6.138478, -0.1037333, 5.555640) \end{pmatrix}. \quad (3.25)$$

This solution is indeed fuzzier than solutions (3.21) and (3.24). It is therefore the solution containing the fuzziest estimates for $\hat{\hat{\beta}}_1$ and $\hat{\hat{\beta}}_2$. Figure 3.5 presents a plot of the fitted values $\hat{y}_i = \hat{\hat{\beta}}_0 + \hat{\hat{\beta}}_1 \otimes \tilde{x}_{i1}^\perp + \hat{\hat{\beta}}_2 \otimes \tilde{x}_{i2}^\perp$ with respect to the fuzziest solution (3.25).

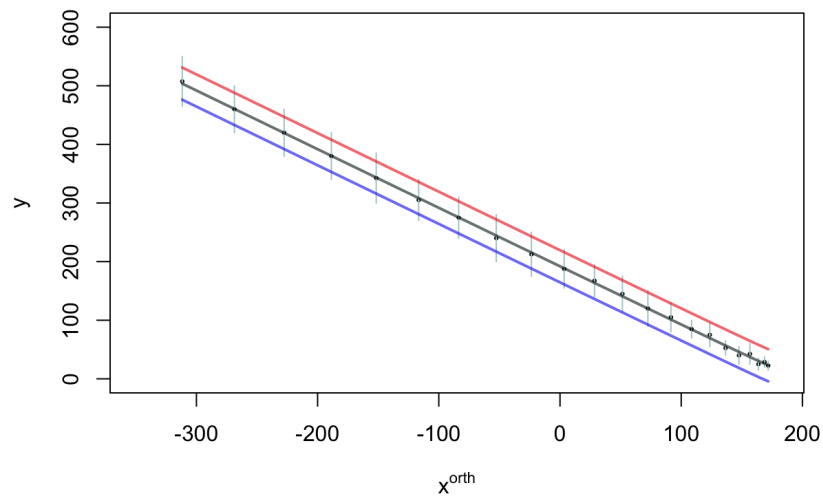


Figure 3.1: Plot of the fitted values $\hat{y}_i = \hat{\beta}_0 + (\hat{\beta}_1 \otimes x_{i1}^{\perp}) + (\hat{\beta}_2 \otimes x_{i2}^{\perp})$ given by the constrained solution (3.20), where $x^{\text{orth}} := x_{i1}^{\perp}$. The left, middle and right regression lines are respectively coloured in blue, dark grey and red. Fuzzy observations of the dependent variable are represented by light blue lines whose vertices are $(x_{i1}^{\perp}, y_i^L)$, $(x_{i1}^{\perp}, y_i^R)$. The observations are from Table 2.1.

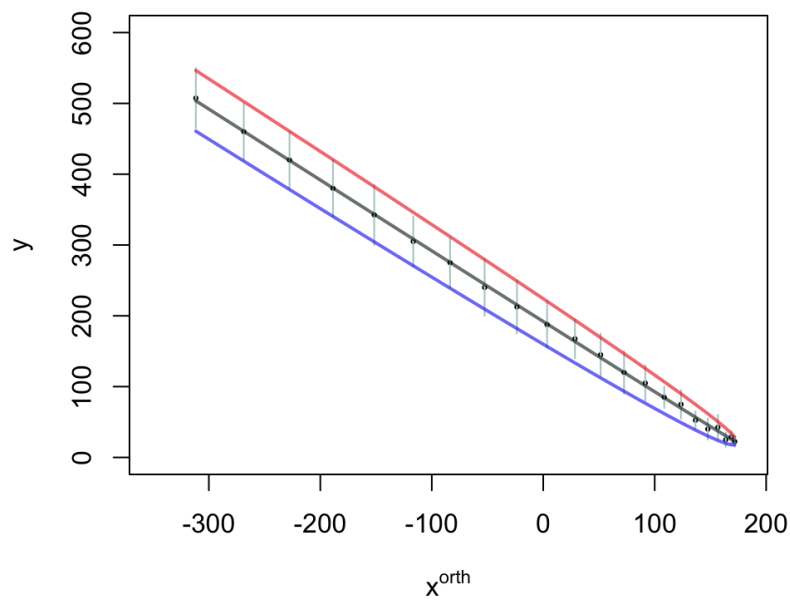


Figure 3.2: Plot of the fitted values $\hat{y}_i = \hat{\beta}_0 + (\hat{\beta}_1 \otimes x_{i1}^\perp) + (\hat{\beta}_2 \otimes x_{i2}^\perp)$ with respect to the unconstrained solution (3.21), where $x^{\text{orth}} := x_{i1}^\perp$. The left, middle and right regression lines are respectively coloured in blue, dark grey and red. Fuzzy observations are represented by light blue lines whose vertices are $(x_{i1}^\perp; y_i^L)$, $(x_{i1}^\perp; y_i^R)$. The observations are from Table 2.1.

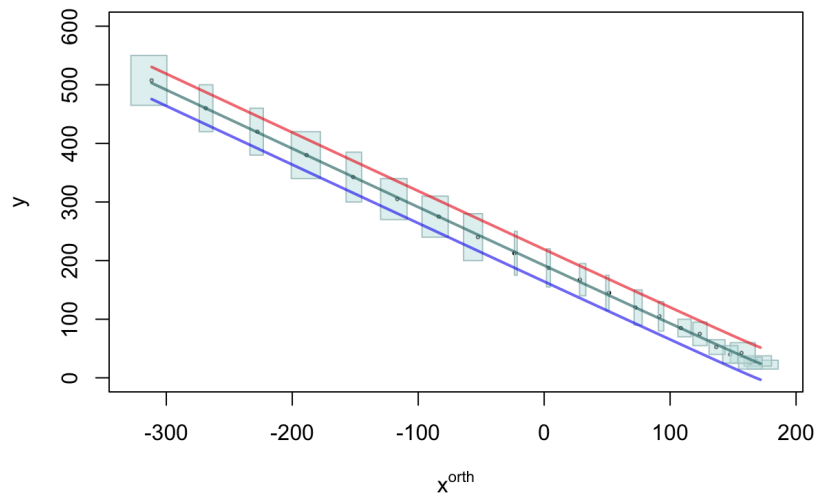


Figure 3.3: Plot of the fitted values $\hat{y}_i = \hat{\beta}_0 + (\hat{\beta}_1 \otimes \tilde{x}_{i1}^\perp) + (\hat{\beta}_2 \otimes \tilde{x}_{i2}^\perp)$ with respect to the constrained solution (3.23), where $x^{\text{orth}} := x_{i1}^\perp$. The left, middle and right regression lines are respectively coloured in blue, dark grey and red. Fuzzy observations of the dependent variable are represented by light blue rectangles whose vertices are $(x_{i1}^{\perp,L}; y_i^L)$, $(x_{i1}^{\perp,L}; y_i^R)$, $(x_{i1}^{\perp,R}; y_i^L)$, $(x_{i1}^{\perp,R}; y_i^R)$. The observations are from Table 2.1.

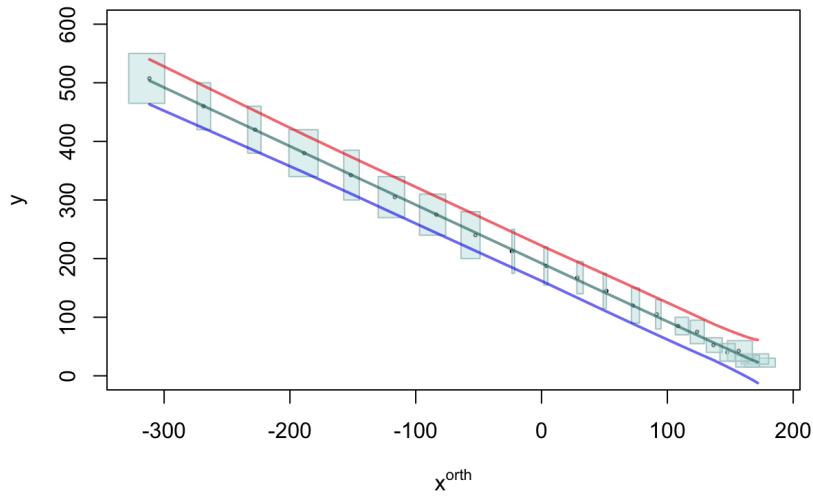


Figure 3.4: Plot of the fitted values $\hat{y}_i = \hat{\beta}_0 + (\hat{\beta}_1 \otimes \tilde{x}_{i1}^\perp) + (\hat{\beta}_2 \otimes \tilde{x}_{i2}^\perp)$ with respect to the unconstrained solution (3.24), where $x^{\text{orth}} := x_{i1}^\perp$. The left, middle and right regression lines are respectively coloured in blue, dark grey and red. Fuzzy observations are represented by light blue rectangles whose vertices are $(x_{i1}^{\perp,L}; y_i^L)$, $(x_{i1}^{\perp,L}; y_i^R)$, $(x_{i1}^{\perp,R}; y_i^L)$, $(x_{i1}^{\perp,R}; y_i^R)$. The observations are from Table 2.1.

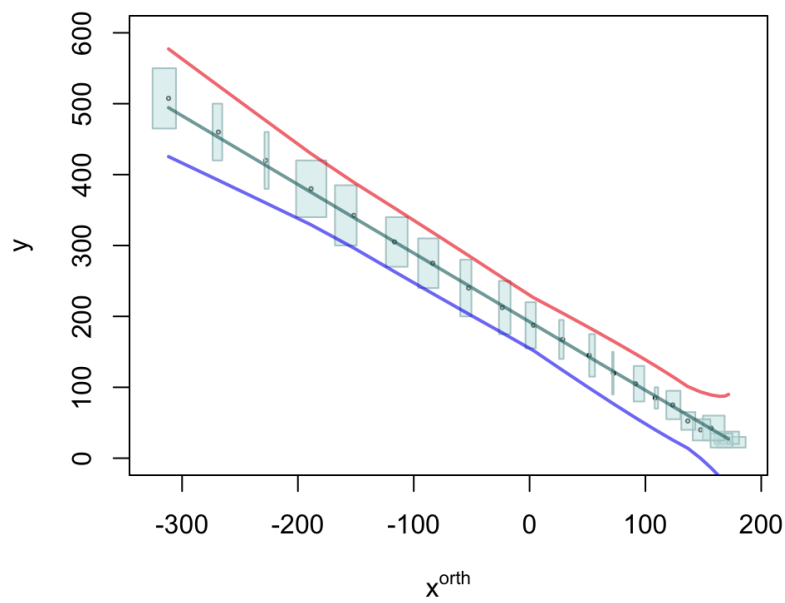


Figure 3.5: Plot of the fitted values $\hat{y}_i = \hat{\beta}_0 + (\hat{\beta}_1 \otimes \tilde{x}_{i1}^\perp) + (\hat{\beta}_2 \otimes \tilde{x}_{i2}^\perp)$ with respect to the solution (3.25), where $x^{\text{orth}} := x_{i1}^\perp$. The left, middle and right regression lines are respectively coloured in blue, dark grey and red. Fuzzy observations are represented by light blue rectangles whose vertices are $(x_{i1}^{\perp,L}; y_i^L)$, $(x_{i1}^{\perp,L}; y_i^R)$, $(x_{i1}^{\perp,R}; y_i^L)$, $(x_{i1}^{\perp,R}; y_i^R)$. The observations are from Table 2.1.

Chapter 4

Fuzzy inference

In statistical analysis, a point estimate of a parameter is typically accompanied by a corresponding measure of significance. However, to properly address this question in fuzzy statistics, we need to extend classical inference methods. Since the parameters are now fuzzy, their distributions cannot be assumed to be crisp any longer. As a result, other important statistical methods, such as confidence intervals and test statistics, will generally not remain crisp either. This chapter is dedicated to explore the necessary concepts for developing fuzzy inference procedures. Specifically, we will begin with an introduction to fuzzy random variables, which will lead us to the definition of empirical fuzzy distributions. This, in turn, allows for the introduction of fuzzy confidence intervals. Following this, we will describe a procedure for fuzzy hypothesis testing. Given the common use of p-values in statistical analysis, we will also define fuzzy p-values. To conclude, we will present a numerical example to compare the results of the fuzzy hypothesis testing procedure with those obtained using fuzzy p-values.

4.1 Fuzzy random variable

The notion of a fuzzy random variable is nuanced in the sense that, one needs to distinguish between the epistemic and ontic view of a fuzzy random variable. The distinction between these two views lies primarily in how uncertainty is perceived and treated. If the uncertainty comes from a lack of certainty about the observed values of the random variable, then we measure the so called *original* values of our random variable, but due to incomplete information, we question these values and because we show some doubt, we decide to fuzzify the realisations of the random variable. The fuzziness here is related to how we perceive and model uncertainty based on lack of information. An epistemic fuzzy random variable represents the fuzzy interpretation of original observations. Essentially, it is a "fuzzified" random variable, designed to capture the uncertainty or lack of clarity regarding the values of a fuzzy random variable. On the other hand, when the uncertainty is ontological, that is, part of the system's nature, the fuzziness arising in the observations is due to the fuzzy behaviour itself

of such system. In the words of Gil, López-Díaz, and Dan A. Ralescu (2006), *ontic fuzzy random variables are considered in the setting of a random experiment to model an essentially fuzzy-valued mechanism, that is, a mechanism associating a fuzzy value with each experimental outcome*. To sum it up, in the epistemic case, the fuzziness comes from what we know (original values) and don't know about (lack of certainty) while, in the ontic case, the fuzziness arises from the nature itself of the system. Another important consideration is that, ontic uncertainty exists regardless of how much we know or understand the system. It is a fundamental property of the system itself.

Before giving the formal definition of a fuzzy random variable with respect to the epistemic and ontic views, let us first recall two metrics, the Hausdorff metric d_H discovered by Hausdorff (1914) and the d_∞ -metric introduced by Puri and Dan A Ralescu (1986)[p.56].

Definition 4.1.1 (Hausdorff Metric).

Let $\mathcal{K}(\mathbb{R}^p)$ denote the family of all nonempty compact subsets of $\mathbb{R}^p$. For $A, B \in \mathcal{K}(\mathbb{R}^p)$, the *Hausdorff metric* is defined as

$$d_H(A, B) = \max \left\{ \sup_{a \in A} \inf_{b \in B} \|a - b\|, \sup_{b \in B} \inf_{a \in A} \|a - b\| \right\}.$$

Definition 4.1.2 (d_∞ -metric).

Let $\mathcal{F}_c(\mathbb{R}^p)$ be the set of all fuzzy subsets of $\mathbb{R}^p$ with compact α -cuts. For $\tilde{\mathbf{x}}, \tilde{\mathbf{y}} \in \mathcal{F}_c(\mathbb{R}^p)$, Puri and Dan A Ralescu (1986)[p.56]'s d_∞ -metric is given by

$$d_\infty(\tilde{\mathbf{x}}, \tilde{\mathbf{y}}) = \sup_{\alpha \in [0,1]} d_H(\tilde{\mathbf{x}}^\alpha, \tilde{\mathbf{y}}^\alpha).$$

Note that Puri and Dan A Ralescu (1986)[p.56]'s d_∞ -metric extends the Hausdorff metric to elements of $\mathcal{F}_c(\mathbb{R}^p)$ by comparing their α -cuts, which are by definition elements of $\mathcal{K}(\mathbb{R}^p)$, across all $\alpha \in [0, 1]$.

With respect to Kwakernaak and Kruse and Meyer (1987)[p.63]'s approach, a fuzzy random variable in the epistemic sense is defined as follow in Gil, López-Díaz, and Dan A. Ralescu 2006.

Definition 4.1.3. Fuzzy random variable (epistemic view)

Given a probability space $(\Omega, \mathcal{A}, P)$, a mapping

$$\tilde{\mathcal{X}} : \Omega \rightarrow \mathcal{F}_c(\mathbb{R})$$

is said to be a **Fuzzy random variable (FRV)** in the epistemic sense if for all $\alpha \in [0, 1]$ the two real valued mappings

$$\inf \tilde{\mathcal{X}}^\alpha : \Omega \longrightarrow \mathbb{R} \quad \sup \tilde{\mathcal{X}}^\alpha : \Omega \longrightarrow \mathbb{R}$$

defined so that for all $\omega \in \Omega$ we have that the left and right components of $\tilde{\mathcal{X}}^\alpha(\omega) = [\inf(\tilde{\mathcal{X}}(\omega))^\alpha, \sup(\tilde{\mathcal{X}}(\omega))^\alpha] = [\mathcal{X}^{L,\alpha}, \mathcal{X}^{R,\alpha}]$ are real valued random variables.

Moreover, Gil, López-Díaz, and Dan A. Ralescu (2006)[p.4] also gave a definition concerning induced fuzzy parameters by fuzzy random variables in the epistemic sense.

Definition 4.1.4. Induced fuzzy parameter (epistemic view)

If $\theta(\mathcal{X})$ is a real valued parameter of a random variable $\mathcal{X} : \Omega \rightarrow \mathbb{R}$ associated with the probability space $(\Omega, \mathcal{A}, P)$ and $\mathcal{E}(\Omega, \mathcal{A}, P)$ denotes the class of all possible "originals" of a fuzzy random variable $\tilde{\mathcal{X}} : \Omega \rightarrow \mathcal{F}_c(\mathbb{R})$ associated with $(\Omega, \mathcal{A}, P)$, then the induced fuzzy parameter of $\tilde{\mathcal{X}}$ corresponds to

$$\theta(\tilde{\mathcal{X}}) : \mathbb{R} \longrightarrow \mathbb{R}$$

such that for all $t \in \mathbb{R}$

$$\tilde{\theta}(\tilde{\mathcal{X}})(t) = \sup_{\mathcal{X} \in \mathcal{E}(\Omega, \mathcal{A}, P) | \theta(\mathcal{X})=t} \inf_{\omega \in \Omega} \{ \tilde{\mathcal{X}}(\omega)(\mathcal{X}(\omega)) \}.$$

If $\theta(\mathcal{X}) = \mathbb{E}(\mathcal{X}|P)$ then $\tilde{\theta}(\tilde{\mathcal{X}})$ corresponds to the so-called fuzzy expected value of $\tilde{\mathcal{X}}$, noted $\tilde{\mathbb{E}}(\tilde{\mathcal{X}})$, which is defined for all $\alpha \in [0, 1]$ by

$$\tilde{\mathbb{E}}(\tilde{\mathcal{X}}) = \left(\inf \mathbb{E}(\mathcal{X}^{L,\alpha}), \sup \mathbb{E}(\mathcal{X}^{R,\alpha}) \right). \quad (4.1)$$

Following the approach of Puri and Dan A Ralescu (1986)[p.63], Gil, López-Díaz, and Dan A. Ralescu (2006)[p. 7] defines a fuzzy random variable **FRV** in the ontic sense as follows.

Definition 4.1.5. Fuzzy random variable in the ontic sense

Given a probability space $(\Omega, \mathcal{A}, P)$, a mapping

$$\tilde{\mathcal{X}} : \Omega \longrightarrow \mathcal{F}(\mathbb{R}^p)$$

is said to be a **fuzzy random variable in the ontic sense** (also called a **random fuzzy set** in Couso, Dubois, and Sánchez (2014)) if for all $\alpha \in [0, 1]$ the set valued mappings

$$\tilde{\mathcal{X}}^\alpha : \Omega \longrightarrow \mathcal{K}(\mathbb{R}^p)$$

defined so that for all $\omega \in \Omega$

$$\tilde{\mathcal{X}}^\alpha(\omega) = (\tilde{\mathcal{X}}(\omega))^\alpha$$

are random sets (that is, Borel measurable mappings with the Borel σ -field generated by the topology associated with the Hausdorff metric d_H on $\mathcal{K}(\mathbb{R}^p)$), see Definition 4.1.1.

Gil, López-Díaz, and Dan A. Ralescu (2006)[p.8] also defined the fuzzy expected value in the ontic sense as follows.

Definition 4.1.6 (Fuzzy expected value in the ontic sense).

Given a probability space $(\Omega, \mathcal{A}, P)$ if the mapping $\tilde{\mathcal{X}} : \Omega \rightarrow \mathcal{F}_c(\mathbb{R}^p)$ is a **FRV** which is integrably bounded in the sense that $d_H(\{0\}, \tilde{\mathcal{X}}_0) \in L^1(\Omega, \mathcal{A}, P)$, then the **fuzzy expected value** of $\tilde{\mathcal{X}}$ corresponds to the unique $\mathbb{E}(\tilde{\mathcal{X}}|P) \in \mathcal{F}(\mathbb{R}^p)$ such that for all $\alpha \in [0, 1]$

$$\begin{aligned} (\tilde{\mathbb{E}}(\tilde{\mathcal{X}}|P))^\alpha &= \text{Aumann's integral of the random set } \tilde{\mathcal{X}}^\alpha \\ &= \{ \tilde{\mathbb{E}}(X|P) | X : \Omega \rightarrow \mathbb{R}^p, X \in L^1(\Omega, \mathcal{A}, P), X \in \tilde{\mathcal{X}}^\alpha \text{ almost surely w.r.t } P \}. \end{aligned} \quad (4.2)$$

By looking closely at Definitions 4.1.3 and 4.1.5 and restricting ourselves to mappings $\tilde{\mathcal{X}} : \Omega \rightarrow \mathcal{F}_c(\mathbb{R})$, we may observe that the ontic and epistemic views seem to be mathematically similar. The following proposition from Gil, López-Díaz, and Dan A. Ralescu (2006)[p.10], Proposition 4.1.1 states that the two views are mathematically equivalent when FRV in the ontic and epistemic sense are $\mathcal{F}_c(\mathbb{R})$ -valued mappings. Naturally, the two views remain conceptually different.

Proposition 4.1.1.

Given a probability space $(\Omega, \mathcal{A}, P)$ and a mapping $\tilde{\mathcal{X}} : \Omega \rightarrow \mathcal{F}_c(\mathbb{R})$, then $\tilde{\mathcal{X}}$ is a FRV in Kwakernakk/Kruse and Meyer's sense (Definition 4.1.3) if and only if it is a FRV in Puri and Ralescu's sense (Definition 4.1.5).

Moreover, Gil, López-Díaz, and Dan A. Ralescu (2006)[p.10] give, with Proposition 4.1.2 an equivalence result concerning the fuzzy expected values of FRVs in both the epistemic and ontic sense.

Proposition 4.1.2.

Given a probability space $(\Omega, \mathcal{A}, P)$ and a mapping $\tilde{\mathcal{X}} : \Omega \rightarrow \mathcal{F}_c(\mathbb{R})$, if the parameter $\theta(X)$ of the original real-valued random variable is the expected value $\mathbb{E}(X|P)$, then the induced parameter $\theta(\tilde{\mathcal{X}})$ in Kruse and Meyer's sense (Definition 4.1.4) coincides with the fuzzy expected value $\mathbb{E}(\mathcal{X}|\mathcal{P})$ in Puri and Ralescu's sense (Definition 4.1.6).

Despite what Propositions 4.1.1 and 4.1.2 tell, do not be fooled into thinking that the two views are equivalent even when we only consider a probability space $(\Omega, \mathcal{A}, P)$ and a mapping $\tilde{\mathcal{X}} : \Omega \rightarrow \mathcal{F}_c(\mathbb{R})$. As a matter of fact, the variance of a FRV in the epistemic sense is fuzzy while the variance of a FRV in the ontic sense is crisp, showing that the ontic and epistemic perceptions indeed differ even under the assumptions of Proposition 4.1.1.

Definition 4.1.7 (Variance of a FRV (epistemic view)).

Consider a fuzzy random variable $\tilde{\mathcal{X}}$ in the epistemic sense defined on the probability space $(\Omega, \mathcal{A}, P)$ by the mapping $\tilde{\mathcal{X}} : \Omega \rightarrow \mathcal{F}_c(\mathbb{R})$. Moreover, let $\mathcal{X} : \Omega \rightarrow \mathbb{R}$ be a random variable associated with the probability space $(\Omega, \mathcal{A}, P)$ and let $\mathcal{E}(\Omega, \mathcal{A}, P)$ denote the class of all possible "originals" of a fuzzy random variable $\tilde{\mathcal{X}} : \Omega \rightarrow \mathcal{F}_c(\mathbb{R})$ associated with $(\Omega, \mathcal{A}, P)$. Then, the variance of $\tilde{\mathcal{X}}$ denoted by $\widetilde{\text{Var}}(\tilde{\mathcal{X}})$ is the fuzzy perception of the variance of X given by

$$\widetilde{\text{Var}}(\tilde{\mathcal{X}})(\sigma^2) = \sup_{\mathcal{X} \in \mathcal{E}(\Omega, \mathcal{A}, P) | \text{Var}(\mathcal{X}) = \sigma^2} \inf_{\omega \in \Omega} \tilde{\mathcal{X}}(\mathcal{X}(\omega)). \quad (4.3)$$

Additionally, its membership function $\mu_{\widetilde{\text{Var}}(\tilde{\mathcal{X}})}(\sigma^2)$ is defined as

$$\mu_{\widetilde{\text{Var}}(\tilde{\mathcal{X}})}(\sigma^2) = \sup\{\mu_{\tilde{\mathcal{X}}}(\omega) \mid \mathcal{X} \in \mathcal{E}(\Omega, \mathcal{A}, P), \text{Var}(\mathcal{X}) = \sigma^2\}, \omega \in \Omega, \sigma \in \mathbb{R}^+. \quad (4.4)$$

Definition 4.1.8 (Variance of a FRV (ontic view)).

Consider a fuzzy random variable $\tilde{\mathcal{X}}$ in the ontic sense defined on the probability space $(\Omega, \mathcal{A}, P)$ by the mapping $\tilde{\mathcal{X}} : \Omega \rightarrow \mathcal{F}_c(\mathbb{R}^p)$. If it exists, the variance of $\tilde{\mathcal{X}}$ is given by the real number $\text{Var}(\tilde{\mathcal{X}})$,

$$\text{Var}(\tilde{\mathcal{X}}) = \mathbb{E} \left([d_\infty(\tilde{\mathcal{X}}, \tilde{\mathbb{E}}(\tilde{\mathcal{X}}))]^2 \right), \quad (4.5)$$

where $\tilde{\mathbb{E}}$ is the fuzzy expected value from Definition 4.1.6 and d_∞ from Definition 4.1.2 is given, for $\tilde{\mathbf{x}}, \tilde{\mathbf{y}} \in \mathcal{F}(\mathbb{R}^p)$, by

$$d(\tilde{\mathbf{x}}, \tilde{\mathbf{y}}) = \sup_{\alpha \in [0,1]} d_H(\tilde{\mathbf{x}}^\alpha, \tilde{\mathbf{y}}^\alpha).$$

Notice that, from the epistemic perspective, fuzzy random variables take values in the space $\mathcal{F}_c(\mathbb{R}^1)$ rather than $\mathcal{F}_c(\mathbb{R}^p)$. In this context, induced fuzzy parameters are mappings from $\mathbb{R}^1 \rightarrow [0, 1]$. On the other hand, the ontic perspective, as developed by Puri and Dan A Ralescu (1986), treats fuzzy random variables as mappings valued in $\mathcal{F}_c(\mathbb{R}^p)$. When $p \geq 2$, these can also be referred to as fuzzy random vectors. However, it is crucial to distinguish between a vector of fuzzy random variables and a fuzzy random vector. This distinction is conceptually similar to the difference between a vector of fuzzy numbers and a fuzzy vector (see Definition 1.3). To clarify, consider a $\mathbb{R}^p$ -valued random vector $\mathcal{X} : \Omega \rightarrow \mathbb{R}^p$, where $\mathcal{X} = (\mathcal{X}_1, \dots, \mathcal{X}_p)'$ whose components $\mathcal{X}_i$ are random variables. These random variables can be fuzzified, yielding a vector of fuzzy random variables $\tilde{\mathcal{X}} = (\tilde{\mathcal{X}}_1, \dots, \tilde{\mathcal{X}}_p)'$. The realizations of such a random vector of fuzzy random variables are vectors of fuzzy numbers.

In contrast, a fuzzy random vector $\tilde{\mathcal{X}}^* : \Omega \rightarrow \mathcal{F}(\mathbb{R}^p)$ has realizations that are fuzzy vectors. While it is possible to construct a random vector of fuzzy random variables in both the epistemic and ontic senses, a fuzzy random vector can only be constructed in the ontic sense. The key difference is that the ontic view allows us to directly consider components that live in the space $\mathcal{F}(\mathbb{R}^p)$, whereas the epistemic view does not.

If we think about the classical case, that is every random vector is a vector whose components are random variables, but not every collection of random variables is necessarily considered or treated as a single random vector unless we specify the joint structure, we could argue that fuzzy random vectors are the natural extension of random vectors.

This distinction shows that, when calculating induced fuzzy parameters, such as the expected value of a fuzzy random vector, one must apply Definition 4.1.6. An example of such a calculation can be found in Gil, López-Díaz, and Dan A. Ralescu (2006) [p. 9].

4.2 Fuzzy distribution

Now that the concept of fuzzy random variable has been outlined, the following question lingers in our mind : " What is the distribution of fuzzy random variables ?" Despite the two different ways to perceive fuzzy random variables, we will restrict ourselves to $\mathcal{F}_c(\mathbb{R})$ -valued

mappings $\tilde{\mathcal{X}} : \Omega \rightarrow \mathcal{F}_c(\mathbb{R})$ so that the two perceptions, namely the ontic and epistemic views, **mathematically** coincide in the sense of proposition 4.1.1. Remember that, although the views "mathematically" coincide when restricted to $\mathcal{F}_c(\mathbb{R})$ -valued mappings $\tilde{\mathcal{X}} : \Omega \rightarrow \mathcal{F}_c(\mathbb{R})$, their induced fuzzy parameters may not. As stated before, under the epistemic's approach, the variance of **Random fuzzy variable (epistemic view) (RFV)** is fuzzy but is crisp if calculated with respect to the ontic approach. Hence, it is important to remember that these two approaches are conceptually different. In the following, since we work with fuzzy triangular observations, the originals are the core of these observations and it seems more reasonable to work with the epistemic view. Since, to our knowledge, theoretical fuzzy distributions have not been defined, we propose constructing fuzzy empirical distributions following Arefi, Viertl, and S. Mahmoud Taheri (2012)'s methodology.

Fuzzy empirical distribution by Arefi, Viertl, and S. Mahmoud Taheri (2012)

Let $\tilde{\mathbf{X}}$ be a collection of n fuzzy observations $\tilde{x}_i = (x_i^L, x_i, x_i^R)$, $i = 1, \dots, n$. Its estimated empirical fuzzy distribution denoted by $\hat{\tilde{F}} = (\hat{F}^L, \hat{F}, \hat{F}^R)$ is computed as follow:

$$\hat{F}^L(s) = \frac{1}{n} \sum_{i=1}^n \mathbb{1}_{\{x_i^L \leq s\}}, \quad (4.6)$$

$$\hat{F}(s) = \frac{1}{n} \sum_{i=1}^n \mathbb{1}_{\{x_i \leq s\}}, \quad (4.7)$$

$$\hat{F}^R(s) = \frac{1}{n} \sum_{i=1}^n \mathbb{1}_{\{x_i^R \leq s\}}, \quad (4.8)$$

where $\mathbb{1}_{\{\cdot\}}$ denotes the indicator function. To construct the empirical fuzzy distribution of a fuzzy estimator corresponding to a fuzzy parameter, a similar approach from Rosset and Donzé (2024d) can be adopted by employing the bootstrap method. Specifically, bootstrap samples are used to generate repeated estimates of the parameter, which are then used to form its empirical fuzzy distribution.

Let $\tilde{T}$ be a fuzzy estimator for a fuzzy parameter $\tilde{\theta}$ and by $\tilde{t}_0$ its estimation. Denote by $\tilde{t}_1^*, \dots, \tilde{t}_k^*$ the k new fuzzy parameters estimated by Bootstrap. Then, the estimated empirical fuzzy distribution noted $\hat{\tilde{F}} = (\hat{F}^L, \hat{F}, \hat{F}^R)$ is built using the vector of the $k + 1$ estimates $\tilde{\mathbf{t}} = (\tilde{t}_0, \tilde{t}_1^*, \dots, \tilde{t}_k^*)$. More specifically, by noting $\mathbf{t}^L = (t_0^L, t_1^{*,L}, \dots, t_k^{*,L})$, $\mathbf{t} = (t_0, t_1^*, \dots, t_k^*)$, $\mathbf{t}^R = (t_0^R, t_1^{*,R}, \dots, t_k^{*,R})$, the fuzzy empirical distribution at a value $s \in \mathbb{R}$, $\hat{\tilde{F}}(s) = (\hat{F}^L(s), \hat{F}(s), \hat{F}^R(s))$ is

calculated as follow:

$$\hat{F}^L(s) = \frac{1}{(k+1)} \sum_{i=0}^k \mathbb{1}_{\{t_i^L \leq s\}}, \quad (4.9)$$

$$\hat{F}(s) = \frac{1}{(k+1)} \sum_{i=0}^k \mathbb{1}_{\{t_i \leq s\}}, \quad (4.10)$$

$$\hat{F}^R(s) = \frac{1}{(k+1)} \sum_{i=0}^k \mathbb{1}_{\{t_i^R \leq s\}}. \quad (4.11)$$

Moreover, we call this distribution “augmented” since one estimate comes from the original data while the k others are from the bootstrapped samples.

4.3 Fuzzy confidence interval

This section aims at discussing different constructions of fuzzy confidence intervals and the required mathematical tools to define them, starting from Kruse and Meyer (1987) traditional approach to more recent ways to generate such confidence intervals proposed by Berkachy and Donzé (2020) and Rosset and Donzé (2024d). Kruse and Meyer (1987) assumed a crisp distribution for the unknown parameter in order to build its fuzzy confidence interval. While this approximation may give good results, it is a rather arbitrary assumption given the fuzzy nature of the observations.

Based on Kruse and Meyer (1987), let us first define a fuzzy confidence interval (see, e.g. Berkachy and Donzé (2020) and Berkachy (2020)). Recall that in a frequentist approach, the bounds of a confidence interval are functions of the sample. We note these functions by $\pi_i(\cdot)$, $i = 1, 2$.

Definition 4.3.1 (Fuzzy Confidence Interval).

Let $[\pi_1, \pi_2]$ be a symmetrical confidence interval for the parameter θ at significance level γ such that π_1 and π_2 are functions of the observations, i.e. $\pi_1 := \pi_1(x_1, \dots, x_n)$ and $\pi_2 := \pi_2(x_1, \dots, x_n)$. A fuzzy confidence interval $\tilde{\Pi}$ is a convex and normal fuzzy set, for which its respective left and right α -cuts $[\Pi^{L,\alpha}, \Pi^{R,\alpha}]$ are defined as

$$\begin{aligned} \Pi^{L,\alpha} &= \inf\{a \in \mathbb{R} : \exists x_i \in \tilde{x}_i^\alpha, \forall i = 1, \dots, n, \text{ such that } \pi_1(x_1, \dots, x_n) \leq a\}, \\ \Pi^{R,\alpha} &= \sup\{a \in \mathbb{R} : \exists x_i \in \tilde{x}_i^\alpha, \forall i = 1, \dots, n, \text{ such that } \pi_2(x_1, \dots, x_n) \geq a\}. \end{aligned}$$

Its membership function $\mu_{\tilde{\Pi}}(x)$ can be written as

$$\mu_{\tilde{\Pi}}(x) = \sup\{\alpha \mathbb{1}_{\{[\Pi^{L,\alpha}, \Pi^{R,\alpha}]\}} : \alpha \in [0, 1]\}.$$

where the indicator function $\mathbb{1}_{\{\tilde{x}^\alpha\}}(x)$ of an α -cut $\tilde{x}^\alpha$ is defined as follows:

$$\mathbb{1}_{\{\tilde{x}^\alpha\}}(x) = \begin{cases} 1 & \text{if } \mu_{\tilde{x}}(x) \geq \alpha, \\ 0 & \text{otherwise.} \end{cases} \quad (4.12)$$

This fuzzy confidence interval is said to belong to the $1 - \gamma$ confidence region such that for all parameter θ , we have that

$$\mathbb{P}(\square^{L,\alpha} \leq \theta \leq \square^{R,\alpha}) \geq 1 - \gamma, \forall \alpha \in [0, 1].$$

Having established the notion of a fuzzy confidence interval, let us introduce two distinct methodologies from Rosset and Donzé (2024d) for its construction. The first method extends the classical framework traditionally employed for confidence interval estimation while the second approach is grounded in the concept of fuzzy quantiles.

4.3.1 Fuzzy confidence interval: Method 1

Let θ be an unknown distribution parameter. Then, let $x_1, \dots, x_n$ be the n realisations of a random sample $X_1, \dots, X_n$. Assuming a fuzzy modelling of the data, note by $\tilde{X}_1, \dots, \tilde{X}_n$, the fuzzy equivalent of the random sample and write its realisations as $\tilde{x}_1, \dots, \tilde{x}_n$. Let $\tilde{T} := \tilde{T}(\tilde{X}_1, \dots, \tilde{X}_n)$ be a fuzzy estimator of θ . An estimation is noted by $\tilde{t} := \tilde{T}(\tilde{x}_1, \dots, \tilde{x}_n)$. Let $\tilde{F}$ be the unknown fuzzy distribution of $\tilde{T}$. Note that, in absence of fuzziness, $\tilde{F}$ simply reduces to a crisp distribution F for all α -cut. By means of Bootstrap, generate k new fuzzy random samples $\tilde{X}_1^{(1)}, \dots, \tilde{X}_n^{(1)}, \dots, \tilde{X}_1^{(k)}, \dots, \tilde{X}_n^{(k)}$. We can then estimate k new parameters $\tilde{t}_1^* = \tilde{T}(\tilde{x}_1^{(1)}, \dots, \tilde{x}_n^{(1)}), \dots, \tilde{t}_k^* = \tilde{T}(\tilde{x}_1^{(k)}, \dots, \tilde{x}_n^{(k)})$ allowing us to generate a fuzzy empirical distribution function as in Definition 4.2, noted by $\hat{\tilde{F}}$.

By using the quantiles of the left and right parts of $\hat{\tilde{F}}$, the left and right parts of a fuzzy confidence interval for $\hat{\theta}$ at a given significance level γ and α -cut are:

$$\hat{\theta}_{\gamma/2}^{L,\alpha} = t^{L,\alpha} - (t_{(k+1)(1-\gamma/2)}^{*,L,\alpha} - t^{L,\alpha}) = 2t^{L,\alpha} - t_{(k+1)(1-\gamma/2)}^{*,L,\alpha}, \quad (4.13)$$

$$\hat{\theta}_{1-\gamma/2}^{R,\alpha} = t^{R,\alpha} - (t_{(k+1)(\gamma/2)}^{*,R,\alpha} - t^{R,\alpha}) = 2t^{R,\alpha} - t_{(k+1)(\gamma/2)}^{*,R,\alpha}, \quad (4.14)$$

where, $t_{(k+1)(1-\gamma/2)}^{*,L,\alpha}$ and $t_{(k+1)(\gamma/2)}^{*,R,\alpha}$ are respectively the $(1 - \gamma/2)$ -quantile, $\gamma/2$ -quantile of $\hat{\tilde{F}}^L$, $\hat{\tilde{F}}^R$. At confidence level $1 - \gamma$, we can thus write the fuzzy confidence interval as

$$\tilde{\square}^\alpha = [\square^{L,\alpha}, \square^{R,\alpha}] = [\hat{\theta}_{\gamma/2}^{L,\alpha}, \hat{\theta}_{1-\gamma/2}^{R,\alpha}]. \quad (4.15)$$

Let us prove that equation (4.15) indeed is a fuzzy confidence interval, defined above in Definition 4.3.1.

Proof.

1. Observe that the quantiles $\hat{\theta}_{\gamma/2}^{L,\alpha}$, $\hat{\theta}_{1-\gamma/2}^{R,\alpha}$ are taken from the left and right parts of a fuzzy distribution which are respectively non-decreasing and non-increasing when seen as a function of α , implying that, for $\alpha_2 < \alpha_1$, $\alpha_1, \alpha_2 \in (0, 1]$,

$$[(\hat{\theta}_{\gamma/2})^{L,\alpha_1}, (\hat{\theta}_{1-\gamma/2})^{R,\alpha_1}] \subset [(\hat{\theta}_{\gamma/2})^{L,\alpha_2}, (\hat{\theta}_{1-\gamma/2})^{R,\alpha_2}]. \quad (4.16)$$

Then, let $\mu_{\tilde{\Gamma}}$ be the membership function of $\tilde{\Gamma}$. Choose $x_1 \in [(\hat{\theta}_{\gamma/2})^{L,\alpha_1}, (\hat{\theta}_{1-\gamma/2})^{R,\alpha_1}] = \tilde{\Gamma}^{\alpha_1}$ and $x_2 \in [(\hat{\theta}_{\gamma/2})^{L,\alpha_2}, (\hat{\theta}_{1-\gamma/2})^{R,\alpha_2}] = \tilde{\Gamma}^{\alpha_2}$. Since $\alpha_2 < \alpha_1$, we have that $\tilde{\Gamma}^{\alpha_1} \subseteq \tilde{\Gamma}^{\alpha_2}$ which means that $\min(\mu_{\tilde{\Gamma}}(x_1)) \geq \min(\mu_{\tilde{\Gamma}}(x_2))$ and in particular, for any point $x_1 \in \tilde{\Gamma}^{\alpha_1}$, we have $\mu_{\tilde{\Gamma}}(x_1) \geq \min(\mu_{\tilde{\Gamma}}(x_1)) \geq \min(\mu_{\tilde{\Gamma}}(x_2))$. Let $x_3 = tx_1 + (1-t)x_2$, $t \in [0, 1]$ be a linear combination of x_1 and x_2 . Thus x_3 belongs in either $\tilde{\Gamma}^{\alpha_1}$ or $\tilde{\Gamma}^{\alpha_1}$, implying $\mu_{\tilde{\Gamma}}(x_3) \geq \min(\mu_{\tilde{\Gamma}}(x_1), \mu_{\tilde{\Gamma}}(x_2))$, ending up proving the convexity of $\tilde{\Gamma}$.

2. Observe that, for all $\alpha \in [0, 1]$, $[(\hat{\theta}_{\gamma/2})^{L,\alpha}, (\hat{\theta}_{1-\gamma/2})^{R,\alpha}] \neq \emptyset$ since $[\hat{\theta}_{\gamma/2}, \hat{\theta}_{1-\gamma/2}] \subseteq [(\hat{\theta}_{\gamma/2})^{L,\alpha}, (\hat{\theta}_{1-\gamma/2})^{R,\alpha}]$ where $\hat{\theta}_{\gamma/2}$, $\hat{\theta}_{1-\gamma/2}$ are the quantiles of $\hat{F}$. Since it holds for any $\alpha \in [0, 1]$, it is in particular true for $\alpha = 1$, meaning that $\sup(\mu_{\tilde{\Gamma}}(x)) = 1$, proving the normality.
3. Finally, by construction, it is evident that $\mathbb{P}(\square^{L,\alpha} \leq \theta \leq \square^{R,\alpha}) \geq 1 - \gamma, \forall \alpha \in [0, 1]$.

□

4.3.2 Fuzzy confidence interval: Method 2

The second method to generate fuzzy confidence intervals is based upon the construction of quantile functions of a fuzzy random variable in the epistemic sense defined by Shvedov (2016). As previously, denote by $\tilde{T}$ a fuzzy estimator of a fuzzy parameter $\tilde{\theta}$ following an unknown fuzzy distribution $\tilde{F}$. Assume that, in the same fashion as in the first method, we built by Bootstrap an empirical fuzzy distribution function $\hat{\tilde{F}}$. Let us remind that, for a fixed α -cut, the fuzzy estimator and its associated distribution are noted as $\tilde{T}^\alpha = (T^{L,\alpha}, T, T^{R,\alpha}) \sim \hat{\tilde{F}}^\alpha = (\hat{F}^{L,\alpha}, \hat{F}, \hat{F}^{R,\alpha})$. Then, with a quantile function denoted by $Q(\cdot)$, the fuzzy quantiles at a given α -cut of $\hat{\tilde{F}}$ are given by

$$\tilde{Q}(p, \hat{\tilde{F}}^\alpha) = \left(Q(p, \hat{F}^{L,\alpha}), Q(p, \hat{F}), Q(p, \hat{F}^{R,\alpha}) \right), \quad \alpha \in (0, 1].$$

Moreover, by construction the following inequality holds,

$$Q(p, \hat{F}^{L,\alpha}) \leq Q(p, \hat{F}) \leq Q(p, \hat{F}^{R,\alpha}). \quad (4.17)$$

Assuming $p < (1 - p)$, with regard to the above property (4.17), we can create three intervals,

$$I_p^{U,\alpha} = [Q(p, \hat{F}^{L,\alpha}); Q(1-p, \hat{F}^{R,\alpha})] \quad (4.18)$$

$$I_p = [Q(p, \hat{F}); Q(1-p, \hat{F})] \quad (4.19)$$

$$I_p^{L,\alpha} = \begin{cases} [Q(p, \hat{F}^{R,\alpha}); Q(1-p, \hat{F}^{L,\alpha})] & \text{if } Q(p, \hat{F}^{R,\alpha}) < Q(1-p, \hat{F}^{L,\alpha}) \\ \emptyset & \text{otherwise.} \end{cases} \quad (4.20)$$

At an α -cut, for a small value p , $p < (1-p)$, the interval $I_p^{U,\alpha}$ is formed by the smallest p quantile, $Q(p, \hat{F}^{L,\alpha})$ and the greatest $1-p$ quantile, $Q(1-p, \hat{F}^{R,\alpha})$. As such, $I_p^{U,\alpha}$ is the widest interval of the three. I_p is generated by the p and $1-p$ quantiles of the crisp distribution $\hat{F}$, hence I_p has the form of a $(1-2p)$ confidence interval. Lastly, $I_p^{L,\alpha}$ consists of the biggest p quantile, $Q(p, \hat{F}^{R,\alpha})$ and the smallest $1-p$ quantile, $Q(1-p, \hat{F}^{L,\alpha})$, meaning it is the smallest interval of the three, sometimes even being empty if $Q(p, \hat{F}^{R,\alpha}) > Q(1-p, \hat{F}^{L,\alpha})$ which can occur for low α -cut values of rather fuzzy distributions. Based on these considerations, it is easy to verify that, under the assumption that $p < (1-p)$,

$$I_p^{U,\alpha} \supseteq I_p \supseteq I_p^{L,\alpha}.$$

A fuzzy $(1-2p)$ confidence interval is then defined as

$$[Q(p, \hat{F}^{L,\alpha}); Q(1-p, \hat{F}^{R,\alpha})], \quad \forall \alpha \in [0, 1]. \quad (4.21)$$

Because it is computationally heavy to calculate many α -cuts, we can also define a fuzzy $(1-2p)$ confidence interval as a trapezoidal fuzzy number:

$$[Q(p, \hat{F}^{L,\alpha=0}); Q(p, \hat{F}); Q(1-p, \hat{F}); Q(1-p, \hat{F}^{R,\alpha=0})], \quad \forall \alpha \in [0, 1]. \quad (4.22)$$

The proof showing that equations (4.21), (4.22) are indeed fuzzy confidence intervals satisfying Definition 4.3.1 is identical to Proof 4.3.1.

4.4 Fuzzy inference for a parameter

Similarly to subsection 4.3.1, let θ be an unknown distribution parameter of an unknown distribution, and let $X_1, \dots, X_n$ represent a random sample drawn from this distribution. Assuming a fuzzy modeling approach for the data, we denote the fuzzy equivalents of the random sample as $\tilde{X}_1, \dots, \tilde{X}_n$. The vector of fuzzy numbers corresponding to the sample is denoted by $\tilde{\mathbf{X}} = (\tilde{X}_1, \dots, \tilde{X}_n)$, with the membership functions of its corresponding components being $\mu_{\tilde{X}_i} : \mathbb{R} \rightarrow [0, 1]$.

Let $\phi : \mathbb{R}^n \rightarrow \mathbb{R}$ be a real-valued function, and assume that ϕ serves as the appropriate test function for the parameter θ . In the case of fuzzy data, this function is applied to the vector of fuzzy numbers, so that $\phi(\tilde{\mathbf{X}}) = \phi(\tilde{X}_1, \dots, \tilde{X}_n)$. The result is a fuzzy number, which we denote as $\tilde{\phi}$, and we refer to its α -cuts by $\tilde{\phi}^\alpha$. Let us define a fuzzy test in the sense of Kruse and Meyer (1987). See also Berkachy (2020, p.122) and Viertl (2011, p.82) for a complete definition of a fuzzy test statistic.

Definition 4.4.1 (Fuzzy Test).

Consider a fuzzy random vector $\tilde{X}_1, \dots, \tilde{X}_n$ related to the original $X_1, \dots, X_n$ in the space Ω . Let δ be a particular value of the interval $[0, 1]$. A function $\tilde{\phi}$ is called a fuzzy test for the null H_0 and the alternative H_1 hypotheses at the significance level δ if

$$\sup_{\alpha \in [0,1]} P_{H_0} \left\{ \omega_i \in \Omega, i = 1, \dots, n : \tilde{\phi}^\alpha \left(\tilde{X}_1(\omega_1), \dots, \tilde{X}_n(\omega_n) \right) \in \{1\} \right\} \leq \delta \quad (4.23)$$

where $\tilde{\phi}^\alpha$ is the α -cut of $\tilde{\phi}$ and P_{H_0} means that the probability is evaluated under the condition of the null hypothesis, i.e. $\tilde{\phi}$ is constructed to test H_0 .

Following the procedure outlined, for instance, in Berkachy (2020), we can propose new decision rules for hypothesis testing at a significance level δ , specifically designed for fuzzy data and fuzzy hypotheses. It is important to note that a fuzzy hypothesis represents the transformation of a crisp hypothesis into a fuzzy number, typically triangular in form. The fuzzy null and alternative hypotheses are denoted as $\tilde{H}_0$ and $\tilde{H}_1$, respectively.

First, let us assume that $\tilde{\pi}$ represents a fuzzy confidence interval for the parameter δ at a confidence level of $1 - \delta$. Below, in Definition 4.4.2, we describe how to construct such a confidence interval.

Definition 4.4.2 (Fuzzy statistical testing procedure). Let $\tilde{\phi}^{\text{rej}}$ and $\tilde{\phi}^{\text{acc}}$, two fuzzy statistical tests on $\mathcal{F}_c^*(\mathbb{R})$, the class of the non-empty compact, convex and normal fuzzy sets on $\mathbb{R}$, such that

$$\tilde{\phi}^{\text{rej}} : (\mathcal{F}_c^*(\mathbb{R}))^n \rightarrow [0, 1], \quad \tilde{\phi}^{\text{acc}} : (\mathcal{F}_c^*(\mathbb{R}))^n \rightarrow [0, 1].$$

For a given α -cut, we assume the following functions:

$$\tilde{\phi}^{\text{rej},\alpha}(\tilde{X}_1, \dots, \tilde{X}_n) = \begin{cases} 0 & \text{if } \tilde{H}_0^\alpha \subseteq \tilde{\pi}^\alpha \text{ or if } \alpha = 1, \\ 1 & \text{if } \tilde{H}_0^\alpha \subseteq (\tilde{\pi}^\alpha)^c, \\ b_\alpha & \text{if } \left(\tilde{H}_0^\alpha \cap (\tilde{\pi}^\alpha)^c \neq \emptyset \text{ and } \neg(\tilde{H}_0^\alpha \supset \tilde{\pi}^\alpha) \text{ and } \alpha < 1 \right), \\ c_\alpha & \text{if } \left(\tilde{H}_0^\alpha \cap (\tilde{\pi}^\alpha)^c \neq \emptyset \text{ and } \tilde{H}_0^\alpha \supset \tilde{\pi}^\alpha \text{ and } \alpha < 1 \right), \\ 0 & \text{otherwise,} \end{cases} \quad (4.24)$$

$$\tilde{\phi}^{\text{acc},\alpha}(\tilde{X}_1, \dots, \tilde{X}_n) = \begin{cases} 0 & \text{if } \left(\tilde{H}_0^\alpha \subseteq (\tilde{\pi}^\alpha)^c \text{ or } \alpha = 1 \right), \\ 1 & \text{if } \tilde{H}_0^\alpha \subseteq \tilde{\pi}^\alpha, \\ d_\alpha & \text{if } \left(\tilde{H}_0^\alpha \cap \tilde{\pi}^\alpha \neq \emptyset \text{ and } \alpha < 1 \right), \\ 0 & \text{otherwise,} \end{cases} \quad (4.25)$$

where $b_\alpha = (\frac{r_R - r_L}{H_0^{R,\alpha} - H_0^{L,\alpha}})$, $c_\alpha = (\frac{r_R - r_L - (\cap^{R,\alpha} - \cap^{L,\alpha})}{H_0^{R,\alpha} - H_0^{L,\alpha}})$ and $d_\alpha = (\frac{a_R - a_L}{H_0^{R,\alpha} - H_0^{L,\alpha}})$. When $\alpha = 1$, $H_0^{R,\alpha} - H_0^{L,\alpha} = 0$, we cannot evaluate b_α , c_α , d_α , hence we have to define a specific rule. The quantities a_L , a_R , r_L and r_R are given by:

$$\begin{aligned} a_L &= \inf(\tilde{H}_0^\alpha \cap \tilde{\Gamma}^\alpha), & a_R &= \sup(\tilde{H}_0^\alpha \cap \tilde{\Gamma}^\alpha), \\ r_L &= \inf(\tilde{H}_0^\alpha \cap (\tilde{\Gamma}^\alpha)^c), & r_R &= \sup(\tilde{H}_0^\alpha \cap (\tilde{\Gamma}^\alpha)^c). \end{aligned}$$

The fuzzy tests $\tilde{\phi}^{\text{rej}}$ and $\tilde{\phi}^{\text{acc}}$ constructed above allow us to propose a set of two new decision rules, as defined in the following:

Definition 4.4.3 (Fuzzy decisions rules). Let

$$L^{\text{rej}} = \int_{[0,1]} \tilde{\phi}^{\text{rej},\alpha} d\alpha \quad \text{and} \quad L^{\text{acc}} = \int_{[0,1]} \tilde{\phi}^{\text{acc},\alpha} d\alpha. \quad (4.26)$$

L^{rej} and L^{acc} are referred to as the levels of rejection and acceptance (non-rejection) of the (fuzzy) null hypothesis $\tilde{H}_0$, respectively.

1. If $L^{\text{rej}} = 1$, then $\tilde{H}_0$ is rejected. In this case, $L^{\text{acc}} = 0$;
2. If $L^{\text{rej}} = 0$, then $\tilde{H}_0$ is not rejected. In this case, $L^{\text{acc}} = 1$;
3. If $0 < L^{\text{rej}} < 1$, the test indicates the strength of rejection of $\tilde{H}_0$. The higher the value of L^{rej} , the stronger the rejection of $\tilde{H}_0$;
4. If $0 < L^{\text{acc}} < 1$, the test indicates the strength of non-rejection of $\tilde{H}_0$. The higher the value of L^{acc} , the stronger the non-rejection of $\tilde{H}_0$;
5. If $L^{\text{rej}} > L^{\text{acc}}$, $\tilde{H}_0$ tends to be rejected;
6. If $L^{\text{rej}} < L^{\text{acc}}$, $\tilde{H}_0$ tends to be not rejected;
7. If $L^{\text{rej}} = L^{\text{acc}}$, no decision can be made.

Geometrically, the idea behind this fuzzy inference is to construct a fuzzy decision rule by comparing the following two ratios: the area of the fuzzy hypothesis $\tilde{H}_0$ inside the rejection domain $\neg\tilde{\Gamma}$, and the area of $\tilde{H}_0$ inside the acceptance domain $\tilde{\Gamma}$, both divided by the total area of $\tilde{H}_0$. These ratios are then compared to make a decision. Additionally, observe that this method is general in the sense that no assumption is made regarding the distribution of the parameter θ . However, the fuzzy confidence interval plays a key role in the values taken by the two fuzzy tests $\tilde{\phi}^{\text{rej}}$ and $\tilde{\phi}^{\text{acc}}$, hence the method chosen to estimate it will impacts the decision taken.

4.5 Fuzzy p-values and inferences

p-values are a central component of statistical inference, and as such, it is natural to examine the p-values that arise from fuzzy testing procedures. Similarly to Chapters 2 and 3 about regression methods, there are several angles to explore the concept of fuzzy p-values. For example, one could consider fuzzy hypotheses with crisp data, crisp hypotheses with fuzzy data, or both. Traditionally, similar to classical statistical methods, fuzzy p-values have been computed assuming a known distribution for the test statistic. As such, this approach has been widely examined in the literature. However, when both the data and possibly the hypotheses are fuzzy, it becomes crucial to account for the fuzzy distribution of the test statistic. As far as we are aware, little research has focused on this particular case.

In the traditional framework, Filzmoser and Viertl (2004) and Viertl (2011) studied fuzzy p-values arising from fuzzy data with crisp hypotheses. They did so by employing a fuzzy test statistic while assuming a crisp distribution for the sample. This setup allowed them to compute a p-value corresponding to each of the left and right α -cuts of the test statistic, which together formed the α -cuts of a fuzzy p-value. In contrast, Parchami, S. Mahmoud Taheri, and Mashinchi (2010) focused on the case where hypotheses are fuzzy and the data are crisp. They introduced the concept of a fuzzy p-value through the application of Zadeh's extension principle. Later, Parchami (2020) extended this work to the scenario where both the hypotheses and the data are fuzzy. He investigated the fuzzy p-values derived from Zadeh's extension principle, assuming a crisp distribution for the test statistic despite the fuzziness of the data. Similarly, Berkachy and Donzé (2017) analyzed the case of fuzzy data and hypotheses while also assuming a crisp distribution for the test statistic.

Additionally, Hryniewicz (2018) proposed a method for deriving fuzzy p-values using fuzzy confidence intervals based on the work of Kruse and Meyer (1987). His approach involves identifying the intersection points between the fuzzy null hypothesis and the fuzzy confidence intervals of the test statistic. While this method is interesting, it has two main limitations. First, it does not account for the fuzzy distribution of the test statistic. Second, when the fuzzy confidence interval becomes crisp, the intersection points (p-values) take values of either 0 or 1, which does not generalize the classical p-value approach.

Given that little research has yet addressed the issue of using fuzzy distributions to compute fuzzy p-values, when dealing with fuzzy data and hypotheses, a significant advancement would be to incorporate the fuzzy distribution of the test statistic in the calculation of fuzzy p-values. Such procedure should also guarantee that, in the case of crisp data and hypotheses, the resulting fuzzy p-values revert to classical p-values. Finally, another important point is to establish a clear decision rule to evaluate fuzzy p-values in order to ensure their interpretability.

Let us start by defining what we mean by a fuzzy hypothesis.

Fuzzy hypothesis

Denote the unknown fuzzy parameter of interest by $\tilde{\theta}$. A fuzzy hypothesis can be expressed by using linguistic terms, for example, by "*The value of the parameter θ is more or less equal to θ_0* ", or rewritten in terms of differences, i.e. "*The difference between the value of the parameter θ and θ_0 is more or less equal to 0*". Thus, a two-sided fuzzy hypothesis takes the following form:

$$\tilde{H}_0 : \theta = \tilde{0} \text{ against } \tilde{H}_1 : \theta \neq \tilde{0}. \quad (4.27)$$

Of course, in the same way, one-sided fuzzy hypotheses could be written as:

$$\tilde{H}_0 : \theta \leq \tilde{0} \text{ against } \tilde{H}_1 : \theta > \tilde{0}, \quad (4.28)$$

or

$$\tilde{H}_0 : \theta \geq \tilde{0} \text{ against } \tilde{H}_1 : \theta < \tilde{0}. \quad (4.29)$$

Remark that the null fuzzy hypothesis $\tilde{H}_0$ or the alternative one $\tilde{H}_1$ can be modelled by triangular numbers, i.e. (H_0^L, H_0, H_0^R) or (H_1^L, H_1, H_1^R) . On the other hand, it is common practice to implement the inference process through a proper test statistic.

To understand how to compute fuzzy p-values, one first needs to have a fuzzy statistic. Let $\tilde{T} = (T^L, T, T^R)$ be a fuzzy statistic to test $\tilde{H}_0$ with its estimations noted by $\tilde{t} = (t^L, t, t^R)$. For example, if we want to test the mean of the distribution, we could use:

$$\tilde{T} := (\tilde{X} - \theta_0) \odot \tilde{\sigma}_{\tilde{X}}, \quad (4.30)$$

where $\odot$ is the fuzzy division operator and $\tilde{\sigma}_{\tilde{X}}$ is the standard-error of $\tilde{X}$.

Assuming that we estimated by bootstrap B statistics $\tilde{t}_1^*, \dots, \tilde{t}_b^*, \dots, \tilde{t}_B^*$ plus the original $\tilde{t}$, we can generate a fuzzy empirical distribution as explained in Definition 4.2.

However, given that we are interested in determining whether or not "*the difference between the value of the parameter θ and θ_0 is more or less equal to 0*", it is more advantageous to consider the centred bootstrapped distribution $\hat{\tilde{F}}$ built from $\tilde{t}_1^* - \tilde{t}^*, \dots, \tilde{t}_b^* - \tilde{t}^*, \dots, \tilde{t}_B^* - \tilde{t}^*$ where $\tilde{t}^* := \frac{1}{B+1} \left(t + \sum_{i=1}^B t_i^* \right)$. Then, by considering the fuzzy observed statistic $\tilde{t} = (t^L, t, t^R)$ and its absolute value:

$$\begin{aligned} |\tilde{t}| &= (t_{abs}^L, t_{abs}, t_{abs}^R) \\ &= (\min(|t^L|, |t|, |t^R|), |t|, \max(|t^L|, |t|, |t^R|)), \end{aligned}$$

we are now able to give the definition of the fuzzy p-value.

Definition 4.5.1. Fuzzy p-value

The fuzzy p-value $\tilde{p} = (p^L, p, p^R)$ of the two-sided hypotheses test (4.27) can be approximated as follow:

$$\begin{aligned} p^L &= 2 \cdot \frac{\#(F^{*,L} \geq t_{abs}^R) + 1}{B + 1}, \\ p &= 2 \cdot \frac{\#(F^* \geq t_{abs}) + 1}{B + 1}, \\ p^R &= \min(2 \cdot \frac{\#(F^{*,R} \geq t_{abs}^L) + 1}{B + 1}, 1), \end{aligned}$$

where $\#$ gives the number of cases for which the condition in parentheses is true. For the one-sided hypotheses test (4.28), the fuzzy p-value $\tilde{p} = (p^L, p, p^R)$ is calculated as:

$$\begin{aligned} p^L &= \frac{\#(F^{*,L} \geq t^R) + 1}{B + 1}, \\ p &= \frac{\#(F^* \geq t) + 1}{B + 1}, \\ p^R &= \min(\frac{\#(F^{*,R} \geq t^L) + 1}{B + 1}, 1), \end{aligned}$$

and for the second one-sided hypotheses test (4.29), we have:

$$\begin{aligned} p^L &= \frac{\#(F^{*,R} \leq t^L) + 1}{B + 1}, \\ p &= \frac{\#(F^* \leq t) + 1}{B + 1}, \\ p^R &= \min(\frac{\#(F^{*,L} \leq t^R) + 1}{B + 1}, 1). \end{aligned}$$

Note that we used the crisp value $\bar{t}^*$ to centre the fuzzy bootstrapped distribution $\hat{\tilde{F}}$. Instead of building it from $\tilde{t}_1^* - \bar{t}^*, \dots, \tilde{t}_b^* - \bar{t}^*, \dots, \tilde{t}_B^* - \bar{t}^*$, one can centre it around the fuzzy empirical mean $\tilde{t}^*$ by constructing it from $\tilde{t}_1^* \ominus \tilde{t}^*, \dots, \tilde{t}_b^* \ominus \tilde{t}^*, \dots, \tilde{t}_B^* \ominus \tilde{t}^*$. Empirically, we noticed that the latter gives a much fuzzier centred bootstrapped distribution. This phenomenon may be attributed to the following consideration: since the fuzziness of the data is already reflected in both the empirical mean $\tilde{t}^*$ and the distribution $\hat{\tilde{F}}$, it is reasonable to centre the distribution around the crisp value $\bar{t}^*$ in order to avoid the redundant propagation of the fuzziness inherent in $\hat{\tilde{F}}$.

The computation steps to find a fuzzy p-value are summarised in algorithm 4.1.

Interpretation of a fuzzy p-value

Fuzzy p-values, denoted as (p^L, p, p^R) , are more challenging to interpret than their crisp counterparts. While an appropriate defuzzification method may facilitate their interpretation, we propose an alternative approach that does not rely on such process by constructing a measure conceived for fuzzy p-values that provides an intuitive understanding.

Definition 4.5.2 (Rejection and acceptance levels).

Let $\gamma \in [0, 1]$ be a chosen significance level and $\tilde{p} = (p^L, p, p^R)$ be a triangular fuzzy p-value with membership function $\mu_{\tilde{p}}(x)$ defined as:

$$\mu_{\tilde{p}}(x) = \begin{cases} \frac{x - p^L}{p - p^L} & \text{if } p^L < x < p, \\ \frac{x - p^R}{p - p^R} & \text{if } p \leq x < p^R, \\ 0 & \text{otherwise.} \end{cases}$$

Define the following integrals:

$$P = \int_0^1 \mu_{\tilde{p}}(x) dx, \quad (4.31)$$

$$A = \int_0^\gamma \mu_{\tilde{p}}(x) dx. \quad (4.32)$$

Then, the *rejection level* and *acceptance level* are defined, respectively, as:

$$r_{\text{rej}} = \frac{A}{P}, \quad (4.33)$$

$$r_{\text{acc}} = 1 - r_{\text{rej}}. \quad (4.34)$$

The value $r_{\text{rej}} \in [0, 1]$ quantifies the tendency to reject the null hypothesis. Conversely, we can define $r_{\text{acc}} = 1 - r_{\text{rej}}$, which represents the tendency to not reject the null hypothesis. Hence, we can enunciate the following decision rules :

- if $r_{\text{rej}} > 0.5$, there is an inclination to reject the null hypothesis;
- if $r_{\text{rej}} < 0.5$, the null hypothesis is retained;
- if $r_{\text{rej}} = 0.5$, no decision can be made.

This allows r_{rej} and r_{acc} to be thought of as weights indicating the inclination of rejecting, respectively not rejecting the null hypothesis. Figure 4.1 illustrates this concept at a significance level $\gamma = 0.05$. The blue region corresponding to $r_{\text{rej}} = 0.22$, represents the tendency to reject the null hypothesis, while the orange region, with an area of $r_{\text{acc}} = 1 - r_{\text{rej}} = 0.78$, reflects the inclination to retain it.

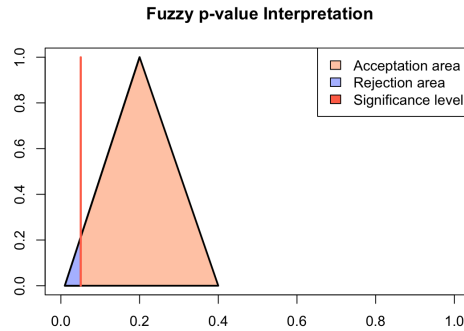


Figure 4.1: Depiction of the fuzzy p-value $\tilde{p} = (p^L, p, p^R) = (0.01, 0.2, 0.4)$. At the significance level $\delta = 0.05$, the tendency to reject the null hypothesis (in blue) is $r_{rej} = 0.22$, and the tendency to retain the null hypothesis (in orange) is $r_{acc} = 1 - r_{rej} = 0.78$

Algorithm 4.1 Fuzzy p-value computation by a bootstrapped fuzzy distribution from Rosset and Donzé (2024a)

- 1: Observe $\tilde{x}_1, \dots, \tilde{x}_n$ a fuzzy realization of the random sample;
 - 2: Compute $\tilde{t} = (t^L, t, t^R)$ and $|\tilde{t}| = (\min(|t^L|, |t|, |t^R|), |t|, \max(|t^L|, |t|, |t^R|))$;
 - 3: **for** $b = 1, \dots, B$ **do**
 - 4: Draw bootstrapped sample $\tilde{s}_b = \{\tilde{x}_1^*, \dots, \tilde{x}_n^*\}$;
 - 5: Compute $\tilde{t}_b^*$;
 - 6: **end for**
 - 7: Compute $\bar{t}^* := \frac{1}{B+1} \left(t + \sum_{b=1}^B t_b^* \right)$;
 - 8: Generate the empirical centred fuzzy distribution $\hat{F}^*$ from $\tilde{t}_1^* - \bar{t}^*, \dots, \tilde{t}_B^* - \bar{t}^*$;
 - 9: Compute the fuzzy p-value $\tilde{p} = (p^L, p, p^R)$ via Definition 4.5.1.
-

4.6 Simulation studies

Let us see through a small numerical application how the notions we introduced interconnect and allow us to perform fuzzy inferences. The idea is the following. First we set the fuzzy null and alternative hypotheses $\tilde{H}_0$ and $\tilde{H}_1$. Then with the help of a proper estimator, with the observed fuzzy data we compute an original estimate. Then, by bootstrap, we can generate more estimates from which we can build an empirical fuzzy distribution. From this distribution we can derive a fuzzy confidence interval. From there, we can either calculate a fuzzy p -value or use the statistical testing procedure from definition 4.24 to take a decision concerning which hypothesis is more inclined to be retained. Consider first the following set of fuzzy observations. Let $X \sim N(2.8, 3)$, $U_1 \sim \text{Unif}(0.1, 0.6)$, $U_2 \sim \text{Unif}(0.4, 0.9)$. Then write by x_i , $u_{1,i}$, $u_{2,i}$ the realisations of the random variables X , U_1 and U_2 respectively. We can then easily create fuzzy observations of the form (x_i^L, x_i, x_i^R) where $x_i^L = x_i - u_{1,i}$, $x_i^R = x_i + u_{2,i}$. With this procedure, we can generate any amount n of fuzzy triangular observations $\tilde{x}_i = (x_i^L, x_i, x_i^R)$, $i = 1, \dots, n$. For this numerical application, we chose $n = 500$. Table 4.1 displays some observations generated via this method.

In order to test the null hypothesis $\tilde{H}_0 = (H_0^L, H_0, H_0^R)$ against the alternative hypothesis $\tilde{H}_1 = (H_1^L, H_1, H_1^R)$, where both $\tilde{H}_0$ and $\tilde{H}_1$ are described by TFNs, we first need to specify a test statistic. Let us perform a test on the mean $\tilde{\theta}$. More specifically, let us consider the following two sided test:

$$\tilde{H}_0 : \tilde{\theta} = \tilde{4} \text{ against } \tilde{H}_1 : \tilde{\theta} \neq \tilde{4},$$

where $\tilde{4} = (3.3, 4, 4.5)$. By using the fuzzy estimator for the fuzzy mean, $\tilde{T}(\tilde{\mathbf{x}}) := \frac{1}{n} \sum_{i=1}^{n=500} \tilde{x}_i$, we find with respect to the set of observations 4.1 the original estimate $\tilde{\bar{x}}_0 = (2.468454, 2.81223, 3.462101)$. Then, by bootstrap, we can find $B = 999$ new estimates $\tilde{\bar{x}}_b^*$, $b = 1, \dots, B$ for the mean, of which 9 are shown in Table 4.2.

Then, with $\tilde{\bar{x}}_0$ and all the $\tilde{\bar{x}}_b^*$, we can build the augmented empirical fuzzy distribution $\hat{\tilde{F}} = (\hat{\tilde{F}}^L, \hat{\tilde{F}}, \hat{\tilde{F}}^R)$ of the fuzzy mean. Figure 4.2 displays in blue $\hat{\tilde{F}}^L$, in black $\hat{\tilde{F}}$ and in red $\hat{\tilde{F}}^R$. Moreover, we can also generate a histogram of the left, centre and right part of the fuzzy mean as depicted in Figure 4.3. The blue, black and red histogram have been generated with respect to the left, centre and right parts of $\tilde{\bar{x}}_0$ and of $\tilde{\bar{x}}_b^*$.

By using equations (4.13) and (4.14), at the significance level $\gamma = 5\%$ and $\alpha = 0$ -cut, one finds the fuzzy confidence interval $(2.2077, 2.5647, 3.0733, 3.7043)$. Similarly, with the technique presented in Section (4.3.2), one finds $(2.2085, 2.5512, 3.0598, 3.7194)$ which only differs slightly. This is due to the fact that the left, centre and right part of the original estimate $\tilde{\bar{x}}_0 = (\bar{x}_0^L, \bar{x}_0, \bar{x}_0^R)$ are already very close to the respective mode of $\hat{\tilde{F}} = (\hat{\tilde{F}}^L, \hat{\tilde{F}}, \hat{\tilde{F}}^R)$ as it can be seen on Figure 4.3.

Now that we've calculated all the necessary quantities, we can finally perform the two statistical inference methods we proposed in Section 4.5 and Subsection 4.5.1. Note that, since the two fuzzy confidence intervals were almost identical, we chose to use $(2.2077, 2.5647, 3.0733, 3.7043)$ only.

1. The fuzzy hypothesis testing procedure from Subsection 4.5, with $\tilde{H}_0 = (3.3, 4, 4.5)$ and $\tilde{H}_1 = (3.3, 4, 4.5)^c$ states that the level of rejecting $\tilde{H}_0$ is $L_{rej} = 1$ and the level of not rejecting (accepting) $\tilde{H}_0$ is $L_{acc} = 0.082$. Thus, in this case, with respect to the decision rules from Definition 4.4.3, the test strongly favours the rejection of the null hypothesis. A geometric depiction of this result is shown by Figure 4.4.
2. Using the algorithm 4.1 with respect to $\tilde{H}_0 = (3.3, 4, 4.5)$ and $\tilde{H}_1 = (3.3, 4, 4.5)^c$ gives a fuzzy p-value $\tilde{p} = (0, 0, 1)$. This extreme value for $\tilde{p}$ is too fuzzy to take a decision with, but if we ignore this fact, $\tilde{p} = (0, 0, 1)$ would tell us that the tendency to retain the null hypothesis is $r_{acc} = 0.9025$ and the tendency to reject it $r_{rej} = 0.0975$, which is in contradiction with the previous result where $L_{acc} = 0.082$ and $L_{rej} = 1$.

We can replicate this testing procedure for various null hypotheses $\tilde{H}_0$ and alternative hypotheses $\tilde{H}_1$. The outcomes, as computed via both the fuzzy testing hypothesis procedure and the fuzzy p-value approach, are reported in Table 4.3.

We begin by noting that Tests 1, 2, 5, 9, 10, and 15 produce consistent decisions across both inference procedures. These cases correspond to situations where the fuzzy null hypothesis $\tilde{H}_0$ is either clearly outside or entirely embedded within the core of the fuzzy confidence interval (2.2077, 2.5647, 3.0733, 3.7043). This alignment in extreme configurations suggests that both methods tend to produce consistent conclusions.

However, discrepancies arise in other instances. In particular, Tests 3, 4, 6, 7, and 8 reveal that the fuzzy p -value approach fails to reject certain null hypotheses even when they lie completely outside the fuzzy confidence interval, situations in which the fuzzy testing procedure returns a full rejection with $L^{\text{rej}} = 1$. This indicates that the fuzzy p -value method may be overly conservative in rejecting $\tilde{H}_0$, requiring more extreme deviations from the fuzzy confidence interval to trigger rejection.

An additional subtlety is highlighted in Tests 13 and 14: although both procedures recommend the same decision, the fuzzy p -value is of the form $(0, 0, 1)$, offering no informative support for the decision. This suggests caution when interpreting such p -values, as they fail to provide discriminatory power.

Looking at Tests 3, 4, 6, 7, 8, and 12, we observe systematic contradictions between the two methods. In these cases, the fuzzy p -value approach tends to retain $\tilde{H}_0$ despite substantial evidence against it provided by the fuzzy confidence interval. This illustrates that the fuzzy p -value method may have a higher threshold for rejecting hypotheses and thus be less sensitive in certain scenarios.

Notably, when both the null hypothesis and the observed data become increasingly "crisp," the behavior of both testing procedures approaches that of classical hypothesis testing. For example, when both $\tilde{H}_0$ and the fuzzy confidence interval $\tilde{\Gamma}$ are crisp, the test outcome becomes binary, either full acceptance or full rejection, as formalised in Section 4.4.2. Similarly, when both the observed statistic $\tilde{t}$ and the bootstrap distribution $\tilde{F}$ are crisp, the fuzzy p -value reduces to a classical one, yielding a deterministic decision, as per Definition 4.5.1.

Focusing on the first 10 tests in Table 4.3, the fuzzy testing procedure appears to deliver nearly binary outcomes. This is primarily because, in those cases, the null hypothesis $\tilde{H}_0$ is usually either fully inside or fully outside the fuzzy confidence interval. However, Tests 11–14 showcase intermediate scenarios where $\tilde{H}_0$ partially overlaps with $\tilde{\Gamma}$, leading to genuinely fuzzy (non-binary) decisions. Test 15 further examines a case where $\tilde{H}_0$ encapsulates the entire fuzzy confidence interval, introducing a distinct configuration. In this latter case, the fuzzy p -value offers little interpretive value, whereas the fuzzy testing procedure provides a more informative and nuanced assessment. These subtleties are visually illustrated in Figures 4.5, 4.6, 4.7, 4.8 and 4.9.

In summary, while both the fuzzy hypothesis testing procedure and the fuzzy p -value approach aim to accommodate fuzziness in statistical inference, they often lead to differing conclusions, particularly in non-extreme or intermediate configurations. The hypothesis testing procedure introduced in Section 4.4 demonstrates greater discriminative power, consistently providing clear decisions even in nuanced scenarios where the null hypothesis $\tilde{H}_0$ only partially overlaps with the fuzzy confidence interval $\tilde{\Gamma}$. In contrast, the fuzzy p -value approach tends to be less decisive and frequently produces values close to $(0, 0, 1)$, offering little to no interpretive value in practice. These limitations highlight that, for more reliable and informative inference, the fuzzy hypothesis testing procedure should be preferred over the fuzzy p -value approach.

i	$\tilde{x}_i$
1	(2.191853914, 2.31189276, 2.99605832)
2	(1.113084025, 1.45380088, 2.11630707)
3	(-1.593999754, -1.27626106, -0.71437472)
4	(-1.837104318, -1.48717425, -0.70604804)
5	(1.909652480, 2.09672666, 2.59067942)
6	(4.649264018, 5.17102544, 5.61976025)
7	(-1.317263910, -0.96710914, -0.35206863)
$\vdots$	$\dots$
498	(-0.593304877, -0.03576033, 0.49703778)
499	(3.964550711, 4.42436589, 5.27998739)
500	(0.458157942, 0.81613583, 1.43783539)

This Table shows the first 9 fuzzy observations $\tilde{x}_i = (x_i^L, x_i, x_i^R)$ generated with the random variables $X \sim N(2.8, 3)$, $U_1 \sim \text{Unif}(0.1, 0.6)$, $U_2 \sim \text{Unif}(0.4, 0.9)$ so that $x_i^L = x_i - u_{1,i}$, $x_i^R = x_i + u_{2,i}$.

Table 4.1: Table displaying 10 fuzzy observations.

b	$\bar{\tilde{x}}_b^*$
1	(2.342722, 2.768636, 3.449368)
2	(2.213637, 3.180347, 3.229983)
3	(2.793774, 2.821751, 3.779562)
4	(2.528093, 2.801062, 3.471481)
5	(2.772593, 2.942546, 3.431178)
6	(2.407227, 2.974958, 3.633329)
$\vdots$	$\dots$
997	(2.534152, 2.914441, 3.382806)
998	(2.339465, 2.859847, 3.387453)
999	(2.445547, 2.833967, 3.522199)

Table of 9 estimated bootstrapped fuzzy means.

Table 4.2: Table displaying 10 bootstrapped fuzzy means.

Tests	$\tilde{H}_0$	L^{acc}	L^{rej}	Fuzzy p-value	r^{acc}	r^{rej}
1	(0.5,1,1.5)	0	1	(0,0,0.022)	0	1
2	(0.6,1.1,1.6)	0	1	(0,0,0.126)	0.36	0.64
3	(0.7,1.2,1.7)	0	1	(0,0,0.38)	0.75	0.25
4	(0.8,1.3,1.8)	0	1	(0,0,1)	0.9025	0.0975
5	(2.7,3,3.3)	1	0.09	(0,0,1)	0.98	0.02
6	(3.3,4,4.5)	0.082	1	(0,0,1)	0.9025	0.0975
7	(4.1,4.6,5.1)	0	1	(0,0,1)	0.9025	0.0975
8	(4.2,4.7,5.2)	0	1	(0,0,0.608)	0.84	0.16
9	(4.3,4.8,5.3)	0	1	(0,0,0.152)	0.46	0.54
10	(4.4,4.9,5.4)	0	1	(0,0,0.04)	0	1
11	(1.9,2.1,2.4)	0.11	1	(0,0,1)	0.9025	0.0975
12	(2.1,2.3,2.5)	0.38	0.90	(0,0,1)	0.9025	0.0975
13	(2.3,2.5,2.7)	0.95	0.31	(0,0,1)	0.9025	0.0975
14	(2.5,2.7,2.9)	1	0.018	(0,0,1)	0.9025	0.0975
15	(1.8,3.2,4.1)	0.7	0.44	(0,0.004, 1)	0.906	0.094

This Table shows the levels of acceptance (L^{acc}) and rejection (L^{rej}) given by the fuzzy testing procedure of section 4.4. Then, the levels of acceptance (r^{acc}) and rejection (r^{rej}) of the associated fuzzy p-values are displayed. The fuzzy trapezoidal number (2.2077, 2.5647, 3.0733, 3.7043) is the computed fuzzy confidence interval at a 95% confidence level for the distribution's mean. The values shown in red are in contradiction with the results of the testing procedure of section 4.4.

Table 4.3: Fuzzy Hypothesis testing at a 5% significant level and fuzzy p-values.

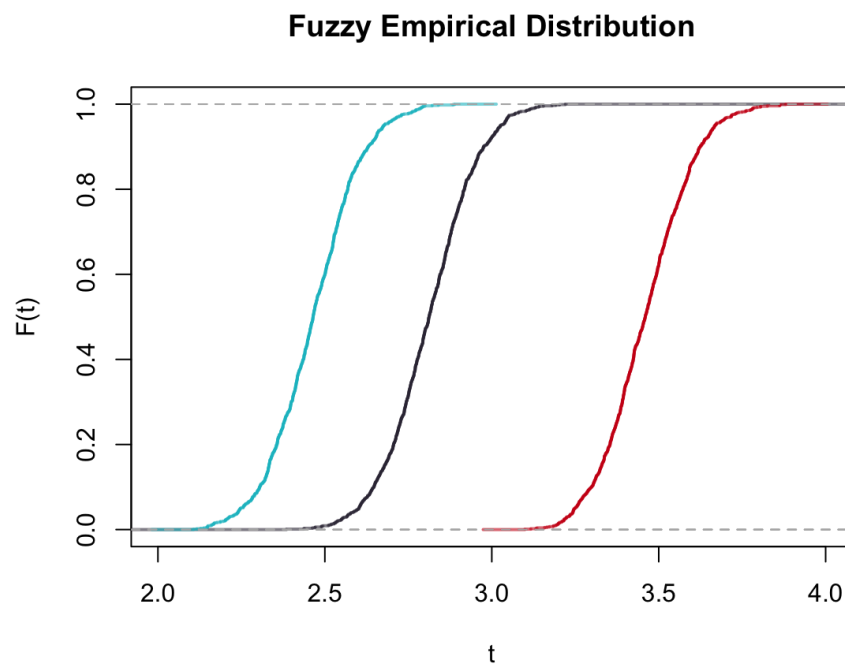


Figure 4.2: In blue, respectively red depicts the left and right parts of the empirical fuzzy distribution of the mean of observations from Table 4.1. The black curve represents the classical empirical fuzzy distribution obtained with the centres of the observations from Table 4.1.

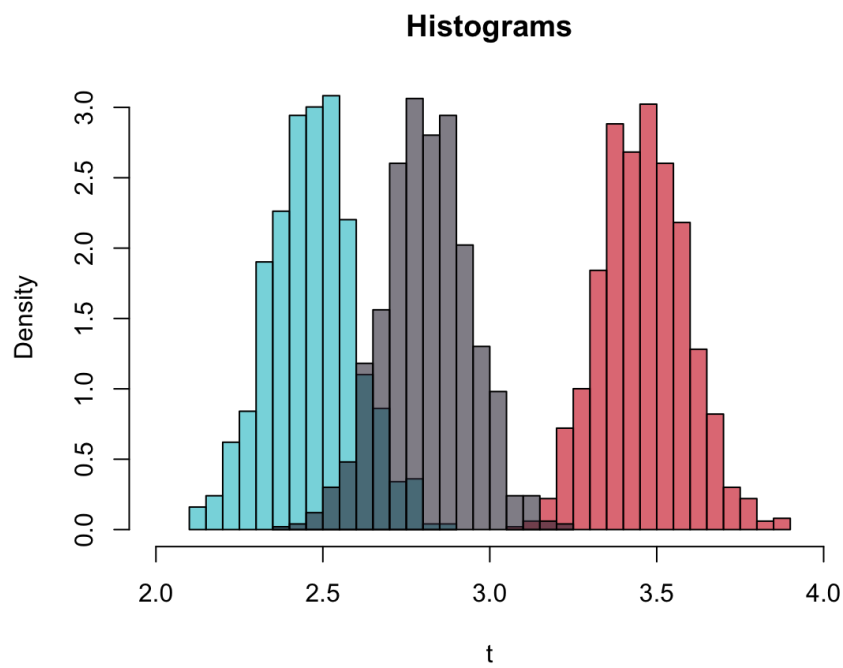


Figure 4.3: Histograms of the left, centre and right part of the estimated fuzzy means generated by means of bootstrap with the observations from table 4.1 as described in the beginning of section 4.6. The blue, dark and red histograms have been created respectively with the left, centre and right parts of the estimated fuzzy means $\tilde{x}_0, \tilde{x}_1^*, \dots, \tilde{x}_b^*, \dots, \tilde{x}_B^*$.

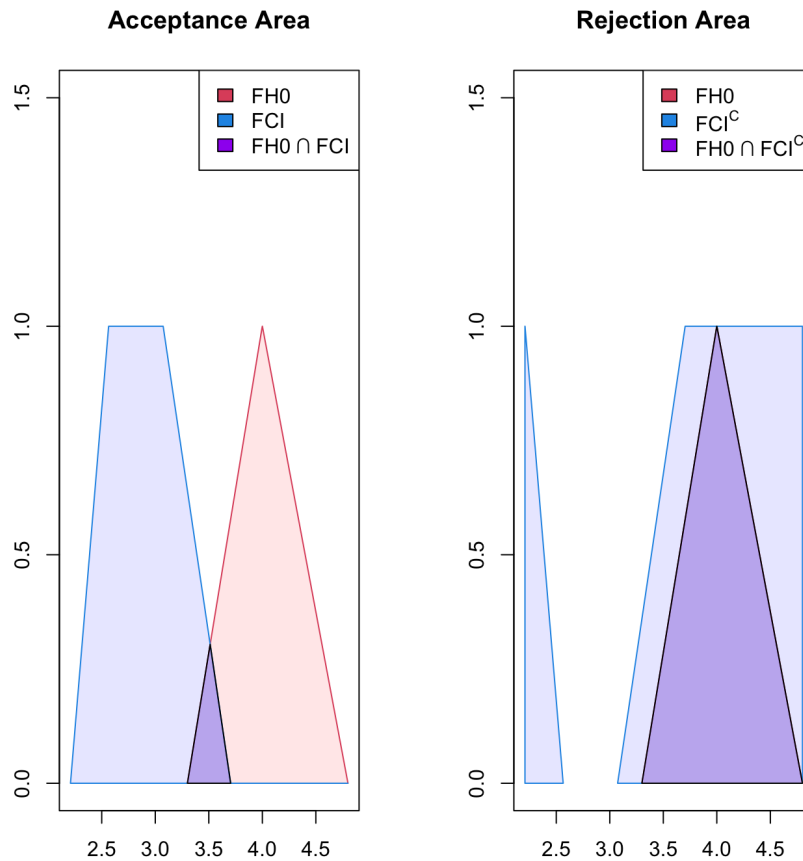


Figure 4.4: Geometric interpretation of the fuzzy hypothesis testing procedure from Definition 4.4.2 where the mean θ is tested with $\tilde{H}_0 : \theta = (3.3, 4, 4.8)$ against $\tilde{H}_1 : \theta \neq (3.3, 4, 4.8)$. FHO stands for $\tilde{H}_0$, FH1 for $\tilde{H}_1$ and FCI for Fuzzy Confidence Interval. In this case, the test strongly favours the rejection of the null hypothesis with $L_{rej} = 1$ while $L_{acc} = 0.082$.

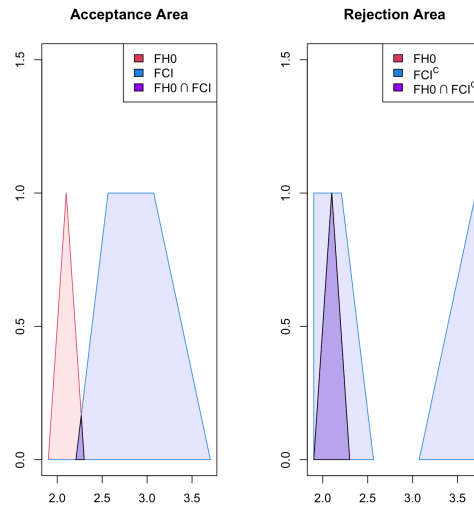


Figure 4.5: Fuzzy test of the mean θ , $\tilde{H}_0 : \theta = (1.9, 2.1, 2.3)$ against $\tilde{H}_1 : \theta \neq (1.9, 2.1, 2.3)$. FH0 stands for $\tilde{H}_0$, and FCI stands for "Fuzzy Confidence Interval". The tendency to not reject the null hypothesis (acceptance area in purple on the left graph) is $L^{acc} = 0.11$. The tendency to reject the null hypothesis (rejection area in purple on the right graph) is $L^{rej} = 1$, the test thus strongly favours the rejection of $\tilde{H}_0$.

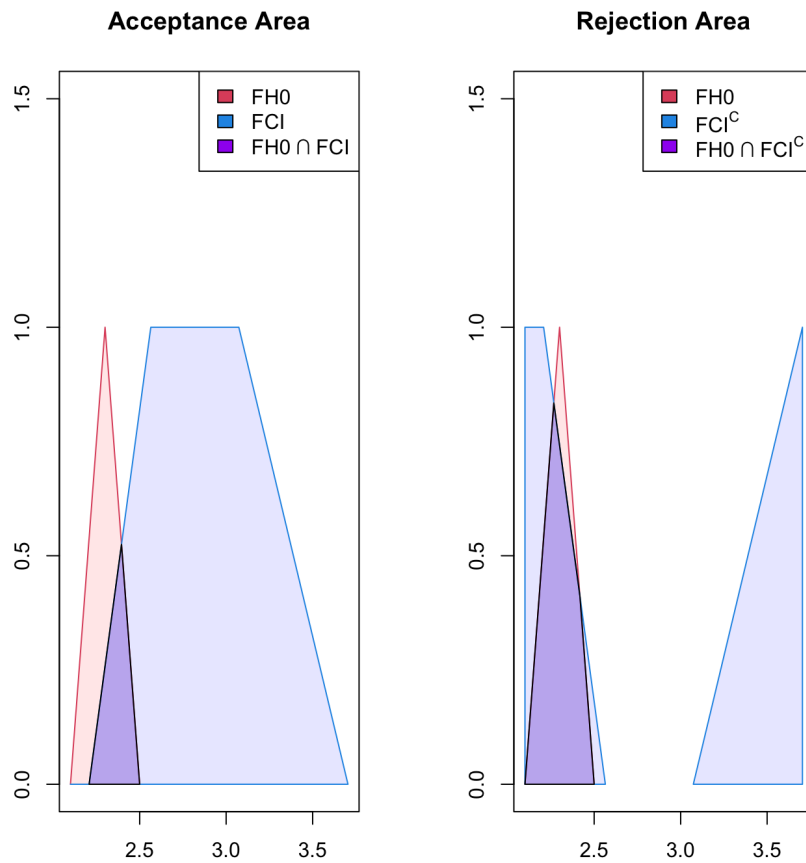


Figure 4.6: Fuzzy test of the mean θ , $\tilde{H}_0 : \theta = (2.1, 2.3, 2.5)$ against $\tilde{H}_1 : \theta \neq (2.1, 2.3, 2.5)$. FH0 stands for $\tilde{H}_0$, and FCI stands for "Fuzzy Confidence Interval". The tendency to not reject the null hypothesis (acceptance area in purple on the left graph) is $L^{acc} = 0.38$. The tendency to reject the null hypothesis (rejection area in purple on the right graph) is $L^{rej} = 0.9$, the test thus favours the rejection of $\tilde{H}_0$.

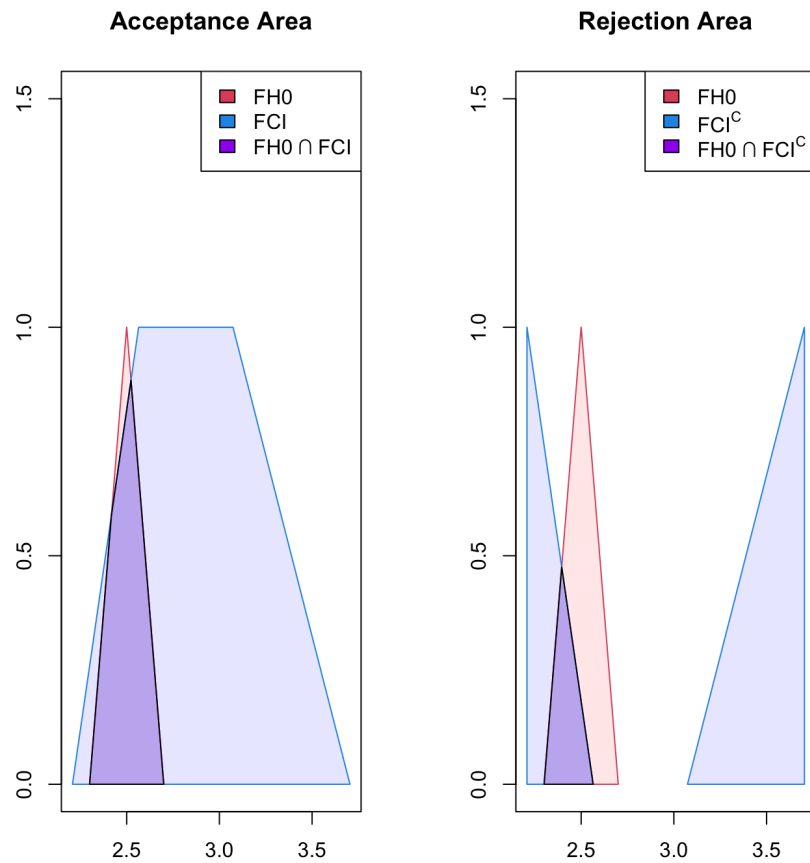


Figure 4.7: Fuzzy test of the mean θ , $\tilde{H}_0 : \theta = (2.3, 2.5, 2.7)$ against $\tilde{H}_1 : \theta \neq (2.3, 2.5, 2.7)$. FH0 stands for $\tilde{H}_0$, and FCI stands for "Fuzzy Confidence Interval". The tendency to not reject the null hypothesis (acceptance area in purple on the left graph) is $L^{acc} = 0.95$. The tendency to reject the null hypothesis (rejection area in purple on the right graph) is $L^{rej} = 0.31$, the test thus favours not to reject $\tilde{H}_0$.

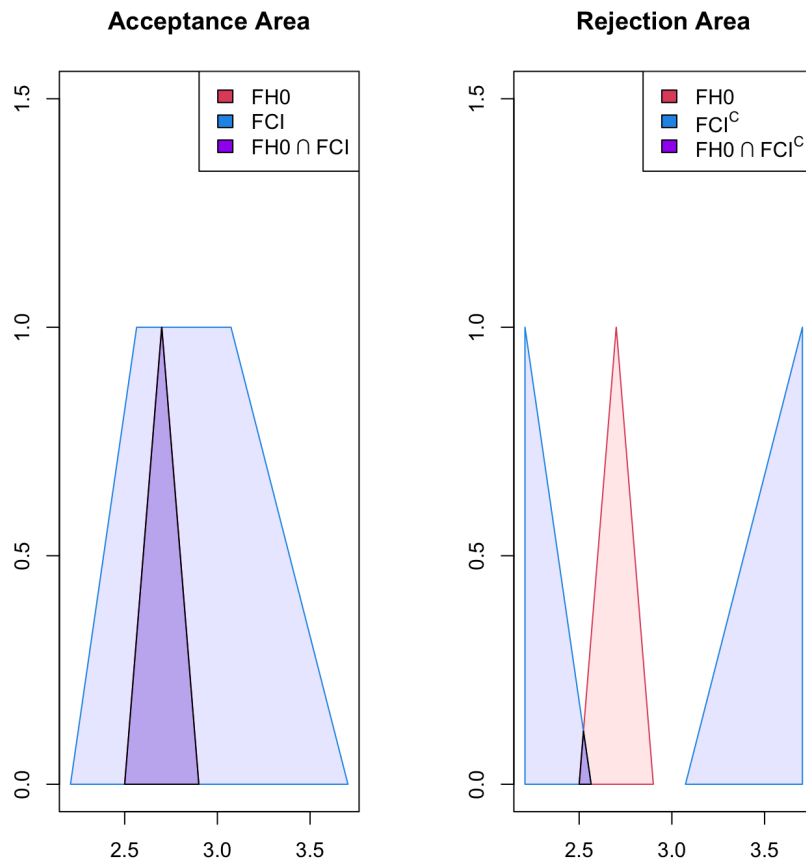


Figure 4.8: Fuzzy test of the mean θ , $\tilde{H}_0 : \theta = (2.5, 2.7, 2.9)$ against $\tilde{H}_1 : \theta \neq (2.5, 2.7, 2.9)$. FH0 stands for $\tilde{H}_0$, and FCI stands for "Fuzzy Confidence Interval". The tendency to not reject the null hypothesis (acceptance area in purple on the left graph) is $L^{acc} = 1$. The tendency to reject the null hypothesis (rejection area in purple on the right graph) is $L^{rej} = 0.018$, the test thus strongly favours not to reject $\tilde{H}_0$.

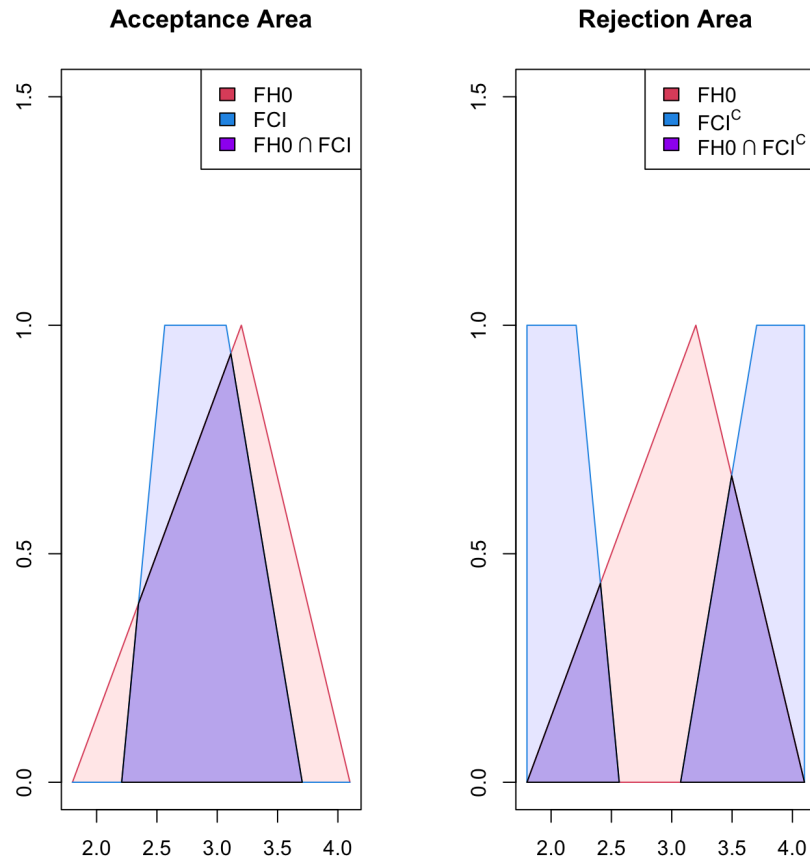


Figure 4.9: Fuzzy test of the mean θ , $\tilde{H}_0 : \theta = (1.8, 3.2, 4.1)$ against $\tilde{H}_1 : \theta \neq (1.8, 3.2, 4.1)$. FH_0 stands for $\tilde{H}_0$, and FCI stands for "Fuzzy Confidence Interval". The tendency to not reject the null hypothesis (acceptance area in purple on the left graph) is $L^{acc} = 0.7$. The tendency to reject the null hypothesis (rejection area in purple on the right graph) is $L^{rej} = 0.44$, the test thus favours not to reject $\tilde{H}_0$.

Chapter 5

Fuzzy numbers ranking

Ranking fuzzy numbers is a fundamental task in decision-making processes involving imprecise, vague, or uncertain data. Since fuzzy numbers are not naturally totally ordered, numerous methods have been developed to impose a ranking, each grounded in distinct mathematical principles—such as defuzzification, distance metrics, dominance relations, or geometric representations.

Among the earliest and most influential ranking methods is the one introduced by Yager (1981), who proposed calculating the integral of the means of all α -cuts, effectively yielding an expected value. This defuzzification-based approach enabled a basic means of comparing fuzzy numbers. Building on this, Liou and M. J. Wang (1992) proposed using the centroid (center of gravity) of a fuzzy number, with the ranking determined by the x -coordinate of the centroid. However, both methods share a common limitation: they are not well-suited for ranking asymmetric fuzzy numbers.

To address this issue, Chu and Tsao (2002) developed a method that separates and weighs the left and right spreads of a fuzzy number, thereby improving sensitivity to asymmetry. Their approach computes a signed area between the centroid and the original reference point, enhancing discrimination among non-symmetric fuzzy numbers. More recently, Prasad and Sinha (2023) proposed a unified distance-based method that combines centroid data with directional spread analysis, thereby increasing discriminatory power by evaluating how much one fuzzy number extends beyond another. Additionally, Ruolan, Bingyi, and Jianfeng (2022) developed a rule-based approach that considers both the horizontal position of the centroid and the area of the fuzzy number, formulating four ranking rules to effectively compare fuzzy numbers.

Concerning distance-based approaches, one can also mention Berkachy and Donzé (2024) who introduced the generalised signed distance approach. Their distance function accounts for asymmetries through analysing direction spreads between fuzzy numbers and allows to rank them accordingly.

Across existing methods, a common limitation remains: the reduction of a fuzzy number to one or several scalar indicators, such as centroids or distance measures. While these scalar summaries are computationally convenient, they inevitably obscure critical geometric information—such as the directionality given by the positions of the fuzzy numbers on the real line and the size of their non-overlapping regions—which can be crucial in distinguishing fuzzy numbers with similar spreads but different shapes.

To address these shortcomings, we propose a method that directly analyzes the spatial configuration of fuzzy numbers by quantifying the directional asymmetry of their non-overlapping regions. For any given pair of fuzzy numbers, our approach identifies the regions that belong to one fuzzy number but not the other, and partitions these non-overlapping areas into left and right components relative to the peak (mode) of the opposing fuzzy number. By integrating the area on each side, the method infers which fuzzy number "dominates" in a direction-sensitive and geometrically meaningful manner. To the best of our knowledge, no existing method considers this comparison from the dual perspective of both fuzzy numbers simultaneously, making this bilateral analysis a distinctive feature of our approach.

Moreover, while our method is particularly intuitive for triangular fuzzy numbers, it is not confined to them. It naturally extends to fuzzy numbers whose membership functions are represented by arbitrary convex polygons, enabling the comparison of more complex fuzzy sets beyond standard linear or trapezoidal forms. This added generality sets our approach apart from many existing methods, which are often restricted to, or optimized for, simpler fuzzy number representations such as triangular or trapezoidal fuzzy numbers.

5.1 Polygon ranking method

As its name suggests, the **Polygon ranking method (PRM)** is a procedure that permits the ranking of convex polygons, or more specifically, generalised fuzzy numbers. To build intuition, we begin with the simplest case of polygons, triangles. Let $\tilde{T}$ denote a generalised triangular fuzzy number as introduced in Definition 1.1.17. In the plane $\mathbb{R}^2$, it can be represented as:

$$\tilde{T} = \left\{ (T^L, 0), (T, h), (T^R, 0) \right\},$$

where $h > 0$ is the vertical height of the triangle. For *normal* fuzzy numbers, this value is typically $h = 1$.

Now consider two triangular generalised fuzzy numbers as described by Definition 1.1.17, $\tilde{T}_1$ and $\tilde{T}_2$, with heights h_1 and h_2 , respectively. Define the following geometric components:

- $\mathcal{I} = \tilde{T}_1 \cap \tilde{T}_2$: the intersection area,
- $\mathcal{A}_1 = \tilde{T}_1 \setminus \mathcal{I}$: the residual (non-overlapping) area of $\tilde{T}_1$,

- $\mathcal{A}_2 = \tilde{T}_2 \setminus \mathcal{I}$: the residual (non-overlapping) area of $\tilde{T}_2$.

Let x_1 and x_2 denote the x-coordinate of the centres of $\tilde{T}_1$ and $\tilde{T}_2$, respectively. Based on these, we define left and right half-planes:

$$\begin{aligned}\mathcal{L}_{x_2} &= \{(x, y) \mid x < x_2\}, & \mathcal{R}_{x_2} &= \{(x, y) \mid x \geq x_2\}, \\ \mathcal{L}_{x_1} &= \{(x, y) \mid x < x_1\}, & \mathcal{R}_{x_1} &= \{(x, y) \mid x \geq x_1\}.\end{aligned}$$

Here, $\mathcal{L}_{x_2}$ and $\mathcal{R}_{x_2}$ correspond to the left and right half-planes relative to $\tilde{T}_2$, while $\mathcal{L}_{x_1}$ and $\mathcal{R}_{x_1}$ are defined relative to $\tilde{T}_1$.

Next, we compute the "left" and "right" residual areas from both perspectives:

$$L_1 = \text{Area}(\mathcal{A}_1 \cap \mathcal{L}_{x_2}), \quad (5.1)$$

$$R_1 = \text{Area}(\mathcal{A}_1 \cap \mathcal{R}_{x_2}), \quad (5.2)$$

$$L_2 = \text{Area}(\mathcal{A}_2 \cap \mathcal{L}_{x_1}), \quad (5.3)$$

$$R_2 = \text{Area}(\mathcal{A}_2 \cap \mathcal{R}_{x_1}). \quad (5.4)$$

There are four distinct point of views (5.5)–(5.8) to assess in order to evaluate if a triangle $\tilde{T}_1$ is greater or lesser than a triangle $\tilde{T}_2$.

$$L_1 \text{ assesses the relation } \tilde{T}_1 < \tilde{T}_2 \quad (\text{left-side residual of } \tilde{T}_1 \text{ relative to } \tilde{T}_2), \quad (5.5)$$

$$R_1 \text{ assesses the relation } \tilde{T}_1 > \tilde{T}_2 \quad (\text{right-side residual of } \tilde{T}_1 \text{ relative to } \tilde{T}_2), \quad (5.6)$$

$$L_2 \text{ assesses the relation } \tilde{T}_2 < \tilde{T}_1 \quad (\text{left-side residual of } \tilde{T}_2 \text{ relative to } \tilde{T}_1), \quad (5.7)$$

$$R_2 \text{ assesses the relation } \tilde{T}_2 > \tilde{T}_1 \quad (\text{right-side residual of } \tilde{T}_2 \text{ relative to } \tilde{T}_1). \quad (5.8)$$

Note that the two pairs of equations (5.5) & (5.8) and (5.6) & (5.7) describe the same phenomenon but from opposite viewpoints. Indeed, observe that L_1 and R_2 are both evaluating if $\tilde{T}_1 < \tilde{T}_2$, (5.5) from $\tilde{T}_1$'s position and (5.8) from $\tilde{T}_2$'s view. Similarly, R_1 and L_2 are assessing if $\tilde{T}_2 < \tilde{T}_1$. This symmetry is essential for a complete comparison.

Importantly, the strength of the relation $\tilde{T}_1 < \tilde{T}_2$ increases with the combined values of L_1 and R_2 , which must be considered together. Conversely, the greater the combined values of L_2 and R_1 , the stronger the relation $\tilde{T}_2 < \tilde{T}_1$ becomes.

We thus propose unifying these four conditions into a single one, as formalised in the following definition:

Definition 5.1.1. Comparison rules

We can compare two generalised fuzzy numbers $\tilde{T}_1$ and $\tilde{T}_2$ using the following criteria:

$$\begin{aligned}\tilde{T}_1 < \tilde{T}_2 & \text{ if } L_1 + R_2 > R_1 + L_2, \\ \tilde{T}_1 = \tilde{T}_2 & \text{ if } L_1 + R_2 = R_1 + L_2, \\ \tilde{T}_1 > \tilde{T}_2 & \text{ if } L_1 + R_2 < R_1 + L_2.\end{aligned}\tag{5.9}$$

In practice, it is useful to define the total left and right residual areas as:

$$L = L_1 + R_2, \quad R = R_1 + L_2.$$

The sums of areas L and R respectively tells us how much $\tilde{T}_1$ is on the left or on the right of $\tilde{T}_2$. Let us emphasize that the order in which one compares two triangles is important. Given a pair $\{\tilde{T}_u, \tilde{T}_v\}$ of two triangles $\tilde{T}_u$ and $\tilde{T}_v$ indexed by u and v , the values L and R tells how much $\tilde{T}_u$ is on the left, respectively on the right of $\tilde{T}_v$, relative to $\tilde{T}_u$'s reference point.

By dividing L and R by the addition of $\tilde{T}_1$ and $\tilde{T}_2$'s surfaces,

$$L\% = \frac{(L_1 + R_2)}{\text{Area}(\tilde{T}_1) + \text{Area}(\tilde{T}_2)}, \quad R\% = \frac{(R_1 + L_2)}{\text{Area}(\tilde{T}_1) + \text{Area}(\tilde{T}_2)},$$

one can then interpret $L\%$ and $R\%$ as the percentage level to which $\tilde{T}_1$ is on the left, respectively on the right of $\tilde{T}_2$, relative to $\tilde{T}_1$'s reference point. If instead $\tilde{T}_2$ is taken as the reference, the roles of $L\%$ and $R\%$ are exchanged, that is $L\%$ becomes $R\%$ and conversely, $R\%$ becomes $L\%$. Importantly, however, the ordering between $\tilde{T}_1$ and $\tilde{T}_2$ remains unchanged.

In practical applications, it is convenient to define the total left and right residual areas as follows:

$$L = L_1 + R_2, \quad R = R_1 + L_2.$$

These aggregate quantities L and R respectively quantify the extent to which triangle $\tilde{T}_1$ lies to the left or to the right of triangle $\tilde{T}_2$. It is important to note that the notions of left and right arising in the comparison between two triangles is order-sensitive. Specifically, given a pair $\{\tilde{T}_u, \tilde{T}_v\}$ of triangles indexed by u and v , the values L and R measure how much $\tilde{T}_u$ lies to the left or right, respectively, of $\tilde{T}_v$, with respect to the reference point of $\tilde{T}_u$.

Additionally, one may define the normalised left and right residual areas as:

$$L\% = \left(100 \cdot \frac{L_1 + R_2}{\text{Area}(\tilde{T}_1) + \text{Area}(\tilde{T}_2)} \right) \%, \quad R\% = \left(100 \cdot \frac{R_1 + L_2}{\text{Area}(\tilde{T}_1) + \text{Area}(\tilde{T}_2)} \right) \%.$$

The quantities $L\%$ and $R\%$ can thus be interpreted as the proportion (in percent) by which $\tilde{T}_1$ lies to the left or right, respectively, of $\tilde{T}_2$, again measured relative to the reference point of $\tilde{T}_1$. Observe that it is sufficient to calculate L and R to compare two triangles, hence why the computation of these values will sometimes be omitted.

To illustrate the application of this method, consider a simple example involving two non-intersecting triangles of height $h = 1$:

$$\tilde{T}_1 = \{(3, 0), (5, 1), (6, 0)\}, \quad \tilde{T}_2 = \{(-2, 0), (-1, 1), (1, 0)\}.$$

Applying the PRM to triangles $\tilde{T}_1$ and $\tilde{T}_2$, one obtains the values L_1 , L_2 , R_1 , and R_2 using equations (5.1)–(5.4), as reported in Table 5.1. Since $L = L_1 + R_2 = 0$ is smaller than $R = R_1 + L_2 = 3$, intuitively, we know that $\tilde{T}_1$ is of $R = 3$ to the right of $\tilde{T}_2$ and of $L = 0$ to the left of $\tilde{T}_2$. With the normalised values $L^\%$ and $R^\%$, this means that $\tilde{T}_1$ is $R^\% = 100\%$ to the right of $\tilde{T}_2$ and $L^\% = 0\%$ to the left of $\tilde{T}_2$. This implies that triangle $\tilde{T}_1$ lies entirely to the right of triangle $\tilde{T}_2$ relative to the reference point of $\tilde{T}_1$. According to the comparison rules established in Definition 5.1.1, the PRM thus yields $\tilde{T}_1 > \tilde{T}_2$. A visual representation of these triangles and the non-zero residual areas L_2 and R_1 arising in their comparison is provided in Figure 5.1.

Area Component	Value
L_1 (left of $\tilde{T}_1$)	0
R_1 (right of $\tilde{T}_1$)	1.5
L_2 (left of $\tilde{T}_2$)	1.5
R_2 (right of $\tilde{T}_2$)	0
$\text{Area}(\tilde{T}_1)$	1.5
$\text{Area}(\tilde{T}_2)$	1.5
$L = L_1 + R_2$	0
$R = R_1 + L_2$	3
$L^\% = \left(100 \cdot \frac{L_1 + R_2}{\text{Area}(\tilde{T}_1) + \text{Area}(\tilde{T}_2)}\right) \%$	0%
$R^\% = \left(100 \cdot \frac{R_1 + L_2}{\text{Area}(\tilde{T}_1) + \text{Area}(\tilde{T}_2)}\right) \%$	100%

Table 5.1: Residual area comparison of two non-intersecting triangular fuzzy numbers.

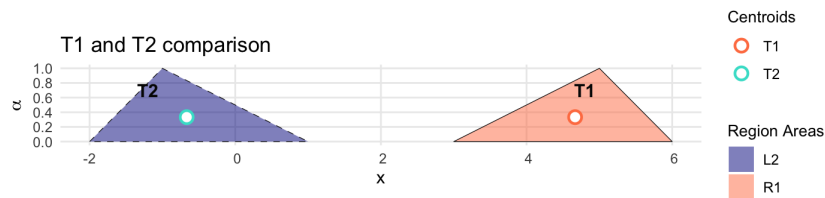


Figure 5.1: Visualisation of $\tilde{T}_1 = \{(3, 0), (5, 1), (6, 0)\}$ and $\tilde{T}_2 = \{(-2, 0), (-1, 1), (1, 0)\}$ with areas $L_1 = 0$, $R_1 = 1.5$, $L_2 = 1.5$, $R_2 = 0$. $L = L_1 + R_2 = 0$, $R = R_1 + L_2 = 3$, thus $\tilde{T}_1 > \tilde{T}_2$.

We now turn our attention to the case of two intersecting triangles of height $h = 100$:

- $\tilde{T}_1 = \{(160.6117, 0), (200.2386, 100), (228.1923, 0)\},$

- $\tilde{T}_2 = \{(170.2094, 0), (180.2094, 100), (235.0998, 0)\}$.

The computed values of the residual areas L_1 , L_2 , R_1 , and R_2 are presented in Table 5.2.

Area Component	Value
L_1 (left of $\tilde{T}_1$)	155.46
R_1 (right of $\tilde{T}_1$)	532.43
L_2 (left of $\tilde{T}_2$)	464.82
R_2 (right of $\tilde{T}_2$)	88.56
Area($\tilde{T}_1$)	3379.03
Area($\tilde{T}_2$)	3244.52
$L = L_1 + R_2$	244.02
$R = R_1 + L_2$	997.25
$L\% = \frac{L_1 + R_2}{\text{Area}(\tilde{T}_1) + \text{Area}(\tilde{T}_2)}$	3.36%
$R\% = \frac{R_1 + L_2}{\text{Area}(\tilde{T}_1) + \text{Area}(\tilde{T}_2)}$	15.05%

Table 5.2: Residual area comparison of two intersecting generalized triangular fuzzy numbers.

Since $L < R$, the PRM yields the ranking $\tilde{T}_1 > \tilde{T}_2$. Moreover, from $\tilde{T}_1$'s perspective, $\tilde{T}_1$ lies $L\% = 3.36\%$ to the left and $R\% = 15.05\%$ to the right of $\tilde{T}_2$. Figure 5.2 provides a visual illustration of this comparison, highlighting the non-overlapping residual areas L_1 , L_2 , R_1 , and R_2 .

To further illustrate different configurations, Figure 5.3 depicts the case in which the two triangles have different heights ($h_1 = 40$, $h_2 = 100$), while Figure 5.4 shows a scenario where one triangle is entirely nested within the other.

5.1.1 Generalisation of the polygon ranking method

The procedure outlined above is, in fact, more general and not limited to the comparison of generalised triangular fuzzy numbers. As we will see, it can be applied to compare any generalised fuzzy number whose membership function is represented by a convex polygon. In fact, by allowing the number of vertices in such a polygon to approach infinity, the method naturally extends to arbitrary generalised fuzzy numbers with convex membership functions. Therefore, rather than restricting ourselves to triangles, we now consider convex polygons.

Let $P_x = \{x_1, x_2, \dots, x_k\}$ be the ordered set of x -coordinates of a convex polygon, with $x_1 \leq x_2 \leq \dots \leq x_k$.

Let $P_y = \{y_1, y_2, \dots, y_k\}$ be the corresponding y -coordinates, such that $y_1 = 0$, $y_k = 0$, and there exists an index $s \in \{1, 2, \dots, k\}$ for which $y_s = h$. For normal fuzzy numbers, $h = 1$.

Moreover, observe that any non-piecewise linear membership function can easily be approximated by such a polygon, whose vertices are $\{(x_1, y_1), (x_2, y_2), \dots, (x_k, y_k)\}$.

A generalised fuzzy number represented by a convex polygon in $\mathbb{R}^2$ is thus defined as:

$$\tilde{P} = \{(x_1, 0), (x_2, y_2), \dots, (x_{k-1}, y_{k-1}), (x_k, 0)\}.$$

These convex polygonal generalised fuzzy numbers can be compared using the same geometric procedure described previously for triangular generalised fuzzy numbers, by replacing triangles $\tilde{T}_1$ and $\tilde{T}_2$ with polygons $\tilde{P}_1$ and $\tilde{P}_2$. Hence, to rank $\tilde{P}_1$ and $\tilde{P}_2$, one uses the same rules (5.9) of the PRM defined above. This ranking process thus again involves calculating the intersection area and residual non-overlapping regions in the left and right half-planes relative to each polygon.

Figure 5.6 illustrates a comparison between two pentagonal generalised fuzzy numbers of equal height h . Figure 5.7 demonstrates the comparison of convex polygons with differing heights. Figure 5.5 shows the comparison in presence of degenerate polygons, i.e polygons with identical x -coordinates.

5.2 Centroid ranking

This section outlines how the centroid of a generalised fuzzy number $\tilde{x}$ has been defined by Hajjari (2012) and how it can be used for ranking purposes.

Let us recall that a *generalised fuzzy number* $\tilde{x}$ is characterised by its membership function $\mu_{\tilde{x}}(x)$, defined as

$$\mu_{\tilde{x}}(x) = \begin{cases} L_{\tilde{x}}(x), & a \leq x \leq b, \\ h, & b \leq x \leq c, \\ R_{\tilde{x}}(x), & c \leq x \leq d, \\ 0, & \text{otherwise,} \end{cases}$$

where $h > 0$ denotes the height of $\tilde{x}$ and the functions

$$L_{\tilde{x}} : [a, b] \rightarrow [0, h], \quad R_{\tilde{x}} : [c, d] \rightarrow [0, h],$$

are strictly monotone and continuous. See Definition 1.1.17 for more details.

Definition 5.2.1 (Centroid).

The **centroid** of a generalised fuzzy number $\tilde{x}$ is defined as the point (x_0, y_0) , where:

$$x_0 = \frac{\int_a^b x L_{\tilde{x}}(x) dx + \int_b^c x dx + \int_c^d x R_{\tilde{x}}(x) dx}{\int_a^b L_{\tilde{x}}(x) dx + \int_b^c dx + \int_c^d R_{\tilde{x}}(x) dx}, \quad (5.10)$$

$$y_0 = h \cdot \frac{\int_0^1 y R_{\tilde{x}}^{-1}(y) dy - \int_0^1 y L_{\tilde{x}}^{-1}(y) dy}{\int_0^1 R_{\tilde{x}}^{-1}(y) dy - \int_0^1 L_{\tilde{x}}^{-1}(y) dy}. \quad (5.11)$$

For a non-normal trapezoidal fuzzy number $\tilde{x} = (a, b, c, d; h)$, the centroid simplifies to:

$$x_0 = \frac{1}{3} \left[a + b + c + d - \frac{dc - ab}{(d + c) - (a + b)} \right], \quad (5.12)$$

$$y_0 = h \cdot \left[1 + \frac{c - d}{(d + c) - (a + b)} \right]. \quad (5.13)$$

In the special case of a non-normal triangular fuzzy number $\tilde{x} = (a, b, d; h)$, the centroid coordinates reduce to:

$$x_0 = \frac{1}{3}(a + b + d), \quad (5.14)$$

$$y_0 = h. \quad (5.15)$$

In most ranking approaches, only the x -coordinate of the centroid is used to compare fuzzy numbers. However, in some contexts, the y -coordinate also carries important information—such as in the method of Ruolan, Bingyi, and Jianfeng (2022), which leverages both dimensions. In the remainder of this work, we will focus on using the x -coordinate of the centroid for fuzzy number ranking.

5.3 Centroid ranking vs polygon ranking method

We now compare the **PRM** with the centroid-based ranking approach. To further illustrate the differences between these methods, consider $\tilde{T}_1 = (1.6, 4, 4.8)$ and $\tilde{T}_2 = (2.2, 3, 5.4)$. The x-coordinates of their centroids are 3.46 and 3.53, respectively, thus $\tilde{T}_1 < \tilde{T}_2$ under the centroid method. However, using **PRM**, we compute:

$$L_1 = 0.1125, \quad R_1 = 0.2083, \quad L_2 = 0.1125, \quad R_2 = 0.225, \\ L = L_1 + R_2 = 0.3375, \quad R = R_1 + L_2 = 0.3208.$$

Since $L > R$, the polygon method concludes $\tilde{T}_1 > \tilde{T}_2$. As seen in Figure 5.8, the centroid values are very close, making it hard to discriminate between the shapes. In contrast, the polygon method produces more distinct values, suggesting stronger discriminative power.

Now consider $\tilde{T}_1 = (1, 1, 4)$ and $\tilde{T}_2 = (0, 3, 3)$. Both have the same centroid, 2, leading to a tie under the centroid method. However, using the polygon comparison:

$$L = L_1 + R_2 = 0.33 + 0.33 = 0.66, \quad R = L_2 + R_1 = 0.166 + 0.166 = 0.332,$$

thus $\tilde{T}_1 < \tilde{T}_2$. Again, the polygon approach reveals a distinction not captured by the centroid method (see Figure 5.9).

Furthermore, when comparing two triangular fuzzy numbers, the comparison is typically performed at the α -cut level of $\alpha = 0$. However, it is worth exploring how varying the α -cut affects the comparison outcome and identifying the threshold value of α at which the decision changes.

Consider the triangular fuzzy numbers $\tilde{T}_1 = (1, 4, 5)$ and $\tilde{T}_2 = (2, 3, 6)$. For a fixed $\alpha \in [0, 1]$, we define the corresponding α -cut triangles as follows:

$$\tilde{T}_1^\alpha = (4 - (1 - \alpha)(4 - 1), 4, 4 + (1 - \alpha)(5 - 4)), \\ \tilde{T}_2^\alpha = (3 - (1 - \alpha)(3 - 2), 3, 3 + (1 - \alpha)(6 - 3)).$$

Using either the **PRM** or the centroid-based method, we can rank $\tilde{T}_1^\alpha$ and $\tilde{T}_2^\alpha$ at any chosen level $\alpha \in [0, 1]$. Table 5.3 presents the ranking results for different α -cuts using both methods.

Notably, the two methods disagree in the interval $\alpha \in [0.10, 0.25]$. Specifically, for $\alpha \in [0.10, 0.24]$, they yield opposite rankings, while at $\alpha = 0.25$, the centroid method considers them equal, whereas the polygon method concludes $\tilde{T}_1 > \tilde{T}_2$.

Moreover, the decision outcome based on the **PRM** changes at $\alpha = 0.10$, whereas for the centroid ranking method, this change occurs at $\alpha = 0.26$. The inflection point, that is, the threshold value of α at which the decision shifts, is considerably lower for the **PRM** than for the centroid-based approach. This indicates that the **PRM** offers greater discriminatory power compared to the centroid ranking method.

α	centroid $\tilde{\tau}_1$	centroid $\tilde{\tau}_2$	Centroid	PRM	Match
0.00	3.333	3.667	$\tilde{\tau}_1 < \tilde{\tau}_2$	$\tilde{\tau}_1 < \tilde{\tau}_2$	✓
⋮	⋮	⋮	⋮	⋮	⋮
0.09	3.393	3.607	$\tilde{\tau}_1 < \tilde{\tau}_2$	$\tilde{\tau}_1 < \tilde{\tau}_2$	✓
0.10	3.400	3.600	$\tilde{\tau}_1 < \tilde{\tau}_2$	$\tilde{\tau}_1 > \tilde{\tau}_2$	✗
⋮	⋮	⋮	⋮	⋮	⋮
0.24	3.493	3.507	$\tilde{\tau}_1 < \tilde{\tau}_2$	$\tilde{\tau}_1 > \tilde{\tau}_2$	✗
0.25	3.500	3.500	$\tilde{\tau}_1 = \tilde{\tau}_2$	$\tilde{\tau}_1 > \tilde{\tau}_2$	✗
0.26	3.507	3.493	$\tilde{\tau}_1 > \tilde{\tau}_2$	$\tilde{\tau}_1 > \tilde{\tau}_2$	✓
⋮	⋮	⋮	⋮	⋮	⋮
1.00	4.000	3.000	$\tilde{\tau}_1 > \tilde{\tau}_2$	$\tilde{\tau}_1 > \tilde{\tau}_2$	✓

Table 5.3: Comparison of centroid-based and polygon-based rankings across different α values. Highlighted rows indicate discrepancies.

Limitations and extensions of the polygon ranking method

While the PRM offers several appealing features, it is not without limitations. Notably, it fails to distinguish between symmetric fuzzy numbers that share the same midpoint. For example, the fuzzy numbers $\tilde{\tau}_1 = (1, 3, 5)$ and $\tilde{\tau}_2 = (2, 3, 4)$, illustrated in Figure 5.11, both yield $L = R$ and are consequently regarded as equivalent. A similar outcome arises when applying the centroid-based method, which also results in a tie.

To formally assess whether such a ranking procedure defines a partial order, let us recall the definition: a binary relation $\leq$ on a set S is called a *partial ordering* if, for all $a, b, c \in S$, it satisfies the following properties:

1. *Reflexivity*: $a \leq a$,
2. *Antisymmetry*: If $a \leq b$ and $b \leq a$, then $a = b$,
3. *Transitivity*: If $a \leq b$ and $b \leq c$, then $a \leq c$.

A set S equipped with such a relation is referred to as a *partially ordered set*, or *poset*.

In the example above, the PRM violates the antisymmetry condition: although $\tilde{\tau}_1 \leq \tilde{\tau}_2$ and $\tilde{\tau}_2 \leq \tilde{\tau}_1$ both hold, we have $\tilde{\tau}_1 \neq \tilde{\tau}_2$. This contradiction confirms that, in its current form, the PRM method does not induce a valid partial ordering. To induce such ordering, the PRM must return that two triangles are equal if and only if they are identical.

Moreover, it is also true that two *asymmetric* triangular fuzzy numbers can satisfy the condition $L_1 + R_2 = L_2 + R_1$. While such configurations do exist, they are rare and not easily constructed by hand. In contrast, symmetric cases satisfying this condition are far more common. One example of two asymmetric triangular fuzzy numbers that meet the condition $L = R$ is given by $\tilde{T}_1 = (1 + t, 1 + t, 4 + t)$ with $t = 0.2426407$, and $\tilde{T}_2 = (0, 3, 3)$. The precise value of t was determined using a root-finding approach based on Newton's method, specifically the `uniroot` function from the R programming language. Figure 5.10 illustrates these two asymmetric triangles, where the equality $L = R$ leads the PRM to conclude $\tilde{T}_1 = \tilde{T}_2$.

To address this limitation in symmetric cases, we propose the introduction of an additional discriminative measure: the *index of fuzziness*, defined as the area under the membership function:

$$\text{Fuzz}(\tilde{x}) = \int_{-\infty}^{\infty} \mu_{\tilde{x}}(x) dx.$$

When $L = R$ in (5.9), the fuzziness index provides a means to further rank fuzzy generalised numbers deemed equal by the PRM. Depending on the application, one may prefer to rank higher the fuzzy number with the larger (or smaller) index of fuzziness. Furthermore, in cases involving shapes of differing complexity (e.g., a triangle versus a pentagon), where both $L = R$ and $\text{Fuzz}(\tilde{x}_1) = \text{Fuzz}(\tilde{x}_2)$, one may adopt a simplicity-based criterion, favouring the shape with fewer vertices.

Ultimately, the choice of such secondary criteria should be guided by the specific goals and requirements of the application domain.

Finally, while there exist asymmetric triangles satisfying the condition $L_1 + R_2 = L_2 + R_1$, identifying such configurations remains quite tedious and a non-trivial task in practice. Given this empirical difficulty, and under the reasonable working assumption that it is extremely unlikely to fall upon two such triangles, the PRM, augmented with the above proposed two corrections, namely, the index of fuzziness and number of vertices, appears to induce a partial ordering in practical settings.

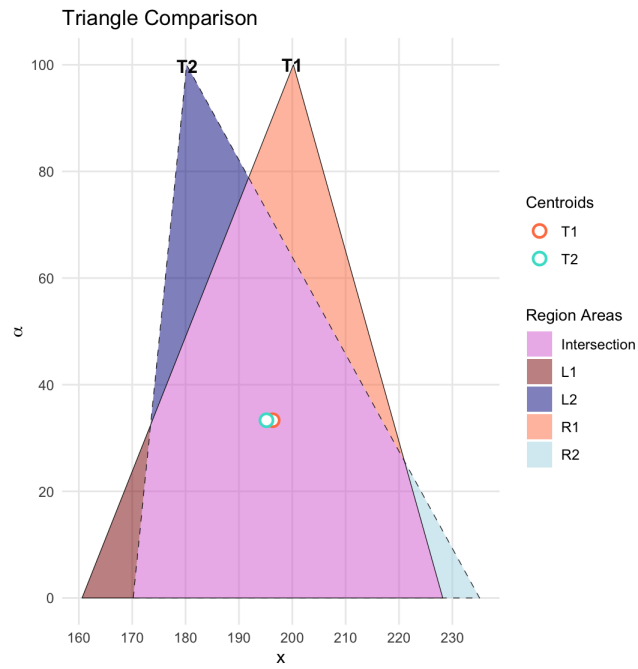


Figure 5.2: Visualization of $\tilde{T}_1 = \{(160.6, 0), (200.2, 100), (228.2, 0)\}$ and $\tilde{T}_2 = \{(170.2, 0), (180.2, 100), (235.1, 0)\}$ with areas $L_1 = 155.46$, $R_1 = 532.43$, $L_2 = 464.82$, $R_2 = 88.56$. $L = L_1 + R_2 = 244.03$, $R = R_1 + L_2 = 997.25$, thus $\tilde{T}_1 > \tilde{T}_2$.

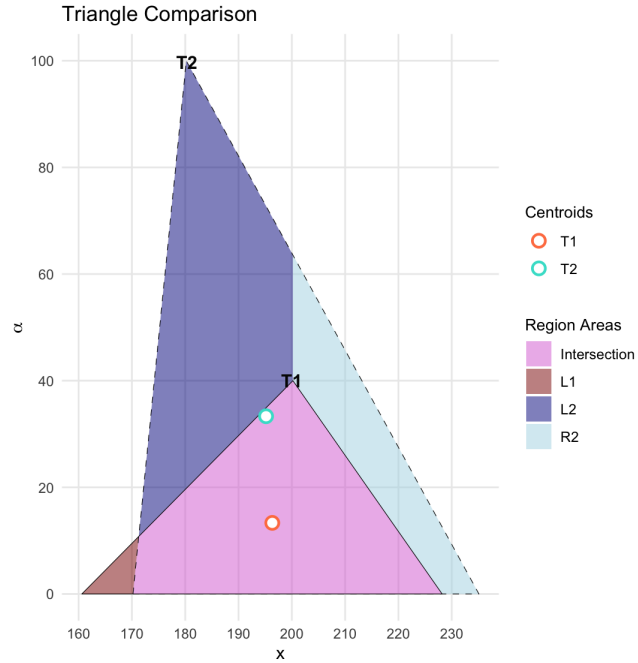


Figure 5.3: Comparison of $\tilde{T}_1 = \{(160.6, 0), (200.2, 40), (228.2, 0)\}$ and $\tilde{T}_2 = \{(170.2, 0), (180.2, 100), (235.1, 0)\}$. Areas: $L_1 = 51.71$, $R_1 = 0$, $L_2 = 1396.66$, $R_2 = 547.95$, $\text{Area}(\tilde{T}_1) = 1352$, $\text{Area}(\tilde{T}_2) = 3245$. Combined values: $L = 599.66$, $R = 1396.66$, $L\% = 13.04\%$ and $R\% = 30.38\%$, implying $\tilde{T}_1 > \tilde{T}_2$.

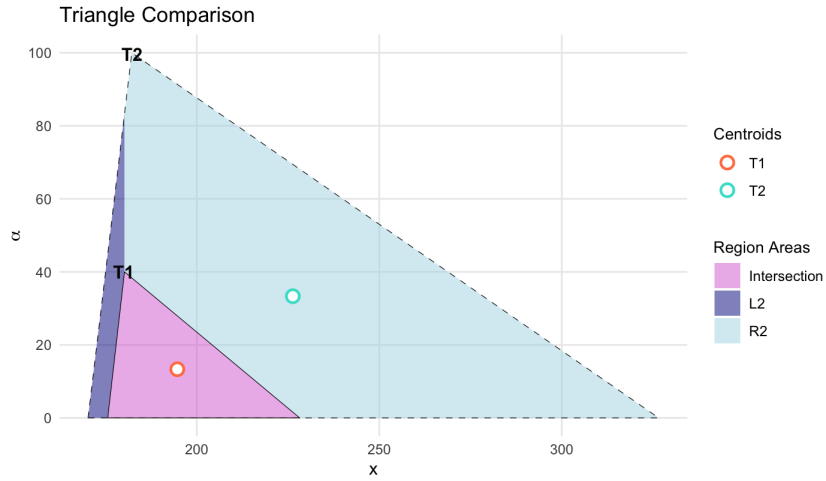


Figure 5.4: Comparison of $\tilde{T}_1 = \{(175.6, 0), (180.2, 40), (228.2, 0)\}$ and $\tilde{T}_2 = \{(170.2, 0), (182.2, 100), (235.1, 0)\}$. Area values: $L_1 = R_1 = 0$, $L_2 = 326.56$, $R_2 = 1866.34$. With $\text{Area}(\tilde{T}_1) = 1052$, $\text{Area}(\tilde{T}_2) = 3245$, we find $L = 1866.34$, $R = 326.56$, $L\% = 43.43\%$ and $R\% = 7.59\%$, yielding $\tilde{T}_1 < \tilde{T}_2$.

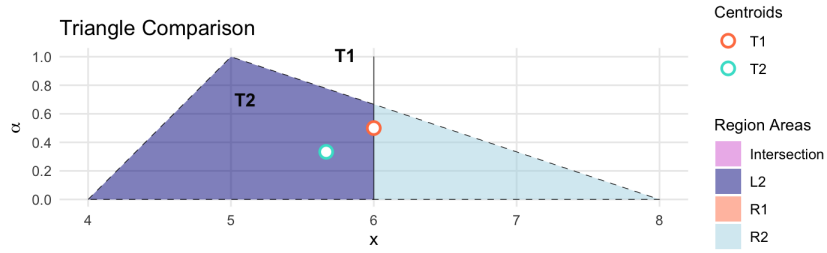


Figure 5.5: Comparison of $\tilde{T}_1 = \{(6, 0), (6, 1), (6, 0)\}$ and $\tilde{T}_2 = \{(4, 0), (5, 1), (8, 0)\}$. Area values: $L_1 = R_1 = 0$, $L_2 = 1.3$, $R_2 = 0.6$. Therefore, $L = 0.6$, $R = 1.3$, $L\% = 33.3\%$, $R\% = 66.6\%$ and $\tilde{T}_1 > \tilde{T}_2$.

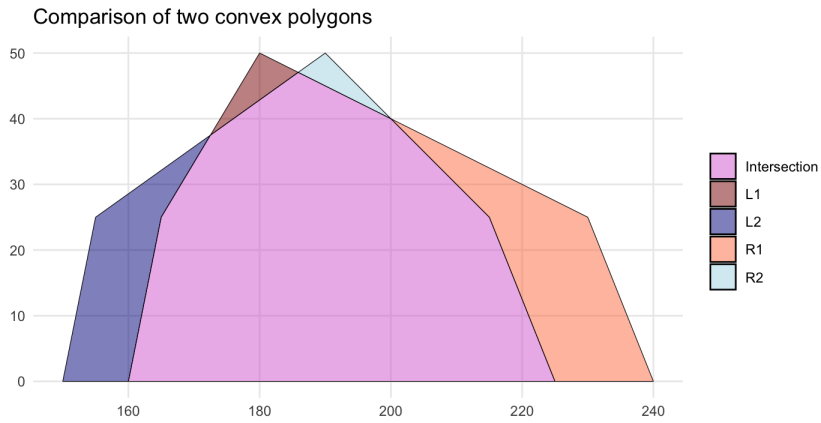


Figure 5.6: Comparison of pentagonal fuzzy numbers $\tilde{P}_1 = \{(160, 0), (165, 25), (180, 50), (230, 25), (240, 0)\}$ and $\tilde{P}_2 = \{(150, 0), (155, 25), (190, 50), (215, 25), (225, 0)\}$. Area values: $L_1 = 47.79$, $R_1 = 487.5$, $L_2 = 312.5$, $R_2 = 35.29$, $\text{Area}(\tilde{P}_1) = 2625$, $\text{Area}(\tilde{P}_2) = 2437.50$. Total: $L = 83.08$, $R = 800$, $L\% = 1.64\%$, $R\% = 15.80\%$ thus $\tilde{P}_1 > \tilde{P}_2$.

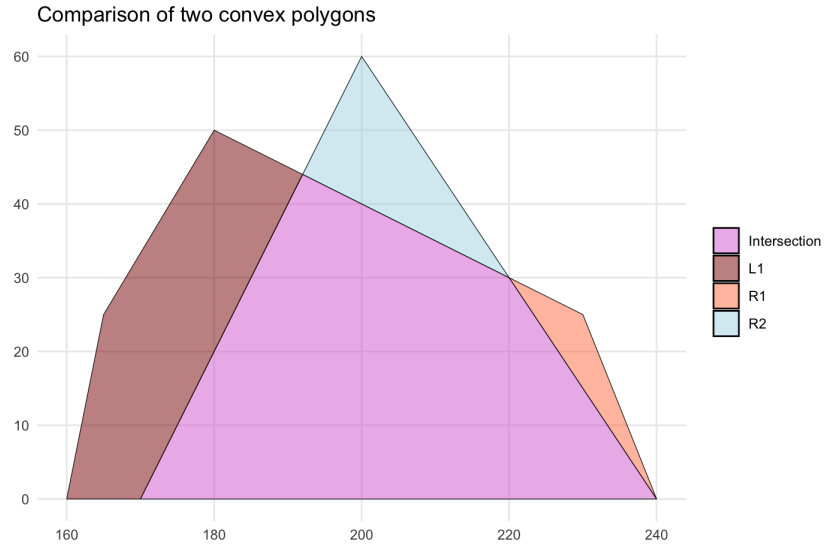


Figure 5.7: Comparison of $\tilde{P}_1 = \{(160,0), (165,25), (180,50), (230,25), (240,0)\}$ and triangular $\tilde{P}_2 = \{(170,0), (200,60), (240,0)\}$. Areas: $L_1 = 705$, $R_1 = 100$, $L_2 = 0$, $R_2 = 280$, $\text{Area}(\tilde{P}_1) = 2625$, $\text{Area}(\tilde{P}_2) = 2100$. Totals: $L = 985$, $R = 100$, $L\% = 20.84\%$, $R\% = 2.11\%$ hence $\tilde{P}_1 < \tilde{P}_2$.

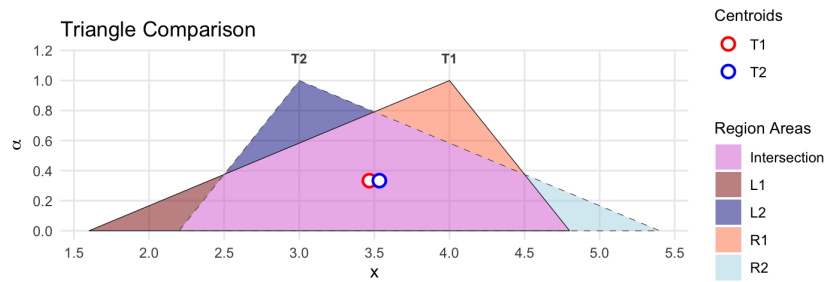


Figure 5.8: Comparison of $\tilde{T}_1 = \{(1.6,0), (4,1), (4.8,0)\}$ and $\tilde{T}_2 = \{(2.2,0), (3,1), (5.4,0)\}$. With centroid $\bar{x}_1 = 3.4\bar{6}$, centroid $\bar{x}_2 = 3.5\bar{3}$, the centroid method ranks $\tilde{T}_1 < \tilde{T}_2$, while the polygon method concludes the opposite: $\tilde{T}_1 > \tilde{T}_2$, with $L = 0.225$ and $R = 0.41\bar{6}$.

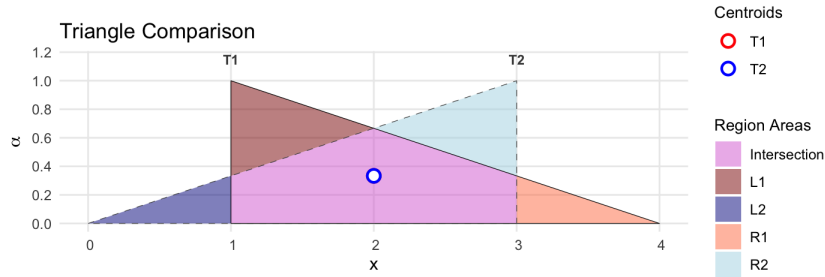


Figure 5.9: Comparison of $\tilde{T}_1 = \{(1, 0), (1, 1), (4, 0)\}$ and $\tilde{T}_2 = \{(0, 0), (3, 1), (3, 0)\}$. While the centroid method finds $\tilde{T}_1 = \tilde{T}_2$, the polygon method indicates $\tilde{T}_1 < \tilde{T}_2$ with $L = L_1 + R_2 = 0.33 + 0.33 = 0.66$ and $R = R_1 + L_2 = 0.16 + 0.16 = 0.32$.

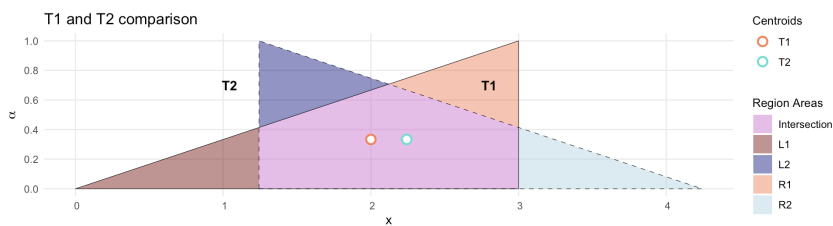


Figure 5.10: Comparison of asymmetric triangles $\tilde{T}_1 = \{(1 + t, 0), (1 + t, 1), (4 + t, 0)\}$, $t = 0.2426407$ and $\tilde{T}_2 = \{(0, 0), (3, 1), (3, 0)\}$. With centroid $\tilde{t}_1 = 2.242641$, centroid $\tilde{t}_2 = 2$, the centroid method ranks $\tilde{T}_1 > \tilde{T}_2$, whereas the polygon method yields $\tilde{T}_1 = \tilde{T}_2$ with equal values: $L = 0.5147186$, $R = 0.5147186$. In details, the four values L_1 , L_2 , R_1 and R_2 are equal to 0.2573593. Additionally, knowing that $\text{Area}(\tilde{T}_1) = \text{Area}(\tilde{T}_2) = 1.5$, one finds $L\% = R\% = 17.15\%$.

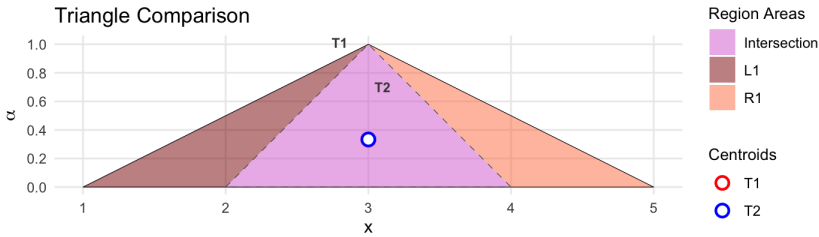


Figure 5.11: Comparison of symmetric triangles with identical midpoints: $\tilde{T}_1 = \{(1, 0), (3, 1), (5, 0)\}$ and $\tilde{T}_2 = \{(2, 0), (3, 1), (4, 0)\}$. Both centroid and polygon methods conclude $\tilde{T}_1 = \tilde{T}_2$. Centroid x -coordinate: 3. Total areas: $L = R = 1.5$.

Chapter 6

Selection of regression models

Linear models form a cornerstone of statistical modelling and inference, widely used due to their simplicity and interpretability. Among these, the *Gaussian linear models* assume that the error terms follow a normal distribution, allowing for closed-form expressions of test statistics and likelihood functions. This Gaussian assumption plays a pivotal role in model selection procedures, particularly through the **AIC** discovered by Akaike (1974), which evaluates models based on their maximum likelihood while penalizing complexity. Its popularity stems from its ease of computation in Gaussian linear models, where the log-likelihood is straightforwardly derived under normality assumptions.

However, the assumption of normally distributed errors may not always hold in practical applications. In such cases, alternative approaches like the *Empirical Likelihood* from Owen (1988, 2001b) offer a nonparametric way to conduct likelihood-based inferences without fully specifying the distribution of errors, enabling robust inference in broader settings.

In this chapter, we take a step further by proposing a fuzzy extension of both the **AIC** and the **Empirical Likelihood Akaike information criterion (ELAIC)**. These model selection criteria are particularly valuable in contexts where data and model parameters are subject to uncertainty or imprecision, conditions naturally addressed by fuzzy set theory. To the best of our knowledge, such an extension has not yet been explored in the literature. By reinterpreting these selection criteria as fuzzy quantities, we enable the direct incorporation of epistemic uncertainty into the model comparison process, improving interpretability when crisp comparisons are infeasible or misleading.

To this end, we develop a framework for defining fuzzy **AIC** and fuzzy **ELAIC**, where the output is no longer a crisp score but a fuzzy number. This allows for a more nuanced comparison between models. We then employ the fuzzy ranking method based on geometric comparison of fuzzy numbers (as described in Section 5.1) to establish a partial order among candidate models.

6.1 Gaussian linear regression model

Linear regression is one of the most common model used by analysts that describes a linear relationship between a dependent variable and one or several independent variables. We will study linear statistical models, that is, the linear relationship between the response variable and the covariates also assumes a stochastic error term written as ε . Hence why we refer to a linear regression model instead of a linear regression method. Moreover, if the errors are independents and normally distributed, the model becomes a Gaussian linear regression model.

Consider the stochastic pair $\{\mathbf{y}, \mathbf{X}\}$, where $\mathbf{y} = (y_1, \dots, y_n)$ is a $n \times 1$ vector of dependent (endogenous) random variables y_i , $i = 1, \dots, n$ and $\mathbf{X}$ is a $n \times (p+1)$ matrix of (exogenous) random variables x_{ij} . The i -th line of $\mathbf{X}$ is denoted by $\mathbf{x}'_i = (x_{i0}, x_{i1}, \dots, x_{ip})$ and its j -th column, $\mathbf{x}^j = (x_{1j}, \dots, x_{nj})'$ represents the j -th covariate, $j = 1, \dots, p$. Moreover, assume that the first covariate $\mathbf{x}^0 = (1, 1, \dots, 1) \in \mathbb{R}^n$ stands for the intercept. Then, a Gaussian linear regression model takes the form :

$$y_i = \mathbf{x}'_i \boldsymbol{\beta} + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma^2), \quad i = 1, \dots, n, \quad (6.1)$$

where $\boldsymbol{\beta} = (\beta_0, \dots, \beta_p) \in \mathbb{R}^{p+1}$ is the vector of unknown regression coefficients and ε_i are independent and identically distributed random error terms drawn from a Gaussian distribution of expected value 0 and variance σ^2 . With a more compact notation, the n equations from (6.1) can be written as

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad (6.2)$$

where $\boldsymbol{\varepsilon} \sim \mathcal{N}_n(\mathbf{0}_n, \mathbf{I}_n \sigma^2)$, $\mathbf{y} \sim \mathcal{N}_n(\mathbf{X}\boldsymbol{\beta}, \mathbf{I}_n \sigma^2)$, with $\mathbf{0}_n$ and $\mathbf{I}_n$ respectively denoting an n -vector of zeros and the identity matrix of size $n \times n$.

It is well known that one can estimate the regression coefficients $\boldsymbol{\beta}$ by the ordinary least squares in minimising the residual sum of squares $\sum_{i=1}^n (y_i - \mathbf{x}'_i \boldsymbol{\beta})^2$ and find $\hat{\boldsymbol{\beta}}_{OLS} = (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}'\mathbf{y}$.

Moreover, given that $y_i \sim \mathcal{N}(\mathbf{x}'_i \boldsymbol{\beta}, \sigma^2)$, the model's likelihood function writes as

$$\begin{aligned} L(\boldsymbol{\beta}, \sigma^2) &= \prod_{i=1}^n \left(\frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(y_i - \sum_j \beta_j x_{ij})^2} \right) \\ &= \frac{1}{(2\pi\sigma^2)^{\frac{n}{2}}} e^{-\frac{1}{2}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})'(\sigma^2 \mathbf{I}_n)^{-1}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})}. \end{aligned} \quad (6.3)$$

From there, one can retrieve the **Maximum likelihood (ML)** estimates $\hat{\boldsymbol{\beta}}_{ML}, \hat{\sigma}_{ML}^2$ by considering the log likelihood function:

$$\begin{aligned} \log L(\boldsymbol{\beta}, \sigma^2) &= -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\sigma^2) - \frac{1}{2}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})'(\sigma^2 \mathbf{I}_n)^{-1}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) \\ &= -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n e_i^2(\boldsymbol{\beta}), \end{aligned} \quad (6.4)$$

where $e_i(\beta) = y_i - \mathbf{x}'_i\beta$. The ML estimates $\hat{\beta}_{ML}, \hat{\sigma}_{ML}^2$ are then the ones which maximise $\log L$. It is easy to see that maximising $\log L$ with respect to β is equivalent to minimising $\sum_{i=1}^n (y_i - \mathbf{x}'_i\beta)^2$, hence $\hat{\beta}_{ML} = \hat{\beta}_{OLS}$. In addition, $\frac{\partial}{\partial \sigma^2} \log L(\hat{\beta}_{ML}, \sigma^2) = 0$ yields

$$\begin{aligned} -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4} \sum_{i=1}^n e_i^2(\hat{\beta}_{ML}) &= 0 \iff \\ -n + \frac{1}{\sigma^2} \sum_{i=1}^n e_i^2(\hat{\beta}_{ML}) &= 0 \iff \\ \sigma^2 &= \frac{1}{n} \sum_{i=1}^n e_i^2(\hat{\beta}_{ML}), \quad \text{thus} \\ \hat{\sigma}_{ML}^2 &= \frac{1}{n} \sum_{i=1}^n e_i^2(\hat{\beta}_{ML}). \end{aligned} \tag{6.5}$$

Knowing the ML estimates, one can also calculate the AIC which allows to assess the quality of a model given a set of data. Although various formulations of the AIC exist, we adopt the original version proposed by Akaike (1974), given by:

$$\text{AIC} = 2k - 2 \log L(\hat{\beta}_{ML}, \hat{\sigma}_{ML}^2), \tag{6.6}$$

where k accounts for the number of parameters. Be aware that the ML estimate $\hat{\sigma}_{ML}^2$ is included in the number of parameters k for the calculation of a Gaussian linear regression model's AIC. In this case one finds

$$\begin{aligned} \text{AIC} &= 2(p+2) + n \log(2\pi) + n \log(\hat{\sigma}_{ML}^2) + \frac{1}{\hat{\sigma}_{ML}^2} \sum_{i=1}^n e_i^2(\hat{\beta}_{ML}) \\ &= 2(p+2) + n[\log(2\pi) + \log(\hat{\sigma}_{ML}^2)] + n \frac{\sum_{i=1}^n e_i^2(\hat{\beta}_{ML})}{\sum_{i=1}^n e_i^2(\hat{\beta}_{ML})} \\ &= 2(p+2) + n[\log(2\pi) + \log(\hat{\sigma}_{ML}^2) + 1], \end{aligned} \tag{6.7}$$

where p is the number of regression coefficients (without the intercept) of the model.

Remark 6.1.1.

Note that since the residuals are given by $e_i(\hat{\beta}) = y_i - \mathbf{x}'_i\hat{\beta}$, the residuals for a nested submodel can be computed using the full design matrix $\mathbf{X}$ by simply setting the coefficients $\beta_k = 0$ for all predictors $\mathbf{x}^k$ that are excluded from the submodel (i.e., those present only in the full model). This yields the same result as computing the residuals using only the reduced set of predictors and corresponding coefficients of the nested submodel. Hence, the AIC of a nested submodel with $0 \leq q < p$ regression coefficients is calculated similarly as

$$\text{AIC}_{\text{submodel}} = 2(q+2) + n[\log(2\pi) + \log(\hat{\sigma}_{ML}^2) + 1].$$

6.2 Empirical likelihood in linear regression

By keeping the notation of the previous section, let us relax a bit the conditions on the error term and consider the following parametric regression model as described in S. X. Chen and Keilegom (2009)[p.416]:

$$y_i = m(\mathbf{x}_i; \beta) + \varepsilon_i, \quad i = 1, \dots, n, \quad (6.8)$$

where $m(\mathbf{x}_i; \beta) = \mathbf{x}_i' \beta$ is the known regression function and ε_i are random independent variables such that $\mathbb{E}(\varepsilon_i | \mathbf{x}_i) = 0$ and $\text{Var}(\varepsilon_i | \mathbf{x}_i) = \sigma^2(\mathbf{x}_i) < \infty$ for some function $\sigma(\cdot)$, meaning that the errors can be heteroscedastic. In addition, the conditional variance of the errors ε_i must be finite.

By denoting as $p_1, p_2, \dots, p_n$ the probability weights associated to the data, the empirical likelihood of β can be expressed as (Owen (2001b)):

$$L_n(\beta) = \prod_{i=1}^n p_i \quad (6.9)$$

under the constraints

$$\sum_{i=1}^n p_i = 1, \quad (6.10)$$

$$\sum_{i=1}^n p_i \frac{\partial m(\mathbf{x}_i; \beta)}{\partial \beta} \{y_i - m(\mathbf{x}_i; \beta)\} = 0. \quad (6.11)$$

The empirical likelihood estimator of β is then obtained by maximising the likelihood $L_n(\beta)$ under constraints (6.10) and (6.11).

With the help of the Lagrangian multipliers $\lambda_0 \in \mathbb{R}$ and $\lambda_1 \in \mathbb{R}^p$, it is possible to rewrite the above constrained optimisation problem as an unconstrained one with the objective function:

$$\begin{aligned} \arg\max_{p, \lambda_0, \lambda_1} T(p, \lambda_0, \lambda_1) &= \sum_{i=1}^n \log(p_i) + \lambda_0 \left(\sum_{i=1}^n p_i - 1 \right) \\ &\quad + \lambda_1' \sum_{i=1}^n p_i \frac{\partial m(\mathbf{x}_i; \beta)}{\partial \beta} \{y_i - m(\mathbf{x}_i; \beta)\}, \end{aligned} \quad (6.12)$$

where $p = (p_1, p_2, \dots, p_n)'$. Then, by calculating the partial derivative for all p_i 's

$$\frac{\partial}{\partial p_i} T(p, \lambda_0, \lambda_1) = \frac{1}{p_i} + \lambda_0 + \lambda_1' \frac{\partial m(\mathbf{x}_i; \beta)}{\partial \beta} \{y_i - m(\mathbf{x}_i; \beta)\}$$

and setting them to 0 yields

$$0 = \frac{1}{p_i} + \lambda_0 + \lambda_1' \frac{\partial m(\mathbf{x}_i; \beta)}{\partial \beta} \{y_i - m(\mathbf{x}_i; \beta)\} \iff$$

$$0 = 1 + p_i \lambda_0 + \lambda_1' p_i \frac{\partial m(\mathbf{x}_i; \beta)}{\partial \beta} \{y_i - m(\mathbf{x}_i; \beta)\} \iff \quad (6.13)$$

$$(6.14)$$

Then, by summing over all i s, (6.14) is equivalent to

$$0 = \sum_{i=1}^n \left[1 + p_i \lambda_0 + \lambda_1' p_i \frac{\partial m(\mathbf{x}_i; \beta)}{\partial \beta} \{y_i - m(\mathbf{x}_i; \beta)\} \right] \iff$$

$$0 = n + \lambda_0 \sum_{i=1}^n p_i + \lambda_1' \sum_{i=1}^n p_i \frac{\partial m(\mathbf{x}_i; \beta)}{\partial \beta} \{y_i - m(\mathbf{x}_i; \beta)\} \quad (6.15)$$

By using constraints (6.10) and (6.11), (6.15) becomes

$$0 = n + \lambda_0 \iff$$

$$\lambda_0 = -n.$$

Thus, the p_i 's can be expressed as

$$0 = 1 - np_i + p_i \lambda_1' \frac{\partial m(\mathbf{x}_i; \beta)}{\partial \beta} \{y_i - m(\mathbf{x}_i; \beta)\} \iff$$

$$p_i = \frac{1}{n} \frac{1}{1 - \frac{1}{n} \lambda_1' \frac{\partial m(\mathbf{x}_i; \beta)}{\partial \beta} \{y_i - m(\mathbf{x}_i; \beta)\}}.$$

Noting $-\frac{1}{n} \lambda_1 = \lambda$ yields

$$p_i = \frac{1}{n} \frac{1}{1 + \lambda' \frac{\partial m(\mathbf{x}_i; \beta)}{\partial \beta} \{y_i - m(\mathbf{x}_i; \beta)\}}. \quad (6.16)$$

By injecting this value of p_i in constraint (6.11), one finds

$$\sum_{i=1}^n \frac{(\frac{\partial m(\mathbf{x}_i; \beta)}{\partial \beta})(y_i - m(\mathbf{x}_i; \beta))}{1 + \lambda' \frac{\partial m(\mathbf{x}_i; \beta)}{\partial \beta} \{y_i - m(\mathbf{x}_i; \beta)\}} = 0. \quad (6.17)$$

Similarly, the likelihood given in (6.9) becomes

$$L_n(\beta) = \prod_{i=1}^n \frac{1}{n} \frac{1}{1 + \lambda' \frac{\partial m(\mathbf{x}_i; \beta)}{\partial \beta} \{y_i - m(\mathbf{x}_i; \beta)\}}, \quad (6.18)$$

and the loglikelihood then writes as

$$\ell_n(\beta) = \log(L_n(\beta))$$

$$= - \sum_{i=1}^n \log \left(1 + \lambda' \frac{\partial m(\mathbf{x}_i; \beta)}{\partial \beta} \{y_i - m(\mathbf{x}_i; \beta)\} \right) - n \log(n). \quad (6.19)$$

This tells us that, in order to find the empirical likelihood at a given β , we first need to solve the nonlinear equation (6.17) depending on β to find the value of λ which then can be used to compute the p_i 's. Doing so is computationally very heavy, it is why Owen (2001b) proposed to consider the dual problem to the optimisation problem (6.9)-(6.11). This dual problem is stated as follows:

Minimise the objective function

$$Q(\lambda) := - \sum_{i=1}^n \log \left\{ 1 + \lambda' \frac{\partial m(\mathbf{x}_i; \beta)}{\partial \beta} (y_i - m(\mathbf{x}_i; \beta)) \right\} \quad (6.20)$$

with respect to the constraints

$$1 + \lambda' \frac{\partial m(\mathbf{x}_i; \beta)}{\partial \beta} (y_i - m(\mathbf{x}_i; \beta)) \geq \frac{1}{n} \quad \forall i = 1, \dots, n. \quad (6.21)$$

Observe that the objective function $Q(\lambda)$ is actually the expression of the loglikelihood in (6.19) without the constant $n \log(n)$.

Moreover, Let us write by

$$\mathcal{D} = \left\{ \lambda : 1 + \lambda' \frac{\partial m(\mathbf{x}_i; \beta)}{\partial \beta} (y_i - m(\mathbf{x}_i; \beta)) \geq \frac{1}{n} \text{ for each } i = 1, \dots, n \right\} \quad (6.22)$$

the set of λ following the constraints (6.21). Remark that the dual problem reduces to minimise Q over $\mathcal{D}$. It is possible to show that the set $\mathcal{D}$ is convex, closed and compact. Thus it guarantees the existence of a unique minimum on $\mathcal{D}$. As suggested by Owen (2001b), the constraints can be removed by considering a modified log function in Q defined as follows:

$$\log^*(x) = \begin{cases} \log x & \text{if } x \geq \frac{1}{n} \\ -\frac{n^2 x^2}{2} + 2nx - \frac{3}{2} - \log(n) & \text{for } x < \frac{1}{n}, \end{cases}$$

where $\log^*(x)$ is a quadratic approximation of $\log(x)$ constructed to match the value of the first and second derivatives of $\log(x)$ at $x = 1/n$. Note that the profile likelihood $\prod_{i=1}^n p_i$ achieves its maximum value n^{-n} when all probability weights p_i are equal, i.e $p_i = 1/n$ for all $i = 1, \dots, n$. Therefore, if there exists a parameter $\hat{\beta}$ that solves the unweighted moment condition

$$\sum_{i=1}^n \frac{\partial m(\mathbf{x}_i; \beta)}{\partial \beta} \{y_i - m(\mathbf{x}_i; \beta)\} = \mathbf{0}, \quad (6.23)$$

then it follows that $\lambda = 0$. This, in turn, implies that $p_i = \frac{1}{n}$, and the empirical likelihood attains its maximum at $\hat{\beta}$, with

$$L_n(\hat{\beta}) = n^{-n}.$$

Let us clarify why the above condition (6.23) leads to $\lambda = \mathbf{0}$. If the unweighted moment condition (6.23) holds, then

$$p_i = \frac{1}{n} \frac{1}{1 + \lambda' \frac{\partial m(\mathbf{x}_i; \beta)}{\partial \beta} \{y_i - m(\mathbf{x}_i; \beta)\}} = \frac{1}{n}.$$

If the constraints are already satisfied by uniform weights, no adjustment via the Lagrange multiplier λ is needed. In the Lagrangian formulation of the empirical likelihood, λ is introduced precisely to enforce the constraints when uniform probability weights do not satisfy it. Thus, when (6.23) holds, the corresponding multiplier is zero: $\lambda = \mathbf{0}$ and the empirical likelihood is maximised with $p_i = \frac{1}{n}$ and equals n^{-n} .

In particular, the least squares estimate $\hat{\beta}$ defined as

$$\hat{\beta} = \arg \min_{\beta} \sum_{i=1}^n (y_i - \mathbf{x}_i' \beta)^2.$$

satisfies the unweighted moment condition (6.23). Indeed, taking the gradient of the residual sum of squares with respect to β gives

$$\frac{\partial}{\partial \beta} \sum_{i=1}^n (y_i - \mathbf{x}_i' \beta)^2 = -2 \sum_{i=1}^n \mathbf{x}_i (y_i - \mathbf{x}_i' \beta) = \mathbf{0}.$$

Setting this derivative to zero and evaluating at $\hat{\beta}$ yields

$$\sum_{i=1}^n \mathbf{x}_i (y_i - \mathbf{x}_i' \hat{\beta}) = \mathbf{0}.$$

Now observe that for the linear model $m(\mathbf{x}_i; \beta) = \mathbf{x}_i' \beta$, the gradient of $m(\cdot, \cdot)$ with respect to β is $\partial m(\mathbf{x}_i; \beta) / \partial \beta = \mathbf{x}_i$, and hence

$$\sum_{i=1}^n \frac{\partial m(\mathbf{x}_i; \beta)}{\partial \beta} \{y_i - m(\mathbf{x}_i; \beta)\} = \sum_{i=1}^n \mathbf{x}_i (y_i - \mathbf{x}_i' \beta).$$

Therefore, the least squares estimator $\hat{\beta}$ satisfies the unweighted moment condition (6.23).

Because $\hat{\beta}$ satisfies this moment condition, the associated Lagrange multiplier in the empirical likelihood formulation must be zero, i.e., $\lambda = \mathbf{0}$. Consequently, the empirical likelihood weights reduces to

$$p_i = \frac{1}{n},$$

and the empirical likelihood is maximised at $\beta = \hat{\beta}$, achieving a value of

$$L_n(\hat{\beta}) = n^{-n}.$$

Algorithm 6.1 summarises the procedure for obtaining the estimator $\hat{\beta}$ that maximises the empirical likelihood.

6.3 Nonlinear regression via empirical likelihood

Consider the nonlinear regression model

$$y_i = m(\mathbf{x}_i; \beta) + \varepsilon_i, \quad i = 1, \dots, n, \quad (6.24)$$

where y_i , $\mathbf{x}_i$ and β are defined as before but now $m(\mathbf{x}_i; \beta)$ is a known nonlinear regression function and ε_i is an unknown stochastic error term satisfying $\mathbb{E}[\varepsilon_i | \mathbf{x}_i] = 0$.

Since $\mathbb{E}[y_i | \mathbf{x}_i] = m(\mathbf{x}_i; \beta)$, we have the following moment condition:

$$\mathbb{E} \left[\frac{\partial m(\mathbf{x}_i; \beta)}{\partial \beta} \cdot (y_i - m(\mathbf{x}_i; \beta)) \right] = \mathbf{0}.$$

Define the vector valued function:

$$\mathbf{g}_i(\beta) := \frac{\partial m(\mathbf{x}_i; \beta)}{\partial \beta} (y_i - m(\mathbf{x}_i; \beta)).$$

As previously, the empirical likelihood method estimates β by finding weights $\{p_i\}_{i=1}^n$ satisfying:

$$\sum_{i=1}^n p_i = 1, \quad \sum_{i=1}^n p_i \mathbf{g}_i(\beta) = \mathbf{0}, \quad p_i > 0, \quad (6.25)$$

and maximising the empirical likelihood:

$$L_n(\beta) = \prod_{i=1}^n p_i.$$

As discussed in the previous section, this optimisation problem is solved by introducing a Lagrange multiplier $\lambda \in \mathbb{R}^p$. The solution to the constrained optimisation yields:

$$p_i(\beta, \lambda) = \frac{1}{n} \cdot \frac{1}{1 + \lambda^\top \mathbf{g}_i(\beta)}, \quad \text{provided } 1 + \lambda^\top \mathbf{g}_i(\beta) > 0 \text{ for all } i.$$

Substituting into the likelihood yields the empirical log-likelihood function:

$$\ell_n(\beta) = - \sum_{i=1}^n \log \left(1 + \lambda^\top \mathbf{g}_i(\beta) \right) - n \log n.$$

The empirical likelihood estimator is then defined as:

$$\hat{\beta} = \arg \max_{\beta} \ell_n(\beta),$$

which is typically solved numerically in two steps as explained in Algorithm 6.1.

AIC estimated by empirical likelihood

In general, the **ELAIC** for a regression model, denoted AIC_{el} , can be estimated as

$$AIC_{el} = 2(p + 1) - 2 \log(L_n(\hat{\beta})), \quad (6.26)$$

where $p + 1$ corresponds to the number of covariates plus the intercept. Note that, unlike classical likelihood-based criteria, the empirical likelihood formulation does not require estimation of the error variance, and thus no such term is included in the expression.

As previously discussed, when there exists a parameter estimate $\hat{\beta}$ that satisfies the moment condition

$$\sum_{i=1}^n \frac{\partial m(\mathbf{x}_i; \beta)}{\partial \beta} \{y_i - m(\mathbf{x}_i; \beta)\} = 0,$$

the empirical likelihood achieves its maximum at $\hat{\beta}$. This occurs, for example, when the **ML** and **Least squares (LS)** estimates coincide. In such situations, the empirical likelihood evaluates to

$$L_n(\hat{\beta}) = n^{-n},$$

and hence the empirical likelihood AIC becomes

$$AIC_{el} = 2(p + 1) - 2 \log(n^{-n}) = 2(p + 1) + 2n \log n.$$

Empirically, such conditions frequently hold, particularly in standard linear models where the **ML** and **LS** estimators are equivalent. However, this formulation presents a critical issue when comparing submodels with $k < p$ regression parameters that share the same **ML** and **LS** estimates. If AIC_{el} is computed as above for all such submodels, the empirical likelihood remains n^{-n} regardless of the number of parameters, and the criterion will always favor the model with the fewest covariates. Consequently, this approach fails to penalize model complexity adequately.

To address this issue, one must be mindful of Remark 6.1.1 and compute the empirical likelihood under nested submodel-specific restrictions. In particular, nested submodels are formed by constraining certain regression coefficients from the full model to be zero. Under these additional constraints, the resulting estimator $\hat{\beta}$ no longer satisfies the moment condition in (6.23), and the corresponding Lagrange multiplier is generally non-zero: $\lambda \neq \mathbf{0}$. This adjustment leads to a genuinely lower empirical likelihood for more constrained models, thereby restoring the discriminatory power of the **ELAIC** in model selection. More formally, let $S \subset \{1, \dots, p\}$, $\#S = k$, then $\beta_i \neq 0$ if $i \in S$, $\beta_i = 0$ if $i \notin S$. The set S corresponds to the positions of the non-zero regression coefficient β_s , $1 \leq s \leq p$ of a submodel of (6.35). Such submodel can be written as

$$y_i = \beta_0 + \sum_{s \in S} \beta_s x_{is} + \varepsilon_i. \quad (6.27)$$

With the empirical likelihood approach described in section 6.2, compute the maximum likelihood estimates $\hat{\beta}_s$, $s \in S$ of (6.27). Then, let $\hat{\beta}_{sub} = \hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_p$ where $\hat{\beta}_s = 0$ if $s \notin S$. With the empirical likelihood function $L_n(\cdot) : \mathbb{R}^{p+1} \rightarrow \mathbb{R}$, the AIC_{el} is then

$$AIC_{el} = 2(k+1) - 2 \log(L_n(\hat{\beta}_{sub})). \quad (6.28)$$

Note that, if $k = p$, $\hat{\beta}_{sub} = \hat{\beta}$ where $\hat{\beta}$ is the vector containing the estimates by maximum empirical likelihood of the full model. Hence, if $\hat{\beta}$ satisfies (6.23), for example if the ML estimates coincide with the LS estimates, then

$$AIC_{el} = 2(p+1) - 2 \log(L_n(\hat{\beta})) = 2(p+1) + 2n \log(n).$$

When defining the AIC_{el} , one must observe that $L_n(\hat{\beta})$ is non-parametric and thus does not share the same properties as a standard parametric likelihood. The validity of the ELAIC is thus not ensured by the Kullback-Leibler (KL) asymptotics that justify the classical AIC as explained for example in Bozdogan (1987).

6.3.1 Fuzzy AIC

Before introducing the concept of a Fuzzy Akaike Information Criterion, we first outline the fuzzy regression model of interest. At a given α – cut $\alpha \in [0, 1]$, consider the stochastic pair $(\tilde{\mathbf{X}}^\alpha, \tilde{\mathbf{y}}^\alpha)$ where $\tilde{\mathbf{X}}^\alpha$ is a matrix whose components are (exogenous) random fuzzy variables (see Definition 4.1.3) $\tilde{x}_{ij}^\alpha = (x_{ij}^{L,\alpha}, x_{ij}, x_{ij}^{R,\alpha})$ and $\tilde{\mathbf{y}}^\alpha$ is a vector of (endogenous) random fuzzy variables $(y_i^{L,\alpha}, y_i, y_i^{R,\alpha})$. Both $\tilde{x}_{ij}^\alpha$ and $\tilde{y}_i^\alpha$ are viewed in the epistemic sense. Additionally, $\mathbf{X}$ and $\mathbf{y}$ denote the matrix and vector of crisp random variables x_{ij} , respectively y_i . Then, consider the following fuzzy regression model

$$\tilde{y}_i^\alpha = \tilde{\beta}_0^\alpha \oplus (\tilde{\beta}_1^\alpha \otimes \tilde{x}_{i1}^\alpha) \oplus \dots \oplus (\tilde{\beta}_p^\alpha \otimes \tilde{x}_{ip}^\alpha) \oplus \varepsilon_i, \quad i = 1, \dots, n, \quad (6.29)$$

where ε_i is an unknown stochastic error term. We aim to compute an AIC for the regression model in (6.29). Since both the observations and the regression parameters are fuzzy, the estimation of an AIC is not straightforward. Nevertheless, following a similar approach as described in Section 3.4, we can approximate a fuzzy AIC by generating multiple observations within the intervals $x_{ij}^{L,\alpha}, x_{ij}^{R,\alpha}$ and $y_i^{L,\alpha}, y_i^{R,\alpha}$. We will see that, depending on the assumptions we will make on the unknown stochastic error terms that two information criteria will emerge: the fuzzy AIC with the strongest assumptions and fuzzy ELAIC with relaxed assumptions.

Consider the indexed matrices $\mathbf{X}_\chi$ and vectors $\mathbf{y}_\chi$ with components x_{ij_χ}, y_{i_χ} bounded by the left and right components of $\tilde{\mathbf{X}}^\alpha, \tilde{\mathbf{y}}^\alpha$, $x_{ij}^{L,\alpha} \leq x_{ij_\chi} \leq x_{ij}^{R,\alpha}$, $y_i^{L,\alpha} \leq y_{i_\chi} \leq y_i^{R,\alpha}$. Additionally, let I_χ be the set containing all the indices χ .

We would be interested in finding regression coefficients $\tilde{\beta}_j^\alpha$, $j = 0, \dots, p$ and the corresponding AIC of (6.29) at any α –cut. To do so, we can use Zadeh's extension principle and consider all regression models of the form

$$y_{i_\chi} = \beta_{0_\chi} + \beta_{1_\chi} x_{i1_\chi} + \dots + \beta_{p_\chi} x_{ip_\chi} + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma^2), \quad i = 1, \dots, n. \quad (6.30)$$

It is well known that one can estimate the regression coefficients by ordinary least squares. Let $\hat{\beta}_{j_\chi}$ be estimated this way for all $j = 0, \dots, p$. Then, let us denote the set containing all the estimated $\hat{\beta}_{j_\chi}$ by $\mathcal{B}_j = \{\hat{\beta}_{j_\chi} | \chi \in I_\chi\}$, where I_χ is the set containing all indices χ . Thus, $\hat{\beta}_j^L = \inf_\chi(\mathcal{B}_j)$, $\hat{\beta}_j^R = \sup_\chi(\mathcal{B}_j)$ and the fuzzy regression coefficients are, for $\alpha \in [0, 1]$,

$$\hat{\beta}_j^\alpha = (\hat{\beta}_j^L, \hat{\beta}_j^R).$$

As explained in 6.1, under the assumptions of the linear regression models (6.30), we know that $\hat{\beta}_{j_\chi}$ are also maximum likelihood estimates. Moreover, with equation (6.5), by noting $e_{i_\chi} = y_{i_\chi} - \mathbf{X}'_{i_\chi} \hat{\beta}_{j_\chi}$, we know that the maximum likelihood estimate of the variance is

$$\hat{\sigma}_{ML_\chi}^2 = \frac{1}{n} \sum_{i=1}^n e_{i_\chi}^2.$$

With (6.7), the AIC is given by

$$AIC_\chi = 2(p+2) + n[\log(2\pi) + \log(\hat{\sigma}_{ML_\chi}^2) + 1], \quad (6.31)$$

where p is the number of regression coefficients without the intercept. By taking the infimum and supremum among all AIC_χ^α , we are able to define the α -cut of a fuzzy AIC by

$$\widetilde{AIC}^\alpha = (\inf(AIC_\chi^\alpha), \sup(AIC_\chi^\alpha)). \quad (6.32)$$

Note that, if $\widetilde{AIC}^{\alpha=1}$ corresponds to the original crisp AIC value of the model and $\widetilde{AIC}^{\alpha=0} = (\widetilde{AIC}^{L,0}, \widetilde{AIC}^{R,0})$ one can approximate $\widetilde{AIC}$ by a triangular fuzzy number

$$\widetilde{AIC} = (\widetilde{AIC}^{L,0}, AIC, \widetilde{AIC}^{R,0}). \quad (6.33)$$

If one only knows $\widetilde{AIC}^{L,0}$ and $\widetilde{AIC}^{R,0}$, it is possible to estimate AIC by the median or mean of all AIC_χ values.

The assumptions that $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$ in (6.30) must be satisfied for all $y_i^L \leq y_{i_\chi} \leq y_i^R$, $x_{ij}^L \leq x_{ij_\chi} \leq x_{ij}^R$ which in practice might not be true. To relax the conditions on the errors, we can consider that the errors ε_i of the regression model

$$\tilde{y}_i^\alpha = \tilde{\beta}_0^\alpha \oplus (\tilde{\beta}_1^\alpha \otimes \tilde{x}_{i1}^\alpha) \oplus \dots \oplus (\tilde{\beta}_p^\alpha \otimes \tilde{x}_{ip}^\alpha) \oplus \varepsilon_i, \quad i = 1, \dots, n. \quad (6.34)$$

are random independent variables such that $\mathbb{E}(\varepsilon_i | \mathbf{x}_i) = 0$ and $\text{Var}(\varepsilon_i | \mathbf{x}_i) = \sigma^2(\mathbf{x}_i) < \infty$ for some function $\sigma(\cdot)$. Then, in the same spirit as before, we need to consider all linear regression models of the form

$$y_{i_\chi} = \beta_{0_\chi} + \beta_{1_\chi} x_{i1_\chi} + \dots + \beta_{p_\chi} x_{ip_\chi} + \varepsilon_i, \quad i = 1, \dots, n. \quad (6.35)$$

Then, as before, with the set $\mathcal{B}_j = \{\hat{\beta}_{j_\chi} | \chi \in I_\chi\}$ containing all the maximum empirical likelihood estimates $\hat{\beta}_{j_\chi}$, we set $\hat{\beta}_j^L = \inf_\chi(\mathcal{B}_j)$, $\hat{\beta}_j^R = \sup_\chi(\mathcal{B}_j)$ and the fuzzy regression coefficients estimated by empirical maximum likelihood are, for $\alpha \in [0, 1]$,

$$\hat{\beta}_j^\alpha = (\hat{\beta}_j^L, \hat{\beta}_j^R).$$

With $S \subset \{1, \dots, p\}$, $\#S = k$, one thus have the vector of maximum empirical likelihood estimates $\hat{\beta}_{sub,\chi}^\alpha = (\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_p)$ where $\hat{\beta}_s = 0$ if $s \notin S$ of any submodel of k parameters of the form

$$y_{i_\chi} = \beta_{0_\chi} + \mathbf{x}'_{i_\chi} \beta_{sub,\chi} + \varepsilon_i. \quad (6.36)$$

Then, with the empirical likelihood function $L_{n,\chi}(\cdot)$ of the full model, one can use (6.26) to compute $AIC_{el,\chi}$ for all submodels given in (6.36) by

$$AIC_{el,\chi} = 2(k+1) - 2 \log(L_n(\hat{\beta}_{sub,\chi})). \quad (6.37)$$

The α -cut of the fuzzy ELAIC is then,

$$\widetilde{AIC}_{el}^\alpha = (\inf_\chi AIC_{el,\chi}^\alpha, \sup_\chi AIC_{el,\chi}^\alpha). \quad (6.38)$$

If $\widetilde{AIC}_{el}^{\alpha=0} = (\widetilde{AIC}_{el}^{L,0}, \widetilde{AIC}_{el}^{R,0})$ and $\widetilde{AIC}_{el}^{\alpha=1} = AIC_{el}$, where AIC_{el} is the original ELAIC value of the crisp model, one can estimate the fuzzy ELAIC by a triangular fuzzy number,

$$\widetilde{AIC}_{el} = (\widetilde{AIC}_{el}^{L,0}, AIC_{el}, \widetilde{AIC}_{el}^{R,0}).$$

If AIC_{el} is not known, one can approximate it by the median or mean of all $AIC_{el,\chi}$ values.

More generally, when one only knows the bounds $y_i^{L,\alpha=0}$, $y_i^{R,\alpha=0}$, $x_{ij}^{L,\alpha=0}$ and $x_{ij}^{R,\alpha=0}$, it is possible to use the quantiles of the distribution of the AIC_χ , respectively $AIC_{el,\chi}$ values to construct a convex polygon as explained in Section 5.1.1. One can then use this polygon to approximate the membership function of $\widetilde{AIC}$, respectively $\widetilde{AIC}_{el}$. The same reasoning can be applied to the estimated fuzzy regression coefficients $\hat{\beta}_j^\alpha$ to approximate their membership function by a triangle or convex polygon, when only their 0-cut is known.

Surprisingly enough, if (6.23) holds for all $\hat{\beta}_{s_\chi}$, in particular, if $\hat{\beta}_{s_\chi}$ coincide with the LS estimates for all χ , then the empirical likelihood achieves its maximum at $\hat{\beta}_\chi$, with $L_{n,\chi}(\hat{\beta}) = n^{-n}$ for all χ . As a result, the ELAIC estimated by empirical likelihood $\widetilde{AIC}_{el}$ is crisp at any $\alpha \in [0, 1]$ for the full model,

$$\widetilde{AIC}_{el}^\alpha = (\inf_\chi AIC_{el,\chi}^\alpha, \sup_\chi AIC_{el,\chi}^\alpha) = (-n \log(n), -n \log(n)) = -n \log(n). \quad (6.39)$$

Note that, in practice, this scenario arises quite often. Moreover, observe that in general $\widetilde{AIC}_{el}$ is fuzzy when calculated for submodels since fixing a regression parameter to 0 enforces the use of non-zero λ in (6.16) as explained in Subsection 6.3.

6.3.2 Numerical example

For this numerical example, we are using the `mtcars` dataset, a well-known dataset in R, containing information on 32 automobiles and including 11 variables. These data were extracted from the 1974 Motor Trend US magazine and are commonly used for regression modelling examples. Below is a description of the dataset variables:

Variable	Description
<code>mpg</code>	Miles per gallon (fuel efficiency)
<code>cyl</code>	Number of cylinders
<code>disp</code>	Displacement (cu. in.)
<code>hp</code>	Gross horsepower
<code>drat</code>	Rear axle ratio
<code>wt</code>	Weight (1000 lbs)
<code>qsec</code>	1/4 mile time (seconds)
<code>vs</code>	Engine (0 = V-shaped, 1 = straight)
<code>am</code>	Transmission (0 = automatic, 1 = manual)
<code>gear</code>	Number of forward gears
<code>carb</code>	Number of carburetors

In our analysis, we consider `mpg` as the independent variable of interest, and focus on the following four covariates: `disp`, `hp`, `wt`, and `qsec`.

We model the fuel efficiency `mpg` using a Gaussian linear regression model with these four selected predictors. The model is given by:

$$\text{mpg}_i = \beta_0 + \beta_1 \text{disp}_i + \beta_2 \text{hp}_i + \beta_3 \text{wt}_i + \beta_4 \text{qsec}_i + \varepsilon_i, \quad i = 1, \dots, 32, \quad (6.40)$$

where we consider ε_i as independent error terms following a normal distribution with mean zero and constant variance. Moreover, with $S \subset \{1, \dots, 4\}$, $\#S = k$, $1 \leq k \leq 4$, then $\beta_i \neq 0$ if $i \in S$, $\beta_i = 0$ if $i \notin S$, and by recalling that $\mathbf{x}^j = (x_{1j}, \dots, x_{4j})'$ represents the j -th covariate, $j = 1, \dots, 4$, we can write $\mathbf{y} = \text{mpg}$, $\mathbf{x}^1 = \text{disp}$, $\mathbf{x}^2 = \text{hp}$, $\mathbf{x}^3 = \text{wt}$, $\mathbf{x}^4 = \text{qsec}$. The different submodels are thus given by

$$y_i = \beta_0 + \sum_{s \in S} \beta_s x_{is} + \varepsilon_i, \quad i = 1, \dots, 32. \quad (6.41)$$

Then, with (6.7), we can compute the standard AIC of the full model and each submodel. Similarly, if we consider the errors ε_i in (6.40) and (6.41) to be independent random variables with mean 0 and a finite non constant variance, we can compute the AIC by the empirical

Model ID	Covariates	AIC (Standard)	AIC (Empirical Likelihood)
1	disp	170.2094	247.5134
2	hp	181.2386	293.2860
3	wt	166.0294	251.5391
4	qsec	204.5881	386.6601
5	disp + hp	168.6186	227.8071
6	disp + wt	164.1678	247.1163
7	disp + qsec	171.8418	248.6423
8	hp + wt	156.6523	231.1510
9	hp + qsec	180.3391	264.4083
10	wt + qsec	156.7205	233.5813
11	disp + hp + wt	158.6430	229.8071
12	disp + hp + qsec	169.8525	229.8071
13	disp + wt + qsec	158.7203	235.5776
14	hp + wt + qsec	157.1426	229.9584
15	disp + hp + wt + qsec	159.0696	231.8071

Table 6.1: Comparison of standard and empirical likelihood AICs for submodels of the `mtcars` Dataset. In blue : Standard AIC's best model. In orange : empirical likelihood AIC's best model.

likelihood approach with equation (6.26) of the full model and each submodel. The results are displayed in Table 6.1.

Analysing the set of 15 linear regression models present in Table 6.1 based on our two criteria for model selection, the standard AIC and the empirical likelihood ELAIC, we can note that the model with the lowest standard AIC is $hp + wt$, with an AIC of 156.65. The model with the lowest ELAIC is $disp + hp$, with an ELAIC of 227.81.

Observe that both criteria select a model with two covariates and that hp is included among the two selected models. The standard AIC favors wt while the ELAIC favors $disp$. Moreover, notice that the values of the standard AIC are ranging from 156.65 to 204.59 whereas the ones of the ELAIC are much widely spread, i.e across 227.81 to 386.66, indicating a higher variability. These differences likely stem from the underlying assumptions of the criterion. The standard AIC is derived from a parametric likelihood while the ELAIC is built from a non parametric likelihood.

If the assumptions of a Gaussian linear regression model are met, Model 8 ($hp + wt$) should be selected. On the other hand, if the residuals exhibits violations of normality or homoscedasticity, Model 5 ($disp + hp$) might be a better choice.

To assess the validity of the assumptions underlying the Gaussian linear regression model, we examined the residuals of Model 5. A Shapiro-Wilk test for normality yielded a p-value of 0.034, indicating a statistically significant deviation from normality at the 5% level. However, a Breusch-Pagan test for homoscedasticity returned a p-value of 0.402, suggesting that the assumption of constant variance is not violated. While the non-normality of the residuals is statistically significant, the central limit theorem provides some robustness in larger samples, mitigating its practical impact. Given that homoscedasticity is preserved and asymptotic normality may still hold, Model 5 remains a preferable choice over Model 8 under the Gaussian linear model framework.

We are now interested in the following fuzzy linear regression model :

$$\widetilde{mpg}_i = \tilde{\beta}_0 \oplus (\tilde{\beta}_1 \otimes \widetilde{disp}_i) \oplus \tilde{\beta}_2 \otimes \widetilde{hp}_i \oplus (\tilde{\beta}_3 \otimes \widetilde{wt}_i) \oplus (\tilde{\beta}_4 \otimes \widetilde{qsec}_i) \oplus \varepsilon_i, \quad i = 1, \dots, 32, \quad (6.42)$$

where we applied a fuzzification procedure described below to both the covariates and the independent variable from the `mtcars` dataset.

- For each covariates, the fuzzified version is given by an interval that spans from a lower bound to an upper bound. Specifically:

$$\tilde{\mathbf{x}}^1 = \widetilde{disp} = (disp - (1 - \alpha) \cdot 0.5 \cdot disp, disp, disp + (1 - \alpha) \cdot 0.7 \cdot disp)$$

$$\tilde{\mathbf{x}}^2 = \widetilde{hp} = (hp - (1 - \alpha) \cdot 0.5 \cdot hp, hp, hp + (1 - \alpha) \cdot 0.7 \cdot hp)$$

$$\tilde{\mathbf{x}}^3 = \widetilde{wt} = (wt - (1 - \alpha) \cdot 0.5 \cdot wt, wt, wt + (1 - \alpha) \cdot 0.7 \cdot wt)$$

$$\tilde{\mathbf{x}}^4 = \widetilde{qsec} = (qsec - (1 - \alpha) \cdot 0.5 \cdot qsec, qsec, qsec + (1 - \alpha) \cdot 0.7 \cdot qsec)$$

Here, each predictor is fuzzified by subtracting and adding a portion of the original value, with the α – *cut* α controlling the level of fuzzification.

- Similarly, for the response variable y (i.e., mpg), the fuzzified version is:

$$\tilde{y} = \widetilde{\text{mpg}} = ((\text{mpg} - (1 - \alpha) \cdot 0.6 \cdot \text{mpg}), \text{mpg}, (\text{mpg} + (1 - \alpha) \cdot 0.8 \cdot \text{mpg})).$$

Setting $\alpha = 0$, with $b = 1, \dots, B$, we drew samples x_{is_b} and y_{i_b} satisfying $x_{is}^L \leq x_{is_b} \leq x_{is}^R$ and $y_i^L \leq y_{i_b} \leq y_i^R$, where x_{is}^L and x_{is}^R are the left and right parts of the covariates $\tilde{x}^1, \dots, \tilde{x}^4$ and y_i^L, y_i^R are the left and right parts of the components of $\tilde{y}$. This allows us to construct $B = 200$ linear (sub)-models of the form

$$y_{i_b} = \beta_{0_b} + \sum_{s \in S} \beta_{s_b} x_{is_b} + \varepsilon_i, \quad (6.43)$$

where $S \subset \{1, \dots, 4\}$, $\#S = k$, $1 \leq k \leq 4$. We can then compute for each model B **AIC** and **ELAIC** values AIC_b and $AIC_{el,b}$. Then, we form fuzzy **AIC** and **ELAIC** as in equations (6.32), respectively (6.39). This yields

$$\begin{aligned} \widetilde{AIC} &= (\min_b(AIC_b), AIC, \max_b(AIC_b)) \\ \widetilde{AIC}_{el} &= ((\min_b AIC_{el,b}), AIC_{el}, \max_b(AIC_{el,b})), \end{aligned}$$

where AIC and AIC_{el} are respectively the original **AIC** and **ELAIC** of the considered model (computed under the original crisp observations at the $\alpha = 1$ α -cut). The results are found in Table 6.3. If AIC and AIC_{el} cannot be computed because one only knows the lower and upper bounds x_{is}^L and x_{is}^R , respectively y_i^L and y_i^R corresponding to the $\alpha = 0$ α -cut. One can approximate AIC and AIC_{el} by the median or mean of the AIC_b and $AIC_{el,b}$ values.

It could be interesting to perform a Shapiro-Wilk and Breusch-Pagan test on each of the B models to find out how often the homoscedasticity and normality of their residuals are violated. Figures 6.1 and 6.2 give the histograms of the p -values obtained for all 15 models with respect to the Breusch-Pagan and Shapiro-Wilk tests. Additionally, Table 6.2 shows the percentages of the p -values less than 0.05 obtained by these two tests. We see that, with the rather low percentage of rejection that most of the time, the homoscedasticity and normality of the residuals of each model are preserved.

Our next objective is to classify the fuzzy **AIC** and **ELAIC** values. To achieve this, we must address the geometric nature of fuzzy **AICs**, which necessitates a method for ordering geometric shapes. This is precisely where the **PRM**, introduced in Sections 5.1 and 5.1.1, becomes relevant.

Triangle The fuzzy **AICs** and **ELAICs** derived from (6.32) and (6.39) can naturally be approximated by triangular fuzzy numbers using their left, center, and right points. Since a fuzzy **AIC** and **ELAIC** are computed for each model, we can rank the models using the triangular ranking method described in Section 5.1. The ranking is performed pairwise: for each comparison where a model has a smaller fuzzy **AIC** or **ELAIC** than another, it receives one point. The model with the highest total score is considered the best. The outcomes of this procedure, applied to the fuzzy **AICs** and **ELAICs** respectively, are

presented in Tables 6.4 and 6.5. These results show that, based on the fuzzy AIC, the best model is $\tilde{\mathbf{x}}^1 + \tilde{\mathbf{x}}^2$, which aligns with the result obtained from crisp data in Table 6.1. However, a notable divergence appears with the fuzzy ELAIC: the best model turns out to be $\tilde{\mathbf{x}}^1 + \tilde{\mathbf{x}}^2 + \tilde{\mathbf{x}}^3 + \tilde{\mathbf{x}}^4$. This model, comprising all predictors, is the most comprehensive—and the most uncertain. When there is no clear information about the covariates' impact, selecting such a full model becomes reasonable to avoid missing any potentially important predictors. Thus, under high fuzziness, the model with the most covariates tends to be favoured in this case.

Pentagon Once all B values of AIC_b and $AIC_{el,b}$ have been computed for each model, we obtain empirical distributions of the AIC and ELAIC values. Figures 6.3 and 6.4 display these distributions. Notably, the original AIC and ELAIC values for each model rarely coincide with the modes of their corresponding distributions. Additionally, we may only know the lower and upper bounds x_{ij}^L , y_i^L , x_{ij}^R and y_i^R . In this case, we do not have access to the original AIC and ELAIC values for each model and cannot infer the location of the central value of $\widetilde{AIC}$ and $\widetilde{AIC}_{el}$. This consideration motivates the use of convex pentagons as an ad-hoc approximation of the membership functions of $\widetilde{AIC}$ and $\widetilde{AIC}_{el}$, derived from the distributions of the sampled values AIC_b and $AIC_{el,b}$.

For each model defined as in (6.43), we calculate the empirical 0%, 25%, 50%, 75%, and 100% quantiles from the B resampled values of AIC_b or $AIC_{el,b}$, using equations (6.31) and (6.26). These quantiles form the x-coordinates of the pentagon. To ensure consistency across models, we fix the y-coordinates to the vector:

$$\mathbf{y} = (0, 0.5, 1, 0.5, 0).$$

It is important to emphasize that this approach does not yield fuzzy AIC or ELAIC values in the usual sense. The original AIC and ELAIC values, which previously served as the centres of the triangular $\widetilde{AIC}$ and $\widetilde{AIC}_{el}$, are no longer part of the construction. Instead, the pentagons serve as representations of the empirical distributions of AIC_b and $AIC_{el,b}$ for each model. These empirical distributions represented by pentagons are approximations of the true unknown membership function of $\widetilde{AIC}$ and $\widetilde{AIC}_{el}$.

Our objective is to use the PRM to rank these pentagon-approximated distributions. As with the triangular case, we apply the same pairwise ranking algorithm to the pentagonal representations of the AIC and ELAIC distributions. The resulting rankings, presented in Tables 6.6 and 6.7, identify the models $\tilde{\mathbf{x}}^2 + \tilde{\mathbf{x}}^3$ and $\tilde{\mathbf{x}}^1 + \tilde{\mathbf{x}}^2 + \tilde{\mathbf{x}}^3 + \tilde{\mathbf{x}}^4$ as the best performers under both pentagonal AIC and ELAIC criteria.

This finding is particularly noteworthy, as it confirms the consistency between rankings derived from fuzzy numbers and those based on the pentagon-approximated distributions. Moreover, the model with the largest number of predictors, selected under the fuzzy ELAIC approach, is again favoured when comparing pentagon-based ELAIC distributions. This supports the hypothesis that, under higher degrees of fuzziness, with respect to the ELAIC criterion, more complex models may be preferred.

Now, suppose we repeat the analysis on the same data but with reduced fuzziness. We redefine the fuzzy observation matrix $\tilde{X}$ and the fuzzy dependent vector $\tilde{y}$ as:

$$\begin{aligned} \mathbf{X}^L &= \mathbf{X} - 0.1 \cdot \mathbf{X}, \\ \mathbf{X}^R &= \mathbf{X} + 0.06 \cdot \mathbf{X}, \\[1ex] \mathbf{y}^L &= \mathbf{y} - 0.07 \cdot \mathbf{y}, \\ \mathbf{y}^R &= \mathbf{y} + 0.12 \cdot \mathbf{y}. \end{aligned} \tag{6.44}$$

The fuzzy **AICs** and **ELAICs** corresponding to this dataset with reduced fuzziness are displayed on Table 6.13. Applying the same analysis to this less fuzzified dataset, we recover the classical crisp **AIC** and **ELAIC** results, as shown in Tables 6.8, 6.9, 6.10, and 6.11, and summarized in Table 6.1.

These numerical experiments demonstrate that, in the ranking of fuzzy **AICs** and **ELAICs**, both the shape of the fuzzy numbers and the degree of fuzziness play a critical role. As the data become increasingly fuzzier, models with greater complexity, that is those including more parameters, tend to be favoured under the fuzzy **ELAIC** criterion. Moreover, while the fuzzy **AIC** values generally form triangles of similar shape across models, the fuzzy **ELAIC** values exhibit greater variability in their triangular representations. This difference may stem from the more flexible and general nature of the **ELAIC**, which can better capture nuanced distinctions between models, as opposed to the **AIC**, which relies on more restrictive assumptions.

Model	Breusch–Pagan < 0.05	Shapiro–Wilk < 0.05
$\tilde{\mathbf{x}}^2 + \tilde{\mathbf{x}}^3 + \tilde{\mathbf{x}}^4$	38%	15.5%
$\tilde{\mathbf{x}}^1 + \tilde{\mathbf{x}}^2 + \tilde{\mathbf{x}}^3 + \tilde{\mathbf{x}}^4$	37%	12%
$\tilde{\mathbf{x}}^1 + \tilde{\mathbf{x}}^2 + \tilde{\mathbf{x}}^3$	36%	12%
$\tilde{\mathbf{x}}^2 + \tilde{\mathbf{x}}^3$	35.5%	18%
$\tilde{\mathbf{x}}^1 + \tilde{\mathbf{x}}^3 + \tilde{\mathbf{x}}^4$	33%	14.5%
$\tilde{\mathbf{x}}^1 + \tilde{\mathbf{x}}^2 + \tilde{\mathbf{x}}^4$	32%	15%
$\tilde{\mathbf{x}}^1 + \tilde{\mathbf{x}}^2$	30.5%	20.5%
$\tilde{\mathbf{x}}^1 + \tilde{\mathbf{x}}^3$	29%	14%
$\tilde{\mathbf{x}}^3 + \tilde{\mathbf{x}}^4$	28.5%	23%
$\tilde{\mathbf{x}}^1 + \tilde{\mathbf{x}}^4$	26%	17.5%
$\tilde{\mathbf{x}}^3$	25.5%	26.5%
$\tilde{\mathbf{x}}^2 + \tilde{\mathbf{x}}^4$	22%	21%
$\tilde{\mathbf{x}}^1$	21%	22%
$\tilde{\mathbf{x}}^2$	18%	25.5%
$\tilde{\mathbf{x}}^4$	8.5%	44%

Table 6.2: Percentage of cases where $p < 0.05$ for Breusch–Pagan and Shapiro–Wilk tests applied to all B estimated models.

Algorithm 6.1 Empirical likelihood optimisation procedure

- 1: **Input:** Data $\{y_i, \mathbf{x}_i\}_{i=1}^n$, linear model: $y_i = m(\mathbf{x}_i, \beta) + \varepsilon_i$
- 2: **Output:** Estimated parameter vector $\hat{\beta}$
- 3: **procedure** EMPIRICAL LIKELIHOOD ESTIMATION
- 4: Initialize $\beta^{(k)}$
- 5: **repeat**
- 6: Compute $\frac{\partial}{\partial \beta} m(\mathbf{x}_i, \beta^{(k)})$ for $i = 1, \dots, n$
- 7: Find $\lambda^{(k)}$ by minimising

$$Q(\lambda^{(k)}) := - \sum_{i=1}^n \log \left\{ 1 + \lambda^{(k)} \frac{\partial m(\mathbf{x}_i; \beta^{(k)})}{\partial \beta} (y_i - m(\mathbf{x}_i; \beta^{(k)})) \right\}$$

with $\lambda^{(k)}$ subject to constraints

$$1 + \lambda^{(k)} \frac{\partial m(\mathbf{x}_i; \beta^{(k)})}{\partial \beta} (y_i - m(\mathbf{x}_i; \beta^{(k)})) \geq \frac{1}{n} \quad \forall i = 1, \dots, n.$$

- 8: Compute empirical likelihood:

$$L_n(\beta^k) = \prod_{i=1}^n \frac{1}{n} \frac{1}{1 + \lambda^{(k)} \frac{\partial m(\mathbf{x}_i; \beta^{(k)})}{\partial \beta} \{y_i - m(\mathbf{x}_i; \beta^{(k)})\}}$$

- 9: Update $\beta^{(k+1)}$ via an optimisation method (e.g. gradient descent) to maximise $L_n(\beta)$.
 - 10: **until** Convergence criterion is met
 - 11: **return** $\hat{\beta} = \beta^{(k+1)}$, $L_n(\hat{\beta})$, $\lambda^{(k+1)}$ and p_i from (6.16) computed with $\lambda^{(k+1)}$ and $\hat{\beta}$.
 - 12: **end procedure**
-

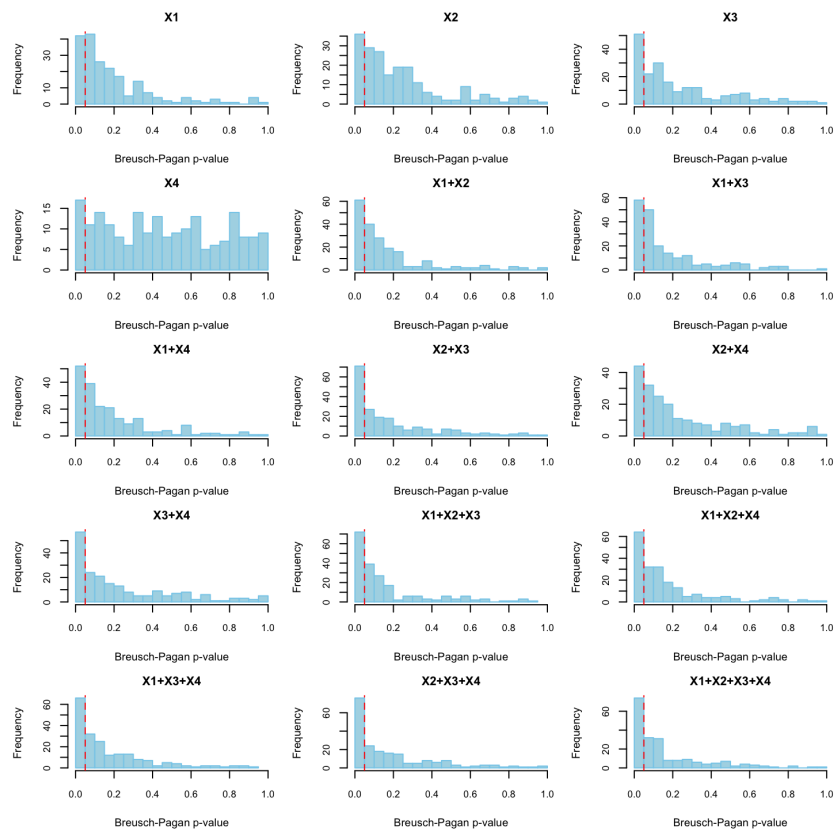


Figure 6.1: Histograms of p-values of Breusch-Pagan tests performed on all sampled model combinations.

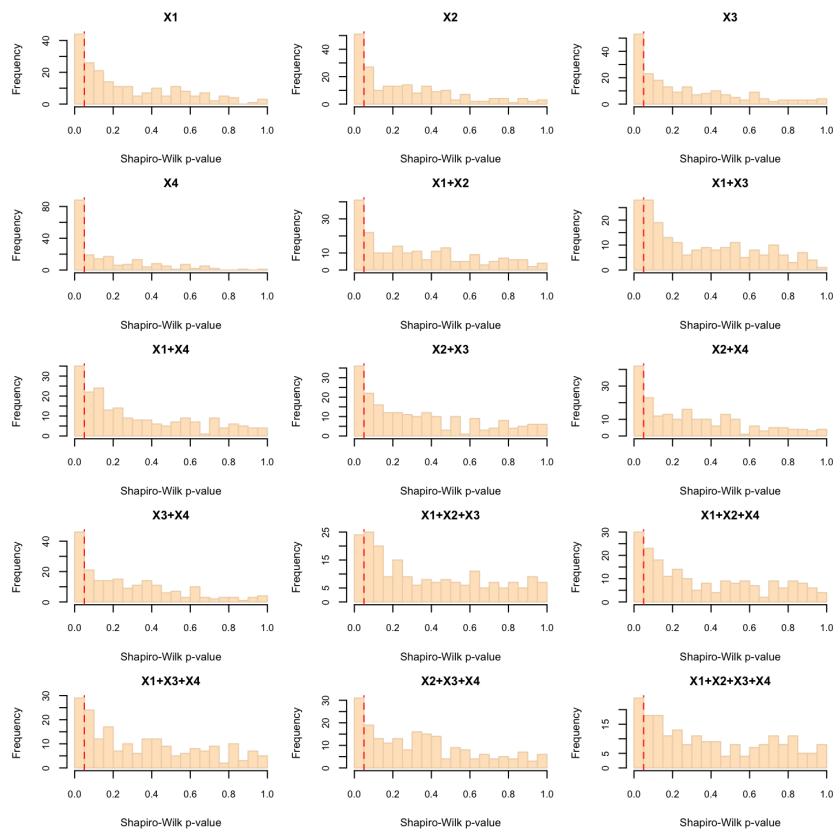


Figure 6.2: Histograms of sampled p-values of Shapiro-Wilk tests performed on all model combinations.

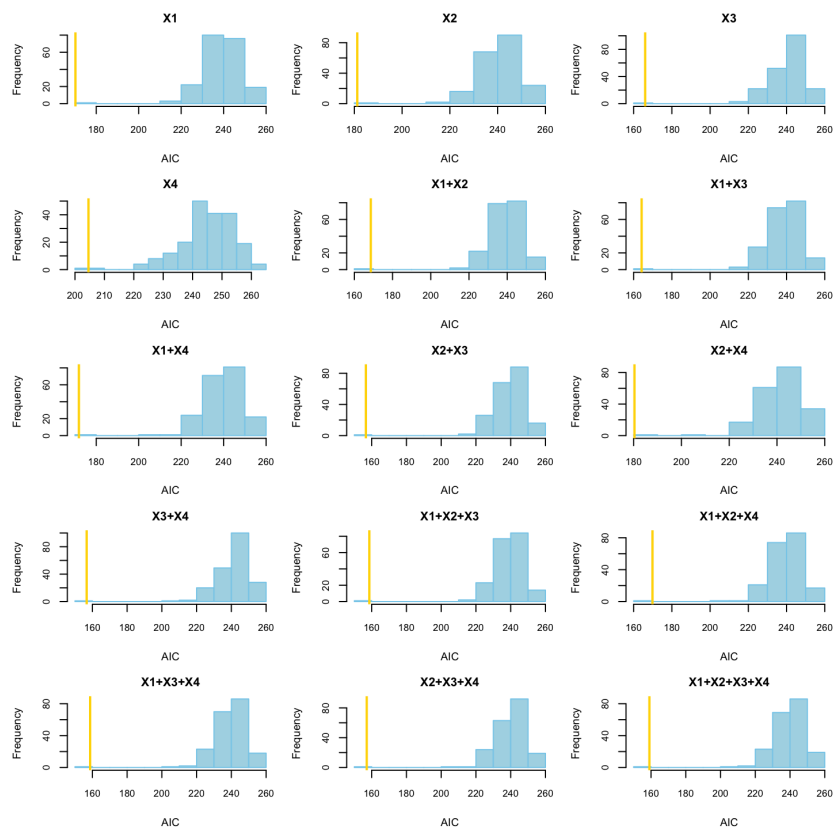


Figure 6.3: Histograms of each model constructed with $B = 200$ sampled AIC_b values.

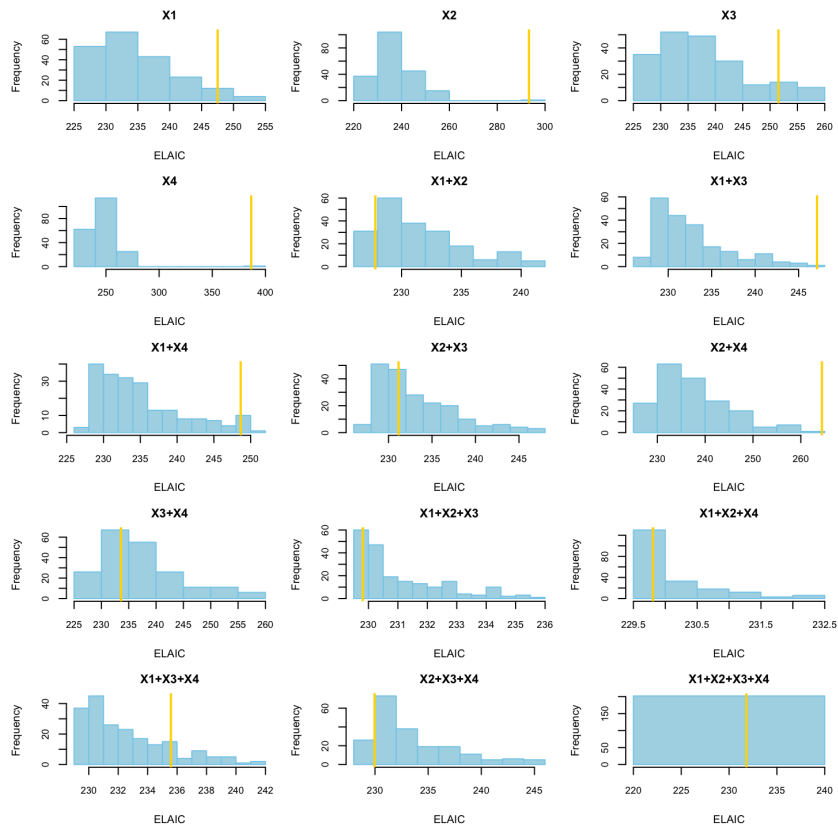


Figure 6.4: Histograms of each model constructed with $B = 200$ sampled $AIC_{el,b}$ values.

Model	AIC^L	AIC	AIC^R	AIC_{el}^L	AIC_{el}	AIC_{el}^R
$\tilde{\mathbf{X}}^1$	170.2094	170.2094	243.3680	225.8889	247.5134	293.7950
$\tilde{\mathbf{X}}^2$	181.2386	181.2386	246.9350	226.4446	293.2860	293.2860
$\tilde{\mathbf{X}}^3$	166.0294	166.0294	247.4612	226.8132	251.5391	288.4799
$\tilde{\mathbf{X}}^4$	204.5881	204.5881	254.8302	229.0727	386.6601	386.6601
$\tilde{\mathbf{X}}^1 + \tilde{\mathbf{X}}^2$	168.6186	168.6186	243.4887	227.8071	227.8071	256.0344
$\tilde{\mathbf{X}}^1 + \tilde{\mathbf{X}}^3$	164.1678	164.1678	243.3757	227.8450	247.1163	282.0051
$\tilde{\mathbf{X}}^1 + \tilde{\mathbf{X}}^4$	171.8418	171.8418	245.1728	227.8071	248.6423	291.8063
$\tilde{\mathbf{X}}^2 + \tilde{\mathbf{X}}^3$	156.6523	156.6523	243.5917	227.8077	231.1510	262.4878
$\tilde{\mathbf{X}}^2 + \tilde{\mathbf{X}}^4$	180.3391	180.3391	248.3481	228.4509	264.4083	294.3391
$\tilde{\mathbf{X}}^3 + \tilde{\mathbf{X}}^4$	156.7205	156.7205	249.4325	228.1751	233.5813	289.7971
$\tilde{\mathbf{X}}^1 + \tilde{\mathbf{X}}^2 + \tilde{\mathbf{X}}^3$	158.6430	158.6430	245.0750	229.8071	229.8071	253.2783
$\tilde{\mathbf{X}}^1 + \tilde{\mathbf{X}}^2 + \tilde{\mathbf{X}}^4$	169.8525	169.8525	245.0168	229.8071	229.8071	245.2684
$\tilde{\mathbf{X}}^1 + \tilde{\mathbf{X}}^3 + \tilde{\mathbf{X}}^4$	158.7203	158.7203	245.0160	229.8075	235.5776	282.3707
$\tilde{\mathbf{X}}^2 + \tilde{\mathbf{X}}^3 + \tilde{\mathbf{X}}^4$	157.1426	157.1426	245.4859	229.8073	229.9584	267.6082
$\tilde{\mathbf{X}}^1 + \tilde{\mathbf{X}}^2 + \tilde{\mathbf{X}}^3 + \tilde{\mathbf{X}}^4$	159.0696	159.0696	246.7087	231.8071	231.8071	231.8071

Table 6.3: Fuzzy AIC and ELAIC values for all combination of models

Model	Score	AIC ^L	AIC	AIC ^R
$\tilde{\mathbf{X}}^2 + \tilde{\mathbf{X}}^3$	14	156.7	156.7	243.6
$\tilde{\mathbf{X}}^2 + \tilde{\mathbf{X}}^3 + \tilde{\mathbf{X}}^4$	13	157.1	157.1	245.5
$\tilde{\mathbf{X}}^3 + \tilde{\mathbf{X}}^4$	12	156.7	156.7	249.4
$\tilde{\mathbf{X}}^1 + \tilde{\mathbf{X}}^2 + \tilde{\mathbf{X}}^3$	11	158.6	158.6	245.1
$\tilde{\mathbf{X}}^1 + \tilde{\mathbf{X}}^3 + \tilde{\mathbf{X}}^4$	10	158.7	158.7	245.0
$\tilde{\mathbf{X}}^1 + \tilde{\mathbf{X}}^2 + \tilde{\mathbf{X}}^3 + \tilde{\mathbf{X}}^4$	9	159.1	159.1	246.7
$\tilde{\mathbf{X}}^1 + \tilde{\mathbf{X}}^3$	8	164.2	164.2	243.4
$\tilde{\mathbf{X}}^3$	7	166.0	166.0	247.5
$\tilde{\mathbf{X}}^1 + \tilde{\mathbf{X}}^2$	6	168.6	168.6	243.5
$\tilde{\mathbf{X}}^1$	5	170.2	170.2	243.4
$\tilde{\mathbf{X}}^1 + \tilde{\mathbf{X}}^2 + \tilde{\mathbf{X}}^4$	4	169.9	169.9	245.0
$\tilde{\mathbf{X}}^1 + \tilde{\mathbf{X}}^4$	3	171.8	171.8	245.2
$\tilde{\mathbf{X}}^2 + \tilde{\mathbf{X}}^4$	2	180.3	180.3	248.3
$\tilde{\mathbf{X}}^2$	1	181.2	181.2	246.9
$\tilde{\mathbf{X}}^4$	0	204.6	204.6	254.8

Table 6.4: Model ranked by scores and fuzzy AICs values for fuzzy linear models.

Model	Score	AIC_{el}^L	AIC	AIC_{el}^R
$\tilde{\mathbf{X}}^1 + \tilde{\mathbf{X}}^2 + \tilde{\mathbf{X}}^3 + \tilde{\mathbf{X}}^4$	14	231.8	231.8	231.8
$\tilde{\mathbf{X}}^1 + \tilde{\mathbf{X}}^2 + \tilde{\mathbf{X}}^4$	13	229.8	229.8	245.3
$\tilde{\mathbf{X}}^1 + \tilde{\mathbf{X}}^2$	12	227.8	227.8	256.0
$\tilde{\mathbf{X}}^1 + \tilde{\mathbf{X}}^2 + \tilde{\mathbf{X}}^3$	11	229.8	229.8	253.3
$\tilde{\mathbf{X}}^2 + \tilde{\mathbf{X}}^3$	10	227.8	231.2	262.5
$\tilde{\mathbf{X}}^2 + \tilde{\mathbf{X}}^3 + \tilde{\mathbf{X}}^4$	9	229.8	230.0	267.6
$\tilde{\mathbf{X}}^1 + \tilde{\mathbf{X}}^3 + \tilde{\mathbf{X}}^4$	8	229.8	235.6	282.4
$\tilde{\mathbf{X}}^3 + \tilde{\mathbf{X}}^4$	7	228.2	233.6	289.8
$\tilde{\mathbf{X}}^1 + \tilde{\mathbf{X}}^3$	6	227.8	247.1	282.0
$\tilde{\mathbf{X}}^1$	5	225.9	247.5	293.8
$\tilde{\mathbf{X}}^1 + \tilde{\mathbf{X}}^4$	4	227.8	248.6	291.8
$\tilde{\mathbf{X}}^3$	3	226.8	251.5	288.5
$\tilde{\mathbf{X}}^2 + \tilde{\mathbf{X}}^4$	2	228.5	264.4	294.3
$\tilde{\mathbf{X}}^2$	1	226.4	293.3	293.3
$\tilde{\mathbf{X}}^4$	0	229.1	386.7	386.7

Table 6.5: Models ranked by their fuzzy triangular ELAIC and their scores.

Model	Score	Q0	Q25	Q50	Q75	Q100
$\tilde{\mathbf{X}}^2 + \tilde{\mathbf{X}}^3$	14	156.7	220.3	226.3	232.4	243.6
$\tilde{\mathbf{X}}^2 + \tilde{\mathbf{X}}^3 + \tilde{\mathbf{X}}^4$	13	157.1	221.2	227.1	232.6	245.5
$\tilde{\mathbf{X}}^1 + \tilde{\mathbf{X}}^2 + \tilde{\mathbf{X}}^3$	12	158.6	220.5	224.9	231.3	245.1
$\tilde{\mathbf{X}}^1 + \tilde{\mathbf{X}}^3 + \tilde{\mathbf{X}}^4$	11	158.7	221.9	226.8	232.2	245.0
$\tilde{\mathbf{X}}^1 + \tilde{\mathbf{X}}^2 + \tilde{\mathbf{X}}^3 + \tilde{\mathbf{X}}^4$	10	159.1	220.8	226.2	232.1	246.7
$\tilde{\mathbf{X}}^3 + \tilde{\mathbf{X}}^4$	9	156.7	223.4	229.5	234.8	249.4
$\tilde{\mathbf{X}}^1 + \tilde{\mathbf{X}}^3$	8	164.2	221.0	225.9	231.2	243.4
$\tilde{\mathbf{X}}^1 + \tilde{\mathbf{X}}^2$	7	168.6	220.4	225.3	231.3	243.5
$\tilde{\mathbf{X}}^1$	6	170.2	221.0	226.2	230.9	243.4
$\tilde{\mathbf{X}}^3$	5	166.0	222.8	228.6	233.8	247.5
$\tilde{\mathbf{X}}^1 + \tilde{\mathbf{X}}^2 + \tilde{\mathbf{X}}^4$	4	169.9	221.2	226.4	232.3	245.0
$\tilde{\mathbf{X}}^1 + \tilde{\mathbf{X}}^4$	3	171.8	221.8	227.2	232.0	245.2
$\tilde{\mathbf{X}}^2$	2	181.2	221.9	227.9	233.2	246.9
$\tilde{\mathbf{X}}^2 + \tilde{\mathbf{X}}^4$	1	180.3	222.7	228.9	234.5	248.3
$\tilde{\mathbf{X}}^4$	0	204.6	226.9	233.8	238.9	254.8

Table 6.6: Models ranked by their fuzzy pentagonal AIC and their scores.

Model	Score	Q0	Q25	Q50	Q75	Q100
$\tilde{\mathbf{X}}^1 + \tilde{\mathbf{X}}^2 + \tilde{\mathbf{X}}^3 + \tilde{\mathbf{X}}^4$	14	231.8	231.8	231.8	231.8	231.8
$\tilde{\mathbf{X}}^1 + \tilde{\mathbf{X}}^2 + \tilde{\mathbf{X}}^4$	13	229.8	229.8	229.9	231.1	245.3
$\tilde{\mathbf{X}}^1 + \tilde{\mathbf{X}}^2$	12	227.8	228.8	230.7	234.1	256.0
$\tilde{\mathbf{X}}^1 + \tilde{\mathbf{X}}^2 + \tilde{\mathbf{X}}^3$	11	229.8	229.9	230.4	231.9	253.3
$\tilde{\mathbf{X}}^2 + \tilde{\mathbf{X}}^3$	10	227.8	230.1	232.9	236.9	262.5
$\tilde{\mathbf{X}}^2 + \tilde{\mathbf{X}}^3 + \tilde{\mathbf{X}}^4$	9	229.8	230.6	232.8	237.3	267.6
$\tilde{\mathbf{X}}^1 + \tilde{\mathbf{X}}^3 + \tilde{\mathbf{X}}^4$	8	229.8	230.3	232.2	235.7	282.4
$\tilde{\mathbf{X}}^3 + \tilde{\mathbf{X}}^4$	7	228.2	233.6	238.7	244.9	289.8
$\tilde{\mathbf{X}}^1 + \tilde{\mathbf{X}}^3$	6	227.8	229.6	232.0	236.8	282.0
$\tilde{\mathbf{X}}^1$	5	225.9	230.4	234.1	240.2	293.8
$\tilde{\mathbf{X}}^1 + \tilde{\mathbf{X}}^4$	4	227.8	231.0	234.8	240.2	291.8
$\tilde{\mathbf{X}}^3$	3	226.8	233.0	237.9	245.3	288.5
$\tilde{\mathbf{X}}^2 + \tilde{\mathbf{X}}^4$	2	228.5	233.9	238.2	243.8	294.3
$\tilde{\mathbf{X}}^2$	1	226.4	232.9	237.6	243.5	293.3
$\tilde{\mathbf{X}}^4$	0	229.1	241.2	250.3	261.3	386.7

Table 6.7: Models ranked by their fuzzy pentagonal ELAIC and their scores.

Model	Score	AIC_{el}^L	AIC	AIC_{el}^R
$\tilde{\mathbf{X}}^2 + \tilde{\mathbf{X}}^3$	14	144.2	156.7	178.8
$\tilde{\mathbf{X}}^2 + \tilde{\mathbf{X}}^3 + \tilde{\mathbf{X}}^4$	13	144.4	157.1	180.8
$\tilde{\mathbf{X}}^1 + \tilde{\mathbf{X}}^2 + \tilde{\mathbf{X}}^3$	12	146.0	158.6	180.5
$\tilde{\mathbf{X}}^3 + \tilde{\mathbf{X}}^4$	11	148.3	156.7	180.5
$\tilde{\mathbf{X}}^1 + \tilde{\mathbf{X}}^2 + \tilde{\mathbf{X}}^3 + \tilde{\mathbf{X}}^4$	10	146.4	159.1	182.5
$\tilde{\mathbf{X}}^1 + \tilde{\mathbf{X}}^3 + \tilde{\mathbf{X}}^4$	9	149.2	158.7	182.0
$\tilde{\mathbf{X}}^1 + \tilde{\mathbf{X}}^3$	8	154.5	164.2	181.7
$\tilde{\mathbf{X}}^3$	7	158.4	166.0	183.8
$\tilde{\mathbf{X}}^1 + \tilde{\mathbf{X}}^2$	6	161.1	168.6	186.2
$\tilde{\mathbf{X}}^1 + \tilde{\mathbf{X}}^2 + \tilde{\mathbf{X}}^4$	5	160.4	169.9	187.8
$\tilde{\mathbf{X}}^1$	4	163.7	170.2	186.6
$\tilde{\mathbf{X}}^1 + \tilde{\mathbf{X}}^4$	3	165.2	171.8	188.6
$\tilde{\mathbf{X}}^2 + \tilde{\mathbf{X}}^4$	2	167.7	180.3	195.9
$\tilde{\mathbf{X}}^2$	1	172.5	181.2	195.9
$\tilde{\mathbf{X}}^4$	0	198.3	204.6	215.0

Table 6.8: Models ranked by their fuzzy triangular AIC and scores, computed with Data from (6.44).

Model	Score	Q0	Q25	Q50	Q75	Q100
$\tilde{\mathbf{X}}^2 + \tilde{\mathbf{X}}^3$	14	144.2	161.3	165.4	168.7	178.8
$\tilde{\mathbf{X}}^2 + \tilde{\mathbf{X}}^3 + \tilde{\mathbf{X}}^4$	13	144.4	162.1	165.7	169.3	180.8
$\tilde{\mathbf{X}}^1 + \tilde{\mathbf{X}}^2 + \tilde{\mathbf{X}}^3$	12	146.0	162.6	167.2	170.5	180.5
$\tilde{\mathbf{X}}^1 + \tilde{\mathbf{X}}^2 + \tilde{\mathbf{X}}^3 + \tilde{\mathbf{X}}^4$	11	146.4	163.7	167.4	170.7	182.5
$\tilde{\mathbf{X}}^3 + \tilde{\mathbf{X}}^4$	10	148.3	163.4	167.9	171.6	180.5
$\tilde{\mathbf{X}}^1 + \tilde{\mathbf{X}}^3 + \tilde{\mathbf{X}}^4$	9	149.2	164.6	168.7	172.1	182.0
$\tilde{\mathbf{X}}^1 + \tilde{\mathbf{X}}^3$	8	154.5	167.5	171.1	174.7	181.7
$\tilde{\mathbf{X}}^3$	7	158.4	169.9	174.1	176.9	183.8
$\tilde{\mathbf{X}}^1 + \tilde{\mathbf{X}}^2$	6	161.1	170.7	175.1	177.8	186.2
$\tilde{\mathbf{X}}^1 + \tilde{\mathbf{X}}^2 + \tilde{\mathbf{X}}^4$	5	160.4	172.0	175.7	179.1	187.8
$\tilde{\mathbf{X}}^1$	4	163.7	173.0	175.8	179.7	186.6
$\tilde{\mathbf{X}}^1 + \tilde{\mathbf{X}}^4$	3	165.2	174.0	177.3	180.9	188.6
$\tilde{\mathbf{X}}^2 + \tilde{\mathbf{X}}^4$	2	167.7	183.3	185.5	188.7	195.9
$\tilde{\mathbf{X}}^2$	1	172.5	183.3	186.1	188.6	195.9
$\tilde{\mathbf{X}}^4$	0	198.3	206.1	208.5	210.5	215.0

Table 6.9: Models ranked by their fuzzy pentagonal AIC and their scores with Data from (6.44).

Model	Score	AIC_{el}^L	AIC_{el}	AIC_{el}^R
$\tilde{X}^1 + \tilde{X}^2$	14	227.8	227.8	230.8
$\tilde{X}^1 + \tilde{X}^2 + \tilde{X}^4$	13	229.8	229.8	229.8
$\tilde{X}^2 + \tilde{X}^3 + \tilde{X}^4$	12	229.8	230.0	231.6
$\tilde{X}^1 + \tilde{X}^2 + \tilde{X}^3$	11	229.8	229.8	233.2
$\tilde{X}^2 + \tilde{X}^3$	10	227.8	231.2	236.3
$\tilde{X}^1 + \tilde{X}^2 + \tilde{X}^3 + \tilde{X}^4$	9	231.8	231.8	231.8
$\tilde{X}^3 + \tilde{X}^4$	8	228.1	233.6	255.3
$\tilde{X}^1 + \tilde{X}^3 + \tilde{X}^4$	7	229.8	235.6	255.4
$\tilde{X}^1 + \tilde{X}^3$	6	232.5	247.1	270.8
$\tilde{X}^1$	5	234.7	247.5	274.5
$\tilde{X}^1 + \tilde{X}^4$	4	236.1	248.6	275.4
$\tilde{X}^3$	3	234.1	251.5	278.3
$\tilde{X}^2 + \tilde{X}^4$	2	242.8	264.4	317.4
$\tilde{X}^2$	1	249.0	293.3	345.6
$\tilde{X}^4$	0	298.9	386.7	649.7

Table 6.10: Models ranked by their fuzzy triangular ELAIC and their scores with Data from (6.44).

Model	Score	Q0	Q25	Q50	Q75	Q100
$\tilde{\mathbf{X}}^1 + \tilde{\mathbf{X}}^2$	14	227.8	227.8	227.8	227.8	230.8
$\tilde{\mathbf{X}}^1 + \tilde{\mathbf{X}}^2 + \tilde{\mathbf{X}}^4$	13	229.8	229.8	229.8	229.8	229.8
$\tilde{\mathbf{X}}^2 + \tilde{\mathbf{X}}^3 + \tilde{\mathbf{X}}^4$	12	229.8	229.9	230.0	230.4	231.6
$\tilde{\mathbf{X}}^2 + \tilde{\mathbf{X}}^3$	11	227.8	228.6	229.7	231.6	236.3
$\tilde{\mathbf{X}}^1 + \tilde{\mathbf{X}}^2 + \tilde{\mathbf{X}}^3$	10	229.8	229.8	229.9	230.7	233.2
$\tilde{\mathbf{X}}^1 + \tilde{\mathbf{X}}^2 + \tilde{\mathbf{X}}^3 + \tilde{\mathbf{X}}^4$	9	231.8	231.8	231.8	231.8	231.8
$\tilde{\mathbf{X}}^3 + \tilde{\mathbf{X}}^4$	8	228.1	232.5	236.5	240.2	255.3
$\tilde{\mathbf{X}}^1 + \tilde{\mathbf{X}}^3 + \tilde{\mathbf{X}}^4$	7	229.8	233.6	236.7	241.4	255.4
$\tilde{\mathbf{X}}^1 + \tilde{\mathbf{X}}^3$	6	232.5	241.1	246.5	253.5	270.8
$\tilde{\mathbf{X}}^1$	5	234.7	244.6	250.6	257.4	274.5
$\tilde{\mathbf{X}}^1 + \tilde{\mathbf{X}}^4$	4	236.1	245.3	250.6	257.5	275.4
$\tilde{\mathbf{X}}^3$	3	234.1	243.5	250.2	257.5	278.3
$\tilde{\mathbf{X}}^2 + \tilde{\mathbf{X}}^4$	2	242.8	259.1	267.6	279.7	317.4
$\tilde{\mathbf{X}}^2$	1	249.0	265.8	277.7	292.0	345.6
$\tilde{\mathbf{X}}^4$	0	298.9	337.1	363.3	396.7	649.7

Table 6.11: Models ranked by fuzzy pentagonal ELAIC values with their associated scores with Data from (6.44).

Model	Score	Q0	Q25	Q50	Q75	Q100
$\tilde{X}^3 + \tilde{X}^4$	14	156.7	293.5	299.5	304.2	315.5
$\tilde{X}^2 + \tilde{X}^3$	13	156.7	294.3	299.8	305.1	316.3
$\tilde{X}^2 + \tilde{X}^3 + \tilde{X}^4$	12	157.1	294.1	300.4	305.6	315.9
$\tilde{X}^1 + \tilde{X}^2 + \tilde{X}^3$	11	158.6	295.2	301.1	305.9	316.2
$\tilde{X}^1 + \tilde{X}^3 + \tilde{X}^4$	10	158.7	294.5	300.6	305.1	316.7
$\tilde{X}^1 + \tilde{X}^2 + \tilde{X}^3 + \tilde{X}^4$	9	159.1	295.2	301.1	306.0	317.0
$\tilde{X}^1 + \tilde{X}^3$	8	164.2	294.6	300.1	304.8	316.3
$\tilde{X}^3$	7	166.0	293.8	298.7	304.2	315.3
$\tilde{X}^1 + \tilde{X}^2$	6	168.6	294.5	299.9	304.7	315.0
$\tilde{X}^1$	5	170.2	293.5	299.5	303.9	314.6
$\tilde{X}^1 + \tilde{X}^2 + \tilde{X}^4$	4	169.9	294.3	300.3	304.9	315.2
$\tilde{X}^1 + \tilde{X}^4$	3	171.8	293.5	299.5	303.8	314.7
$\tilde{X}^2$	2	181.2	293.7	298.4	303.7	314.5
$\tilde{X}^2 + \tilde{X}^4$	1	180.3	293.5	299.3	304.2	315.4
$\tilde{X}^4$	0	204.6	292.8	298.8	303.4	313.7

Table 6.12: Fuzzy pentagonal AIC values ranked by score from highly fuzzy data.

Model	AIC^L	AIC	AIC^R	AIC_{el}^L	AIC_{el}	AIC_{el}^R
$\tilde{\mathbf{X}}^1$	163.7	170.2	186.6	234.7	247.5	274.5
$\tilde{\mathbf{X}}^2$	172.5	181.2	195.9	249.0	293.3	345.6
$\tilde{\mathbf{X}}^3$	158.4	166.0	183.8	234.1	251.5	278.3
$\tilde{\mathbf{X}}^4$	198.3	204.6	215.0	298.9	386.7	649.7
$\tilde{\mathbf{X}}^1 + \tilde{\mathbf{X}}^2$	161.1	168.6	186.2	227.8	227.8	230.8
$\tilde{\mathbf{X}}^1 + \tilde{\mathbf{X}}^3$	154.5	164.2	181.7	232.5	247.1	270.8
$\tilde{\mathbf{X}}^1 + \tilde{\mathbf{X}}^4$	165.2	171.8	188.6	236.1	248.6	275.4
$\tilde{\mathbf{X}}^2 + \tilde{\mathbf{X}}^3$	144.2	156.7	178.8	227.8	231.2	236.3
$\tilde{\mathbf{X}}^2 + \tilde{\mathbf{X}}^4$	167.7	180.3	195.9	242.8	264.4	317.4
$\tilde{\mathbf{X}}^3 + \tilde{\mathbf{X}}^4$	148.3	156.7	180.5	228.1	233.6	255.3
$\tilde{\mathbf{X}}^1 + \tilde{\mathbf{X}}^2 + \tilde{\mathbf{X}}^3$	146.0	158.6	180.5	229.8	229.8	233.2
$\tilde{\mathbf{X}}^1 + \tilde{\mathbf{X}}^2 + \tilde{\mathbf{X}}^4$	160.4	169.9	187.8	229.8	229.8	229.8
$\tilde{\mathbf{X}}^1 + \tilde{\mathbf{X}}^3 + \tilde{\mathbf{X}}^4$	149.2	158.7	182.0	229.8	235.6	255.4
$\tilde{\mathbf{X}}^2 + \tilde{\mathbf{X}}^3 + \tilde{\mathbf{X}}^4$	144.4	157.1	180.8	229.8	230.0	231.6
$\tilde{\mathbf{X}}^1 + \tilde{\mathbf{X}}^2 + \tilde{\mathbf{X}}^3 + \tilde{\mathbf{X}}^4$	146.4	159.1	182.5	231.8	231.8	231.8

Table 6.13: AIC and ELAIC values for each model combination for fuzzy data from (6.44).

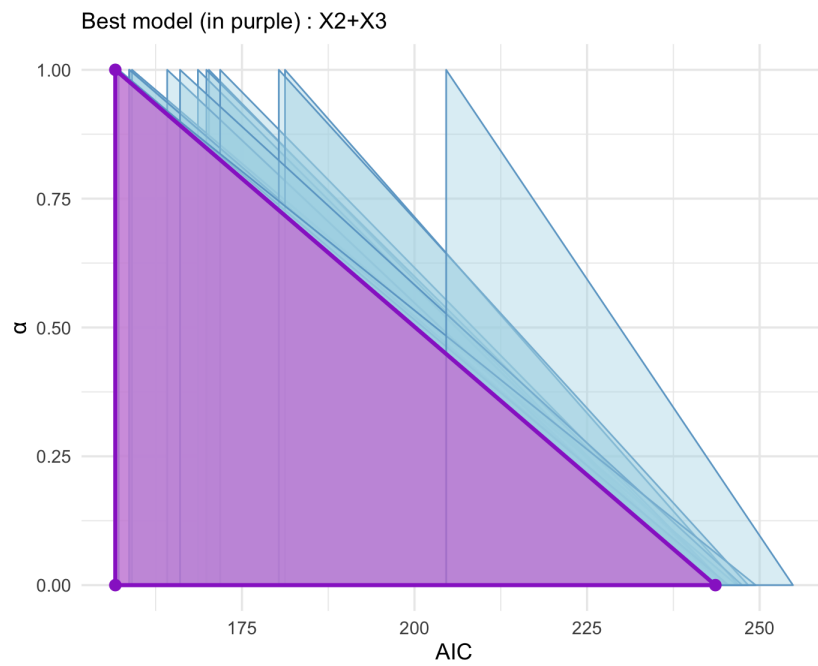


Figure 6.5: Visual depiction of ranked fuzzy triangular AICs from 6.4.

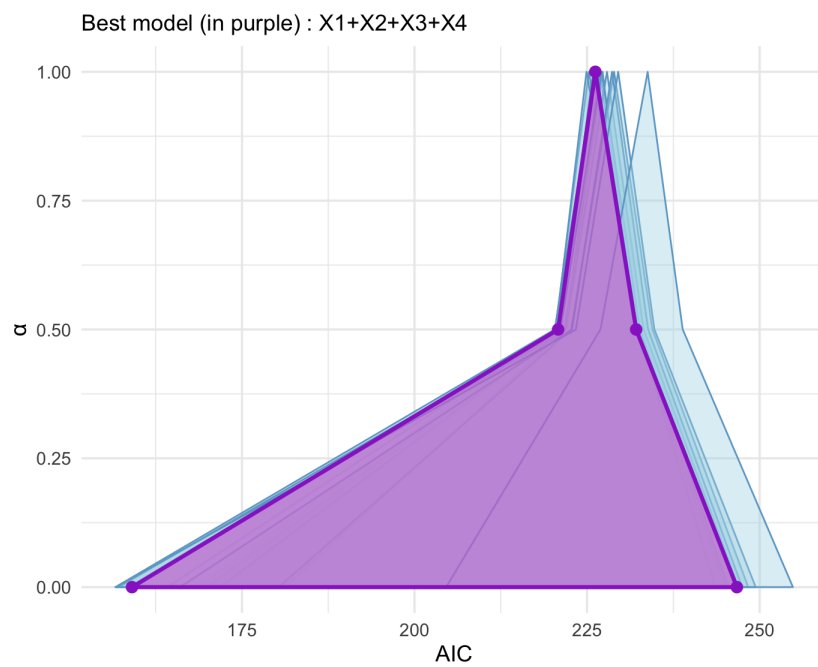


Figure 6.6: Visual depiction of ranked fuzzy pentagonal AICs from 6.6.

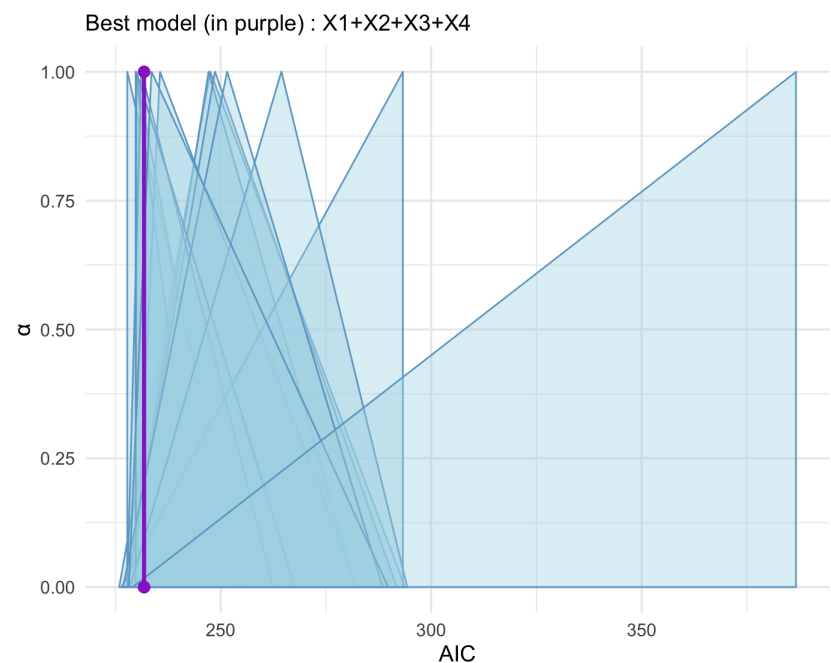


Figure 6.7: Visual depiction of ranked fuzzy triangular ELAICs from 6.5.

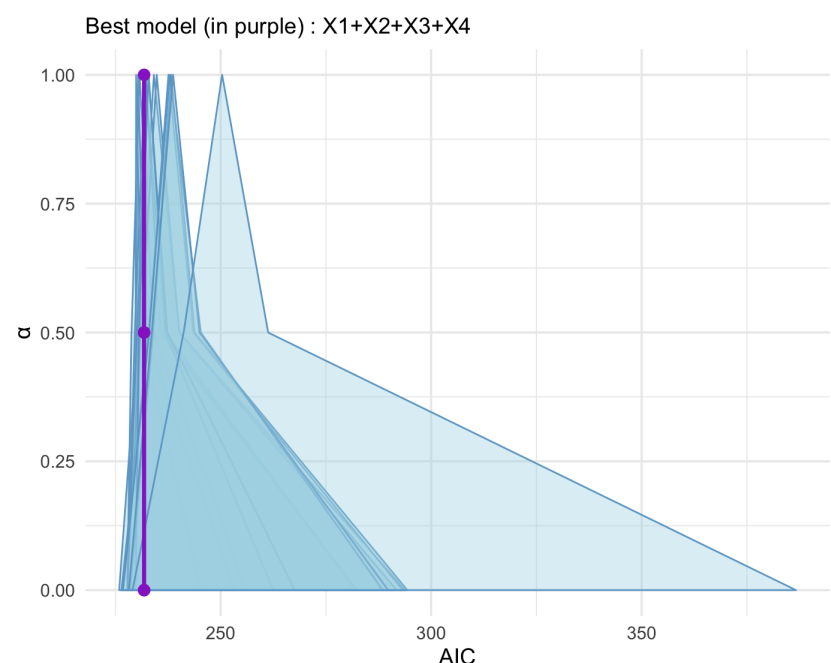


Figure 6.8: Visual depiction of ranked fuzzy pentagonal ELAICs from 6.7.

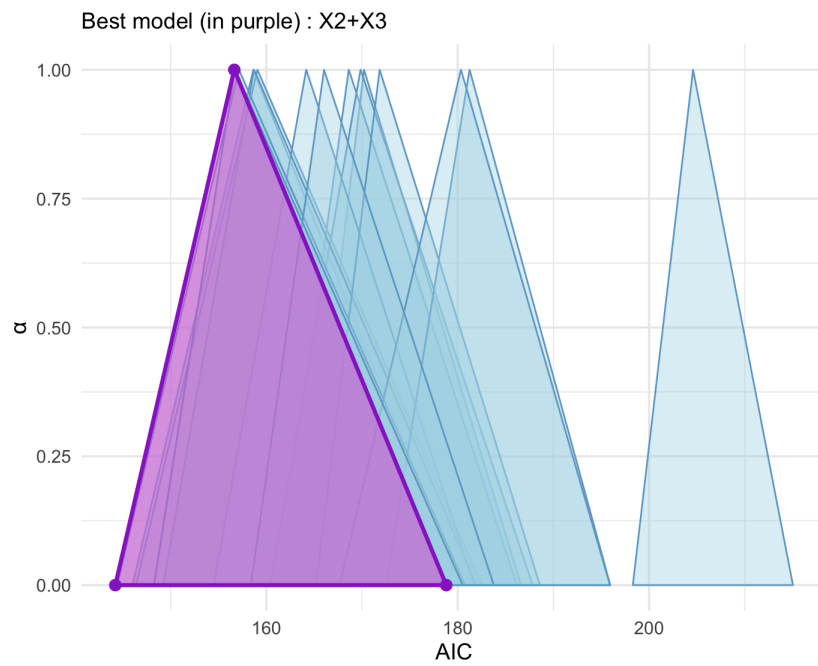


Figure 6.9: Visual depiction of ranked fuzzy triangular AICs from 6.8.

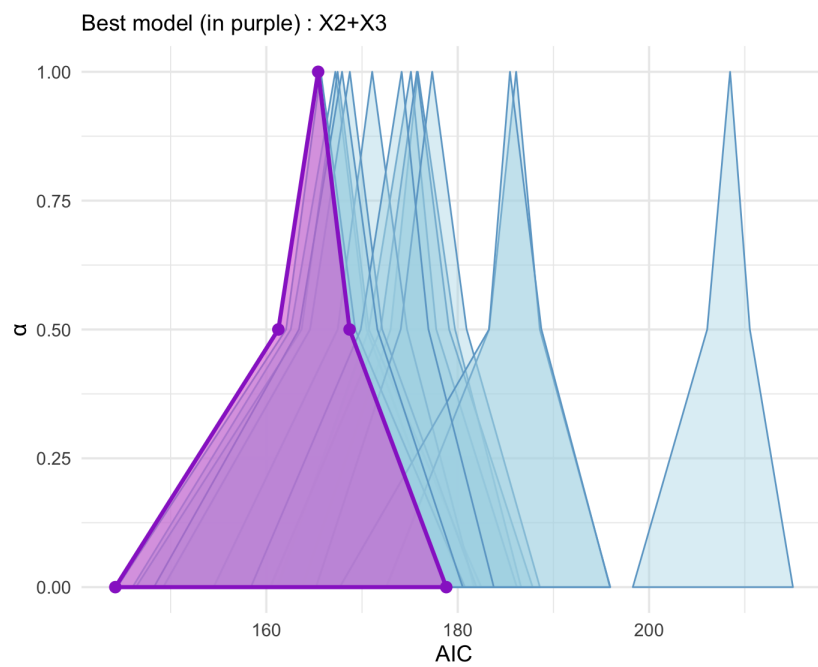


Figure 6.10: Visual depiction of ranked fuzzy pentagonal AICs from 6.9.

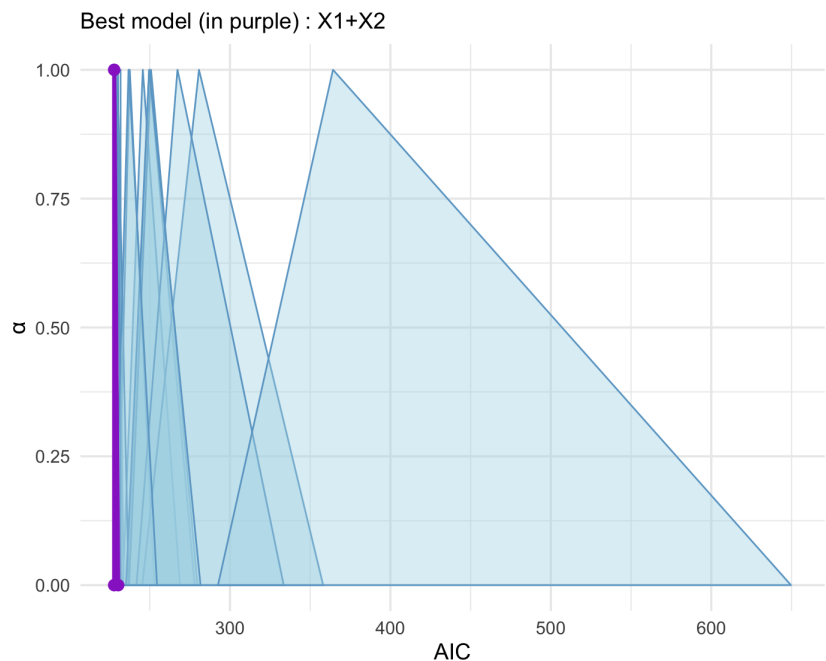


Figure 6.11: Visual depiction of ranked fuzzy triangular ELAICs from 6.10.

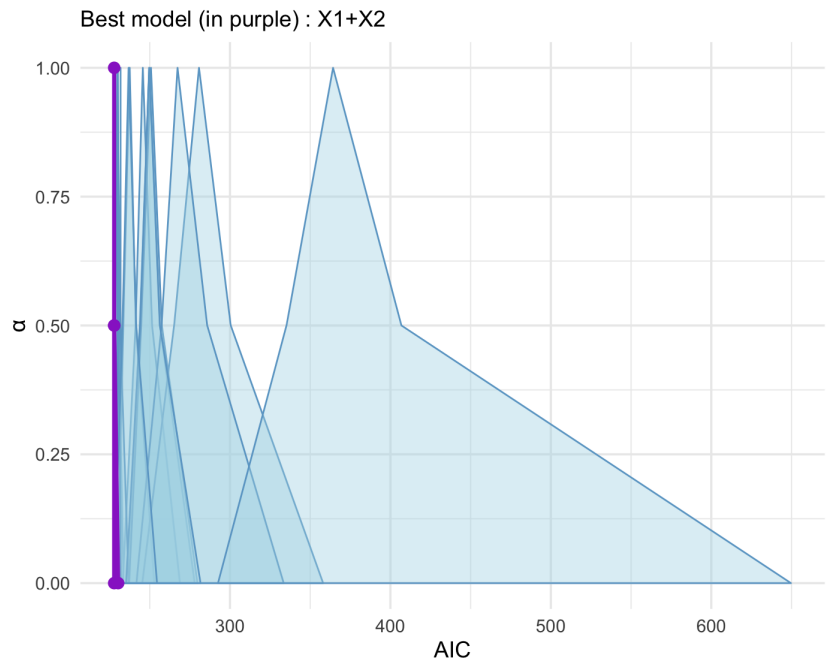


Figure 6.12: Visual depiction of ranked fuzzy pentagonal ELAICs from 6.11.

Conclusion

The theory of fuzzy sets, introduced by Zadeh in 1965, provides a formal framework to handle imprecision, vagueness, and incomplete information, phenomena that frequently arise in real-world data. This thesis contributes to the field of fuzzy statistical modelling by exploring and extending methods of fuzzy linear regression, statistical inference under fuzziness assumption of observations, ranking fuzzy numbers and fuzzy models selection.

In the beginning, we lay the theoretical foundation necessary for all subsequent developments. We revisit core concepts of fuzzy logic, such as fuzzy sets, fuzzy numbers, and their key properties. Logical operations on fuzzy sets are defined through t-norms and t-conorms, and the arithmetic of fuzzy numbers is introduced via Zadeh's extension principle. These formal definitions and conventions serve as a rigorous basis for the upcoming material.

One is fuzzy regression analysis. We first recall two foundational approaches: the fuzzy least squares method by Diamond (1988), which minimises the squared distance between fuzzy numbers using a carefully defined metric; and the possibilistic regression method of Tanaka, Hayashi, and Watada (1989), which minimizes the spread of fuzzy output variables under constraint-based formulations. These two philosophies were later unified by Tanaka and Lee (1997), though limited to crisp covariates.

We extend these models in two main directions. First, we study fuzzy least squares regression to account for both fuzzy dependent and independent variables as demonstrated by Rosset and Donzé (2023). Second, we explored orthogonality in the fuzzy context, introducing two orthogonal regression procedures: one rooted in Itoh (2017)'s definition of orthogonality for fuzzy numbers, and the other, more general, derived via Zadeh's extension principle, providing a more accurate approximation of fuzzy orthogonality, as outlined by Rosset and Donzé (2024c). The motivation for introducing orthogonality was to demonstrate that, under orthogonal fuzzy observations, fuzzy regression parameters can be computed independently, i.e., by regressing each parameter over a single covariate while yielding the same results as when regressed jointly over all covariates.

Moving beyond parameter estimation, we also study fuzzy inference, an area where classical methods must be rethought due to the fuzziness of the parameters themselves. We introduce the concept of fuzzy random variables and define empirical fuzzy distributions, allowing us to construct fuzzy confidence intervals and perform fuzzy hypothesis testing. We also extend the notion of p -values into the fuzzy setting, providing tools to interpret statistical evidence

under epistemic uncertainty. A numerical illustration highlights the operational equivalence and interpretational differences between fuzzy test procedures and fuzzy p -value analyses.

Some effort was also placed on ranking fuzzy numbers, which underpins both decision-making and model comparison. We examine existing methods, such as centroid-based and distance-based rankings and highlight their limitations. To overcome these shortcomings, we introduce a novel ranking procedure based on directional non-overlapping regions: the **PRM**. This bilateral comparison method captures both shape and location asymmetries in a geometrically intuitive way. While initially tailored to triangular fuzzy numbers, **PRM** extends naturally to generalised fuzzy numbers, setting it apart from methods restricted to specific shapes.

Finally, we addressed the problem of model selection in fuzzy settings. Building on classical information criteria such as **AIC**, we propose fuzzy versions of the **AIC** and its empirical likelihood analogue **ELAIC**. The absence of a clear distributional assumption in fuzzy models renders standard likelihood-based approaches hard to justify. To bypass this, we rely on empirical likelihood Owen (1988, 2001b) and extend it using Zadeh's principle, enabling the construction of fuzzy likelihoods. By resampling from fuzzy inputs, we generate fuzzy **ELAIC** values and rank candidate models using the **PRM** method introduced earlier. These criteria allow for model selection when crisp comparisons are inappropriate or misleading due to data uncertainty.

This thesis opens several avenues for future research. First, it would be valuable to investigate whether an optimal number of resamples exists for the sampling procedures and the methods that rely on them, such as the *k-th fuzziest estimate by OLS* and *empirical likelihood for fuzzy regression models*. Identifying such an optimal resampling strategy could significantly improve the scalability of these approaches for larger datasets. Second, the framework for fuzzy inference could be broadened to accommodate more complex test statistics. Third, it would be valuable to further investigate the discriminative power of the **PRM**, particularly through comparative studies with other ranking methods such as the signed distance, first introduced by Dubois and Prade (1978). Finally, a natural direction for future work lies in extending fuzzy model selection criteria to fuzzy generalized linear models and fuzzy non-linear models, and in exploring their practical application in real-world decision-making settings.

In summary, this thesis illustrates the versatility and power of fuzzy methods in statistical modelling, inference, ranking and selection. From foundational theory to new methodologies, we present an integrated set of tools that rigorously address the presence of fuzziness in both data and models. We hope that these contributions will serve as a solid foundation for continued research in fuzzy statistics and its applications in vague environments, highlighting the importance of embracing uncertainty not as a limitation, but as a fundamental characteristic of complex, real-world data.

Appendix A

In Appendix A, we provide a summary of the construction of the L^2 -metric as defined by Puri and Dan A Ralescu (1986), and subsequently utilised by Diamond (1988). In addition, regarding Tanaka and Lee (1997) and Rosset and Donzé (2023)'s approaches, the use of the quadratic programming to obtain estimates for a linear fuzzy regression problem is presented.

A.1 An L^2 -metric

The fuzzy least squares regression method proposed by Diamond (1988) is based on an L^2 -metric used to compute the distance between fuzzy numbers. This distance plays a crucial role since it allows the calculation of the sum of squares appearing in fuzzy linear least squares regression.

Puri and Dan A Ralescu (1986) defined such L^2 -metric on the space $\mathcal{F}_c(\mathbb{R}^n)$ by embedding it in a Banach space. Their idea was the following. Since the α -cuts of the elements of $\mathcal{F}_c(\mathbb{R}^n)$ are elements of the space $\mathcal{K}^n$, it can be shown that one can embed these elements in a Banach space via the support function

$$S : S^{n-1} \times \mathcal{K}^n \rightarrow \mathbb{R}^+$$

defined by

$$S(\lambda, \tilde{x}^{*,\alpha}) := \sup \left\{ \langle \lambda, x \rangle, \lambda \in S^{n-1}, x \in \tilde{x}^{*,\alpha} \right\}. \quad (\text{A.45})$$

where $S^{n-1} \subseteq \mathbb{R}^n$ is the $n - 1$ sphere, $S^{n-1} = \{\mathbf{u} \in \mathbb{R}^n \mid \|\mathbf{u}\| = 1\}$, $\lambda \in S^{n-1}$, $\tilde{x}^{*,\alpha} \in \mathcal{K}^n$ and $\langle \cdot, \cdot \rangle$ is the euclidean inner product.

This construction allows us to write the following L^2 -metric

$$D^2 : \mathcal{K}^n \times \mathcal{K}^n \rightarrow \mathbb{R}^+ \quad (\text{A.46})$$

$$A \times B \mapsto \int_{S^{n-1}} (S(\lambda, A) - S(\lambda, B))^2 d\lambda. \quad (\text{A.47})$$

The distance D^2 between two fuzzy numbers $\tilde{x}, \tilde{y} \in \mathcal{F}_c(\mathbb{R})$ at a given α -cut can be calculated and is given by

$$D^2(\tilde{x}^\alpha, \tilde{y}^\alpha) = (x^{R,\alpha} - y^{R,\alpha})^2 - (x^{L,\alpha} - y^{L,\alpha})^2. \quad (\text{A.48})$$

Based on Puri and Dan A Ralescu (1986)'s D^2 metric, Diamond (1988) then defined another metric,

$$d^2(\tilde{x}, \tilde{y}) = D^2(\text{Supp}(\tilde{x}), \text{Supp}(\tilde{y})) + (m(\tilde{x}) - m(\tilde{y}))^2 \quad (\text{A.49})$$

where $m(\cdot)$ is the modal value of a fuzzy number, for example the centre of a triangular fuzzy number. In particular, for TFNs $\tilde{x} = (x^L, x, x^R)$ and $\tilde{y} = (y^L, y, y^R)$, by noting that $\text{Supp}(\tilde{x}) = [x^L, x^R] = \tilde{x}^{\alpha=0}$, $\text{Supp}(\tilde{y}) = [y^L, y^R] = \tilde{y}^{\alpha=0}$, one finds

$$d^2(\tilde{x}, \tilde{y}) = (x^R - y^R)^2 + (x^L - y^L)^2 + (x - y)^2. \quad (\text{A.50})$$

A.2 Quadratic Programming

In this section, we will show how to construct the matrices $\mathbf{P}$, $\mathbf{G}$ and the vectors $\mathbf{q}$, $\mathbf{h}$ of (2.25) associated to the minimisation problems of sections 2.4, 2.5.

- A The fuzzy regression problem from section 2.4 can be solved by a quadratic solver as follows. With respect to subsection 2.3.2, to find the vector of solutions $(\beta_1, \dots, \beta_m, \gamma_1, \dots, \gamma_m)'$ of the minimisation problem (2.32), we need to find the matrices $\mathbf{P}$, $\mathbf{G}$ and the vectors $\mathbf{q}$, $\mathbf{h}$ of (2.25) that satisfy (2.32). These are written as

$$\mathbf{P} = \begin{bmatrix} \mathbf{X}'\mathbf{X} & \mathbf{0}_{m \times m} \\ \mathbf{0}_{m \times m} & \mathbf{X}'\mathbf{X} \end{bmatrix} \quad \mathbf{q} = \begin{bmatrix} -\mathbf{X}'\mathbf{y} \\ \mathbf{0}_m \end{bmatrix}. \quad (\text{A.51})$$

where $\mathbf{X}$ is an $n \times m$ matrix containing the observations x_{ij} , $i = 1, \dots, n$, $j = 1, \dots, m$, $\mathbf{0}_{m \times m}$ is an $m \times m$ matrix of zeros, $\mathbf{y} = (y_1, \dots, y_n)'$ and $\mathbf{0}_m$ is an m -vector of zeros. Concerning the constraints, one can write

$$\mathbf{G} = \begin{bmatrix} \mathbf{X} & \mathbf{X} \\ \mathbf{X} & -\mathbf{X} \\ \mathbf{0}_{m \times m} & -\mathbf{I}_{m \times m} \end{bmatrix} \quad \mathbf{h} = \begin{bmatrix} \mathbf{y} \\ \mathbf{y} \\ \mathbf{0}_m \end{bmatrix}. \quad (\text{A.52})$$

- B In the same way as above, to find the vector of solutions $(\beta_1, \dots, \beta_m, \gamma_1, \dots, \gamma_m)'$ that is minimising (2.40), under the constraints 2.41 we need the matrices $\mathbf{P}$, $\mathbf{G}$ and the vectors $\mathbf{q}$, $\mathbf{h}$ of (2.25) written as

$$\mathbf{P} = \begin{bmatrix} \mathbf{X}'\mathbf{X} & \mathbf{0}_{m \times m} & \mathbf{0}_{m \times m} \\ \mathbf{0}_{m \times m} & \mathbf{X}'\mathbf{X} & \mathbf{0}_{m \times m} \\ \mathbf{0}_{m \times m} & \mathbf{0}_{m \times m} & \mathbf{X}'\mathbf{X} \end{bmatrix} \quad \mathbf{q} = \begin{bmatrix} -\mathbf{X}'\mathbf{y}^L \\ -\mathbf{X}'\mathbf{y} \\ -\mathbf{X}'\mathbf{y}^R \end{bmatrix}, \quad (\text{A.53})$$

$$\mathbf{G} = \begin{bmatrix} -\mathbf{I}_{m \times m} & \mathbf{I}_{m \times m} & \mathbf{0}_{m \times m} \\ \mathbf{0}_{m \times m} & -\mathbf{I}_{m \times m} & \mathbf{I}_{m \times m} \end{bmatrix} \quad \mathbf{h} = \begin{bmatrix} \mathbf{0}_m \\ \mathbf{0}_m \end{bmatrix}, \quad (\text{A.54})$$

where $\mathbf{X}$ is an $n \times m$ matrix composed of the observations x_{ij} . $\mathbf{y}^L$, $\mathbf{y}$, $\mathbf{y}^R$ are respectively n -vectors of observations y_i^L , y_i , y_i^R , $\mathbf{0}_{m \times m}$ is an $m \times m$ matrix of zeros, $\mathbf{I}_{m \times m}$ is the $m \times m$ identity matrix and $\mathbf{0}_m$ is an m -vector of zeros. The $3m$ -vector β^{sol} containing the solutions for the regression coefficients is of the form

$$\beta^{sol} = \begin{bmatrix} \beta^L \\ \beta \\ \beta^R \end{bmatrix} \quad (\text{A.55})$$

where $\beta^L = (\beta_1^L, \dots, \beta_m^L)$, $\beta = (\beta_1, \dots, \beta_m)$ and $\beta^R = (\beta_1^R, \dots, \beta_m^R)$.

C The minimisation problem (2.45) can be easily solved by writing $\mathbf{X}^{\min} = \mathbf{X}^L$ and $\mathbf{X}^{\max} = \mathbf{X}^R$. Then we have

$$\mathbf{P} = \begin{bmatrix} \mathbf{X}^{L'} \mathbf{X}^L & \mathbf{0}_{m \times m} & \mathbf{0}_{m \times m} \\ \mathbf{0}_{m \times m} & \mathbf{X}' \mathbf{X} & \mathbf{0}_{m \times m} \\ \mathbf{0}_{m \times m} & \mathbf{0}_{m \times m} & \mathbf{X}^{R'} \mathbf{X}^R \end{bmatrix} \quad \mathbf{q} = \begin{bmatrix} -\mathbf{X}^{L'} \mathbf{y}^L \\ -\mathbf{X}' \mathbf{y} \\ -\mathbf{X}^{R'} \mathbf{y}^R \end{bmatrix}, \quad (\text{A.56})$$

$$\mathbf{G} = \begin{bmatrix} -\mathbf{I}_{m \times m} & \mathbf{I}_{m \times m} & \mathbf{0}_{m \times m} \\ \mathbf{0}_{m \times m} & -\mathbf{I}_{m \times m} & \mathbf{I}_{m \times m} \end{bmatrix} \quad \mathbf{h} = \begin{bmatrix} \mathbf{0}_m \\ \mathbf{0}_m \end{bmatrix}, \quad (\text{A.57})$$

where $\mathbf{X}^L$, $\mathbf{X}$, $\mathbf{X}^R$ are $n \times m$ matrices respectively composed of observations $x_{ij}^{\min}$, x_{ij} , $x_{ij}^{\max}$. $\mathbf{y}^L$, $\mathbf{y}$, $\mathbf{y}^R$ are respectively n -vectors containing the observed y_i^L , y_i , y_i^R . $\mathbf{0}_{m \times m}$ is an $m \times m$ matrix of zeros, $\mathbf{I}_{m \times m}$ is the $m \times m$ identity matrix and $\mathbf{0}_m$ is an m -vector of zeros. The $3m$ -vector β^{sol} containing the solutions for the regression coefficients is of the form

$$\beta^{\text{sol}} = \begin{bmatrix} \beta^L \\ \beta \\ \beta^R \end{bmatrix} \quad (\text{A.58})$$

where $\beta^L = (\beta_1^{\min}, \dots, \beta_m^{\min})$, $\beta = (\beta_1, \dots, \beta_m)$ and $\beta^R = (\beta_1^{\max}, \dots, \beta_m^{\max})$.

Remark A.2.1.

Observe that we can write the objective function (2.43) in term of a fixed α -cut, $\alpha \in [0, 1]$,

$$J = k_1 \sum_{i=1}^n (y_i - \mathbf{x}_i' \beta)^2 + k_2 \sum_{i=1}^n (y_i^{L,\alpha} - \mathbf{x}_i^{\min'} \beta^{\min,\alpha})^2 + k_3 \sum_{i=1}^n (y_i^{R,\alpha} - \mathbf{x}_i^{\max'} \beta^{\max,\alpha})^2. \quad (\text{A.59})$$

Where $\mathbf{x}_i^{\min'}$ and $\mathbf{x}_i^{\max'}$ are the i -th line of observation of the matrices $\mathbf{X}^{\min}$, $\mathbf{X}^{\max}$ respectively. Then, as before, to find the regression coefficients $\tilde{\beta}$, solving the following minimisation problem :

$$\min_{\beta^{\min,\alpha}, \beta^\alpha, \beta^{\max,\alpha}} J = \sum_{i=1}^n \left(k_1 (y_i^\alpha - \sum_{j=1}^m \beta_j^\alpha x_{ij}^\alpha)^2 + k_2 (y_i^{L,\alpha} - \sum_{j=1}^m \beta_j^{\min,\alpha} x_{ij}^{\min,\alpha})^2 + k_3 (y_i^{R,\alpha} - \sum_{j=1}^m \beta_j^{\max,\alpha} x_{ij}^{\max,\alpha})^2 \right)$$

under the constraints

$$(\beta_j - \beta_j^{\min,\alpha}), (\beta_j^{\max,\alpha} - \beta_j) \geq 0 \text{ for } j = 1, \dots, m.$$

We can solve this minimisation problem by quadratic programming in the same way as before, by using the matrices $\mathbf{X}^{\min,\alpha} = \mathbf{X}^{L,\alpha}$, $\mathbf{X}^{\max,\alpha} = \mathbf{X}^{R,\alpha}$ of observations $x_{ij}^{\min,\alpha}$, $x_{ij}^{\max,\alpha}$

at a given α -cut and the observed vectors of dependent variables $y_i^{L,\alpha}$, $y_i^{R,\alpha}$ denoted by $\mathbf{y}^{L,\alpha}$, $\mathbf{y}^{R,\alpha}$ respectively. This gives

$$\mathbf{P} = \begin{bmatrix} \mathbf{X}^{L,\alpha'} \mathbf{X}^{L,\alpha} & \mathbf{0}_{m \times m} & \mathbf{0}_{m \times m} \\ \mathbf{0}_{m \times m} & \mathbf{X} \mathbf{X}' & \mathbf{0}_{m \times m} \\ \mathbf{0}_{m \times m} & \mathbf{0}_{m \times m} & \mathbf{X}^{R,\alpha'} \mathbf{X}^{R,\alpha} \end{bmatrix} \quad \mathbf{q} = \begin{bmatrix} -\mathbf{X}^{L,\alpha'} \mathbf{y}^{L,\alpha} \\ -\mathbf{X}' \mathbf{y} \\ -\mathbf{X}^{R,\alpha'} \mathbf{y}^{R,\alpha} \end{bmatrix}, \quad (\text{A.60})$$

$$\mathbf{G} = \begin{bmatrix} -\mathbf{I}_{m \times m} & \mathbf{I}_{m \times m} & \mathbf{0}_{m \times m} \\ \mathbf{0}_{m \times m} & -\mathbf{I}_{m \times m} & \mathbf{I}_{m \times m} \end{bmatrix} \quad \mathbf{h} = \begin{bmatrix} \mathbf{0}_m \\ \mathbf{0}_m \end{bmatrix}. \quad (\text{A.61})$$

However, if we seek solutions of the form of fuzzy triangular numbers $\tilde{\beta}_j = (\beta_j^L, \beta_j, \beta_j^R)$, $j = 1, \dots, m$ then we only need to solve for $\alpha = 0$ since triangular fuzzy numbers are uniquely determined by their centre and left/right endpoints.

Acronyms

AIC	Akaike information criterion xvii, xx, 3, 117, 119, 126, 127, 129, 131–134, 158
ELAIC	Empirical Likelihood Akaike information criterion 117, 125, 126, 128, 131–134, 158
FRV	Fuzzy random variable 70–73
KL	Kullback-Leibler 126
LS	Least squares 125, 126, 128
ML	Maximum likelihood 118, 119, 125, 126
PRM	Polygon ranking method xvii, xix, xx, 100, 103–105, 107–109, 132, 133, 158
RFV	Random fuzzy variable (epistemic view) 74
TFN	Triangular fuzzy number 8, 10, 19, 20, 22–24, 28, 43, 55, 86, 160
TRFN	Trapezoidal fuzzy number 10, 11, 19–22

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Declaration of Honour

I hereby declare that I wrote this thesis entitled "Inference and model selection for fuzzy regression methods" on my own and followed the principles of scientific integrity. I affirm that I have not plagiarised, nor have I used any data or research results that have been obtained through fraudulent or unethical means.

I further confirm that this work or parts thereof have not been submitted in this form elsewhere for an examination.

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Parcours professionnel

Julien Rosset est docteur en philosophie (Ph. D.) de l'Université de Fribourg. Après avoir étudié la physique à l'EPFL, il s'est orienté vers les mathématiques à l'Université de Fribourg, où il a obtenu un diplôme de master avec un mémoire intitulé *Chemical Boltzmann Machines*. Sa recherche doctorale, menée au sein du groupe de recherche ASAM, porte sur la logique floue, l'inférence statistique floue et la sélection de modèles, notamment pour les méthodes de régression floue.

	État civil	
Date de naissance	22 avril 1992	
Nationalité	Suisse	
Réseaux sociaux	On peut me retrouver sur :	
	○ ResearchGate	○ gitlab
	○ ORCID	○ LinkedIn

Connaissances et aptitudes professionnelles

Langues

Français	Langue maternelle
Anglais	oral : excellent ; écrit : excellent
Japonais	oral : moyen ; écrit : moyen
Allemand	oral : pas très bon ; écrit : moyen
Italien	Compréhension passive

Programmation et connaissance de logiciels

Langages de programmation	○ Au fil des années : Python, R. ○ L^AT_EX, etc.
Logiciels	○ Logiciels statistiques spécifiques : SPSS, R.
Plates-formes	○ Sites Internet et plate-forme d'enseignement : moodle.
Hobbys	○ Piano et peinture/dessin.

Éducation

Collège

09/2008 – 06/2012	Maturité (Option scientifique), <i>Collège St-Michel</i> , Fribourg, Suisse.
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Université

09/2012 – 06/2014	Etudes en physique, <i>EPFL</i> , Lausanne, Suisse.
09/2014 – 06/2019	Master en mathématiques , <i>University of Fribourg</i> , Fribourg, Suisse, Note : 5.5. ○ Majeur : Mathématiques ○ mineur : Physique (90 ECTS)

Divers

10/2019 – 04/2020	Séjour linguistique au Japon, Fukuoka, Japon.
03/2021 – 12/2021	Didactique universitaire, <i>Centre de didactique universitaire</i> , <i>Université de Fribourg</i> , Fribourg, Suisse.

Emplois

Universités et instituts

08/2020 – 07/2025

Assistant diplômé, doctorant, auprès du Prof. Dr Laurent Donzé, professeur aux Universités de Fribourg et Neuchâtel (et research professor à l'ETH), Université de Fribourg, Fribourg, Suisse.

Domaine de spécialisation : économétrie et statistique.

09/2016 – 06/2018

Assistant (Tutorat), University of Fribourg, Université de Fribourg, Fribourg, Suisse.

Enseignement

Bachelor

09/2020 – 07/2025

Statistique 1.

Assistant pour un cours BA, 1^e année, semestre d'automne.

02/2021 – 02/2025

Statistique approfondissement.

Assistant pour un cours BA, 2^e année, semestre de printemps.

Master

09/2020 – 02/2025

Inférence, évaluation et sélection de modèles.

Assistant pour cours MA, 1 semestre, donné aux semestres d'automne entre 2020-2021, 2022-2023 et 2024-2025.

02/2021 – 07/2025

Méthodes de classifications.

Assistant pour cours MA, 1 semestre, donné aux semestres de printemps 2021, 2023 et 2025.

09/2021 – 02/2024

Thèmes choisis de statistique multivariée.

Assistant pour cours MA, 1 semestre, donné aux semestres d'automne entre 2021-2022 et 2023-2024.

02/2022 – 08/2024

Statistique bayésienne.

Assistant pour cours MA, 1 semestre, donné aux semestres de printemps 2022 et 2024.

Publications, conférences, etc.

Livres et chapitres de livres

Rosset, Julien (10 août 2019). « Chemical Boltzmann machines ». Anglais. Mém. de mast. University of Fribourg.

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Randomness and Dependence (SMPS 2024). Salzburg, Austria : Springer. DOI : [10.1007/978-3-031-65993-5_50](https://doi.org/10.1007/978-3-031-65993-5_50). URL : https://doi.org/10.1007/978-3-031-65993-5_50.

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— (2025c). « Fuzzy Confidence Intervals by Bootstrapped Fuzzy Distributions ». In : *Information Processing and Management of Uncertainty in Knowledge-Based Systems, 20th International Conference, IPMU 2024, Lisbon, Portugal, July 22-26, 2024, Proceedings, Volume 3*. Springer. DOI : [10.1007/978-3-031-73997-2_21](https://doi.org/10.1007/978-3-031-73997-2_21). URL : https://doi.org/10.1007/978-3-031-73997-2_21.

— (2026a). « Selection of Fuzzy Regression Models by AIC ». In : *Advances in Computational Intelligence*. Springer. DOI : [10.1007/978-3-032-15632-7_34](https://doi.org/10.1007/978-3-032-15632-7_34). URL : https://doi.org/10.1007/978-3-032-15632-7_34.

Proceedings et articles

Rosset, Julien et Laurent DONZÉ (2023b). « Fuzzy Least Squares and Fuzzy Orthogonal Least Squares Linear Regressions. » In : *Proceedings of the 15th International Joint Conference on Computational Intelligence (IJCCI 2023)*. SCITEPRESS – Science et Technology Publications, Lda. Under CC license (CC BY-NC-ND 4.0), p. 359-368. ISBN : 978-989-758-674-3. DOI : [10.5220/0012182700003595](https://doi.org/10.5220/0012182700003595). URL : <https://doi.org/10.5220/0012182700003595>.

— (2024a). « Approximated Fuzzy p-values by Bootstrapped Fuzzy Distributions and Fuzzy Hypotheses Testing ». In : *Proceedings of the 16th International Joint Conference on Computational Intelligence (IJCCI 2024), Volume 1 : FCTA*. Porto, Portugal : SCITEPRESS. DOI : [10.5220/0012888300003837](https://doi.org/10.5220/0012888300003837). URL : <https://doi.org/10.5220/0012888300003837>.

— (22 juin 2024e). « Fuzzy Confidence Intervals by Bootstrapped Fuzzy Distributions ». In : *Information Processing and Management of Uncertainty in Knowledge-Based Systems, 20th International Conference, IPMU 2024, Lisbon, Portugal, July 22-26, 2024, Proceedings, Volume 3*. IPMU 2024 LISBOA, 20th International Conference on Information Processing and Management of Uncertainty in Knowledge-Based Systems (22 juin 2024). Lisbon, Portugal : Springer. ISBN : 9783031739965. DOI : [10.1007/978-3-031-73997-2_21](https://doi.org/10.1007/978-3-031-73997-2_21). URL : https://doi.org/10.1007/978-3-031-73997-2_21.

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Conférences

Rosset, Julien et Laurent DONZÉ (15 nov. 2023a). *Fuzzy least squares and fuzzy orthogonal least squares linear regressions*. In : FCTA 2023, 15th International Conference on Fuzzy Computation Theory and Applications. Rome, Italy.

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